GaussDB
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API Reference

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1 Before You Start

GaussDB is a distributed relational database developed by Huawei. It uses a distributed architecture and features hybrid transactional/analytical processing (HTAP) capabilities. GaussDB supports intra-city deployment across AZs for zero data loss, petabytes of storage, and more than 1,000 nodes. It is highly available, reliable, secure, and scalable, and provides key capabilities including quick deployment, backup, restoration, monitoring, and alarm reporting for enterprises.

This document describes how to use APIs to query monitoring metrics and O&M operations. For details about all supported operations, see **API Types**.

API Calling

GaussDB supports Representational State Transfer (REST) APIs, allowing you to call APIs using HTTPS. For details about API calling, see **Calling APIs**.

Asynchronous API Description

Some APIs provided by GaussDB execute asynchronous tasks and return **job_id**. You can use **job_id** to query task status by referring to **Querying Task Status by Job ID**.

Endpoints

An endpoint is the request address for calling an API. You can use any node where GaussDB Management Platform (TPOPS) is deployed as the endpoint. The IP port of the endpoint is the request address of the API.

Constraints

For details, see the description of each API.

Concepts

User: A user is created to use the API service. Each user has its own identity credentials (password and access keys) and permissions to access monitored instances.

2 API Types

You can use GaussDB APIs to create and upgrade instances.

Table 2-1 API types

Type	Subtype	Remarks
GaussDB APIs	Authentication Management	Generate a token, log in to the system, redirect a login request, log out of the system, query users, add a user, delete a user, query basic information about a user, update information about a user, check whether a username is valid, let a user reset their own password, let a user reset another user's password, query instances allocated to a user, allocate instances to a user, query roles of a user, assign roles to a user, query user groups that a user has been added to, add a user to user groups, lock users, unlock users, check whether a password is a weak password, query user groups, add a user group, update information about a user group, delete a user group, query user roles of a user group, update user role information about a user group, query users in a user group, update information about users in a user group, query roles, add a role, update information about a role, delete a role, query permissions of a role, and query permissions.

Type	Subtype	Remarks
	Backup Management	Configure an automated backup policy, stop a backup, change backup medium of an instance, configure an extended backup policy, query an automated backup policy, query an extend backup policy, create a manual backup, delete a manual backup, query information about the original instance based on a specific point in time or a backup file, query instances that can be used for backups and restorations, query the restoration time range, query differential backups, restore data to an existing instance, confirm data integrity after a backup is restored, query backups, query the archive log path of an instance, synchronize metadata using storage medium, delete a backup, and install an SSL certificate.
	Parameter Configuration	Obtain the parameter template of a specified instance and modify parameters of a specified instance.
	Database and User Management	Create a database, create a database schema, query database schemas, create a database user, delete a database user, grant permissions to or revoke permissions from a database user, modify attributes of a database user, initialize the password of a built-in database user, reset the password of a database user, lock a database user, unlock a database user, query databases, query database users, query connected users, applications, and table statistics and lock information of a database, flash back a table, create a database role, delete a database role, and grant permissions to or revoke permissions from a database user or role.
	Database Diagnostic and Optimization	Query node and component information of database key views, query real-time sessions of a component on a node, kill specified sessions, kill all idle sessions, query table analysis information, query index analysis information, query WDR snapshot status, enable or disable WDR snapshots, query the collection result of a WDR snapshot, generate a WDR report immediately, collect a WDR snapshot, download a WDR snapshot, query slow SQL execution nodes, query slow SQL sets, query slow SQL details, obtain top SQL statements, create SQL patches, query SQL patches, and enable or disable SQL Patch or delete a SQL patch.
	Intelligent O&M	Manage an instance or cancel management for an instance, query instances that have been managed or can be managed by a DBMind instance, and query DBMind instances.
	SQL Throttling	Create a throttling task, delete a throttling task, query throttling tasks based on search criteria, and query SQL template of a specified node.

Type	Subtype	Remarks
	Host Key Management	Generate a key pair for host connection, query key pair information, modify key information, delete a key pair, and download a public key.
	Database Center Management	Query hosts, add a host, modify static host information, delete a host, query the standardization check result of a host, download a host import template, Batch import host files for check, query available host specifications, query data centers, add a data center, edit data center information, and delete a data center.
	Routine Inspection	Query inspection tasks, create an inspection task, delete inspection tasks in batches, execute an inspection task, query details of an inspection task, modify an inspection task, query inspection result statistics, query execution details of inspection items, query inspection items, query inspection records, delete inspection records in batches, query inspection scenarios, add an inspection scenario, delete an inspection scenario, modify an inspection scenario.
	Instance Management	Configure extended information for an instance, query extended information of an instance, query components of an instance, perform a primary/standby DN switchover, delete an instance, create an instance, restore data to a new instance using backups, query instances or instance details, stop an instance or node, reboot an instance, start an instance or node, check whether hosts are associated to an instance to be managed, manage an instance, rectify an AZ fault, add a node, replace a node, query versions that an instance can be upgraded to, upgrade an instance, repair a node, change deployment model, enable port for M compatibility, modify or disable port for M compatibility, perform follow-up operations after specifications are changed, and change disk size.
	Engine Versions and Specifications	Query DB engine versions.
	Microservice Upgrade	Query instance Agents, query Agent information of a specified instance, upgrade the Agent of an instance, and upgrade Agents in batches.

Type	Subtype	Remarks
	Alarm Management	Test the alarm interconnection, add alarm interconnection configurations, modify alarm interconnection configurations, view alarm interconnection configurations and alarm results, query the number of alarms, and configure instance alarms in batches.
	Monitoring Dashboards	Query monitoring metrics and query monitoring metric group.
	Task Center	Query task status by job ID.
	DR Management	Modify the DR task name, query the DR relationships, query historical DR operation records, query the real-time DR monitoring status of instances, create a DR task, promote a DR instance to primary, stop a DR task, perform a DR switchover, enable DR drill, disable DR drill, enable log cache, disable log cache of the primary instance, manually delete failed DR tasks, promote a DR instance to primary or demote primary instance to DR after a DR switchover failure, query instances that can be used as a DR instance in a DR relationship, delete a DR record, change the primary instance of a DR relationship, and pre-check DR operations.
	License Management	Obtain a revocation code.
	Log Management	Query operation logs.

3 Calling APIs

3.1 Making an API Request

This section describes the structure of a RESTful API request, and explains how to obtain a user token to call an API. The obtained token can then be used to authenticate the calling of other APIs.

Request URI

A request URI consists of the following:

{URI-scheme}://{Endpoint}/rds/{resource-path}?{query-string}

Although a request URI is included in a request header, most programming languages or frameworks require the request URI to be separately transmitted, rather than being conveyed in a request message.

Table 3-1 Parameters in a URI

Parameter	Description
URI-scheme	Protocol used to transmit requests. All APIs use HTTPS.
Endpoint	*IP address*:*port* of any node where GaussDB Management Platform (TPOPS) is deployed. The default port is **8002**.
resource-path	Access path of an API. Obtain the path from the URI of an API. For example, the **resource-path** of the API used to obtain a user token is **/v3/auth/mini-token**.
query-string	Query parameter, which is optional. Ensure that a question mark (?) is included before each query parameter that is in the format of "Parameter name=Parameter value". For example, **? limit=10** indicates that a maximum of 10 data records will be displayed.

Request Methods

The HTTP protocol defines the following request methods that can be used to send a request to the server.

Table 3-2 HTTP methods

Method	Description
GET	Requests the server to return the specified resources.
PUT	Requests the server to update specified resources.
POST	Requests the server to add a resource or perform special operations.
DELETE	Requests the server to delete a specified resource (for example, an object).

For example, in the case of the API used to obtain a user token, the request method is POST. The request is as follows:

```
POST https://{{Endpoint}}/rds/v3/auth/mini-token
```

Request Header

You can also add additional header fields to a request, such as the fields required by a specified URI or HTTP method. For example, to request authentication information, add **Content-Type**, which specifies the request body type.

The following table lists common request header fields.

Field	Description	Mandatory	Example Value
Host	Requested server information, which can be obtained from the URL of the service API. The value is in the format of *hostname[:port]*. If the port number is not specified, the default port is used. The default port number for HTTPS is **443**.	No	code.test.com or code.test.com:443
Content-Type	MIME type of the request body. You are advised to use the default value **application/json**. For APIs used to upload objects or images, the value varies depending on the flow type.	Yes	application/json
Content-Length	Length of a request body, in bytes.	No	3495

Field	Description	Mandatory	Example Value
X-Mini-Token	User token. The user token is a response to the API used to obtain a user token. This API is the only one that does not require authentication. After a request is processed, the value of **X-Mini-Token** in the header is the token value.	No This parameter is mandatory for token-based authentication.	The following is part of an example token: MIIPAgYJKoZIhvcNAQcCo...ggg1BBIINPXsidG9rZ

The API used to obtain a user token does not require authentication. Therefore, only the **Content-Type** field needs to be added to requests for calling the API. An example of such requests is as follows:

```
POST https://{{Endpoint}}/rds/v3/auth/mini-token
Content-Type: application/json
```

Request Body (Optional)

This part is optional. The request body is often sent in a structured format (for example, JSON or XML) as specified in the **Content-Type** header field. If the request body contains full-width characters, these characters must be coded in UTF-8.

The request body varies according to the APIs. Certain APIs do not require the request body, such as the GET and DELETE APIs.

For an API used to obtain a user token, you can view its request parameters and parameter description in the API request. The following provides an example request with a body. Replace *username* and *password* with actual values.

```
POST https://{{Endpoint}}/rds/v3/auth/mini-token
{
      "username" : "test",
      "password": "***"
}
```

If all data required for the API request is available, you can send the request to call the API through curl, Postman, or coding. For the API of obtaining a user token, **X-Mini-Token** in the response header is the required user token. This token can then be used to authenticate the calling of other APIs.

3.2 Authentication

Token authentication must be performed to call APIs.

A token specifies temporary permissions in a computer system. During API authentication using a token, the token is added to requests to get permissions for calling the API.

After a token is obtained, add field **X-Mini-Token** to the request header to specify the token when other APIs are called. For example, if the token is **ABCDEFG....**, add **X-Mini-Token: ABCDEFG....** to a request as follows:

```
POST https://{{Endpoint}}/rds/v3/interface
Content-Type: application/json
X-Mini-Token: ABCDEFG....
```

3.3 Response

Status Code

After sending a request, you will receive a response, including the status code, response header, and response body.

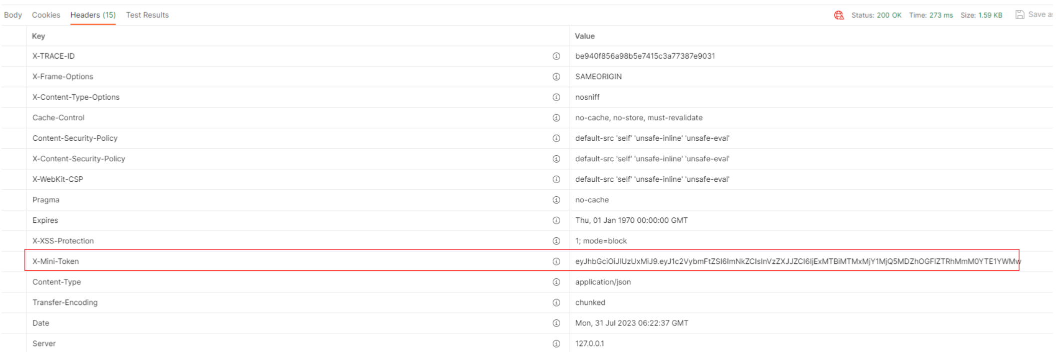
A status code is a group of digits ranging from 1xx to 5xx. It indicates the status of a response. For more information, see **Status Codes**.

For example, if status code **200** is returned for calling the API used to obtain a user token, the request is successful.

Response Header

Similar to a request, a response also has a header, for example, **Content-Type**.

The following figure shows the response header fields for the API used to obtain a user token. The **X-Mini-Token** header field is the desired user token. This token can then be used to authenticate the calling of other APIs.



(Optional) Response Body

This part is optional. The body of a response is often returned in structured format (for example, JSON or XML) as specified in the **Content-Type** header field. The response body transfers content except the response header.

The following is part of the response body for the API used to obtain a user token. **x_mini_token** is the desired user token and its value is the same as the value of **X-Mini-Token** in the response header. The following describes only part of the request body.

```
{
    "user_id": "***",
    "login_ip": "***",
    "login_failed_attempts": 0,
    "is_first_time_login": false,
```

```
    "last_login_statistics": {
        "last_login_ip": "***",
        "last_login_failed_ip": null,
        "last_login_time": "2023-07-31T02:00:15+0000",
        "last_login_failed_time": ""
    },
    "session_timeout": 1440,
    "password_expiration_days": 180,
    "password_expired_remaining_days": 169,
    "password_expired_reminder": false,
    "x_mini_token": "ABCDEFJ...."
}
```


If an error occurs during API calling, an error code and a message will be displayed. The following shows an error response body.


```
{
    "error_msg": "***",
    "error_code": "***"
}
```


In the preceding information, **error_code** is an error code, and **error_msg** describes the error message.

4 APIs

4.1 Platform User Management

4.1.1 Logins

4.1.1.1 Generating a Token

Function

This API is used to generate a token.

URI

POST /v3/auth/mini-token

Request Parameters

Table 4-1 Request body parameters

Parameter	Mandatory	Type	Description
username	Yes	String	Username. Minimum length: 3 characters Maximum length: 64 characters
password	Yes	String	Password. Minimum length: 8 characters Maximum length: 32 characters

Response Parameters

Status code: 200

Table 4-2 Response header parameter

Parameter	Type	Description
X-Mini-Token	String	Token information.

Table 4-3 Response body parameters

Parameter	Type	Description
user_id	String	User ID.
login_ip	String	IP address of the access client.
is_first_time_login	Boolean	Whether the login is the first login.
login_failed_attempts	String	Number of failed login attempts.
last_login_statistics	Table 4-4 object	Last login statistics.
session_timeout	Integer	Session expiration time, in minutes.
password_expiration_remaining_days	String	Remaining password validity period, in days.
password_expiration_days	String	Password validity period, in days.
password_expiration_reminder	Boolean	An alert that notifies you that the password is about to expire. If the remaining validity period of the password is less than or equal to 15 days, set the value to **true**. Otherwise, set the value to **false**.
x_mini_token	String	Token, which is the same as that in the header parameter.

Table 4-4 last_login_statistics

Parameter	Type	Description
last_login_failed_ip	String	Last failed login address.

Parameter	Type	Description
last_login_failed_time	String	Last time at which the login failed.
last_login_ip	String	Last successful login address.
last_login_time	String	Last successful login time.

Status code: 400

Table 4-5 Response header parameter

Parameter	Type	Description
X-Mini-Token	String	Obtained token.

Table 4-6 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

```
POST https://{{Endpoint}}/rds/v3/auth/mini-token

{
   "username": "test",
   "password": "***"
}
```

Example Response

Status code: 200

The token is obtained successfully.

```
{
  "user_id" : "wertsdfg-wrey-gfhj-sdrf-sdfgawerszdf",
  "login_ip" : "192.168.0.*",
  "login_failed_attempts" : 0,
  "is_first_time_login" : false,
  "last_login_statistics" : {
    "last_login_ip" : "10.182.*.*",
    "last_login_failed_ip" : "192.168.0.*",
    "last_login_time" : "2023-05-17T09:00:00Z",
    "last_login_failed_time" : "2023-05-16T09:00:00Z"
  },
  "session_timeout" : 1440,
  "password_expiration_days" : 90,
```

```
  "password_expiration_remaining_days" : 56,
  "password_expiration_reminder" : false,
  "x_mini_token" : "ABCDEFG..."
}
```


Status Code

Status Code	Description
200	Normal
400	Abnormal

Error Code

For details, see **Error Codes**.

4.1.1.2 Logging In to the System

Function

This API is used to log in to the system.

URI

POST /v3/login

Request Parameters

Table 4-7 Request header parameters

Parameter	Mandatory	Type	Description
X-Language	No	String	Language.

Table 4-8 Request body parameters

Parameter	Mandatory	Type	Description
username	Yes	String	Username.
password	Yes	String	Password.
captcha	No	String	Verification code.

Response Parameters

Status code: 200

Table 4-9 Response body parameters

Parameter	Type	Description
user_id	String	User ID.
login_ip	String	IP address of the accessed client.
is_first_time_login	Boolean	Whether the login is the first login.
login_failed_attempts	String	Number of failed login attempts.
last_login_statistics	**last_login_statistics** object	Last login statistics.
session_timeout	Integer	Session expiration time, in minutes.
password_expiration_remaining_days	String	Remaining password validity period, in days.
password_expiration_days	String	Password validity period, in days.

Table 4-10 last_login_statistics

Parameter	Type	Description
last_login_failed_ip	String	Last failed login address.
last_login_failed_time	String	Time at which the last login failed.
last_login_ip	String	Last successful login address.
last_login_time	String	Last successful login time.

Status code: default

Table 4-11 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

Login

```
POST https://{endpoint}/rds/v3/login

{
  "username" : "username",
  "password" : "password"
}
```

Example Response

Status code: 200

Normal

```
{
  "user_id" : "wertsdfg-wrey-gfhj-sdrf-sdfgawerszdf",
  "login_ip" : "192.168.0.***",
  "login_failed_attempts" : 0,
  "is_first_time_login" : false,
  "last_login_statistics" : {
    "last_login_ip" : "10.182.***.***",
    "last_login_failed_ip" : "192.168.0.***",
    "last_login_time" : "2023-05-17T09:00:00Z",
    "last_login_failed_time" : "2023-05-16T09:00:00Z"
  },
  "session_timeout" : 10,
  "password_expiration_days" : 90,
  "password_expiration_remaining_days" : 56
}
```

Status Codes

Status Code	Description
200	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.1.1.3 Redirecting a Login Request

Function

This API is used to redirect a login request from an external platform to TPOPS.

URI

GET /v3/login/redirect

Table 4-12 Query parameters

Parameter	Mandatory	Type	Description
token	Yes	String	Token generated after a successful login.
ip	Yes	String	EIP of TPOPS.
port	Yes	String	Port corresponding to port **8002** mapped to the EIP of TPOPS.

Request Parameters

Table 4-13 Request header parameters

Parameter	Mandatory	Type	Description
X-Language	No	String	Language.

Response Parameters

Status code: 400

Table 4-14 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

Redirecting a login request

```
GET https://{endpoint}/rds/v3/login/redirect?token=token1&ip=ip1&port=port1
```

Example Response

None

Status Code

Status Code	Description
200	Normal

Status Code	Description
400	Error response

Error Code

For details, see **Error Codes**.

4.1.1.4 Logging Out of the System

Function

This API is used to log out of the system.

URI

POST /v3/logout

Request Parameters

Table 4-15 Request header parameters

Parameter	Mandatory	Type	Description
X-Language	No	String	Language.
X-Mini-Token	Yes	String	Request token.

Table 4-16 Request body parameters

Parameter	Mandatory	Type	Description
mode	Yes	String	Whether the logout is manual or automatic.

Response Parameters

Status code: default

Table 4-17 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

Logging out of the system

```
POST https://{endpoint}/rds/v3/logout

{
  "mode" : "manual"
}
```

Example Response

None

Status Codes

Status Code	Description
200	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.1.2 User Management

4.1.2.1 Querying Users

Function

This API is used to query all users that meet the query conditions.

URI

GET /v3/users

Table 4-18 Query parameters

Parameter	Mandatory	Type	Description
limit	No	Integer	Number of records to be queried. The default value is **100**. The value cannot be a negative number. The minimum value is **1**, and the maximum value is **100**.

Parameter	Mandatory	Type	Description
offset	No	Integer	Index offset. If **offset** is set to *N*, the resource query starts from the *N*+1 data entry. The default value is **0**, indicating that the query starts from the first data entry. The value must be a number but cannot be a negative number.
sort_key	No	String	Sorting key. ● created_at ● updated_at ● name ● last_login_time
sort_order	No	String	Sorting order, which can be: ● **asc**: ascending order ● **desc**: descending order
username	No	String	Username. Enter 3 to 64 characters. Only letters (case-sensitive), digits, and underscores (_) are allowed.
user_id	No	String	User ID.

Request Parameters

Table 4-19 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	Request token.
X-Language	No	String	Language.

Response Parameters

Status code: 200

Table 4-20 Response body parameters

Parameter	Type	Description
total_count	Integer	Total number of records.

Parameter	Type	Description
users	Array of **users** objects	User information list.

Table 4-21 users

Parameter	Type	Description
last_login_time	String	Last login time.
updated_at	String	Last update time.
is_locked	Boolean	Lock status. ● true ● false
lock_reason	String	Reason that the user is locked.
password_expiration_remaining_days	Integer	Remaining password validity period, in days.
password_expiration_days	Integer	Password validity period, in days.
created_at	String	Time when the user was created.
id	String	User ID.
name	String	Username.

Status code: default

Table 4-22 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

Querying users

```
GET https://{endpoint}/rds/v3/users?sort_key=name&sort_order=asc&limit=10&offset=0
```

Example Response

Status code: 200

Normal


```
{
  "total_count" : 100,
  "users" : [ {
    "id" : "179a755507c841b2bef7906f50d001ee",
    "updateEnableFlag" : true,
    "super_admin_role_flag" : true,
    "name" : "admin",
    "created_at" : "2023-10-24T19:40:13+0000",
    "updated_at" : "2024-02-05T07:55:37+0000",
    "is_locked" : false,
    "lock_reason" : null,
    "last_login_time" : "2024-03-21T08:28:40+0000",
    "password_expiration_remaining_days" : 21,
    "password_expiration_days" : 90,
    "has_instance_permission" : true
  } ]
}
```


Status Codes

Status Code	Description
200	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.1.2.2 Adding a User

Function

This API is used to add a user.

URI

POST /v3/users

Request Parameters

Table 4-23 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	Request token.
X-Language	No	String	Language.

Table 4-24 Request body parameters

Parameter	Mandatory	Type	Description
username	Yes	String	Username. Enter 3 to 64 characters. Only letters (case-sensitive), digits, and underscores (_) are allowed.
password	Yes	String	Password, which must meet the following requirements: ● Consists of 8 to 32 characters. ● Includes at least four of the following types: uppercase letters, lowercase letters, digits, spaces, and special characters ~`!?,.:;-_"'(){}[]/<>$#%@^&*+|= ● Cannot be the username or the username spelled backwards. ● Cannot contain more than three identical consecutive characters.
password_expiration_days	Yes	Integer	Password validity period, in days. The value cannot be greater than 180. The options are as follows: ● 30 ● 60 ● 90 ● 120 ● 150 ● 180
instance_ids	No	Array of strings	User instance IDs.
role_ids	No	Array of strings	Role IDs.
user_group_ids	No	Array of strings	User group IDs.

Response Parameters

Status code: default

Table 4-25 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

Adding a user

```
POST https://{endpoint}/rds/v3/users

{
  "username" : "test",
  "password" : "Gauss_369",
  "password_expiration_days" : 1,
  "instance_ids" : [ ],
  "role_ids" : [ "2" ],
  "user_group_ids" : [ "2" ]
}
```

Example Response

None

Status Codes

Status Code	Description
200	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.1.2.3 Deleting a User

Function

This API is used to delete a user by user ID.

URI

DELETE /v3/users/{user_id}

Table 4-26 URI parameter

Parameter	Mandatory	Type	Description
user_id	Yes	String	User ID.

Request Parameters

Table 4-27 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	Request token.
X-Language	No	String	Language.

Response Parameters

Status code: default

Table 4-28 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

Deleting a user

```
DELETE https://{endpoint}/rds/v3/users/1fcb3f283774469aaf013d75e8ea75cd
```

Example Response

None

Status Codes

Status Code	Description
200	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.1.2.4 Querying Basic Information About a User

Function

This API is used to query basic information about a user by user ID.

URI

GET /v3/users/{user_id}

Table 4-29 URI parameter

Parameter	Mandatory	Type	Description
user_id	Yes	String	User ID.

Request Parameters

Table 4-30 Request header parameters

Parameter	Mandatory	Type	Description
X-Language	No	String	Language.

Response Parameters

Status code: 200

Table 4-31 Response body parameters

Parameter	Type	Description
username	String	Username.
create_at	String	Creation time.
password_expiration_remaining_days	Integer	Remaining password validity period, in days.

Status code: default

Table 4-32 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

Querying basic information about a user

```
GET   https://{endpoint}/rds/v3/users/ef23bd5fdd074924905cd46a8bbb2bf4
```

Example Response

Status code: 200

Normal

```
{
  "username" : "testt",
  "create_at" : "2024-03-12T03:02:17+0000",
  "password_expiration_remaining_days" : 22
}
```

Status Codes

Status Code	Description
200	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.1.2.5 Updating Information About a User

Function

This API is used to update information about a user.

URI

PUT /v3/users/{user_id}

Table 4-33 URI parameter

Parameter	Mandatory	Type	Description
user_id	Yes	String	User ID.

Request Parameters

Table 4-34 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	Request token.
X-Language	No	String	Language.

Table 4-35 Request body parameters

Parameter	Mandatory	Type	Description
password_expiration_days	No	Integer	Password validity period, in days.
instances	No	**instances** object	Information about the instances to be changed.
roles	No	**roles** object	Information about the roles to be changed.
user_groups	No	**user_groups** object	Information about the user groups to be changed.

Table 4-36 instances

Parameter	Mandatory	Type	Description
add	Yes	Array of strings	Instances to add.
del	Yes	Array of strings	Instances to delete.

Table 4-37 roles

Parameter	Mandatory	Type	Description
add	Yes	Array of strings	Roles to add.

Parameter	Mandatory	Type	Description
del	Yes	Array of strings	Roles to delete.

Table 4-38 user_groups

Parameter	Mandatory	Type	Description
add	Yes	Array of strings	User groups to add.
del	Yes	Array of strings	User groups to delete.

Response Parameters

Status code: default

Table 4-39 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

Updating information about a user

```
PUT https://{endpoint}/rds/v3/users/{user_id}

{
  "password_expiration_days" : 90,
  "instances" : {
    "add" : [ "3255fd92560c40ae8fc1b2695b19fdd7in14" ],
    "del" : [ ]
  },
  "roles" : {
    "add" : [ "2" ],
    "del" : [ "3" ]
  },
  "user_groups" : {
    "add" : [ ],
    "del" : [ ]
  }
}
```

Example Response

None

Status Codes

Status Code	Description
200	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.1.2.6 Checking Whether a Username Is Valid

Function

This API is used to check whether a username is valid.

URI

POST /v3/users/name-validation

Request Parameters

Table 4-40 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	Request token.
X-Language	No	String	Language.

Table 4-41 Request body parameters

Parameter	Mandatory	Type	Description
name	Yes	String	Username.

Response Parameters

Status code: 200

Table 4-42 Response body parameters

Parameter	Type	Description
is_valid	Boolean	Whether the username is valid.

Status code: default

Table 4-43 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

Checking whether a username is valid

```
POST https://{endpoint}/rds/v3/users/name-validation

{
  "name" : "test"
}
```

Example Response

Status code: 200

Normal

```
{
  "is_valid" : true
}
```

Status Code

Status Code	Description
200	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.1.2.7 Letting a User Reset Their Own Password

Function

This API is used to let a user reset their own password.

URI

PUT /v3/users/{user_id}/password

Table 4-44 URI parameter

Parameter	Mandatory	Type	Description
user_id	Yes	String	User ID.

Request Parameters

Table 4-45 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	Request token.
X-Language	No	String	Language.

Table 4-46 Request body parameters

Parameter	Mandatory	Type	Description
user_id	Yes	String	User ID.
password	Yes	String	New password.
old_password	Yes	String	Old password.

Response Parameters

Status code: default

Table 4-47 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

Letting a user reset their own password

```
PUT https://{endpoint}/rds/v3/users/{user_id}/password

{
  "user_id" : "3a97afeffa5e4fa089497f4dbaec4211",
  "password" : "Gauss_3004577",
  "old_password" : "Gauss_3004599"
}
```

Example Response

None

Status Codes

Status Code	Description
200	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.1.2.8 Letting a User Reset Another User's Password

Function

This API is used to let a user reset the password of another user.

URI

PUT /v3/users/{user_id}/password/others

Table 4-48 URI parameter

Parameter	Mandatory	Type	Description
user_id	Yes	String	User ID for obtaining a token.

Request Parameters

Table 4-49 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	Request token.
X-Language	No	String	Language.

Table 4-50 Request body parameters

Parameter	Mandatory	Type	Description
user_id	Yes	String	ID of a user whose password is to be reset.
password	Yes	String	New password.

Response Parameters

Status code: default

Table 4-51 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

Letting a user reset another user's password

```
PUT https://{endpoint}/rds/v3/users/{user_id}/password/others

{
  "user_id" : "b70ec6b35325421a8087ef2ba4d1fa00",
  "password" : "Gauss_33test"
}
```

Example Response

None

Status Codes

Status Code	Description
200	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.1.2.9 Querying the Instances Allocated to a User

Function

This API is used to query the instances allocated to a user.

URI

GET /v3/users/{user_id}/instances

Table 4-52 URI parameter

Parameter	Mandatory	Type	Description
user_id	Yes	String	User ID.

Table 4-53 Query parameters

Parameter	Mandatory	Type	Description
sort_key	No	String	Sorting key. **sort_key** and **sort_order** must be transferred together. Value: ● name
sort_order	No	String	Sorting order. **sort_key** and **sort_order** must be transferred together. The value can be: ● **asc**: ascending order ● **desc**: descending order

Parameter	Mandatory	Type	Description
limit	No	Integer	Number of records to be queried. The default value is **100**. The value cannot be a negative number. The minimum value is **1**, and the maximum value is **100**.
offset	No	Integer	Index offset. If **offset** is set to *N*, the resource query starts from the *N*+1 data entry. The default value is **0**, indicating that the query starts from the first data entry. The value must be a number but cannot be a negative number.

Request Parameters

Table 4-54 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	Request token.
X-Language	No	String	Language.

Response Parameters

Status code: 200

Table 4-55 Response body parameters

Parameter	Type	Description
total_count	Integer	Total number of records.
instances	Array of **instances** objects	Instance information about the user.

Table 4-56 instances

Parameter	Type	Description
id	String	Instance ID.
name	String	Instance name.

Parameter	Type	Description
mode	String	Instance deployment mode.
updateEnableFlag	Boolean	Label.

Status code: default

Table 4-57 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

Querying the instances allocated to a user

```
GET https://{endpoint}/rds/v3/users/dbb117e119dd4f81b6518cc50f97d634/instances
```

Example Response

Status code: 200

Normal

```
{
  "total_count" : 2,
  "instances" : [ {
    "updateEnableFlag" : true,
    "id" : "c66c9a776d66460a929f8ffa5b598939in14",
    "name" : "para3",
    "mode" : "Ha"
  }, {
    "updateEnableFlag" : true,
    "id" : "52d8d44080084e8db82665e209a97650in14",
    "name" : "zl_123",
    "mode" : "Ha"
  } ]
}
```

Status Codes

Status Code	Description
200	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.1.2.10 Allocating Instances to a User

Function

This API is used to allocate instances to a user.

URI

PUT /v3/users/{user_id}/instances

Table 4-58 URI parameter

Parameter	Mandatory	Type	Description
user_id	Yes	String	User ID.

Request Parameters

Table 4-59 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	Request token.
X-Language	No	String	Language.

Table 4-60 Request body parameters

Parameter	Mandatory	Type	Description
add	No	Array of strings	IDs of new instances.
del	No	Array of strings	IDs of deleted instances.

Response Parameters

Status code: default

Table 4-61 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

Allocating instances to a user

```
PUT https://{endpoint}/rds/v3/users/{user_id}/instances

{
  "add" : [ "a354a6423f8e4c4f8cf2da8f10798f47in14" ],
  "del" : [ ]
}
```

Example Response

None

Status Codes

Status Code	Description
200	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.1.2.11 Querying Roles of a User

Function

This API is used to query roles of a user.

URI

GET /v3/users/{user_id}/roles

Table 4-62 URI parameter

Parameter	Mandatory	Type	Description
user_id	Yes	String	User ID.

Table 4-63 Query parameters

Parameter	Mandatory	Type	Description
limit	No	Integer	Number of records to be queried. The default value is **100**. The value cannot be a negative number. The minimum value is **1**, and the maximum value is **100**.
sort_key	No	String	Sorting key.
sort_order	No	String	Sorting order, which can be: ● **asc**: ascending order ● **desc**: descending order
offset	No	Integer	Index offset. If **offset** is set to *N*, the resource query starts from the *N*+1 data entry. The default value is **0**, indicating that the query starts from the first data entry. The value must be a number but cannot be a negative number.

Request Parameters

Table 4-64 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	Request token.
X-Language	No	String	Language.

Response Parameters

Status code: 200

Table 4-65 Response body parameters

Parameter	Type	Description
total_count	Integer	Total number of records.
roles	Array of **roles** objects	Role information about the user.

Table 4-66 roles

Parameter	Type	Description
id	String	Role ID.
name	String	Role name.
type	String	Role type.
description	String	Role description.
updateEnableFlag	Boolean	Label.
created_at	String	Creation time.

Status code: default

Table 4-67 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

Querying roles of a user

```
GET https://{endpoint}/rds/v3/users/dbb117e119dd4f81b6518cc50f97d634/roles
```

Example Response

Status code: 200

Normal

```
{
  "total_count" : 1,
  "roles" : [ {
    "updateEnableFlag" : true,
    "id" : "2",
    "name" : "cluster_role",
    "type" : "system",
    "description" : "cluster role",
    "created_at" : "2024-03-22 09:57:50.6646"
  } ]
}
```

Status Codes

Status Code	Description
200	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.1.2.12 Assigning Roles to a User

Function

This API is used to assign roles to a user.

URI

PUT /v3/users/{user_id}/roles

Table 4-68 URI parameter

Parameter	Mandatory	Type	Description
user_id	Yes	String	User ID.

Request Parameters

Table 4-69 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	Request token.
X-Language	No	String	Language.

Table 4-70 Request body parameters

Parameter	Mandatory	Type	Description
add	No	Array of strings	IDs of roles to add.
del	No	Array of strings	IDs of roles to delete.

Response Parameters

Status code: default

Table 4-71 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

Assigning roles to a user

```
PUT https://{endpoint}/rds/v3/users/{user_id}/roles

{
  "add" : [ "3" ],
  "del" : [ "2" ]
}
```

Example Response

None

Status Codes

Status Code	Description
200	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.1.2.13 Querying User Groups That a User Has Been Added To

Function

This API is used to query the user groups that a user has been added to.

URI

GET /v3/users/{user_id}/user-groups

Table 4-72 URI parameter

Parameter	Mandatory	Type	Description
user_id	Yes	String	User ID.

Table 4-73 Query parameters

Parameter	Mandatory	Type	Description
sort_key	No	String	Sorting key. **sort_key** and **sort_order** must be transferred together. Value: ● name ● created_at
sort_order	No	String	Sorting order. **sort_key** and **sort_order** must be transferred together. The value can be: ● **asc**: ascending order ● **desc**: descending order
limit	No	Integer	Number of records to be queried. The default value is **100**. The value cannot be a negative number. The minimum value is **1**, and the maximum value is **100**.
offset	No	Integer	Index offset. If **offset** is set to *N*, the resource query starts from the *N*+1 data entry. The default value is **0**, indicating that the query starts from the first data entry. The value must be a number but cannot be a negative number.

Request Parameters

Table 4-74 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	Request token.
X-Language	No	String	Language.

Response Parameters

Status code: 200

Table 4-75 Response body parameters

Parameter	Type	Description
total_count	Integer	Total number of records.
user_groups	Array of **user_groups** objects	User group information about the user.

Table 4-76 user_groups

Parameter	Type	Description
id	String	User group ID.
name	String	User group name.
description	String	User group description.
updateEnableFlag	Boolean	Label.
create_at	String	Creation time.

Status code: default

Table 4-77 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

Querying user groups that a user has been added to

```
GET https://{endpoint}/rds/v3/users/dbb117e119dd4f81b6518cc50f97d634/user-groups
```

Example Response

Status code: 200

Normal

```
{
  "total_count" : 1,
```

```
"user_groups" : [ {
  "description" : "cluster group",
  "updateEnableFlag" : true,
  "id" : "2",
  "name" : "cluster_group",
  "type" : "system",
  "created_at" : "2024-03-22T08:25:32+0000"
 } ]
}
```


Status Codes

Status Code	Description
200	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.1.2.14 Adding a User to User Groups

Function

This API is used to add a user to user groups.

URI

PUT /v3/users/{user_id}/user-groups

Table 4-78 URI parameter

Parameter	Mandatory	Type	Description
user_id	Yes	String	User ID.

Request Parameters

Table 4-79 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	Request token.
X-Language	No	String	Language.

Table 4-80 Request body parameters

Parameter	Mandatory	Type	Description
add	No	Array of strings	IDs of user groups to add.
del	No	Array of strings	IDs of user groups to delete.

Response Parameters

Status code: default

Table 4-81 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

Adding a user to user groups

```
PUT https://{endpoint}/rds/v3/users/{user_id}/user-groups

{
  "add" : [ "2" ],
  "del" : [ "3" ]
}
```

Example Response

None

Status Codes

Status Code	Description
200	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.1.2.15 Locking Users

Function

This API is used to lock users.

URI

POST /v3/users/lock

Request Parameters

Table 4-82 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	Request token.
X-Language	No	String	Language.

Table 4-83 Request body parameters

Parameter	Mandatory	Type	Description
user_ids	Yes	Array of objects	IDs of the users to be locked.
lock_reason	Yes	String	Reason that the users are to be locked. The value can contain 1 to 160 characters.

Response Parameters

Status code: default

Table 4-84 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

Locking users

```
POST https://{endpoint}/rds/v3/users/lock
```

```
{
  "user_ids" : [ "dbb117e119dd4f81b6518cc50f97d634" ],
  "lock_reason" : "vvv"
}
```

Example Response

None

Status Codes

Status Code	Description
200	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.1.2.16 Unlocking Users

Function

This API is used to unlock users.

URI

POST /v3/users/unlock

Request Parameters

Table 4-85 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	Request token.
X-Language	No	String	Language.

Table 4-86 Request body parameters

Parameter	Mandatory	Type	Description
user_ids	Yes	Array of objects	IDs of users to be unlocked.

Response Parameters

Status code: default

Table 4-87 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

Unlocking users

```
POST https://{endpoint}/rds/v3/users/unlock

{
  "user_ids" : [ "dbb117e119dd4f81b6518cc50f97d634" ]
}
```

Example Response

None

Status Codes

Status Code	Description
200	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.1.2.17 Checking Whether a Password Is a Weak Password

Function

This API is used to check whether a password is a weak password.

URI

POST /v3/users/weak-password-verification

Request Parameters

Table 4-88 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	Request token.
X-Language	No	String	Language.

Table 4-89 Request body parameters

Parameter	Mandatory	Type	Description
password	Yes	Map<String,String>	User password.

Response Parameters

Status code: 200

Table 4-90 Response body parameters

Parameter	Type	Description
is_weak_password	String	Whether the password is a weak password.

Status code: default

Table 4-91 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

Checking whether a password is a weak password

```
POST https://{endpoint}/rds/v3/users/weak-password-verification

{
  "password" : "test_pwd"
}
```

Example Response

Status code: 200

Normal


```
{
  "is_weak_password" : false
}
```


Status Codes

Status Code	Description
200	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.1.3 User Group Management

4.1.3.1 Querying User Groups

Function

This API is used to query information about all user groups.

URI

GET /v3/user-groups

Table 4-92 Query parameters

Parameter	Mandatory	Type	Description
user_id	No	String	User ID.
limit	No	Integer	Number of records to be queried. The default value is **100**. The value cannot be a negative number. The minimum value is **1**, and the maximum value is **100**.

Parameter	Mandatory	Type	Description
offset	No	Integer	Index offset. If **offset** is set to *N*, the resource query starts from the *N*+1 data entry. The default value is **0**, indicating that the query starts from the first data entry. The value must be a number but cannot be a negative number.
sort_key	No	String	Sorting key.
sort_order	No	String	Sorting order, which can be: ● **asc**: ascending order ● **desc**: descending order
group_name	No	String	User group name. The value can contain 3 to 64 characters. Only letters (case-sensitive), digits, and underscores (_) are allowed.
user_group_id	No	String	User group ID.

Request Parameters

Table 4-93 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	Request token.
X-Language	No	String	Language.

Response Parameters

Status code: 200

Table 4-94 Response body parameters

Parameter	Type	Description
total_count	Integer	Total number of records.
user_groups	Array of **user_groups** objects	User group information.

Table 4-95 user_groups

Parameter	Type	Description
id	String	User group ID.
name	String	User group name.
description	String	User group description.
user_count	Integer	Number of users.
created_at	String	Creation time.
updateEnableFlag	Boolean	Label.
type	String	Type

Status code: 400

Table 4-96 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

Querying user groups

```
GET https://{endpoint}/rds/v3/user-groups?sort_key=name&sort_order=asc&offset=0&limit=10
```

Example Response

Status code: 200

Normal


```
{
  "total_count" : 2,
  "user_groups" : [ {
    "description" : "cluster group",
    "updateEnableFlag" : false,
    "id" : "2",
    "name" : "cluster_group",
    "type" : "system",
    "created_at" : "2023-10-24T19:40:15+0000",
    "user_count" : 55
  }, {
    "description" : "system group",
    "updateEnableFlag" : false,
    "id" : "3",
    "name" : "system_group",
    "type" : "system",
    "created_at" : "2023-10-24T19:40:15+0000",
    "user_count" : 58
```

```
  }]
}
```

Status Codes

Status Code	Description
200	Normal
400	Abnormal

Error Code

For details, see **Error Codes**.

4.1.3.2 Adding a User Group

Function

This API is used to add a user group.

URI

POST /v3/user-groups

Request Parameters

Table 4-97 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	Request token.
X-Language	No	String	Language.

Table 4-98 Request body parameters

Parameter	Mandatory	Type	Description
name	Yes	String	User group name. The value can contain 3 to 64 characters. Only letters (case-sensitive), digits, and underscores (_) are allowed.
description	No	String	User group description. The value can contain 0 to 160 characters.

Parameter	Mandatory	Type	Description
role_ids	No	Array of strings	Role IDs.
user_ids	No	Array of strings	User IDs.

Response Parameters

Status code: default

Table 4-99 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

Adding a user group

```
POST https://{endpoint}/rds/v3/user-groups

{
  "name" : "group2",
  "description" : "new group",
  "role_ids" : [ "2" ],
  "user_ids" : [ "3a97afeffa5e4fa089497f4dbaec4211" ]
}
```

Example Response

None

Status Codes

Status Code	Description
200	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.1.3.3 Updating Information About a User Group

Function

This API is used to update information about a user group.

URI

PUT /v3/user-groups/{user_group_id}

Table 4-100 URI parameter

Parameter	Mandatory	Type	Description
user_group_id	Yes	String	User group ID.

Request Parameters

Table 4-101 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	Request token.
X-Language	No	String	Language.

Table 4-102 Request body parameters

Parameter	Mandatory	Type	Description
name	No	String	User group name.
description	No	String	User group description.
users	No	**users** object	Information about the users to be changed.
roles	No	**roles** object	Information about the roles to be changed.

Table 4-103 users

Parameter	Mandatory	Type	Description
add	No	Array of strings	Users to add.
del	No	Array of strings	Users to delete.

Table 4-104 roles

Parameter	Mandatory	Type	Description
add	No	Array of strings	Roles to add.
del	No	Array of strings	Roles to delete.

Response Parameters

Status code: default

Table 4-105 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

Updating information about a user group

```
PUT https://{endpoint}/rds/v3/user-groups/{user_group_id}

{
  "name" : "group1",
  "description" : "group",
  "users" : {
    "add" : [ "f02a41abc91b42d29db96e7d3a465955" ],
    "del" : [ "5dc8bd179594493581d80f86bfeaf386" ]
  },
  "roles" : {
    "add" : [ "1217c8dc6ded4c95aa95a88e63d04392" ],
    "del" : [ "2" ]
  }
}
```

Example Response

None

Status Codes

Status Code	Description
200	Normal

Status Code	Description
default	Abnormal

Error Code

For details, see **Error Codes**.

4.1.3.4 Deleting a User Group

Function

This API is used to delete a user group that does not include any users. Before deleting a user group, remove all users from the group first.

URI

DELETE /v3/user-groups/{user_group_id}

Table 4-106 URI parameter

Parameter	Mandatory	Type	Description
user_group_id	Yes	String	User group ID.

Request Parameters

Table 4-107 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	Request token.
X-Language	No	String	Language.

Response Parameters

Status code: default

Table 4-108 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

Deleting a user group

```
DELETE https://{endpoint}/rds/v3/user-groups/fc124fc187cc4911a85fad59cfedb33d
```

Example Response

None

Status Codes

Status Code	Description
200	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.1.3.5 Querying User Roles of a User Group

Function

This API is used to query user role information about a user group by user group ID.

URI

GET /v3/user-groups/{user_group_id}/roles

Table 4-109 URI parameter

Parameter	Mandatory	Type	Description
user_group_id	Yes	String	User group ID.

Table 4-110 Query parameters

Parameter	Mandatory	Type	Description
sort_key	No	String	Sorting key. **sort_key** and **sort_order** must be transferred together. Value: • name • created_at
sort_order	No	String	Sorting order. **sort_key** and **sort_order** must be transferred together. The value must be: • **asc**: ascending order • **desc**: descending order
limit	No	Integer	Number of records to be queried. The default value is **100**. The value cannot be a negative number. The minimum value is **1**, and the maximum value is **100**.
offset	No	Integer	Index offset. If **offset** is set to *N*, the resource query starts from the *N*+1 data entry. The default value is **0**, indicating that the query starts from the first data entry. The value must be a number but cannot be a negative number.

Request Parameters

Table 4-111 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	Request token.
X-Language	No	String	Language.

Response Parameters

Status code: 200

Table 4-112 Response body parameters

Parameter	Type	Description
total_count	Integer	Quantity.
roles	Array of **roles** objects	Role information.

Table 4-113 roles

Parameter	Type	Description
id	String	Role ID.
name	String	Role name.
description	String	Role description.
updateEnableFlag	Boolean	Whether information about the user can be modified.
type	String	Type.
created_at	String	Creation time.

Status code: default

Table 4-114 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

Querying roles of a user group

```
GET https://{endpoint}/rds/v3/user-groups/fc124fc187cc4911a85fad59cfedb33d/roles
```

Example Response

Status code: 200

Normal

```
{
  "total_count" : 3,
  "roles" : [ {
    "updateEnableFlag" : true,
    "id" : "3",
```

```
    "name" : "system_role",
    "type" : "system",
    "description" : "system role",
    "created_at" : "2024-03-22 09:31:13.312442"
  }, {
    "updateEnableFlag" : true,
    "id" : "4",
    "name" : "view_role",
    "type" : "system",
    "description" : "view role",
    "created_at" : "2024-03-22 09:20:34.068015"
  }, {
    "updateEnableFlag" : true,
    "id" : "1217c8dc6ded4c95aa95a88e63d04392",
    "name" : "test_XBSA",
    "type" : "custom",
    "created_at" : "2024-03-24 01:24:47.372187"
  } ]
}
```


Status Codes

Status Code	Description
200	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.1.3.6 Updating Role Information About a User Group

Function

This API is used to update role information about a user group.

URI

PUT /v3/user-groups/{user_group_id}/roles

Table 4-115 URI parameter

Parameter	Mandatory	Type	Description
user_group_id	Yes	String	User group ID.

Request Parameters

Table 4-116 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	Request token.
X-Language	No	String	Language.

Table 4-117 Request body parameters

Parameter	Mandatory	Type	Description
add	No	Array of strings	IDs of roles to add.
del	No	Array of strings	IDs of roles to delete.

Response Parameters

Status code: default

Table 4-118 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

Updating role information about a user group

```
PUT https://{endpoint}/rds/v3/user-groups/{user_group_id}/roles

{
  "add" : [ "3" ],
  "del" : [ "1217c8dc6ded4c95aa95a88e63d04392" ]
}
```

Example Response

None

Status Codes

Status Code	Description
200	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.1.3.7 Querying Users in a User Group

Function

This API is used to query user information about a user group by user group ID.

URI

GET /v3/user-groups/{user_group_id}/users

Table 4-119 URI parameter

Parameter	Mandatory	Type	Description
user_group_id	Yes	String	User group ID.

Table 4-120 Query parameters

Parameter	Mandatory	Type	Description
sort_key	No	String	Sorting key.
sort_order	No	String	Sorting order, which can be: ● **asc**: ascending order ● **desc**: descending order
limit	No	Integer	Number of records to be queried. The default value is **100**. The value cannot be a negative number. The minimum value is **1**, and the maximum value is **100**.

Parameter	Mandatory	Type	Description
offset	No	Integer	Index offset. If **offset** is set to *N*, the resource query starts from the *N*+1 data entry. The default value is **0**, indicating that the query starts from the first data entry. The value must be a number but cannot be a negative number.

Request Parameters

Table 4-121 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	Request token.
X-Language	No	String	Language.

Response Parameters

Status code: 200

Table 4-122 Response body parameters

Parameter	Type	Description
total_count	Integer	Quantity.
users	Array of **users** objects	User information.

Table 4-123 users

Parameter	Type	Description
id	String	User ID.
name	String	Username.
updateEnableFlag	Boolean	Whether information about the user can be modified.
super_admin_role_flag	Boolean	Whether the user is a super administrator.
created_at	String	Creation time.

Parameter	Type	Description
update_at	String	Update time.
is_locked	Boolean	Whether the user has been locked.
lock_reason	String	Reason why the user was locked.
platform_name	String	Name of the platform that the user belongs to.
last_login_time	String	Last login time.
password_expiration_remaining_days	Integer	Remaining password validity period, in days.
password_expiration_days	Integer	Password validity period, in days.
has_instance_permission	Boolean	Whether the user has required permissions for instances.

Status code: default

Table 4-124 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

Querying users in a user group

```
GET https://{endpoint}/rds/v3/user-groups/fc124fc187cc4911a85fad59cfedb33d/users
```

Example Response

Status code: 200

Normal

```
{
  "total_count" : 2,
  "users" : [ {
    "updateEnableFlag" : true,
    "id" : "f02a41abc91b42d29db96e7d3a465955",
    "super_admin_role_flag" : false,
    "name" : "auto_007",
    "created_at" : "2024-03-24 01:24:47.383435",
    "updated_at" : null,
    "is_locked" : null,
    "lock_reason" : null,
```

```
    "last_login_time" : null,
    "password_expiration_remaining_days" : 0,
    "password_expiration_days" : 0,
    "has_instance_permission" : true
  }, {
    "updateEnableFlag" : true,
    "id" : "25d37c29dfaf4cd2b59240570d53fed5",
    "super_admin_role_flag" : false,
    "name" : "auto_1709886138",
    "created_at" : "2024-03-22 09:35:30.656533",
    "updated_at" : null,
    "is_locked" : null,
    "lock_reason" : null,
    "last_login_time" : null,
    "password_expiration_remaining_days" : 0,
    "password_expiration_days" : 0,
    "has_instance_permission" : true
  } ]
}
```

Status Codes

Status Code	Description
200	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.1.3.8 Updating User Information About a User Group

Function

This API is used to update user information about a user group.

URI

PUT /v3/user-groups/{user_group_id}/users

Table 4-125 URI parameter

Parameter	Mandatory	Type	Description
user_group_id	Yes	String	User group ID.

Request Parameters

Table 4-126 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	Request token.
X-Language	No	String	Language.

Table 4-127 Request body parameters

Parameter	Mandatory	Type	Description
add	Yes	Array of strings	IDs of users to add.
del	Yes	Array of strings	IDs of users to delete.

Response Parameters

Status code: default

Table 4-128 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

Updating user information about a user group

```
PUT https://{endpoint}/rds/v3/user-groups/{user_group_id}/users

{
  "add" : [ "25d37c29dfaf4cd2b59240570d53fed5" ],
  "del" : [ "f02a41abc91b42d29db96e7d3a465955" ]
}
```

Example Response

None

Status Code

Status Code	Description
200	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.1.4 Role Management

4.1.4.1 Querying Roles

Function

This API is used to query information about all roles.

URI

GET /v3/roles

Table 4-129 Query parameters

Parameter	Mandatory	Type	Description
limit	No	Integer	Number of records to be queried. The default value is **100**. The value cannot be a negative number. The minimum value is **1**, and the maximum value is **100**.
offset	No	Integer	Index offset. If **offset** is set to *N*, the resource query starts from the *N*+1 data entry. The default value is **0**, indicating that the query starts from the first data entry. The value must be a number but cannot be a negative number.
sort_key	No	String	Sorting key.
sort_order	No	String	Sorting order, which can be: • **asc**: ascending order • **desc**: descending order

Parameter	Mandatory	Type	Description
role_name	No	String	Role name.
role_id	No	String	Role ID.

Request Parameters

Table 4-130 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	Request token.
X-Language	No	String	Language.

Response Parameters

Status code: 200

Table 4-131 Response body parameters

Parameter	Type	Description
total_count	Integer	Total number of records.
roles	Array of **roles** objects	Role information list returned.

Table 4-132 roles

Parameter	Type	Description
updateEnableFlag	Boolean	Whether information about the user can be modified.
id	String	Role ID.
name	String	Role name.
type	String	Role type.
description	String	Role description.
created_at	String	Creation time.

Status code: 400

Table 4-133 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

Querying roles

```
GET https://{endpoint}/rds/v3/roles
```

Example Response

Status code: 200

Normal

```
{
  "total_count" : 2,
  "roles" : [ {
    "updateEnableFlag" : true,
    "id" : "1",
    "name" : "super_role",
    "type" : "system",
    "description" : "super role",
    "created_at" : "2023-10-24T19:40:14+0000"
  }, {
    "updateEnableFlag" : true,
    "id" : "2338214009d947b2a368ec48e9a34d36",
    "name" : "role_test01",
    "type" : "custom",
    "description" : "aaabb",
    "created_at" : "2024-03-22T09:40:01+0000"
  } ]
}
```

Status Codes

Status Code	Description
200	Normal
400	Abnormal

Error Code

For details, see **Error Codes**.

4.1.4.2 Adding a Role

Function

This API is used to add a role.

URI

POST /v3/roles

Request Parameters

Table 4-134 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	Request token.
X-Language	No	String	Language.

Table 4-135 Request body parameters

Parameter	Mandatory	Type	Description
name	Yes	String	Role name. The value can contain 3 to 64 characters. Only letters (case-sensitive), digits, and underscores (_) are allowed.
permission_ids	Yes	Array of strings	Permission IDs.
description	No	String	Role description. The value can contain 0 to 160 characters.

Response Parameters

Status code: default

Table 4-136 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

Adding a role

```
POST https://{endpoint}/rds/v3/roles

{
  "name" : "role_test01",
  "permission_ids" : [ "10002", "10003", "2000" ],
  "description" : "aaabb"
}
```

Example Response

None

Status Codes

Status Code	Description
200	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.1.4.3 Updating Information About a Role

Function

This API is used to update information about a role.

URI

PUT /v3/roles/{role_id}

Table 4-137 URI parameter

Parameter	Mandatory	Type	Description
role_id	Yes	String	Role ID.

Request Parameters

Table 4-138 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	Request token.
X-Language	No	String	Language.

Table 4-139 Request body parameters

Parameter	Mandatory	Type	Description
id	Yes	String	Role ID.
name	Yes	String	Role name.
permission_ids	Yes	Array of strings	Permission IDs.
description	Yes	String	Role description.

Response Parameters

Status code: default

Table 4-140 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

Updating information about a role

```
PUT https://{endpoint}/rds/v3/roles/{role_id}

{
  "name" : "role_test01",
  "permission_ids" : [ "10002", "10003", "9005", "9007" ],
  "description" : "abb"
}
```

Example Response

None

Status Codes

Status Code	Description
200	Success
default	Abnormal

Error Code

For details, see **Error Codes**.

4.1.4.4 Deleting a Role

Function

This API is used to delete a role.

URI

DELETE /v3/roles/{role_id}

Table 4-141 URI parameter

Parameter	Mandatory	Type	Description
role_id	Yes	String	Role ID.

Request Parameters

Table 4-142 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	Request token.
X-Language	No	String	Language.

Response Parameters

Status code: default

Table 4-143 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

Deleting a role

```
DELETE https://{endpoint}/rds/v3/roles/20f52075432b4153a594d584fcff8170
```

Example Response

None

Status Codes

Status Code	Description
200	Success
default	Abnormal

Error Code

For details, see **Error Codes**.

4.1.4.5 Querying the Permissions of a Role

Function

This API is used to query permissions of a role.

URI

GET /v3/roles/{role_id}/permissions

Table 4-144 URI parameter

Parameter	Mandatory	Type	Description
role_id	Yes	String	Role ID.

Table 4-145 Query parameters

Parameter	Mandatory	Type	Description
name	No	String	Permission name.
operation_type	No	String	Operation type.
function_type	No	String	Function type.

Request Parameters

Table 4-146 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	Request token.
X-Language	No	String	Language.

Response Parameters

Status code: 200

Table 4-147 Response body parameters

Parameter	Type	Description
operate	**operate** object	Operation permissions.
view	**view** object	Query permissions.

Table 4-148 operate

Parameter	Type	Description
backup	Array of **PermissionItem** objects	Backup operation permissions.
disaster	Array of **PermissionItem** objects	DR operation permissions.
param	Array of **PermissionItem** objects	Parameter-related operation permissions.

Parameter	Type	Description
instance	Array of **PermissionItem** objects	Instance-related operation permissions.
alarm	Array of **PermissionItem** objects	Alarm-related operation permissions.
monitor	Array of **PermissionItem** objects	Monitoring operation permissions.
user	Array of **PermissionItem** objects	User-related operation permissions.

Table 4-149 view

Parameter	Type	Description
backup	Array of **PermissionItem** objects	Backup-related query permissions.
disaster	Array of **PermissionItem** objects	DR-related query permissions.
param	Array of **PermissionItem** objects	Parameter-related query permissions.
instance	Array of **PermissionItem** objects	Instance-related query permissions.
alarm	Array of **PermissionItem** objects	Alarm-related query permissions.
monitor	Array of **PermissionItem** objects	Monitoring-related query permissions.
audit	Array of **PermissionItem** objects	Audit-related query permissions.
task	Array of **PermissionItem** objects	Task center-related query permissions.

Parameter	Type	Description
user	Array of **PermissionItem** objects	User-related query permissions.

Table 4-150 PermissionItem

Parameter	Type	Description
id	String	Role ID.
name	String	Role name.
description	String	Role description.
function_type	String	Function type.
operation_type	String	Operation type.
type	String	Role type.

Status code: default

Table 4-151 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

Querying the permissions of a role

```
GET https://{endpoint}/rds/v3/roles/90278893478b4599b9d6b096fac528e0/permissions
```

Example Response

Status code: 200

Success

```
{
  "operate" : {
    "Instances" : [ {
      "description" : "upgrade agent",
      "id" : "4053",
      "name" : "gaussdb:agent:upgrade",
      "operation_type" : "operate",
      "function_type" : "Instances"
    }, {
```

```
      "description" : "upload certificate",
      "id" : "a5005d74-3190-d3db-f5c2-00a4b0bad7f0",
      "name" : "gaussdb:certificate:upload",
      "operation_type" : "operate",
      "function_type" : "Instances"
    } ]
  }
}
```

Status Codes

Status Code	Description
200	Success
default	Abnormal

Error Code

For details, see **Error Codes**.

4.1.4.6 Querying Permissions

Function

This API is used to query permissions.

URI

GET /v3/permissions

Table 4-152 Query parameters

Parameter	Mandatory	Type	Description
name	No	String	Permission name.
operation_type	No	String	Operation type.
function_type	No	String	Function type.

Request Parameters

Table 4-153 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	Request token.

Parameter	Mandatory	Type	Description
X-Language	No	String	Language.

Response Parameters

Status code: 200

Table 4-154 Response body parameters

Parameter	Type	Description
operate	**operate** object	Operation permissions.
view	**view** object	Query permissions.

Table 4-155 operate

Parameter	Type	Description
backup	Array of **PermissionItem** objects	Backup operation permissions.
disaster	Array of **PermissionItem** objects	DR operation permissions.
param	Array of **PermissionItem** objects	Parameter-related operation permissions.
instance	Array of **PermissionItem** objects	Instance-related operation permissions.
alarm	Array of **PermissionItem** objects	Alarm-related operation permissions.
monitor	Array of **PermissionItem** objects	Monitoring operation permissions.
user	Array of **PermissionItem** objects	User-related operation permissions.

Table 4-156 view

Parameter	Type	Description
backup	Array of **PermissionItem** objects	Backup-related query permissions.
disaster	**PermissionItem** object	DR-related query permissions.
param	Array of **PermissionItem** objects	Parameter-related query permissions.
instance	Array of **PermissionItem** objects	Instance-related query permissions.
alarm	Array of **PermissionItem** objects	Alarm-related query permissions.
monitor	Array of **PermissionItem** objects	Monitoring-related query permissions.
audit	Array of **PermissionItem** objects	Audit-related query permissions.
task	Array of **PermissionItem** objects	Task center-related query permissions.
user	Array of **PermissionItem** objects	User-related query permissions.

Table 4-157 PermissionItem

Parameter	Type	Description
id	String	Role ID.
name	String	Role name.
description	String	Role description.
function_type	String	Function type.
operation_type	String	Operation type.
type	String	Role type.

Status code: default

Table 4-158 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

Querying permissions

```
GET https://{endpoint}/rds/v3/permissions
```

Example Response

Status code: 200

Success


```
{
  "operate" : {
    "Instances" : [ {
      "description" : "upgrade agent",
      "id" : "4053",
      "name" : "gaussdb:agent:upgrade",
      "operation_type" : "operate",
      "function_type" : "Instances"
    }, {
      "description" : "upload certificate",
      "id" : "a5005d74-3190-d3db-f5c2-00a4b0bad7f0",
      "name" : "gaussdb:certificate:upload",
      "operation_type" : "operate",
      "function_type" : "Instances"
    }, {
      "description" : "add dataCenter",
      "id" : "4045",
      "name" : "gaussdb:dataCenter:add",
      "operation_type" : "operate",
      "function_type" : "Instances"
    } ]
  }
}
```


Status Codes

Status Code	Description
200	Success
default	Abnormal

Error Code

For details, see **Error Codes**.

4.2 Backup Management

4.2.1 Configuring an Automated Backup Policy

Function

- This API is used to configure an automated backup policy.
- If optional parameters are not configured, the original values are used.
- Instances using XBSA backups are not supported.

URI

PUT /gaussdb/v3/instances/{instance_id}/backups/policy

Table 4-159 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-160 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Table 4-161 Request body parameters

Parameter	Mandatory	Type	Description
backup_policy	Yes	Table 4-162 object	Backup policy objects, including the backup retention period (days) and start time.

Table 4-162 BackupPolicy

Parameter	Mandatory	Type	Description
keep_days	Yes	Integer	Backup retention days. Value range: **1** to **732**
start_time	Yes	String	Backup time window. The creation of an automated backup will be triggered during the backup time window. The value cannot be empty. It must be a valid value in the "hh:mm-HH:MM" format. The current time is in the UTC format. The value of **HH** must be 1 greater than the value of **hh**. The values of **mm** and **MM** must be the same and must be **00**. Example value: If the local backup time is 05:00-06:00 and the time zone is UTC+08:00, set the parameter value to **21:00-22:00**. If the time zone is UTC+04:00, set the value to **01:00-02:00**.

Parameter	Mandatory	Type	Description
period	Yes	String	Full backup period. Data will be automatically backed up on the selected days every week. The value is a number consisting of digits separated by commas (,), indicating the days of the week. Each digit ranges from **1** to **7**. For example, the value **1,2,3,4** indicates that the backup period is every Monday, Tuesday, Wednesday, and Thursday. Note: This parameter indicates the date (in UTC format) on which backup is performed. For example, if the local backup time is 05:00–06:00 on Monday and Tuesday, and the time zone is UTC+08:00, set the parameter value to **1,7**. If the time zone is UTC+04:00, set the value to **1,2**.
differential_period	Yes	String	Interval for automated differential backups. Its value can be **15**, **30**, **60**, **180**, **360**, **720**, **1440**, **2880**, or **4320**. The unit is minute. Example value: **30**
rate_limit	No	Integer	Maximum speed at which backup data is uploaded, in MB/s. The default value is **75**. **0** indicates that the speed is not limited. The upload speed is related to the bandwidth. Value range: **0** to **1024** Minimum value: **0**

Parameter	Mandatory	Type	Description
prefetch_block	No	Integer	Number of prefetch pages from the modified pages in the disk table file during a differential backup. When modified pages are adjacent (for example, with a bulk data load), you can set this parameter to a large value. When modified pages are scattered (for example, random update), you can set this parameter to a small value. The default value is **64**. Value: **1** to **8192** Minimum value: **1** Maximum value: **8192**
file_split_size	No	Integer	Size by which full and differential backup files are split, in GB. The value is from **0** to **1024**, but it must be a multiple of 4. The default value is **4**. **0** indicates the size is not limited. Value: **0-1024** Minimum value: **0** Maximum value: **1024**
enable_standby_backup	No	Boolean	Whether to enable backup on a standby node. It is not suitable for single-node instances and instances earlier than V2.0-3.100.0. Value: **true** or **false** Default value: **false**

Parameter	Mandatory	Type	Description
backup_parallel_degree	No	Integer	Number of concurrent backup tasks. A backup task is split into multiple subtasks to improve backup efficiency. The default value is **1**. Value: **1**, **2**, **4**, or **8** **NOTE** • This parameter is only available for instances whose DB engine version is V2.0-9.0.0.SPC0100 or later. • This parameter is only available for an instance having at least 4 vCPUs. • This parameter is only available when an instance-level full backup is performed.

Response Parameters

None

Example Request

Enabling or modifying the automated backup policy

```
PUT https://{Endpoint}/rds/gaussdb/v3/instances/dsfae23fsfdsae3435in14/backups/policy

{
  "backup_policy" : {
    "keep_days" : 7,
    "start_time" : "19:00-20:00",
    "period" : "1,2,3,4,5",
    "differential_period" : "30",
    "rate_limit" : 75,
    "prefetch_block" : 64,
    "file_split_size" : 4,
    "enable_standby_backup" : "false",
    "backup_parallel_degree": 4
  }
}
```

Example Response

None

Status Code

Status Code	Description
200	Normal

Error Code

For details, see **Error Codes**.

4.2.2 Stopping a Backup Task

Function

This API is used to stop a backup task, which can be an instance- or table-level full or differential backup task or a log backup task using the XBSA backup media.

Constraints

A full or differential backup task can be stopped only when the DB engine version is V2.0-2.8 or later. A log backup task can be stopped only when the DB engine version is V2.0-9.0.0 or later.

URI

POST /gaussdb/v3/instances/{instance_id}/backups/stop

Table 4-163 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-164 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token. Minimum length: 0 character Maximum length: 32768 characters
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Table 4-165 Request body parameters

Parameter	Mandatory	Type	Description
backup_type	No	String	Type of the backup task to be stopped. Value: **Log** **Log**: an ongoing log backup task to be stopped. The backup task must use the XBSA backup media. If this parameter is not specified or is left empty, an instance- or table-level full or differential backup task is to be stopped.

Response Parameters

Status code: 202

Table 4-166 Response body parameter

Parameter	Type	Description
job_id	String	Job ID.

Example Request

- Stopping an ongoing full backup task:

```
POST  https://{Endpoint}/rds/gaussdb/v3/instances/dsfae23fsfdsae3435in14/backups/stop
```

- Stopping an ongoing XBSA log backup task:

```
POST  https://{Endpoint}/rds/gaussdb/v3/instances/dsfae23fsfdsae3435in14/backups/stop

{
    "backup_type": "Log"
}
```

Example Response

Status code: 202

Success.

```
{
  "job_id" : "dff1d289-4d03-4942-8b9f-463ea07c000d"
}
```

Status Code

Status Code	Description
202	Normal

Error Code

For details, see **Error Codes**.

4.2.3 Changing Backup Medium of an Instance

Function

- This API is used to change the backup medium of an instance.
- The backup medium of instances cannot be changed from XBSA to NAS.

URI

PUT /gaussdb/v3/instances/{instance_id}/change-backup-media

Table 4-167 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-168 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token. Minimum length: 0 character Maximum length: 32768 characters
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Table 4-169 Request body parameter

Parameter	Mandatory	Type	Description
type	Yes	String	Changed backup type. Enumerated value: **xbsa**

Response Parameters

Status code: 202

Table 4-170 Response body parameters

Parameter	Type	Description
job_id	String	Job ID.

Example Request

Changing the backup medium of an instance.

```
PUT https://{Endpoint}/rds/gaussdb/v3/instances/dsfae23fsfdsae3435in14/change-backup-media

{
  "type" : "xbsa"
}
```

Example Response

```
{
  "job_id" : "a03b1b8a-b756-467c-8a49-38720c3d23ec"
}
```

Status Code

Status Code	Description
202	Normal

Error Code

For details, see **Error Codes**.

4.2.4 Configuring an Extended Backup Policy

Function

This API is used to configure an extended backup policy, including backup flow control, number of differential prefetch pages, whether to enable standby node backup, and whether to disable backup compression for instances using XBSA backups.

Constraints

Snapshot backup is not supported.

URI

PUT /gaussdb/v3/instances/{instance_id}/backups/config

Table 4-171 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-172 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Table 4-173 Request body parameters

Parameter	Mandatory	Type	Description
rate_limit	No	Integer	Maximum speed at which backup data is uploaded, in MB/s. The default value is **75**. **0** indicates that the speed is not limited. The upload speed is related to the bandwidth. Value range: **0** to **1024** Minimum value: **0**

Parameter	Mandatory	Type	Description
prefetch_block	No	Integer	Number of prefetch pages from the modified pages in the disk table file during a differential backup. When modified pages are adjacent (for example, with a bulk data load), you can set this parameter to a large value. When modified pages are scattered (for example, random update), you can set this parameter to a small value. The default value is **64**. Value: **1** to **8192** Minimum value: **1** Maximum value: **8192**
enable_standby_backup	No	Boolean	Whether to enable backup on a standby node. It is not suitable for single-node instances and instances earlier than V2.0-3.100.0. Value: **true** or **false** Default value: **false**
close_compression	No	Boolean	Whether to disable backup compression. This parameter is only suitable for instances with XBSA storage media. Value: **true** or **false** **true**: This function is disabled. **false**: This function is enabled. Default value: **false**

Parameter	Mandatory	Type	Description
backup_parallel_degree	No	Integer	Number of concurrent backup tasks. A backup task is split into multiple subtasks to improve backup efficiency. The default value is **1**. Value: **1**, **2**, **4**, or **8** **NOTE** • This parameter is only available for instances whose DB engine version is V2.0-9.0.0.SPC0100 or later. • This parameter is only available for an instance having at least 4 vCPUs. • This parameter is only available when an instance-level full backup is performed.

Response Parameters

None

Example Request

Setting **rate_limit** to **76** and **prefetch_block** to **65**, enabling standby node backup, and disabling backup compression

```
PUT https://{Endpoint}/rds/gaussdb/v3/instances/dsfae23fsfdsae3435in14/backups/config
{
   "rate_limit" : 76,
   "prefetch_block" : 65,
   "enable_standby_backup" : true,
   "close_compression" : true,
   "backup_parallel_degree": 4
 }
```

Example Response

Status code: 200

API response succeeded.

```
SUCCESS
```

Status Code

Status Code	Description
200	Normal

Error Code

For details, see **Error Codes**.

4.2.5 Querying an Automated Backup Policy

Function

- This API is used to query an automated backup policy.
- Instances using XBSA backups are not supported.

URI

GET /gaussdb/v3/instances/{instance_id}/backups/policy

Table 4-174 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-175 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Response Parameters

Status code: 200

Table 4-176 Response body parameter

Parameter	Type	Description
backup_policy	**Table 4-177** object	Backup policy information.

Table 4-177 ShowBackupPolicy

Parameter	Type	Description
keep_days	Integer	Full backup retention days. Minimum value: **1** Maximum value: **732**
start_time	String	Full backup time window. The creation of an automated backup will be triggered during the backup time window. The value must be a valid value in the "hh:mm-HH:MM" format. The current time is in the UTC format. ● The **HH** value must be 1 greater than the **hh** value. ● The values of **mm** and **MM** must be the same and must be set to **00**.
period	String	Full backup period. After a backup cycle is specified, data will be automatically backed up on the selected days every week. The value is a number consisting of digits separated by commas (,), indicating the days of the week. Each digit ranges from **1** to **7**. For example, the value **1,2,3,4** indicates that the backup period is every Monday, Tuesday, Wednesday, and Thursday.
differential_period	String	Differential backup period. An automated differential backup will be performed on the specified minutes.
rate_limit	Integer	Bandwidth for uploading backup data during backup, in MB/s. Value range: **0** to **1024**. The default value is **75**. **0** indicates the bandwidth is not limited.
status	String	Status of an automated full backup. **Active**: An automated full backup is active and being performed based on a backup policy. **Abort**: An automated full backup is suspended. **Disabled**: An automated full backup is disabled.
prefetch_block	Integer	Number of prefetch pages from the modified pages in the disk table file during a differential backup. Value range: **1-8192**. Default value: **64**.

Parameter	Type	Description
file_split_size	Integer	Size by which full and differential backup files are split, in GB. The value is from **0** to **1024**, but it must be a multiple of 4. The default value is **4**. **0** indicates the size is not limited. Value range: **0-1024** Minimum value: **0** Maximum value: **1024**
enable_standby_backup	Boolean	Whether to enable backup on a standby node. Value: **true** or **false** Default value: **false**
backup_parallel_degree	Integer	Number of concurrent backup tasks. A backup task is split into multiple subtasks to improve backup efficiency. The default value is **1**. Value: **1**, **2**, **4**, or **8**

Example Request

```
GET https://{Endpoint}/rds/gaussdb/v3/instances/3d39c18788b54a919bab633874c159dfin14/backups/policy
```

Example Response

Status code: 200

Success.


```
{
  "backup_policy" : {
    "keep_days" : 7,
    "start_time" : "19:00-20:00",
    "period" : "1,2",
    "differential_period" : 30,
    "rate_limit" : 75,
    "status" : "Active",
    "prefetch_block" : 64,
    "file_split_size" : 4,
    "enable_standby_backup" : false,
    "backup_parallel_degree": 4
  }
}
```


Status Code

Status Code	Description
200	Normal

Error Code

For details, see **Error Codes**.

4.2.6 Querying an Extended Backup Policy

Function

This API is used to query an extended backup policy.

URI

GET /gaussdb/v3/instances/{instance_id}/backups/config

Table 4-178 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-179 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Response Parameters

Status code: 200

Table 4-180 Response body parameters

Parameter	Type	Description
rate_limit	Integer	Backup flow control.
prefetch_block	Integer	Number of differential prefetch pages.

Parameter	Type	Description
enable_standby_backup	Boolean	Whether to enable standby node backup.
close_compression	Boolean	Whether to disable backup compression.
default_backup_media_type	String	Backup media. • **NAS** • **XBSA**
default_backup_method	String	Backup method. • **PHYSICAL_BACKUP**: physical backup
backup_parallel_degree	Integer	Number of concurrent backup tasks. A backup task is split into multiple subtasks to improve backup efficiency. The default value is **1**. Value: **1**, **2**, **4**, or **8**

Example Request

Querying an extended backup policy

```
GET https://{Endpoint}/rds/gaussdb/v3/instances/dsfae23fsfdsae3435in14/backups/config
```

Example Response

Status code: 200

```
{
  "rate_limit" : 76,
  "prefetch_block" : 65,
  "enable_standby_backup" : true,
  "close_compression" : true,
  "default_backup_media_type" : "PHYSICAL_BACKUP",
  "default_backup_method" : "NAS",
  "backup_parallel_degree": 4
}
```

Status Code

Status Code	Description
200	Normal

Error Code

For details, see **Error Codes**.

4.2.7 Creating Manual Backups

Function

This API is used to create manual backups.

Constraints

You can simultaneously send a maximum of 60 requests of creating manual backups per minute by calling this API.

URI

POST /gaussdb/v3/backups

Request Parameters

Table 4-181 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Table 4-182 Request body parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.
name	Yes	String	Backup name. It must contain 4 to 64 characters and start with a letter. Only letters (case-sensitive), digits, hyphens (-), and underscores (_) are allowed. Minimum length: 4 characters Maximum length: 64 characters

Parameter	Mandatory	Type	Description
description	No	String	Backup description. It can contain up to 256 characters and cannot contain special characters (>!<"&'=). Maximum length: 256 characters
backup_type	No	String	Backup type. The default value is **completed**. Enumerated values: • **completed**: full backup • **differential**: differential backup (Only instances using XBSA backups are supported.) • **database_differential**: table-level differential backup. (Only instances using XBSA backups are supported.) • **log**: log backup. (It is suitable for instances V2.0-3.301 or later that use XBSA backups.) Log backups cannot be generated for single-replica DB instances. **NOTE** • If the replica consistency protocol of the instance is **Paxos**, the instance version must be V2.0-8.1.0 or later. • If the replica consistency protocol of the instance is **Quorum**, the instance version must be V2.0-3.301 or later.

Parameter	Mandatory	Type	Description
depend_backupIds	No	Array of strings	IDs of full backups and differential backups on which log backup depends. This parameter is only suitable for instances using XBSA backups. **NOTICE** ● The PITR restoration time is maintained by the third party, so the value of **depend_backupIds** is calculated by the third party. ● If logs are backed up for the first time, you need to enter the IDs of all full backups and differential backups. If operations such as upgrading kernel versions, adding nodes, restoring data to the original instance, and redistributing data after the scale-out, are performed on the instance, the IDs of all full backups and differential backups must be configured again. ● If logs are not backed up for the first time, you need to enter the ID of the latest full backup (or differential backup) on which the last log backup depends and the IDs of all full backups and differential backups generated after the last log backup.

Parameter	Mandatory	Type	Description
table_list	No	Array of **Table 4-183**	Table-level backup information. Table-level backups are created only if this parameter is configured. **Constraints**: ● Only instances later than V2.0-3.200 support NAS-based table-level backups. Only instances of the V2.0-8.200 version and later support XBSA-based table-level backups. ● Up to 100 databases or tables can be backed up at the same time. If the number of databases or tables exceeds 100, instance-level backups are recommended. ● This parameter is required for creating a table-level backup. You do not need to set it for a table-level differential backup.

Table 4-183 Data structure description of the TableList field

Parameter	Mandatory	Type	Description
db_name	Yes	String	**Explanation**: Database name. **Constraints**: The following template databases and system users cannot be used: ● Template databases: **postgres**, **template0**, **templatem**, and **template1**. ● System users: **rdsAdmin**, **rdsMetric**, **rdsBackup**, and **rdsRepl**. **Value range**: N/A **Default value**: N/A
schema_name	Yes	String	**Explanation**: Schema name. **Constraints**: The names of system-level schemas, such as **dbe_application_info** and **rdsRepl**, cannot be specified. **Value range**: N/A **Default value**: N/A
table_name	Yes	String	**Explanation**: Table name. **Constraints**: N/A **Value range**: N/A **Default value**: N/A

Response Parameters

Status code: 202

Table 4-184 Response body parameters

Parameter	Type	Description
backup	Table 4-185 object	Backup Information.
job_id	String	Job ID.

Table 4-185 BackupInfo

Parameter	Type	Description
id	String	Backup ID.
name	String	Backup name.
description	String	Backup description.
begin_time	String	Backup start time in the "yyyy-mm-ddThh:mm:ssZ" format. **T** is the separator between calendar and hourly notation of time. **Z** indicates the time zone offset. For example, in the Beijing time zone, the offset is **+0800**.
status	String	Backup status. Value: **BUILDING** (Backup in progress)
type	String	Backup type. **manual** indicates manual full backup and manual differential backup. **Log_Xbsa** indicates XBSA log backup.
instance_id	String	Instance ID.

Example Request

- Creating a manual full backup
```
POST https://{Endpoint}/rds/gaussdb/v3/backups

{
  "instance_id" : "d8e6ca5a624745bcb546a227aa3ae1cfin01",
  "name" : "backup",
  "description" : "manual backup"
}
```
- Creating a manual table-level backup
```
POST https://{Endpoint}/rds/gaussdb/v3/backups

{
  "instance_id": "87225be9a74b4c3b9241388940197737in14",
  "name": "backup-87225be9a74b4c3b9241388940197737in14-test",
```

```
"backup_type": "completed",
"description": "111",
"table_list": [
    {
        "db_name": "core1",
        "schema_name": "my_schema1",
        "table_name": "test1"
    }
  ]
}
```


Example Response

Status code: 202

Success.


```
{
  "backup": {
    "id": "e76112bfb2074871bf54cb8df5af7f64br14",
    "name": "backupwqwq32",
    "description": "manual backup",
    "status": "BUILDING",
    "type": "manual",
    "begin_time": "2022-05-09T18:02:31+0800",
    "instance_id": "fd26e3bf26e5467587eec857e4f66ef0in14"
  },
  "job_id": "e4733090-b2c8-4ea7-a33a-f55f65723fb3"
}
```


Status Code

Status Code	Description
202	Normal

Error Code

For details, see **Error Codes**.

4.2.8 Deleting a Manual Backup

Function

This API is used to delete a manual backup. Deleting XBSA backups only delete the metadata of backups. Data stored on the XBSA device cannot be deleted.

URI

DELETE /gaussdb/v3/backups/{backup_id}

Table 4-186 URI parameter

Parameter	Mandatory	Type	Description
backup_id	Yes	String	Backup ID.

Request Parameters

Table 4-187 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Response Parameters

Parameter	Type	Description
backup_id	String	Backup ID.
backup_name	String	Name of a backup file.

Example Request

```
DELETE https://{Endpoint}/rds/gaussdb/v3/backups/afbf982b46ca421e9118f7db916058b5br14
```

Example Response

```
{
"backup_id": "a5ad6dde4dc54d7cbf095b45c89d87a8br14",
"backup_name": "backup-4b52"
}
```

Status Code

Status Code	Description
202	Normal

Error Code

For details, see **Error Codes**.

4.2.9 Querying Information About the Original Instance Based on a Specific Point in Time or a Backup File

Function

This API is used to query the information of the original instance based on a specific point in time or a backup file.

Constraints

- The parameters **instance_id** and **restore_time** are mutually exclusive with the parameters **instance_id** and **backup_id**.
- Either the parameters **instance_id** and **restore_time** or parameters **instance_id** and **backup_id** must be specified.

URI

GET /gaussdb/v3/instance-snapshot

Table 4-188 Query parameters

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Original instance ID. If **restore_time** is specified, **instance_id** is mandatory. (**instance_id** and **restore_time** are used together.) (**instance_id** and **backup_id** are used together.)
restore_time	No	String	Instance information at a time point in the UNIX timestamp format, in milliseconds. The time zone is UTC. (**instance_id** and **restore_time** are used together.)
backup_id	No	String	Backup ID. (**instance_id** and **backup_id** are used together.) If the backup ID is not empty, the timestamp is not required.

Parameter	Mandatory	Type	Description
schema_type	No	String	Backup level. This parameter is mandatory when a DB instance is restored to a specific point of time. This parameter cannot be configured when a DB instance is restored using a backup ID. Enumerated values: ● **INSTANCE** (default value): instance-level backup ● **DATABASE_TABLE**: table-level backup

Request Parameters

Table 4-189 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Response Parameters

Status code: 200

Table 4-190 Response body parameters

Parameter	Type	Description
cluster_mode	String	Instance deployment model. For a centralized instance, its value is **Ha** (centralized deployment). For a distributed instance, its value is **Combined** (combined deployment). Enumerated values: ● **Ha** ● **Combined**

Parameter	Type	Description
instance_mode	String	Instance edition. For enterprise edition, its value is **enterprise**. Value: **enterprise**
data_volume_size	String	Storage space, in GB.

Parameter	Type	Description
solution	String	Instance type. Centralized: ● **single**: single-replica ● **double**: 2 nodes (1 primary + 1 standby + 1 log) ● **logger**: 3 nodes (1 primary + 1 standby + 1 log) ● **triset**: 3 nodes (1 primary + 2 standby) ● **quadruset**: 5 nodes (1 primary + 3 standby, 1 arbitration node) ● **pentaset**: 5 nodes (1 primary + 4 standby) ● **doublenoetcd**: 2 nodes (1 primary + 1 standby + 1 log) (without ETCD) ● **loggernoetcd**: 3 nodes (1 primary + 1 standby + 1 log) (without ETCD) ● **trisetnoetcd**: 3 nodes (1 primary + 2 standby, without ETCD) ● **quadrusetnoetcd**: 5 nodes (1 primary + 3 standby, 1 arbitration node) (without ETCD) ● **pentasetnoetcd**: 5 nodes (1 primary + 4 standby, without ETCD) Distributed deployment: ● **hcs1**: 9 nodes. There are 4 shards and each shard contains one primary DN and three standby DNs. ● **hcs2**: 3 nodes. There are 3 shards and each shard contains one primary DN and two standby DNs. ● **hcs3**: 4 nodes. There are 4 shards and each shard contains one primary DN and three standby DNs. ● **hcs4**: 5 nodes. There are 4 shards and each shard contains one primary DN and three standby DNs. ● **hcs6**: 1 node. There are 3 shards and each shard contains one primary DN. ● **hcs9**: 1 node. There are 4 shards and each shard contains one primary DN. Enumerated values: ● single ● double ● logger ● triset

Parameter	Type	Description
		• quadruset • pentaset • doublenoetcd • loggernoetcd • trisetnoetcd • quadrusetnoetcd • pentasetnoetcd • hcs1 • hcs2 • hcs3 • hcs4 • hcs6 • hcs9
node_num	Integer	Number of nodes.
coordinator_num	Integer	Number of CNs.
sharding_num	Integer	Number of shards.
replica_num	Integer	Number of replicas.
engine_version	String	DB engine version.
lower_case_table_names	Integer	Case sensitivity of M-compatible table names. Value: • **0** (default value): The names are case-sensitive. • **1**: The names are case-insensitive.
database_table_info	**Table 4-191** object	Table information.

Table 4-191 DatabaseTableInfo

Parameter	Type	Description
databases	Array of **Table 4-192** objects	Database.
total_dbs	Integer	Number of databases.

Table 4-192 Database

Parameter	Type	Description
name	String	Database name.
total_schems	Integer	Number of schemas.
schemas	Array of **Table 4-193** objects	SCHEMA

Table 4-193 Schema

Parameter	Type	Description
name	String	Schema name.
total_tables	Integer	Number of tables.
tables	Array of **Table 4-194** objects	Table.

Table 4-194 Table

Parameter	Type	Description
name	String	Table name.

Example Request

Querying historical information about an instance

```
GET https://{Endpoint}/rds/gaussdb/v3/instance-snapshot?
instance_id=b043c90a9b5a403f960456ff08ec75d6in14&backup_id=cc94568cb5a54e4a8ab5dff95e64a5e0br1
4&restore_time=1665943652000
```

Example Response

Status code: 200

Success.

```
{
  "cluster_mode" : "Ha",
  "instance_mode" : "enterprise",
  "data_volume_size" : "200",
  "solution" : "triset",
  "node_num" : 3,
  "coordinator_num" : 0,
  "sharding_num" : 3,
  "replica_num" : 3,
  "engine_version" : "2.2.90",
  "database_table_info": {
       "databases": [
```

```
        {
            "name": "table_backup_db",
            "totalSchemas": 1,
            "schemas": [
                {
                    "name": "myschema1",
                    "totalTables": 1,
                    "tables": [
                        {
                            "name": "test"
                        }
                    ]
                }
            ]
        }
    ],
    "total_dbs": 1
  }
}
```

Status Code

Status Code	Description
200	Normal

Error Code

For details, see **Error Codes**.

4.2.10 (Recommended) Querying Information About the Original Instance Based on a Specific Point in Time or a Backup File

Function

This API is used to query the information of the original instance based on a specific point in time or a backup file.

Constraints

- The parameters **instance_id** and **restore_time** are mutually exclusive with the parameters **instance_id** and **backup_id**.
- Either the parameters **instance_id** and **restore_time** or parameters **instance_id** and **backup_id** must be specified.

URI

GET /gaussdb/v3.1/instance-snapshot

Table 4-195 Query parameters

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Original instance ID. If **restore_time** is specified, **instance_id** is mandatory. (**instance_id** and **restore_time** are used together.) (**instance_id** and **backup_id** are used together.)
restore_time	No	String	Instance information at a time point in the UNIX timestamp format, in milliseconds. The time zone is UTC. (**instance_id** and **restore_time** are used together.)
backup_id	No	String	Backup ID. (**instance_id** and **backup_id** are used together.) If the backup ID is not empty, the timestamp is not required.
schema_type	No	String	Backup level. This parameter is mandatory when a DB instance is restored to a specific point of time. This parameter cannot be configured when a DB instance is restored using a backup ID. Enumerated values: • **INSTANCE** (default value): instance-level backup • **DATABASE_TABLE**: table-level backup

Request Parameters

Table 4-196 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.

Parameter	Mandatory	Type	Description
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Response Parameters

Status code: 200

Table 4-197 Response body parameters

Parameter	Type	Description
cluster_mode	String	Instance deployment model. For a centralized instance, its value is **Ha** (centralized deployment). For a distributed instance, its value is **Combined** (combined deployment). Enumerated values: ● Ha ● Combined
instance_mode	String	Instance edition. For enterprise edition, its value is **enterprise**. Value: **enterprise**
data_volume_size	String	Storage space, in GB.

Parameter	Type	Description
solution	String	Instance type. Centralized: • **single**: single-replica • **double**: 2 nodes (1 primary + 1 standby + 1 log) • **logger**: 3 nodes (1 primary + 1 standby + 1 log) • **triset**: 3 nodes (1 primary + 2 standby) • **quadruset**: 5 nodes (1 primary + 3 standby, 1 arbitration node) • **pentaset**: 5 nodes (1 primary + 4 standby) • **doublenoetcd**: 2 nodes (1 primary + 1 standby + 1 log) (without ETCD) • **loggernoetcd**: 3 nodes (1 primary + 1 standby + 1 log) (without ETCD) • **trisetnoetcd**: 3 nodes (1 primary + 2 standby, without ETCD) • **quadrusetnoetcd**: 5 nodes (1 primary + 3 standby, 1 arbitration node) (without ETCD) • **pentasetnoetcd**: 5 nodes (1 primary + 4 standby, without ETCD) Distributed deployment: • **hcs1**: 9 nodes. There are 4 shards and each shard contains one primary DN and three standby DNs. • **hcs2**: 3 nodes. There are 3 shards and each shard contains one primary DN and two standby DNs. • **hcs3**: 4 nodes. There are 4 shards and each shard contains one primary DN and three standby DNs. • **hcs4**: 5 nodes. There are 4 shards and each shard contains one primary DN and three standby DNs. • **hcs6**: 1 node. There are 3 shards and each shard contains one primary DN. • **hcs9**: 1 node. There are 4 shards and each shard contains one primary DN. Enumerated values: • single • double • logger • triset

Parameter	Type	Description
		• quadruset • pentaset • doublenoetcd • loggernoetcd • trisetnoetcd • quadrusetnoetcd • pentasetnoetcd • hcs1 • hcs2 • hcs3 • hcs4 • hcs6 • hcs9
node_num	Integer	Number of nodes.
coordinator_num	Integer	Number of CNs.
sharding_num	Integer	Number of shards.
replica_num	Integer	Number of replicas.
engine_version	String	DB engine version.
lower_case_table_names	Integer	Case sensitivity of M-compatible table names. Value: • **0** (default value): The names are case-sensitive. • **1**: The names are case-insensitive.
database_table_info	**Table 4-198** object	Table information.

Table 4-198 DatabaseTableInfo

Parameter	Type	Description
databases	Array of **Table 4-199** objects	Database.
total_dbs	Integer	Number of databases.

Table 4-199 Database

Parameter	Type	Description
name	String	Database name.
total_schems	Integer	Number of schemas.
schemas	Array of **Table 4-200** objects	SCHEMA

Table 4-200 Schema

Parameter	Type	Description
name	String	Schema name.
total_tables	Integer	Number of tables.
tables	Array of **Table 4-201** objects	Table.

Table 4-201 Table

Parameter	Type	Description
name	String	Table name.

Example Request

Querying historical information about an instance

```
GET https://{Endpoint}/rds/gaussdb/v3.1/instance-snapshot?
instance_id=b043c90a9b5a403f960456ff08ec75d6in14&backup_id=cc94568cb5a54e4a8ab5dff95e64a5e0br1
4&restore_time=1665943652000
```

Example Response

Status code: 200

Success.

```
{
  "cluster_mode" : "Ha",
  "instance_mode" : "enterprise",
  "data_volume_size" : "200",
  "solution" : "triset",
  "node_num" : 3,
  "coordinator_num" : 0,
  "sharding_num" : 3,
  "replica_num" : 3,
  "engine_version" : "V2.0-2.2.90",
  "database_table_info": {
       "databases": [
```

```
        {
          "name": "table_backup_db",
          "totalSchemas": 1,
          "schemas": [
            {
              "name": "myschema1",
              "totalTables": 1,
              "tables": [
                {
                  "name": "test"
                }
              ]
            }
          ]
        }
      ],
      "total_dbs": 1
    }
}
```

Status Code

Status Code	Description
200	Normal

Error Code

For details, see **Error Codes**.

4.2.11 Querying Instances That Can Be Used for Backups and Restorations

Function

- This API is used to query the instances that can be used for backups and restorations. The instance information must meet the backup conditions.
- **source_instance_id** and **backup_id** cannot be both left blank.
- If **backup_id** is not left blank, **source_instance_id** is optional.
- If **backup_id** is left blank and **restore_time** is not left blank, **source_instance_id** is mandatory.

URI

GET /gaussdb/v3/restorable-instances

Table 4-202 Query parameters

Parameter	Mandatory	Type	Description
source_instance_id	Yes	String	ID of the instance to be restored.

Parameter	Mandatory	Type	Description
backup_id	No	String	Instance backup ID. You can use the backup ID to query the instance topology information and filter the queried instances (The number of nodes and replicas of instances are displayed). If this parameter is left empty, **restore_time** is used.
restore_time	No	String	Specific point of time. If the backup ID is empty, this parameter is used to query the instance topology information and filter the queried instances.
offset	No	Integer	Index offset. If **offset** is set to *N*, the resource query starts from the *N*+1 piece of data. The value is **0** by default, indicating that the query starts from the first piece of data. The value must be a number but cannot be a negative number. Minimum value: **0** Maximum value: **999999999**
limit	No	Integer	Number of records to be queried. The default value is **100**. The value cannot be a negative number. The minimum value is **1** and the maximum value is **100**.

Request Parameters

Table 4-203 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.

Parameter	Mandatory	Type	Description
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Response Parameters

Status code: 200

Table 4-204 Response body parameters

Parameter	Type	Description
instances	Array of **Table 4-205** objects	Instances that can be used for backups.
total_count	Integer	Total number of queried instances.

Table 4-205 InstancesResult

Parameter	Type	Description
instance_name	String	Instance name.
instance_id	String	Instance ID.
volume_type	String	Storage type.
data_volume_size	Long	Storage space, in bytes.
version	String	Instance version.
mode	String	Instance deployment model. For a centralized instance, its value is **Ha** (centralized deployment). For a distributed instance, its value is **Combined** (combined deployment). Enumerated values: ● **Ha** ● **Combined**
instance_mode	String	Instance edition. For enterprise edition, its value is **enterprise**. Value: **enterprise**

Example Request

```
GET https://{Endpoint}/rds/gaussdb/v3/restorable-instances?
source_instance_id=88efb3753dc844829c380edff7798eecin14&backup_id=d3f223e9c35d450ea0692bdbff686
e45br14
```

Example Response

Status code: 200

Success.


```
{
  "instances" : [ {
    "instance_name" : "gaussdb_hzx",
    "instance_id" : "3ea6d6463c9a4baf9a47c5b74464307cin14",
    "volume_type" : "ULTRAHIGH",
    "data_volume_size" : 500,
    "version" : 1.3,
    "mode" : "Ha",
    "instance_mode" : "enterprise"
  } ],
  "total_count" : 1
}
```


Status Code

Status Code	Description
200	Normal

Error Code

For details, see **Error Codes**.

4.2.12 (Recommended) Querying Instances That Can Be Used for Backups and Restorations

Function

- This API is used to query the instances that can be used for backups and restorations. The instance information must meet the backup conditions.
- **source_instance_id** and **backup_id** cannot be both left blank.
- If **backup_id** is not left blank, **source_instance_id** is optional.
- If **backup_id** is left blank and **restore_time** is not left blank, **source_instance_id** is mandatory.

URI

GET /gaussdb/v3.1/restorable-instances

Table 4-206 Query parameters

Parameter	Mandatory	Type	Description
source_instance_id	Yes	String	ID of the instance to be restored.
backup_id	No	String	Instance backup ID. You can use the backup ID to query the instance topology information and filter the queried instances (The number of nodes and replicas of instances are displayed). If this parameter is left empty, **restore_time** is used.
restore_time	No	String	Specific point of time. If the backup ID is empty, this parameter is used to query the instance topology information and filter the queried instances.
offset	No	Integer	Index offset. If **offset** is set to *N*, the resource query starts from the *N*+1 piece of data. The value is **0** by default, indicating that the query starts from the first piece of data. The value must be a number but cannot be a negative number. Minimum value: **0** Maximum value: **999999999**
limit	No	Integer	Number of records to be queried. The default value is **100**. The value cannot be a negative number. The minimum value is **1** and the maximum value is **100**.

Request Parameters

Table 4-207 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.

Parameter	Mandatory	Type	Description
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Response Parameters

Status code: 200

Table 4-208 Response body parameters

Parameter	Type	Description
instances	Array of **Table 4-209** objects	Instances that can be used for backups.
total_count	Integer	Total number of queried instances.

Table 4-209 InstancesResult

Parameter	Type	Description
instance_name	String	Instance name.
instance_id	String	Instance ID.
volume_type	String	Storage type.
data_volume_size	Long	Storage space, in bytes.
version	String	Instance version.
mode	String	Instance deployment model. For a centralized instance, its value is **Ha** (centralized deployment). For a distributed instance, its value is **Combined** (combined deployment). Enumerated values: ● Ha ● Combined
instance_mode	String	Instance edition. For enterprise edition, its value is **enterprise**. Value: **enterprise**

Example Request

```
GET https://{Endpoint}/rds/gaussdb/v3.1/restorable-instances?
source_instance_id=88efb3753dc844829c380edff7798eecin14&backup_id=d3f223e9c35d450ea0692bdbff686
e45br14
```

Example Response

Status code: 200

Success.


```
{
  "instances" : [ {
    "instance_name" : "gaussdb_hzx",
    "instance_id" : "3ea6d6463c9a4baf9a47c5b74464307cin14",
    "volume_type" : "ULTRAHIGH",
    "data_volume_size" : 500,
    "version" : V2.0-1.3,
    "mode" : "Ha",
    "instance_mode" : "enterprise"
  } ],
  "total_count" : 1
}
```


Status Code

Status Code	Description
200	Normal

Error Code

For details, see **Error Codes**.

4.2.13 Querying the Restoration Time Range

Function

This API is used to query the restoration time range of an instance. If backups are stored for a long time, you are advised to configure the parameter **date**.

Instances using XBSA backups are not supported.

URI

GET /gaussdb/v3/instances/{instance_id}/restore-time

Table 4-210 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Table 4-211 Query parameter

Parameter	Mandatory	Type	Description
date	Yes	String	Date to be queried. The value is in the "yyyy-mm-dd" format, and the time zone is UTC.

Request Parameters

Table 4-212 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Response Parameters

Status code: 200

Table 4-213 Response body parameters

Parameter	Type	Description
restore_time	Array of **Table 4-214** objects	Restoration time ranges.

Table 4-214 restore_time

Parameter	Type	Description
start_time	Long	Start time of the restoration time range in the UNIX timestamp format. The unit is millisecond and the time zone is UTC.
end_time	Long	End time of the restoration time range in the UNIX timestamp format. The unit is millisecond and the time zone is UTC.

Example Request

```
GET https://{Endpoint}/rds/gaussdb/v3/instances/3d39c18788b54a919bab633874c159dfin14/restore-time?
date=2022-01-01
```

Example Response

Status code: 200

Success.

```
{
  "restore_time" : [ {
    "start_time" : 1532001446987,
    "end_time" : 1532742139000
  } ]
}
```

Status Code

Status Code	Description
200	Normal

Error Code

For details, see **Error Codes**.

4.2.14 Querying Differential Backups

Function

This API is used to query differential backups. Cross-cloud backups of instances can be queried.

URI

GET /gaussdb/v3/instances/{instance_id}/differential-backups

Table 4-215 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Table 4-216 Query parameters

Parameter	Mandatory	Type	Description
limit	No	String	Number of records to be queried. The default value is **10**. The value cannot be a negative number. The minimum value is **1** and the maximum value is **100**. Minimum value: **1** Maximum value: **100**
offset	No	String	Index offset. If **offset** is set to *N*, the resource query starts from the *N*+1 piece of data. The value is **0** by default, indicating that the query starts from the first piece of data. The value must be a number but cannot be a negative number. Minimum value: **0** Maximum value: **999999999**
begin_time	No	String	Query start time. The default value is the current time of the previous day. The format is "yyyy-mm-ddThh:mm:ssZ". **T** is the separator between calendar and hourly notation of time. **Z** indicates the time zone offset. For example, in the Beijing time zone, the offset is **+0800**.
end_time	No	String	Query end time. The format is "yyyy-mm-ddThh:mm:ssZ" and the end time must be later than the start time. **T** is the separator between calendar and hourly notation of time. **Z** indicates the time zone offset. For example, in the Beijing time zone, the offset is **+0800**.

Parameter	Mandatory	Type	Description
backup_type	No	String	Backup type. Value: • **INSTANCE**: instance-level backup • **DatabaseTable**: database- or table-level backup **Default value**: INSTANCE

Request Parameters

Table 4-217 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Response Parameters

Status code: 200

Table 4-218 Response body parameters

Parameter	Type	Description
differential_backups	**Table 4-219** object	Information of differential backups in the list.
total_count	Integer	Total number of differential backups.

Table 4-219 DifferentialBackupResult

Parameter	Type	Description
id	String	Backup ID.
name	String	Backup name.

Parameter	Type	Description
begin_time	String	Backup start time in the "yyyy-mm-ddThh:mm:ssZ" format. **T** is the separator between calendar and hourly notation of time. **Z** indicates the time zone offset. For example, in the Beijing time zone, the offset is shown as **+0800**.
end_time	String	Backup end time in the "yyyy-mm-ddThh:mm:ssZ" format. **T** is the separator between calendar and hourly notation of time. **Z** indicates the time zone offset. For example, in the Beijing time zone, the offset is shown as **+0800**.
status	String	Backup status. Enumerated values: **BUILDING**: Backup in progress. **COMPLETED**: Backup completed. **FAILED**: Backup failed.
size	Double	Size, in MB.

Example Request

```
GET https://{Endpoint}/rds/gaussdb/v3/instances/3d39c18788b54a919bab633874c159dfin14/differential-backups
```

Example Response

Status code: 200

```
{
  "differential_backups" : [ {
    "id" : "964896a1aabf4f90ae0a8884b4a4d5e6br14",
    "name" : "testbackup",
    "begin_time" : "2022-05-09T16:01:10+0800",
    "end_time" : "2022-05-09T16:04:31+0800",
    "status" : "FAILED",
    "size" : 0
  } ],
  "total_count" : 1
}
```

Status Code

Status Code	Description
200	Normal

Error Code

For details, see **Error Codes**.

4.2.15 Restoring Data to the Original Instance or an Existing Instance

Function

This API is used to restore data to the original instance or an existing instance.

Constraints

- After the restoration is complete, you must confirm the data.
- Instances using XBSA backups support IDs of full backups and differential backups. Other instances support IDs of full backups only.

URI

POST /gaussdb/v3/instances/recovery

Request Parameters

Table 4-220 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Table 4-221 Request body parameters

Parameter	Mandatory	Type	Description
source	Yes	**Table 4-222** object	Source backup object information, including the backup ID and instance ID.
target	Yes	**Table 4-224** object	Target backup object information, including the instance ID. Currently, you can only restore data to the original instance or an existing instance.

Parameter	Mandatory	Type	Description
xbsa_extension_infos	No	Array of **Table 5 XbsaExtensionReq** objects	Parameter information used for XBSA-based restoration, including **XbsaRoute** and **BackupSet**. This parameter is mandatory in cross-cloud restoration scenarios.
restore_config	No	**Table 4-226**object	Restoration configurations.

Table 4-222 RecoveryBackupSource

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID. Currently, you can only restore data to the original instance or an existing instance.
type	No	String	Backup restoration type. The value can be **backup** or **timestamp**. • **backup** (default value): Data is restored using backups. In this case, **backup_id** is mandatory. • **timestamp**: Data is restored using point-in-time recovery (PITR). In this case, **restore_time** is mandatory. This type is available only to instance-level restoration. Database- and table-level restoration is not supported.
backup_id	Yes	String	ID of the backup to be restored.
restore_time	Yes	String	Timestamp for restoration by time point. For details about how to obtain the timestamp, see **Querying the Restoration Time Range**. For XBSA-based restoration, the available time range can be obtained by parsing third-party files.

Parameter	Mandatory	Type	Description
table_list	No	Array of **Table 4-223**	Table-level backup information. This parameter is required only when databases and tables are restored. **Constraints**: • Only instances later than V2.0-3.200 support NAS-based table-level backups. Only instances of the V2.0-8.200 version and later support XBSA-based table-level backups. • A maximum of 100 databases or tables can be restored at the same time. If there are more than 100 databases or tables, you are advised to use instance-level restoration.
schema_type	No	String	**Explanation**: Source backup type. This parameter is mandatory for PITR restoration and determines whether to use a full backup or a table-level backup for restoration. During restoration using a backup file, the backup type is automatically obtained based on the backup ID, and you do not need to set this parameter. **Constraints**: N/A **Value range**: • **INSTANCE**: instance-level backup • **DATABASE_TABLE**: table-level backup **Default value**: INSTANCE

Table 4-223 Data structure description of the TableList field

Parameter	Mandatory	Type	Description
db_name	Yes	String	**Explanation**: Name of the database to be restored. **Constraints**: ● M-compatible databases do not support database- and table-level backups and restorations. ● The following template databases and system users cannot be restored: – Template databases: **postgres**, **template0**, **templatem**, and **template1**. – System users: **rdsAdmin**, **rdsMetric**, **rdsBackup**, and **rdsRepl**. **Value range**: N/A **Default value**: N/A
schema_name	No	String	**Explanation**: Name of the schema to be restored. **Constraints**: System-level schemas, such as **dbe_application_info** and **rdsRepl**, cannot be restored. **Value range**: N/A **Default value**: N/A

Parameter	Mandatory	Type	Description
table_name	No	String	**Explanation**: Name of the table to be restored. **Constraints**: N/A **Value range**: N/A **Default value**: N/A
new_db_name	No	String	**Explanation**: Name of the new database. **Constraints**: ● The name cannot be the same as that of existing databases in the original or existing instance. ● It must be different from template databases. Template databases: **postgres**, **template0**, **templatem**, **template1**, and **templatea**. The database name cannot be the system users **rdsAdmin**, **rdsMetric**, **rdsBackup**, and **rdsRepl**. **Value range**: A database name can contain 1 to 63 characters. Only letters, digits, and underscores (_) are allowed. It cannot start with **pg** or a digit. **Default value**: N/A

Parameter	Mandatory	Type	Description
new_schema_name	No	String	**Explanation**: Name of the new schema. **Constraints**: ● The name cannot be the same as that of existing schemas in the original or existing instance. ● The name cannot be the same as that of a template database or an existing schema. Template databases include **postgres**, **template0**, **template1**, **templatem**, and **templatea**. Existing schemas include **public** and **information_schema**. In addition, system-level schemas such as **dbe_application_info** and **rdsRepl** cannot be restored. **Value range**: A schema name can contain 1 to 63 characters. Only letters, digits, and underscores (_) are allowed. It cannot start with **pg** or a digit. **Default value**: N/A

Parameter	Mandatory	Type	Description
new_table_name	No	String	**Explanation**: Name of the new table. **Constraints**: The name cannot be the same as that of existing tables in the original or existing instance. **Value range**: A schema name can contain 1 to 63 characters. Only letters, digits, and underscores (_) are allowed. It cannot start with **pg** or a digit. **Default value**: N/A

Table 4-224 RecoveryBackupTarget

Parameter	Mandatory	Type	Description
instance_id	Yes	String	ID of the original instance or an existing instance on which data is restored to.

Table 4-225 XbsaExtensionReq

Parameter	Mandatory	Type	Description
key	Yes	String	Extended parameter name. The value can be **BackupSet** or **XbsaRoute**.
value	Yes	String	Value of the extended parameter. The value can be the path of **BackupSet** or the address used by BackupAgent of **XbsaRoute**.

Table 4-226 RestoreConfig

Parameter	Mandatory	Type	Description
restore_parallel_degree	No	String	Number of concurrent restoration tasks. A restoration task is split into multiple subtasks to improve restoration efficiency. The default value is **1**. Value: **1**, **2**, **4**, or **8** **NOTE** • This parameter is only available for instances whose DB engine version is V2.0-9.0.0.SPC0100 or later. • This parameter is only available for an instance having at least 4 vCPUs. • This parameter is only available when an instance-level restoration is performed.

Response Parameters

Status code: 202

Table 4-227 Response body parameters

Parameter	Type	Description
job_id	String	Job ID for restoring data to the original instance or an existing instance.

Example Request

- Restoring data to an existing instance using a backup file

```
POST https://{Endpoint}/rds/gaussdb/v3/instances/recovery

{
 "source": {
  "instance_id": "52b401689f984cd8b12b18c889fc2120in14",
  "backup_id": "09e0c95b347049dcb27ff9ce12fee005br14",
  "type": "backup"
 },
 "target": {
  "instance_id": "52b401689f984cd8b12b18c889fc2120in14"
 }
}
```

- Restoring a XBSA-based instance backup file to an existing instance

```
POST https://{Endpoint}/rds/gaussdb/v3/instances/recovery

{
 "source": {
```

```
  "instance_id": "52b401689f984cd8b12b18c889fc2120in14",
  "backup_id": "09e0c95b347049dcb27ff9ce12fee005br14",
  "type": "backup"
 },
 "target": {
  "instance_id": "52b401689f984cd8b12b18c889fc2120in14"
 },
 "xbsa_extension_infos": [
 {
    "key" : "BackupSet",
    "value" : "/opt/xbsa/conf/xbsa_gaussdb_restore.conf"
  },
 {
    "key": "XbsaRoute",
    "value": "{\"port\":\"5000\",\"ip\":\"{{backupAgentIp}}\"}"
  }
 ]
}
```

- Restoring a table-level backup to an existing instance using PITR

```
POST https://{Endpoint}/rds/gaussdb/v3/instances/recovery

{
   "source" : {
     "instance_id" : "be028a90767144f1a1f42d48085ce45cin14",
     "type" : "timestamp",
     "restore_time" : "1752820257000",
     "schema_type":"DATABASE_TABLE",
     "table_list": [
        {
            "db_name": "core",
            "schema_name": "my_schema",
            "table_name": "test",
            "new_db_name": "core1",
            "new_schema_name": "my_schema1",
            "new_table_name": "test1"
        }
    ]
   },
   "target" : {
     "instance_id" : "5c5013552fef4dc1b17ae93e2da5f73ain14"
   }
 }
```

- Restoring a full table-level backup to an existing instance using PITR

```
POST https://{Endpoint}/rds/gaussdb/v3/instances/recovery

{
   "source" : {
     "instance_id" : "be028a90767144f1a1f42d48085ce45cin14",
     "type" : "timestamp",
     "restore_time" : "1752820257000",
     "schema_type":"INSTANCE",
     "table_list": [
        {
            "db_name": "core",
            "schema_name": "my_schema",
            "table_name": "test",
            "new_db_name": "core1",
            "new_schema_name": "my_schema1",
            "new_table_name": "test1"
        }
    ]
   },
   "target" : {
     "instance_id" : "5c5013552fef4dc1b17ae93e2da5f73ain14"
   }
 }
```

- Restoring a full database-level backup to an existing instance using PITR

```
POST https://{Endpoint}/rds/gaussdb/v3/instances/recovery

{
  "source" : {
    "instance_id" : "be028a90767144f1a1f42d48085ce45cin14",
    "type" : "timestamp",
    "restore_time" : "1752820257000",
    "schema_type":"INSTANCE",
    "table_list": [
       {
          "db_name": "core",
          "new_db_name": "core1"
       }
    ]
  },
  "target" : {
    "instance_id" : "5c5013552fef4dc1b17ae93e2da5f73ain14"
  }
}
```

Example Response

Status code: 202

```
{
  "job_id" : "dff1d289-4d03-4942-8b9f-463ea07c000d"
}
```

Status Code

Status Code	Description
202	Normal

Error Code

For details, see **Error Codes**.

4.2.16 Confirming Data Integrity After a Backup Is Restored

Function

This API is used to confirm data integrity after a backup is restored. This API can be called only after data is restored to the original instance. Once data integrity has been confirmed, any logs archived after the point in time data was restored from will be lost, but normal log archiving will be restored.

Constraints

Instances using XBSA backups are not supported.

URI

POST /gaussdb/v3/instances/{instance_id}/confirm-restore-data

Table 4-228 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-229 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Response Parameters

Status code: 202

Table 4-230 Response body parameters

Parameter	Type	Description
job_id	String	Job ID for querying task information.

Example Request

```
POST https://{Endpoint}/rds/gaussdb/v3/instances/a48e43ff268f4c0e879652d65e63d0fbin14/confirm-restore-data
```

Example Response

Status code: 202

All data restored.

```
{
  "job_id": "b0a39f5b-6572-42bf-99b8-3f79ad4b59a2"
}
```

Status Code

Status Code	Description
202	Normal

Error Code

For details, see **Error Codes**.

4.2.17 Querying Backups

Function

This API is used to query backups.

Constraints

This API can be used to query only manual and automated full backups of GaussDB instances, including XBSA manual full backups, manual log backups, XBSA cross-cloud full backups, and XBSA cross-cloud log backups.

URI

GET /gaussdb/v3.1/backups

Table 4-231 Query parameters

Parameter	Mandatory	Type	Description
instance_id	No	String	Instance ID.
backup_id	No	String	Backup ID.

Parameter	Mandatory	Type	Description
backup_type	No	String	Backup type. Value: • **auto**: instance-level automated full backup • **manual**: instance-level manual full backup • **Snapshot_Xbsa**: XBSA manual full backup • **Log_Xbsa**: XBSA log backup • **OffSite_Snapshot_Xbsa**: XBSA cross-cloud full backup • **OffSite_Log_Xbsa**: XBSA cross-cloud log backup • **Snapshot_DatabaseTable_Xbsa**: XBSA-based manual table-level backup • **Offsite_Snapshot_DatabaseTable_Xbsa**: XBSA-based manual cross-cloud table-level backup **NOTICE** • If **backup_type** is set to **Snapshot_Xbsa** or **Log_Xbsa**, instance version must be V2.0-3.301 or later. • If **backup_type** is left blank, the query results do not contain XBSA backups.
offset	No	Integer	Index offset. If **offset** is set to *N*, the resource query starts from the *N*+1 piece of data. The value is **0** by default, indicating that the query starts from the first piece of data. The value must be a number but cannot be a negative number. Minimum value: **0** Maximum value: **999999999**

Parameter	Mandatory	Type	Description
limit	No	Integer	Number of records to be queried. The default value is **100**. The value cannot be a negative number. The minimum value is **1** and the maximum value is **100**. Minimum value: **1** Maximum value: **100**
begin_time	No	String	Query start time in the "yyyy-mm-ddThh:mm:ssZ" format. **T** is the separator between calendar and hourly notation of time. **Z** indicates the time zone offset. For example, in the Beijing time zone, the offset is **+0800**. This parameter must be used together with **end_time**.
end_time	No	String	Query end time. The format is "yyyy-mm-ddThh:mm:ssZ" and the end time must be later than the start time. **T** is the separator between calendar and hourly notation of time. **Z** indicates the time zone offset. For example, in the Beijing time zone, the offset is **+0800**. This parameter must be used together with **begin_time**.

Request Parameters

Table 4-232 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Response Parameters

Status code: 200

Table 4-233 Response body parameters

Parameter	Type	Description
backups	Array of **Table 4-234** objects	Backup information.
total_count	Long	Total number of backup files.

Table 4-234 BackupsResult

Parameter	Type	Description
id	String	Backup ID.
name	String	Backup name.
description	String	Description of the backup file.
begin_time	String	Backup start time in the "yyyy-mm-ddThh:mm:ssZ" format. **T** is the separator between calendar and hourly notation of time. **Z** indicates the time zone offset. For example, in the Beijing time zone, the offset is shown as **+0800**.
end_time	String	Backup end time in the "yyyy-mm-ddThh:mm:ssZ" format. **T** is the separator between calendar and hourly notation of time. **Z** indicates the time zone offset. For example, in the Beijing time zone, the offset is shown as **+0800**.
status	String	Backup status. Enumerated values: ● **BUILDING**: Backup in progress ● **COMPLETED**: Backup completed ● **FAILED**: Backup failed
size	Double	Backup size (or disk size if the snapshot backup mode is used), in MB.

Parameter	Type	Description
type	String	Backup type. Enumerated values: ● **auto**: automated full backup ● **manual**: manual full backup ● **Log_Xbsa**: log backup
detail_message	String	Backup details. If the backup is created, the value is **success** or left blank. If a backup fails to be created, an error message is displayed. This parameter is specified only when log backups are created. For other backup types, this parameter is left blank.
datastore	Table 4-235 object	Database information.
instance_id	String	Instance ID.
file_name	String	Backup file path. Only NAS paths can be displayed. Minimum length: 0 character Maximum length: 256 characters
backup_method	String	**Explanation**: Backup method. **Value range**: ● **Physics**: physical backup

Table 4-235 GaussDatastoreResult

Parameter	Type	Description
type	String	DB engine. The value is case-insensitive and can be: **GaussDB** Value: **GaussDB**
version	String	DB engine version.

Example Request

```
GET https://{Endpoint}/rds/gaussdb/v3.1/backups?instance_id=a48e43ff268f4c0e879652d65e63d0fbin14
```

Example Response

Status code: 200

Success.


```
{
 "backups": [{
  "id": "09e0c95b347049dcb27ff9ce12fee005br14",
  "name": "GaussDB-Node-Replace-wanglin-20230919024605633",
  "status": "COMPLETED",
  "size": 50.268553734375,
  "type": "auto",
  "begin_time": "2023-09-19T10:46:05+0800",
  "end_time": "2023-09-19T10:49:10+0800",
  "instance_id": "52b401689f984cd8b12b18c889fc2120in14",
  "file_name": "/52b401689f984cd8b12b18c889fc2120in14/Db_Nas/1695091565640",
  "backup_method": "Physics",
  "datastore": {
   "type": "GaussDB",
   "version": "8.0.1"
  }
 }],
 "total_count": 1
}
```


Status Code

Status Code	Description
200	Normal

Error Code

For details, see **Error Codes**.

4.2.18 (Recommended) Querying Backups

Function

This API is used to query backups.

Constraints

- This API can be used to query only manual and automated full backups of GaussDB instances, including XBSA manual full backups, manual log backups, XBSA cross-cloud full backups, and XBSA cross-cloud log backups.
- You can simultaneously send a maximum of 80 requests of querying backups per second by calling this API.

URI

GET /gaussdb/v3.2/backups

Table 4-236 Query parameters

Parameter	Mandatory	Type	Description
instance_id	No	String	Instance ID.
backup_id	No	String	Backup ID.
backup_type	No	String	Backup type. Value: • **auto**: instance-level automated full backup • **manual**: instance-level manual full backup • **Snapshot_Xbsa**: XBSA manual full backup • **Log_Xbsa**: XBSA log backup • **OffSite_Snapshot_Xbsa**: XBSA cross-cloud full backup • **OffSite_Log_Xbsa**: XBSA cross-cloud log backup • **Snapshot_DatabaseTable_Xbsa**: XBSA-based manual table-level backup • **Offsite_Snapshot_DatabaseTable_Xbsa**: XBSA-based manual cross-cloud table-level backup **NOTICE** • If **backup_type** is set to **Snapshot_Xbsa** or **Log_Xbsa**, instance version must be V2.0-3.301 or later. • If **backup_type** is left blank, the query results do not contain XBSA backups.
offset	No	Integer	Index offset. If **offset** is set to *N*, the resource query starts from the *N*+1 piece of data. The value is **0** by default, indicating that the query starts from the first piece of data. The value must be a number but cannot be a negative number. Minimum value: **0** Maximum value: **999999999**

Parameter	Mandatory	Type	Description
limit	No	Integer	Number of records to be queried. The default value is **100**. The value cannot be a negative number. The minimum value is **1** and the maximum value is **100**. Minimum value: **1** Maximum value: **100**
begin_time	No	String	Query start time in the "yyyy-mm-ddThh:mm:ssZ" format. **T** is the separator between calendar and hourly notation of time. **Z** indicates the time zone offset. For example, in the Beijing time zone, the offset is **+0800**. This parameter must be used together with **end_time**.
end_time	No	String	Query end time. The format is "yyyy-mm-ddThh:mm:ssZ" and the end time must be later than the start time. **T** is the separator between calendar and hourly notation of time. **Z** indicates the time zone offset. For example, in the Beijing time zone, the offset is **+0800**. This parameter must be used together with **begin_time**.

Request Parameters

Table 4-237 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Response Parameters

Status code: 200

Table 4-238 Response body parameters

Parameter	Type	Description
backups	Array of **Table 4-239** objects	Backup information.
total_count	Long	Total number of backup files.

Table 4-239 BackupsResult

Parameter	Type	Description
id	String	Backup ID.
name	String	Backup name.
description	String	Description of the backup file.
begin_time	String	Backup start time in the "yyyy-mm-ddThh:mm:ssZ" format. **T** is the separator between calendar and hourly notation of time. **Z** indicates the time zone offset. For example, in the Beijing time zone, the offset is shown as **+0800**.
end_time	String	Backup end time in the "yyyy-mm-ddThh:mm:ssZ" format. **T** is the separator between calendar and hourly notation of time. **Z** indicates the time zone offset. For example, in the Beijing time zone, the offset is shown as **+0800**.
status	String	Backup status. Enumerated values: ● **BUILDING**: Backup in progress ● **COMPLETED**: Backup completed ● **FAILED**: Backup failed
size	Double	Backup size in MB.
type	String	Backup type. Enumerated values: ● **auto**: automated full backup ● **manual**: manual full backup ● **Log_Xbsa**: log backup

Parameter	Type	Description
detail_message	String	Backup details. If the backup is created, the value is **success** or left blank. If a backup fails to be created, an error message is displayed. This parameter is specified only when log backups are created. For other backup types, this parameter is left blank.
datastore	Table 4-240 object	Database information.
instance_id	String	Instance ID.
file_name	String	Backup file path. Only NAS paths can be displayed. Minimum length: 0 character Maximum length: 256 characters
backup_method	String	**Explanation**: Backup method. **Value range**: • **Physics**: physical backup

Table 4-240 GaussDatastoreResult

Parameter	Type	Description
type	String	DB engine. The value is case-insensitive and can be: GaussDB Value: **GaussDB**
version	String	DB engine version.

Example Request

```
GET https://{Endpoint}/rds/gaussdb/v3.2/backups?instance_id=a48e43ff268f4c0e879652d65e63d0fbin14
```

Example Response

Status code: 200

Success.

```
{
 "backups": [{
  "id": "09e0c95b347049dcb27ff9ce12fee005br14",
  "name": "GaussDB-Node-Replace-wanglin-20230919024605633",
  "status": "COMPLETED",
  "size": 50.268553734375,
```

```
  "type": "auto",
  "begin_time": "2023-09-19T10:46:05+0800",
  "end_time": "2023-09-19T10:49:10+0800",
  "instance_id": "52b401689f984cd8b12b18c889fc2120in14",
  "file_name": "/52b401689f984cd8b12b18c889fc2120in14/Db_Nas/1695091565640",
  "backup_method": "Physics",
  "datastore": {
   "type": "GaussDB",
   "version": "V2.0-8.0.1"
  }
 }],
 "total_count": 1
}
```


Status Code

Status Code	Description
200	Normal

Error Code

For details, see **Error Codes**.

4.2.19 Querying the Archive Log Path of an Instance

Function

This API is used to query the archive log path of an instance. Instances using XBSA backups are not supported.

URI

GET /gaussdb/v3/instances/{instance_id}/archive-log-path

Table 4-241 URI parameters

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-242 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.

Parameter	Mandatory	Type	Description
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Response Parameters

Status code: 200

Table 4-243 Response body parameters

Parameter	Type	Description
path	String	Archive log path. If an empty string is returned, database archive is disabled for the instance.

Example Request

Querying the archive log path of an instance

```
GET  https://{Endpoint}/rds/gaussdb/v3/instances/d8e6ca5a624745bcb546a227aa3ae1cfin14/archive-log-path
```

Example Response

Status code: 200

Success.

```
{
  "path" : "fdf495076c1e4fadb3c6eadfc41246ff_967a31db0adf41d9b7121858e0a6424bin14/Log"
}
```

Status Code

Status Code	Description
200	Normal

Error Code

For details, see **Error Codes**.

4.2.20 Synchronizing Metadata Using Storage Media

Function

This API is used to synchronize metadata using the storage medium.

Constraints

- **gaussdb_feature_supportSyncBackupCrossSite** must be enabled in the feature whitelist.
- There is an available instance using XBSA backups on the destination end. The extended information of the instance has been configured, and the third-party SSL certificate has been installed.

URI

POST /gaussdb/v3/backups/sync-metadata

Request Parameters

Table 4-244 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Table 4-245 Request body parameters

Parameter	Mandatory	Type	Description
source_instance_id	Yes	String	ID of the instance to be synchronized. The backups for the instances have been created.

Parameter	Mandatory	Type	Description
prepared_instance_id	Yes	String	ID of the XBSA storage-based instance that performs synchronization at the destination end. Extended information such as the IP address and port number must be configured in advance. For details, see **Configuring Extended Information for an Instance**.
media_type	Yes	String	Storage medium. Only XBSA is supported.
start_time	Yes	String	Start timestamp of the metadata to be synchronized. The value is accurate to milliseconds. The granularity of timestamp is the hour during the synchronization. The maximum synchronization interval is 100 days.
end_time	Yes	String	End timestamp of the metadata to be synchronized. The value is accurate to milliseconds. The granularity of timestamp is the hour during the synchronization. The maximum synchronization interval is 100 days.

Response Parameters

Status code: 202

Table 4-246 Response body parameters

Parameter	Type	Description
job_id	String	Job ID.

Status code: default

Table 4-247 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

Synchronizing metadata to storage media

```
POST  https://{Endpoint}/rds/gaussdb/v3/backups/sync-metadata

{
  "start_time" : "1702365380000",
  "end_time" : "1702375380000",
  "media_type" : "XBSA",
  "prepared_instance_id" : "d8e6ca5a624745bcb546a227aa3ae1cfin14",
  "source_instance_id" : "1de27464abdf48519f8574c97a8f0625in14"
}
```

Example Response

Status code: 202

Response body

```
{
  "job_id" : "a12022b8-cf02-4d36-9fd1-147ece691ca1"
}
```

Status Code

Status Code	Description
202	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.2.21 Deleting a Backup

Function

This API is used to delete XBSA-based backups.

URI

DELETE /gaussdb/v3/backups/batch-delete

Request Parameters

Table 4-248 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Table 4-249 Request body parameters

Parameter	Mandatory	Type	Description
backup_ids	No	Array of strings	IDs of the backups to be deleted. To view backup IDs, see **Querying Backups**. Up to 100 IDs are supported. If this parameter is set to an empty array or is left blank, no data is deleted.
instance_id	Yes	String	Instance ID.

Response Parameters

Status code: 200

Table 4-250 Response body parameters

Parameter	Type	Description
delete_results	Array of **Table 4-251** objects	Deletion results.

Table 4-251 DeleteBackupResult

Parameter	Type	Description
backup_id	String	ID of the backup to be deleted.
error_code	String	Error code.

Parameter	Type	Description
error_msg	String	Error message.

Example Request

Deleting synchronized metadata records

```
DELETE  https://{Endpoint}/rds/gaussdb/v3/backups/batch-delete

{
  "backup_ids" : [ "9ef7d7c4b5f511ed8815fa163e061a84br14" ],
  "instance_id" : "1de27464abdf48519f8574c97a8f0625in14"
}
```

Example Response

Status code: 200

Success.

```
{
  "delete_results" : [ {
    "backup_id" : "9ef7d7c4b5f511ed8815fa163e061a84br14",
    "error_code" : "0",
    "error_msg" : "Success"
  } ]
}
```

Status Code

Status Code	Description
200	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.2.22 Installing an SSL Certificate

Function

This API is used to install a certificate. The certificate parameters are transferred in Base64 encoding format by default, and the task flow ID is returned.

URI

POST /gaussdb/v3/instances/{instance_id}/backups/ssl-certs

Table 4-252 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-253 Request body parameters

Parameter	Mandatory	Type	Description
ca_cert_pem	Yes	String	Root certificate file, which is encoded using Base64. Minimum length: 0 character Maximum length: 65536 characters
client_crt	Yes	String	User certificate, which is encoded using Base64. Minimum length: 0 character Maximum length: 65536 characters
client_key	Yes	String	Private key, which is encoded using Base64. Minimum length: 0 character Maximum length: 65536 characters
rand_pass	Yes	String	Client private key password, which is encoded using Base64. Minimum length: 0 character Maximum length: 65536 characters

Response Parameters

Status code: 202

Table 4-254 Response body parameters

Parameter	Type	Description
job_id	String	Job ID in UUID format. Minimum length: 2 characters Maximum length: 128 characters

Status code: 400

Table 4-255 Response body parameters

Parameter	Type	Description
error_code	String	Error code. Minimum length: 8 characters Maximum length: 36 characters
error_msg	String	Error message. Minimum length: 2 characters Maximum length: 512 characters

Example Request

```
POST https://{Endpoint}/rds/gaussdb/v3/instances/43e4feaab48f11e89039fa163ebaa7e4in02/backups/ssl-certs

{
  "ca_cert_pem" : "******",
  "client_crt" : "******",
  "client_key" : "*****",
  "rand_pass" : "******"
}
```

Example Response

Status code: 202

Request for installing the backup certificate succeeded.

```
{
  "job_id" : "a03b1b8a-b756-467c-8a49-38720c3d23ec"
}
```

Status Code

Status Code	Description
202	Normal
400	Abnormal

Error Code

For details, see **Error Codes**.

4.2.23 Changing a Backup Method

Function

- This API is used to change a backup method when XBSA is used as backup media. Physical backup can be changed to snapshot backup, and the reverse also applies.
- Only centralized instances can use snapshot backup.
- The instance version must be later than V2.0-2.7.0.
- The disk type must be Dorado.

URI

PUT /gaussdb/v3.1/instances/{instance_id}/change-backup-media

Table 4-256 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-257 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token. Minimum length: 0 character Maximum length: 32768 characters
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Table 4-258 Request body parameters

Parameter	Mandatory	Type	Description
default_backup_method	Yes	String	Backup method. • **PHYSICAL_BACKUP**: physical backup • **EBACKUP**: snapshot backup
default_backup_media_type	Yes	String	Backup media. • **XBSA**: The backup method can be changed only when XBSA is used.

Response Parameters

Status code: 202

Table 4-259 Response body parameters

Parameter	Type	Description
job_id	String	Job ID.

Example Request

Changing a backup method:

```
PUT https://{Endpoint}/rds/gaussdb/v3.1/instances/d8e6ca5a624745bcb546a227aa3ae1cfin14/change-backup-media

{
   "default_backup_method" : "EBACKUP",
   "default_backup_media_type": "XBSA"
}
```

Example Response

```
{
  "job_id" : "a03b1b8a-b756-467c-8a49-38720c3d23ec"
}
```

Status Code

Status Code	Description
202	Normal

Error Code

For details, see **Error Codes**.

4.2.24 Querying Backup Details

Function

This API is used to query backup details based on the workflow ID (**job_id**) or backup ID (**backup_id**).

URI

GET /gaussdb/v3/instances/{instance_id}/backups/detail

Table 4-260 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-261 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token. Minimum length: 0 character Maximum length: 32768 characters
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Table 4-262 Request body parameters

Parameter	Mandatory	Type	Description
job_id	No	String	Workflow ID used to query backup details. This parameter is mandatory when **backup_id** is left empty.

Parameter	Mandatory	Type	Description
backup_id	No	String	Backup ID used to query backup details. This parameter is mandatory when **job_id** is left empty.

Response Parameters

Status code: 200

Table 4-263 Response body parameters

Parameter	Type	Description
instance_id	String	Instance ID.
status	String	Backup status. ● **BUILDING**: Backup in progress ● **COMPLETED**: Backup completed ● **FAILED**: backup failed
backup_node_infos	Array of **Table 4-264**	Information about the node and disks to be backed up. If **volumes** is not left empty, you can create a snapshot for each data volume of the node. Consistency snapshots must be created for all data volumes of the node.

Table 4-264 NodeInfo

Parameter	Type	Description
node_id	String	ID of the data node to be backed up.
volumes	Array of **Table 4-265**	Information about volumes to be backed up.

Table 4-265 VolumeInfo

Parameter	Type	Description
extra_iaas_id	String	Volume ID.

Example Request

Querying backup details:

```
GET https://{Endpoint}/rds/gaussdb/v3/instances/d8e6ca5a624745bcb546a227aa3ae1cfin14/backups/detail?
job_id=0b9928e1-a51c-4588-90d8-0d886bfbc7c4
```

Example Response

```
{
   "instance_id": "d8e6ca5a624745bcb546a227aa3ae1cfin14",
   "status": "COMPLETED",
   "backup_node_infos": [
      {
         "node_id": "c05b01bdd34b4b3e824cabc445436e4cno14",
         "volumes": [
            {
               "extra_iaas_id": "630e98e100e191e10a34d9d600000065"
            }
         ]
      }
   ]
}
```

Status Code

Status Code	Description
200	Normal

Error Code

For details, see **Error Codes**.

4.2.25 Querying Restoration Details

Function

This API is used to query restoration details based on the workflow ID (**job_id**).

URI

GET /gaussdb/v3/instances/{instance_id}/recovery/detail

Table 4-266 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-267 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token. Minimum length: 0 character Maximum length: 32768 characters
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Table 4-268 Request body parameters

Parameter	Mandatory	Type	Description
job_id	No	String	Workflow ID returned when a restoration task was initiated. You can query restoration details based on the workflow ID.

Response Parameters

Status code: 200

Table 4-269 Response body parameters

Parameter	Type	Description
instance_id	String	Instance ID.
backup_id	String	ID of the full backup used during the restoration.
status	String	Restoration status. The value can be: ● **Running**: The restoration is in progress. ● **Failed**: The restoration fails. ● **Completed**: The restoration is complete.
restore_node_infos	Array of **Table 4-270**	Information about the node to be restored. If the value of **volumes** is returned, the restoration can be performed.

Table 4-270 NodeInfo

Parameter	Type	Description
node_id	String	ID of a DN.
volumes	Array of **Table 4-271**	Information about volumes to be backed up.

Table 4-271 VolumeInfo

Parameter	Type	Description
extra_iaas_id	String	Volume ID.

Example Request

Querying restoration details:

```
GET https://{Endpoint}/rds/gaussdb/v3/instances/d8e6ca5a624745bcb546a227aa3ae1cfin14/recovery/detail?
job_id=0b9928e1-a51c-4588-90d8-0d886bfbc7c4
```

Example Response

```
{
    "instance_Id": "d8e6ca5a624745bcb546a227aa3ae1cfin14",
    "backup_id": "34df435b4c764a56a01009a9e06eb9e0br14",
    "status": "Completed",
    "restore_node_infos": [
        {
            "node_id": "83583af97bd4494294360e9bcad5d678no14",
            "volumes": [
                {
                    "extra_iaas_id": "630e98e100e191e10a34cf3900000064"
                }
            ]
        },
        {
            "node_id": "c05b01bdd34b4b3e824cabc445436e4cno14",
            "volumes": [
                {
                    "extra_iaas_id": "630e98e100e191e10a34d9d600000065"
                }
            ]
        },
        {
            "node_id": "fe22f037b1a5456087f342410d07e44bno14",
            "volumes": [
                {
                    "extra_iaas_id": "630e98e100e191e10a34c49200000063"
                }
            ]
        }
    ]
}
```


Status Code

Status Code	Description
200	Normal

Error Code

For details, see **Error Codes**.

4.2.26 Sending Backup or Restoration Results

Function

This API is used to connect to an external storage system and send the operation result of snapshot creation, copy, or restoration from the external system to TPOPS.

URI

PUT /gaussdb/v3/instances/{instance_id}/backups/notify

Table 4-272 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-273 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token. Minimum length: 0 character Maximum length: 32768 characters
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Table 4-274 Request body parameters

Parameter	Mandatory	Type	Description
job_id	Yes	String	Workflow ID
action	Yes	String	Backup or restoration action. Enumerated values: ● **snapshot**: Create a snapshot. ● **checkPoint**: Copy a snapshot. ● **restore**: Restore data.
status	Yes	String	Backup or restoration status. Enumerated values: ● **Running**: The backup or restoration task is running. ● **Active**: The backup or restoration task is successful. ● **Failed**: The backup or restoration task fails.

Response Parameters

Status code: 200

Example Request

Sending a restoration result:

```
PUT https://{Endpoint}/rds/gaussdb/v3/instances/d8c8371b418b445bb2994d8810c53ce2in14/backups/notify
{
    "job_id": "c20722df-108d-449d-856d-110ac427d119",
    "action": "restore",
    "status": "Active"
}
```

Example Response

None

Status Code

Status Code	Description
200	Normal

Error Code

For details, see **Error Codes**.

4.2.27 Restoring Data from a Backup Set

Function

This API is used to restore data from a backup set to an existing instance.

Constraints

- The database type of the destination instance must be the same as that of the source instance.
- Only restoration to existing instances is supported.
- Currently, only NAS backup files can be restored. Backup sets stored in different storage media cannot be restored. For example, XBSA backup sets cannot be restored on NAS devices.
- Restoration using full, differential, or database- and table-level backup sets is supported. PITR using log backups is not supported.
- When data is restored using a differential backup or table-level differential backup, the backup set path must contain a full backup or table-level full backup required.
- You can only select the instance with the same engine type, engine version, OS, CPU architecture, transaction consistency (for a distributed instance), replica consistency (for a centralized instance), and case sensitivity of M-compatible table names (for a centralized instance) as the original instance that the backup set belongs to. Additionally, it must have sufficient compute resources and have the same number of shards and at least as much storage space as the original instance. It cannot be a DR instance. After the restoration is complete, all archive logs of the target instance are deleted.

URI

POST /gaussdb/v3/instances/file-restore

Request Parameters

Table 4-275 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-token	No	String	User token. Minimum length: 0 character Maximum length: 32768 characters

Parameter	Mandatory	Type	Description
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Table 4-276 Request body parameters

Parameter	Mandatory	Type	Description
source	No	**Table 4-277** object	Instance from which the backup was created.
target	No	**Table 4-280** object	Instance to which the backup is restored.

Table 4-277 SourceInstanceInfo

Parameter	Mandatory	Type	Description
table_list	No	Array of **Table 4-278** objects	Table-level backup information. This parameter is required only when databases and tables are restored.
schema_type	No	String	Backup type of the original instance. The value can be: • **INSTANCE** (default value): instance-level backup • **DATABASE_TABLE**: table-level backup
backup_set	Yes	**Table 4-279** object	Information about the device and directory of the backup set.

Table 4-278 Data structure description of the TableList field

Parameter	Mandatory	Type	Description
db_name	Yes	String	**Explanation**: Name of the database to be restored. **Constraints**: ● M-compatible databases do not support database- and table-level backups and restorations. ● The following template databases and system users cannot be restored: – Template databases: **postgres**, **template0**, **templatem**, and **template1**. – System users: **rdsAdmin**, **rdsMetric**, **rdsBackup**, and **rdsRepl**. **Value range**: N/A **Default value**: N/A
schema_name	No	String	**Explanation**: Name of the schema to be restored. **Constraints**: System-level schemas, such as **dbe_application_info** and **rdsRepl**, cannot be restored. **Value range**: N/A **Default value**: N/A

Parameter	Mandatory	Type	Description
table_name	No	String	**Explanation**: Name of the table to be restored. **Constraints**: N/A **Value range**: N/A **Default value**: N/A
new_db_name	No	String	**Explanation**: Name of the new database. **Constraints**: ● The name cannot be the same as that of an existing database in an existing instance. ● It must be different from template databases. Template databases: **postgres**, **template0**, **templatem**, **template1**, and **templatea**. The database name cannot be the system users **rdsAdmin**, **rdsMetric**, **rdsBackup**, and **rdsRepl**. **Value range**: A database name can contain 1 to 63 characters. Only letters, digits, and underscores (_) are allowed. It cannot start with **pg** or a digit. **Default value**: N/A

Parameter	Mandatory	Type	Description
new_schema_name	No	String	**Explanation**: Name of the new schema. **Constraints**: ● The name cannot be the same as that of an existing schema in an existing instance. ● The name cannot be the same as that of a template database or an existing schema. Template databases include **postgres**, **template0**, **template1**, **templatem**, and **templatea**. Existing schemas include **public** and **information_schema**. In addition, system-level schemas such as **dbe_application_info** and **rdsRepl** cannot be restored. **Value range**: A schema name can contain 1 to 63 characters. Only letters, digits, and underscores (_) are allowed. It cannot start with **pg** or a digit. **Default value**: N/A

Parameter	Mandatory	Type	Description
new_table_name	No	String	**Explanation**: Name of the new table. **Constraints**: The name cannot be the same as that of an existing table in an existing instance. **Value range**: A schema name can contain 1 to 63 characters. Only letters, digits, and underscores (_) are allowed. It cannot start with **pg** or a digit. **Default value**: N/A

Table 4-279 BackupSet

Parameter	Mandatory	Type	Description
device_id	Yes	String	ID of the device where the backup set is located.
media_type	No	String	Backup media type. The default value is **NAS**.
backup_file_path	Yes	String	Directory of the backup set.

Table 4-280 TargetInstanceInfo

Parameter	Mandatory	Type	Description
instance_id	Yes	String	ID of the instance the backup will be restored to.

Response Parameters

Status code: 202

Table 4-281 Response body parameters

Parameter	Type	Description
job_id	String	Task ID.

Example Request

Restoring data from a backup set

```
https://{Endpoint}/rds/gaussdb/v3/instances/file-restore
{
  "source": {
    "backup_set": {
      "device_id": "57101c4f-660d-4bbc-b379-8a1b7de9b957",
      "backup_file_path": "/4fda527b7d5f46d2b86007bb3e228298in14/Snapshot_Nas/20251010113750"
    },
    "schema_type": "INSTANCE",
    "table_list": [
      {
        "db_name": "test",
        "schema_name": "testschema",
        "table_name": "testtable",
        "new_table_name": "newtesttable"
      }
    ]
  },
  "target": {
    "instance_id": "4fda527b7d5f46d2b86007bb3e228298in14"
  }
}
```

Example Response

Status code: 202

```
{
  "job_id" : "a03b1b8a-b756-467c-8a49-38720c3d23ec"
}
```

Status Codes

Status Code	Description
202	Normal

Error Codes

For details, see **Error Codes**.

4.3 Parameter Configuration

4.3.1 Obtaining the Parameter Template of a Specified Instance

Function

This API is used to obtain parameters of a specified instance.

URI

GET /gaussdb/v3.1/instances/{instance_id}/configurations

Table 4-282 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-283 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Response Parameters

Status code: 200

Table 4-284 Response body parameters

Parameter	Type	Description
datastore_version	String	DB engine version.
datastore_name	String	Engine name.
created	String	Creation time in the "yyyy-MM-dd HH:mm:ss" format.

Parameter	Type	Description
updated	String	Update time in the "yyyy-MM-ddHH:mm:ss" format.
configuration_parameters	Array of **Table 4-285** objects	Parameter object.

Table 4-285 configuration_parameters

Parameter	Type	Description
name	String	Parameter name.
value	String	Parameter value.
restart_required	Boolean	Whether a reboot is required.
value_range	String	Parameter value range.
type	String	Parameter type.
description	String	Parameter description.
is_risk_parameter	Boolean	Whether it is a high-risk parameter.
risk_description	String	Risks and impacts of improper settings.

Status code: default

Table 4-286 Response body parameters

Parameter	Type	Description
error_code	String	Error code. Minimum length: 8 characters Maximum length: 36 characters
error_msg	String	Error message. Minimum length: 2 characters Maximum length: 512 characters

Example Request

Obtaining parameters of an instance

```
GET https://{Endpoint}/rds/gaussdb/v3.1/instances/3d39c18788b54a919bab633874c159dfin14/
configurations
```

Example Response

Status code: 200

Success.


```
{
  "datastore_version" : "1.4",
  "datastore_name" : "GaussDB",
  "created" : "2023-06-11 11:40:44",
  "updated" : "2023-06-11 11:40:44",
  "configuration_parameters" : [ {
    "name" : "cn:enable_tsdb",
    "value" : "on",
    "restart_required" : false,
    "value_range" : "on,off",
    "type" : "integer",
    "description" : "whether to enable the time series database feature."
    "is_risk_parameter": "0",
    "risk_description":""
  } ]
}
```


Status Code

Status Code	Description
200	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.3.2 (Recommended) Obtaining the Parameter Template of a Specified Instance

Function

This API is used to obtain parameters of a specified instance.

URI

GET /gaussdb/v3.2/instances/{instance_id}/configurations

Table 4-287 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-288 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Response Parameters

Status code: 200

Table 4-289 Response body parameters

Parameter	Type	Description
datastore_version	String	DB engine version.
datastore_name	String	Engine name.
created	String	Creation time in the "yyyy-MM-dd HH:mm:ss" format.
updated	String	Update time in the "yyyy-MM-ddHH:mm:ss" format.
configuration_parameters	Array of **Table 4-290** objects	Parameter object.

Table 4-290 configuration_parameters

Parameter	Type	Description
name	String	Parameter name.
value	String	Parameter value.
restart_required	Boolean	Whether a reboot is required.
value_range	String	Parameter value range.
type	String	Parameter type.

Parameter	Type	Description
description	String	Parameter description.
is_risk_param eter	Boolean	Whether it is a high-risk parameter.
risk_descriptio n	String	Risks and impacts of improper settings.

Status code: default

Table 4-291 Response body parameters

Parameter	Type	Description
error_code	String	Error Code Minimum length: 8 characters Maximum length: 36 characters
error_msg	String	Error message. Minimum length: 2 characters Maximum length: 512 characters

Example Request

Obtaining parameters of an instance

```
GET https://{Endpoint}/rds/gaussdb/v3.2/instances/3d39c18788b54a919bab633874c159dfin14/
configurations
```

Example Response

Status code: 200

Success.

```
{
  "datastore_version" : "V2.0-1.4",
  "datastore_name" : "GaussDB",
  "created" : "2023-06-11 11:40:44",
  "updated" : "2023-06-11 11:40:44",
  "configuration_parameters" : [ {
    "name" : "cn:enable_tsdb",
    "value" : "on",
    "restart_required" : false,
    "value_range" : "on,off",
    "type" : "integer",
    "description" : "whether to enable the time series database feature."
    "is_risk_parameter": "0",
    "risk_description":""
  } ]
}
```

Status Code

Status Code	Description
200	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.3.3 Modifying Parameters of a Specified Instance

Function

This API is used to modify parameters of a specified instance.

Constraints

The parameter values to be modified must be within the default value range of the specified DB engine version.

URI

PUT /gaussdb/v3/instances/{instance_id}/configurations

Table 4-292 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-293 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Table 4-294 Request body parameter

Parameter	Mandatory	Type	Description
values	Yes	Table 4-295 object	Parameter values defined by users based on a default parameter template.
async	No	boolean	Whether to enable the asynchronous mode. The default value is **false**.

Table 4-295 values

Parameter	Mandatory	Type	Description
map	Yes	Map<String,String>	Parameter name and value.

Response Parameters

Status code: 200

Table 4-296 Response body parameters

Parameter	Type	Description
restart_required	Boolean	Whether a reboot is required.
job_id	String	Job ID.

Status code: default

Table 4-297 Response body parameters

Parameter	Type	Description
error_code	String	Error code. Minimum length: 8 characters Maximum length: 36 characters
error_msg	String	Error message. Minimum length: 2 characters Maximum length: 512 characters

Example Request

Changing the value of **failed_login_attempts** to **4**

```
PUT https://{Endpoint}/rds/gaussdb/v3/instances/dsfae23fsfdsae3435in14/configurations

{
  "async": true,
  "values" : {
    "failed_login_attempts" : "4"
  }
}
```

Example Response

Status code: 200

Success.

```
{
  "restart_required" : false,
  "job_id" : "assakjdjsdhshdhjsdg"
}
```

Status Code

Status Code	Description
200	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.4 Database and User Management

4.4.1 Creating a Database

Function

This API is used to create a database in a specified instance.

Constraints

This operation cannot be performed when the instance is in any of the following statuses: creating, changing instance specifications, changing port, frozen, or abnormal. If you create a database using other methods instead of invoking a v3 API, for example, logging in to a node or using a client tool, the database name verification rule is inconsistent with that of the v3 API. As a result, the v3 API may fail to be invoked to perform operations on the database.

URI

POST /gaussdb/v3/instances/{instance_id}/database

Table 4-298 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-299 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Table 4-300 Request body parameters

Parameter	Mandatory	Type	Description
name	Yes	String	The database name can contain 1 to 63 characters. Only letters, digits, and underscores (_) are allowed. It cannot start with **pg** or a digit, and must be different from GaussDB template library names (**postgres**, **template0**, **templatem**, and **template1**).

Parameter	Mandatory	Type	Description
character_set	No	String	Character encoding. If this parameter is not specified, the character encoding of the new database is the same as that of the template library. For MySQL-compatible databases using the **templatem** template, only UTF8, GBK, GB18030, and SQL_ASCII character sets are supported. For MySQL-compatible databases created for DB instance 8.300 or later, the Latin1 character set is also supported.
owner	No	String	Database owner. The name of a database user must be that of an existing database user and cannot be the built-in users, including **rdsAdmin**, **rdsRepl**, **rdsBackup**, **rdsMetric**, **dbmind_monitor**, **dbmind_monitor_agent** or **hadr_disaster**. The default username is **root**.
template	No	String	Name of the database template. The available database templates are **postgres**, **template0**, **template1**, and **templatem**.
lc_collate	No	String	Database collocation. The default value is **C**.
lc_ctype	No	String	Database classification. The default value is **C**.

Response Parameters

Status code: default

Table 4-301 Response body parameters

Parameter	Type	Description
error_code	String	Error code.

Parameter	Type	Description
error_msg	String	Error message.

Example Request

```
POST https://{Endpoint}/rds/gaussdb/v3/instances/d8e6ca5a624745bcb546a227aa3ae1cfin14/database
{
  "name": "gauss_test",
  "owner": "test",
  "template": "template0",
  "character_set": "UTF8",
  "lc_collate": "C",
  "lc_ctype": "C"
}
```

Example Response

None

Status Code

Status Code	Description
200	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.4.2 Creating a Schema

Function

This API is used to create a database schema in a specified instance.

Constraints

This operation cannot be performed when the instance is in any of the following statuses: creating, changing instance specifications, changing port, frozen, or abnormal.

URI

POST /gaussdb/v3/instances/{instance_id}/schema

Table 4-302 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-303 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Table 4-304 Request body parameters

Parameter	Mandatory	Type	Description
db_name	Yes	String	Database name. It can contain 1 to 63 characters. Only letters, digits, and underscores (_) are allowed. It cannot start with **pg** or a digit, and must be different from GaussDB template library names (**postgres**, **template0**, **template1**, and **templatem**).
schemas	Yes	Array of **Table 4-305** objects	Schemas. Each element is the schema information associated with the database. A single request supports a maximum of 20 elements.

Table 4-305 GaussDBCreateSchemaReq

Parameter	Mandatory	Type	Description
name	Yes	String	Schema name. The value can contain 1 to 63 characters. Only letters, digits, and underscores (_) are allowed. It cannot start with **pg_**, **gs_role_**, or a digit, and must be different from initial user **rdsAdmin**, **rdsRepl**, **rdsBackup**, or **rdsMetric**.
owner	Yes	String	Owner of a schema. The value contains 1 to 63 characters. It cannot start with **pg** or a digit and must be different from system usernames. System users include **rdsAdmin**, **rdsRepl**, **rdsBackup**, **rdsMetric**, **root**, **dbmind_monitor**, **dbmind_monitor_agent**, and **hadr_disaster**.

Response Parameters

Status code: default

Table 4-306 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

```
POST https://{Endpoint}/rds/gaussdb/v3/instances/d8e6ca5a624745bcb546a227aa3ae1cfin14/schema

{
  "db_name" : "rds_test",
  "schemas" : [ {
    "name" : "rds",
    "owner" : "teste123"
  }, {
    "name" : "rds001",
    "owner" : "teste123"
  } ]
}
```


Example Response

None

Status Code

Status Code	Description
200	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.4.3 Querying Database Schemas

Function

This API is used to query database schemas of a specified instance.

Constraints

- This operation cannot be performed when the instance is in the abnormal or frozen state.
- The database schemas of read replicas cannot be queried.

URI

GET /gaussdb/v3/instances/{instance_id}/schemas

Table 4-307 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Table 4-308 Query parameters

Parameter	Mandatory	Type	Description
db_name	Yes	String	Database name.
offset	No	Integer	Offset, which is the position where the query starts. The value must be greater than or equal to 0.

Parameter	Mandatory	Type	Description
limit	No	Integer	Number of records to be queried. The default value is **10**. The value cannot be a negative number. The minimum value is **1** and the maximum value is **100**.

Request Parameters

Table 4-309 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Response Parameters

Status code: 200

Table 4-310 Response body parameters

Parameter	Type	Description
database_schemas	Array of **Table 4-311** objects	Each element in the list indicates a database schema.
total_count	Integer	Total number of database schemas.

Table 4-311 GaussDBDatabaseForListSchema

Parameter	Type	Description
schema_name	String	Schema name.
owner	String	Schema owner.

Status code: default

Table 4-312 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

```
GET https://{Endpoint}/rds/gaussdb/v3/instances/d8e6ca5a624745bcb546a227aa3ae1cfin14/schemas?
db_name=rds_test&offset=0&limit=10
```

Example Response

Status code: 200

Success.


```
{
  "database_schemas" : [ {
    "schema_name" : "rds-test",
    "owner" : "root"
  }, {
    "schema_name" : "testdb1",
    "owner" : "root"
  } ],
  "total_count" : 2
}
```


Status Code

Status Code	Description
200	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.4.4 Querying Database Tables

Function

This API is used to query the database tables of a specified instance.

Constraints

- This operation cannot be performed when the instance is in the abnormal or frozen state.

- The database tables of read replicas cannot be queried.

URI

GET /gaussdb/v3/instances/{instance_id}/tables

Table 4-313 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Table 4-314 Query parameters

Parameter	Mandatory	Type	Description
db_name	Yes	String	Database name.
schema_name	Yes	String	Database schema.
offset	No	Integer	Offset, which is the position where the query starts. The value must be greater than or equal to 0.
limit	No	Integer	Number of records to be queried. The default value is **10**. The value cannot be a negative number. The minimum value is **1**, and the maximum value is **100**.

Request Parameters

Table 4-315 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Response Parameters

Status code: 200

Table 4-316 Response body parameters

Parameter	Type	Description
database_tables	Array of **Table 4-317**objects	Database tables. Each element in the list indicates a database table.
total_count	Integer	Total number of database tables.

Table 4-317 GaussDBDatabaseForListTable

Parameter	Type	Description
table_name	String	Table name.

Status code: default

Table 4-318 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

```
GET https://{Endpoint}/rds/gaussdb/v3/instances/d8e6ca5a624745bcb546a227aa3ae1cfin14/tables?
db_name=core1&schema_name=my_schema1&offset=0&limit=10
```

Example Response

Status code: 200

Success.

```
{
   "database_tables": [
      {
         "table_name": "test1"
      }
   ],
   "total_count": 1
}
```

Status Codes

Status Code	Description
200	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.4.5 Creating a Database User

Function

This API is used to create a database user for a specified instance.

URI

POST /gaussdb/v3/instances/{instance_id}/db-user

Table 4-319 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-320 Request body parameters

Parameter	Mandatory	Type	Description
name	Yes	String	Database username. It can contain 1 to 63 characters. Only letters, digits, and underscores (_) are allowed. It cannot start with **pg** or a digit, and must be different from built-in database users (**rdsAdmin**, **rdsRepl**, **rdsBackup**, **rdsMetric**, **root**, **dbmind_monitor**, **dbmind_monitor_agent** or **hadr_disaster**).

Parameter	Mandatory	Type	Description
password	Yes	String	Password of the database user. The value: ● Can consist of 8 to 32 characters and contain at least three types of the following characters: uppercase letters, lowercase letters, digits, and special characters (~!@#%^*-_=+?). ● Cannot be the same as the **name** value or the **name** value in reverse order. Enter a strong password to improve security, preventing security risks such as brute force cracking.
is_login_only	No	boolean	Whether the user only has the login permission. The value **true** indicates that the user only has the login permission. The value **false** indicates that the user has the Login, Create Role, and Create DB permissions. The default value is **false**.

Response Parameters

Status code: 400

Table 4-321 Response body parameters

Parameter	Type	Description
error_code	String	Error code. Minimum length: 8 characters Maximum length: 36 characters
error_msg	String	Error message. Minimum length: 2 characters Maximum length: 512 characters

Example Request

```
https://{Endpoint}/rds/gaussdb/v3/instances/{instance_id}/db-user
{
"name":"suntao",
"password":"***"
}
```

Example Response

None

Status Code

Status Code	Description
200	Normal
400	Abnormal

Error Code

For details, see **Error Codes**.

4.4.6 Deleting a Database User

Function

This API is used to delete a database user from a specified instance.

URI

DELETE /gaussdb/v3/instances/{instance_id}/db-user

Table 4-322 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-323 Request body parameters

Parameter	Mandatory	Type	Description
name	Yes	String	Database username. The name of a database user must be that of an existing database user and cannot be the built-in users, including **rdsAdmin**, **rdsRepl**, **rdsBackup**, **rdsMetric**, **root**, **dbmind_monitor**, **dbmind_monitor_agent** or **hadr_disaster**.

Response Parameters

Status code: 400

Table 4-324 Response body parameters

Parameter	Type	Description
error_code	String	Error code. Minimum length: 8 characters Maximum length: 36 characters
error_msg	String	Error message. Minimum length: 2 characters Maximum length: 512 characters

Example Request

None

Example Response

None

Status Code

Status Code	Description
200	Normal

Status Code	Description
400	Abnormal

Error Code

For details, see **Error Codes**.

4.4.7 Granting Permissions to and Revoking Permissions from a Database User

Function

This API is used to grant permissions to or revoke permissions from a single database user on a specified instance.

URI

POST /gaussdb/v3/instances/{instance_id}/db-user/privileges

Table 4-325 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-326 Request body parameter

Parameter	Mandatory	Type	Description
privileges	Yes	Array of objects **Table 3 Privileges**	Permission description object.

Table 4-327 privilege

Parameter	Mandatory	Type	Description
name	Yes	String	Database username. The name of a database user must be that of an existing database user and cannot be the built-in users, including **rdsAdmin**, **rdsRepl**, **rdsBackup**, **rdsMetric**, **root**, **dbmind_monitor**, **dbmind_monitor_agent** or **hadr_disaster**.
operation	Yes	String	Permission operation. The value can be **grant** or **revoke**.
scope	Yes	String	Permission scope. Value: **table**, **table_column**, **database**, **schema**, **sequence**, **function**, **procedure**, or **tablespace**.

Parameter	Mandatory	Type	Description
privilege	Yes	String	Permission name. When **scope** is set to **table**, the value can be **SELECT**, **INSERT**, **UPDATE**, **DELETE**, **TRUNCATE**, **REFERENCES**, **TRIGGER**, **ALTER**, **DROP**, **COMMENT**, **INDEX**, or **VACUUM**. When **scope** is set to **table_column**, the value can be **SELECT**, **INSERT**, **UPDATE**, **REFERENCES**, or **COMMENT**. When **scope** is set to **database**, the value can be **CREATE**, **ALTER**, **DROP**, or **COMMENT**. When **scope** is set to **schema**, the value can be **CREATE**, **USAGE**, **ALTER**, **DROP**, or **COMMENT**. When **scope** is set to **sequence**, the value can be **SELECT**, **UPDATE**, **USAGE**, **ALTER**, **DROP**, or **COMMENT**. When **scope** is set to **function**, the value can be **ALTER**, **DROP**, or **COMMENT**. When **scope** is set to **procedure**, the value can be **ALTER**, **DROP**, or **COMMENT**. When **scope** is set to **tablespace**, the value can be **CREATE**, **ALTER**, **DROP**, or **COMMENT**.
schema_name	No	String	Schema name. This parameter is valid only when **scope** is set to **table**, **table_column**, **sequence**, **function**, **procedure**, or **schema**.
table_name	No	String	Table or view name. This parameter is valid only when **scope** is set to **table** or **table_column**.
column_name	No	String	Field name. This parameter is valid only when **scope** is set to **table_column**.

Parameter	Mandatory	Type	Description
database_name	No	String	Database name. This parameter takes effect only when **scope** is set to **table**, **table_column**, **database**, **schema**, **sequence**, **function**, or **procedure**.
sequence_name	No	String	Sequence name. This parameter is valid only when **scope** is set to **sequence**.
function_name	No	String	Function name. This parameter is valid only when **scope** is set to **function**.
procedure_name	No	String	Stored procedure name. This parameter is valid only when **scope** is set to **procedure**.
tablespace_name	No	String	Tablespace name. This parameter is valid only when **scope** is set to **tablespace**.
arg_type	No	String	All parameter types of a function or stored procedure. This parameter is valid only when **scope** is set to **function** or **procedure**. You need to configure all parameter types according to the sequence of functions or stored procedures. Use commas (,) to separate parameter types.

Response Parameters

Status code: 400

Table 4-328 Response body parameters

Parameter	Type	Description
error_code	String	Error code. Minimum length: 8 characters Maximum length: 36 characters
error_msg	String	Error message. Minimum length: 2 characters Maximum length: 512 characters

Example Request

```
https://{Endpoint}/rds/gaussdb/v3/instances/eac0c5157f0b43ecae51a56047f72776in14/db-user/privileges

{
  "privileges" : [ {
    "name" : "Einstein",
    "operation" : "grant",
    "scope" : "table_column",
    "privilege" : "REFERENCES",
    "table_name" : "name_list",
    "schema_name" : "famous_figures",
    "database_name" : "articles",
    "column_name" : "age"
  } ]
}
```

Example Response

None

Status Code

Status Code	Description
200	Normal
400	Abnormal

Error Code

For details, see **Error Codes**.

4.4.8 Modifying Attributes of a Database User

Function

This API is used to modify the attributes of a single database user in a specified instance.

URI

POST /gaussdb/v3/instances/{instance_id}/db-user/attributes

Table 4-329 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-330 Request body parameters

Parameter	Mandatory	Type	Description
name	Yes	String	Database username. The name of a database user must be that of an existing database user and cannot be the built-in users, including **rdsAdmin**, **rdsRepl**, **rdsBackup**, **rdsMetric**, **root**, **dbmind_monitor**, **dbmind_monitor_agent** or **hadr_disaster**.
attributes	Yes	Table 4-331 object	Database user permissions.

Table 4-331 attributes

Parameter	Mandatory	Type	Description
sysadmin_enable	No	Boolean	Whether the user is a database administrator.
createdb_enable	No	Boolean	Whether the user has permissions to create databases.
createrole_enable	No	Boolean	Whether the user has permissions to create users.
auditadmin_enabled	No	Boolean	Whether the user is an audit administrator.
monadmin_enabled	No	Boolean	Whether the user is a monitor administrator.
opradmin_enabled	No	Boolean	Whether the user is an operation administrator.
poladmin_enabled	No	Boolean	Whether the user is a security policy administrator.
replication_enabled	No	Boolean	Whether streaming replication is supported.
login_enabled	No	Boolean	Whether the login is allowed.

Response Parameters

Status code: 400

Table 4-332 Response body parameters

Parameter	Type	Description
errCode	String	Error code. Minimum length: 8 characters Maximum length: 36 characters
externalMessage	String	Error message. Minimum length: 2 characters Maximum length: 512 characters

Example Request

Modifying attributes of a database user (Currently, only the nine **attributes** parameters in the example are supported.)

```
https://{Endpoint}/rds/gaussdb/v3/instances/{instance_id}/db-user/attributes

{
  "name" : "zjqtest",
  "attributes" : {
    "sysadmin_enabled" : false,
    "createdb_enabled" : true,
    "createrole_enabled" : true,
    "auditadmin_enabled" : true,
    "monadmin_enabled" : false,
    "opradmin_enabled" : true,
    "poladmin_enabled" : true,
    "replication_enabled" : false,
    "login_enabled" : true
  }
}
```

Example Response

None

Status Code

Status Code	Description
200	Normal
400	Abnormal

Error Code

For details, see **Error Codes**.

4.4.9 Initializing the Password of a Built-in Database User

Function

This API is used to initialize the password of the built-in database user **rdsAdmin**, **rdsMetric**, **rdsRepl**, **rdsBackup**, or **root** on an instance.

Constraints

If you initialize the password of a built-in database user for a new or original primary instance after a DR instance is promoted to primary, you cannot re-establish the DR relationship.

URI

PUT /gaussdb/v3/instances/{instance_id}/db-user/init-password

Table 4-333 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-334 Request body parameters

Parameter	Mandatory	Type	Description
admin_name	No	String	Name of the **admin** user.
admin_password	No	String	New password of the **admin** user. The value: • Can consist of 8 to 32 characters and contain at least three types of the following characters: uppercase letters, lowercase letters, digits, and special characters (~!@#%^*-_=+?). • Cannot be the same as the **name** value or the **name** value in reverse order. Enter a strong password to improve security, preventing security risks such as brute force cracking.

Parameter	Mandatory	Type	Description
metric_name	No	String	Name of the **metric** user.
metric_password	No	String	New password of the **metric** user. The value: • Can consist of 8 to 32 characters and contain at least three types of the following characters: uppercase letters, lowercase letters, digits, and special characters (~!@#%^*-_=+?). • Cannot be the same as the **name** value or the **name** value in reverse order. Enter a strong password to improve security, preventing security risks such as brute force cracking.
repl_name	No	String	Name of the **repl** user.
repl_password	No	String	New password of the **repl** user. The value: • Can consist of 8 to 32 characters and contain at least three types of the following characters: uppercase letters, lowercase letters, digits, and special characters (~!@#%^*-_=+?). • Cannot be the same as the **name** value or the **name** value in reverse order. Enter a strong password to improve security, preventing security risks such as brute force cracking.
backup_name	No	String	Name of the **backup** user.

Parameter	Mandatory	Type	Description
backup_password	No	String	New password of the **backup** user. The value: • Can consist of 8 to 32 characters and contain at least three types of the following characters: uppercase letters, lowercase letters, digits, and special characters (~!@#%^*-_=+?). • Cannot be the same as the **name** value or the **name** value in reverse order. Enter a strong password to improve security, preventing security risks such as brute force cracking.
root_name	No	String	Name of the **root** user.
root_password	No	String	New password of the **root** user. The value: • Can consist of 8 to 32 characters and contain at least three types of the following characters: uppercase letters, lowercase letters, digits, and special characters (~!@#%^*-_=+?). • Cannot be the same as the **name** value or the **name** value in reverse order. Enter a strong password to improve security, preventing security risks such as brute force cracking.

Response Parameters

Status code: 400

Table 4-335 Response body parameters

Parameter	Type	Description
error_code	String	Error code. Minimum length: 8 characters Maximum length: 36 characters
error_msg	String	Error message. Minimum length: 2 characters Maximum length: 512 characters

Example Request

```
https://{Endpoint}/rds/gaussdb/v3/instances/{instance_id}/db-user/init-password
{
   "metric_name" : "rdsMetric",
   "metric_password" : "***"
}
```

Example Response

None

Status Code

Status Code	Description
200	Normal
400	Abnormal

Error Code

For details, see **Error Codes**.

4.4.10 Resetting the Password of a Database User

Function

This API is used to reset the password of a database user on a specified instance.

URI

PUT /gaussdb/v3/instances/{instance_id}/db-user/password

Table 4-336 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-337 Request body parameters

Parameter	Mandatory	Type	Description
name	Yes	String	Database username. The name of a database user must be that of an existing database user and cannot be the built-in users, including **rdsAdmin**, **rdsRepl**, **rdsBackup**, **rdsMetric**, **root**, **dbmind_monitor**, **dbmind_monitor_agent** or **hadr_disaster**.
password	Yes	String	Password of the new database user. The value: • Can consist of 8 to 32 characters and contain at least three types of the following characters: uppercase letters, lowercase letters, digits, and special characters (~!@#%^*-_=+?). • Cannot be the same as the **name** value or the **name** value in reverse order. Enter a strong password to improve security, preventing security risks such as brute force cracking.

Response Parameters

Status code: 400

Table 4-338 Response body parameters

Parameter	Type	Description
error_code	String	Error code. Minimum length: 8 characters Maximum length: 36 characters
error_msg	String	Error message. Minimum length: 2 characters Maximum length: 512 characters

Example Request

```
https://{Endpoint}/rds/gaussdb/v3/instances/{instance_id}/db-user/password

{
  "name" : "zjqtest",
  "password" : "***"
}
```

Example Response

None

Status Code

Status Code	Description
200	Normal
400	Abnormal

Error Code

For details, see **Error Codes**.

4.4.11 Locking a Database User

Function

This API is used to lock a database user of an instance.

URI

PUT /gaussdb/v3/instances/{instance_id}/db-user/lock

Table 4-339 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-340 Request body parameters

Parameter	Mandatory	Type	Description
name	Yes	String	Database username. The name of a database user must be that of an existing database user and cannot be the built-in users, including **rdsAdmin**, **rdsRepl**, **rdsBackup**, **rdsMetric**, **root**, **dbmind_monitor**, **dbmind_monitor_agent** or **hadr_disaster**.

Response Parameters

Status code: 400

Table 4-341 Response body parameters

Parameter	Type	Description
error_code	String	Error code. Minimum length: 8 characters Maximum length: 36 characters
error_msg	String	Error message. Minimum length: 2 characters Maximum length: 512 characters

Example Request

```
https://{Endpoint}/rds/gaussdb/v3/instances/{instance_id}/db-user/lock

{
  "name" : "zjqtest"
}
```

Example Response

None

Status Code

Status Code	Description
200	Normal
400	Abnormal

Error Code

For details, see **Error Codes**.

4.4.12 Unlocking a Database User

Function

This API is used to unlock a database user on a specified instance.

URI

PUT /gaussdb/v3/instances/{instance_id}/db-user/unlock

Table 4-342 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-343 Request body parameters

Parameter	Mandatory	Type	Description
name	Yes	String	Database username. The name of a database user must be that of an existing database user and cannot be the built-in users, including **rdsAdmin**, **rdsRepl**, **rdsBackup**, **rdsMetric**, **root**, **dbmind_monitor**, **dbmind_monitor_agent** or **hadr_disaster**.

Response Parameters

Status code: 400

Table 4-344 Response body parameters

Parameter	Type	Description
error_code	String	Error code. Minimum length: 8 characters Maximum length: 36 characters
error_msg	String	Error message. Minimum length: 2 characters Maximum length: 512 characters

Example Request

```
https://{Endpoint}/rds/gaussdb/v3/instances/{instance_id}/db-user/unlock

{
  "name" : "zjqtest"
}
```

Example Response

None

Status Code

Status Code	Description
200	Normal
400	Abnormal

Error Code

For details, see **Error Codes**.

4.4.13 Querying Databases

Function

This API is used to query databases.

URI

GET /gaussdb/v3/instances/{instance_id}/databases

Table 4-345 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Table 4-346 Query parameters

Parameter	Mandatory	Type	Description
limit	No	Integer	Number of records to be queried. The default value is **10**. The value cannot be a negative number. The minimum value is **1**, and the maximum value is **1000**.
offset	No	Integer	Index offset. If **offset** is set to *N*, the resource query starts from the *N*+1 piece of data. The value is **0** by default, indicating that the query starts from the first piece of data. The value must be a number but cannot be a negative number.

Request Parameters

Table 4-347 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	Request token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Response Parameters

Status code: 200

Table 4-348 Response body parameters

Parameter	Type	Description
databases	Array of **Table 4-349** objects	Database object details.
total_count	Integer	Total number of databases.

Table 4-349 databases

Parameter	Type	Description
name	String	Database name.
owner	String	Database owner.
size	String	Database size.
character_set	String	Database character set.
collate_set	String	Database collocation.
datctype	String	Character type used by the database.

Status code: 400

Table 4-350 Response body parameters

Parameter	Type	Description
error_code	String	Error code. Minimum length: 8 characters Maximum length: 36 characters
error_msg	String	Error message. Minimum length: 2 characters Maximum length: 512 characters

Example Request

```
https://{Endpoint}/rds/gaussdb/v3/instances/c95385c0b04f4e2cb2af3e5ae0dc5c7bin14/databases
```

Example Response

Status code: 200

Databases queried.

```
{
  "databases": [
```

```
  {
    "name": "test",
    "owner": "test",
    "size": "33 MB",
    "character_set": "UTF8",
    "collate_set": "C",
    "datctype": "C"
  }
 ],
 "total_count": 1
}
```

Status Code

Status Code	Description
200	Normal
400	Abnormal

Error Code

For details, see **Error Codes**.

4.4.14 Querying Database Users

Function

This API is used to query database users.

URI

GET /gaussdb/v3/instances/{instance_id}/db-users

Table 4-351 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Table 4-352 Query parameters

Parameter	Mandatory	Type	Description
limit	No	String	Number of records to be queried. The default value is **10**. The value cannot be a negative number. The minimum value is **1** and the maximum value is **100**.

Parameter	Mandatory	Type	Description
offset	No	String	Index offset. If **offset** is set to *N*, the resource query starts from the *N*+1 piece of data. The value is **0** by default, indicating that the query starts from the first piece of data. The value must be a number but cannot be a negative number.

Request Parameters

Table 4-353 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	Request token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Response Parameters

Status code: 200

Table 4-354 Response body parameters

Parameter	Type	Description
users	Array of **Table 4-355** objects	Locked users.
total_count	Integer	Number of database users in an instance.

Table 4-355 users

Parameter	Type	Description
name	String	Database username.
lock_status	Boolean	Lock status of the database user.

Parameter	Type	Description
attribute	Table 4-356 object	Permission attribute of a user.
memberof	String	Default permissions of a user.

Table 4-356 userAttributes

Parameter	Type	Description
rolsuper	Boolean	Whether the user has the administrator permissions. The value can be **true** or **false**.
rolinherit	Boolean	Whether the user automatically inherits the permissions of the role to which the user belongs. The value can be **true** or **false**.
rolcreaterole	Boolean	Whether the user can create other sub-users. The value can be **true** or **false**.
rolcreatedb	Boolean	Whether the user can create a database. The value can be **true** or **false**.
rolcanlogin	Boolean	Whether the user can log in to the database. The value can be **true** or **false**.
rolconnlimit	Integer	Maximum number of concurrent connections to an instance. The value **-1** indicates that there are no limitations on the number of concurrent connections.
rolreplication	Boolean	Whether the user is a replication role. The value can be **true** or **false**.
rolbypassrls	Boolean	Whether the user bypasses each row-level security policy. The value can be **true** or **false**.
rolpassworddeadline	String	Remaining time before the password expires. The value returned by this parameter is affected by the value of the GUC parameter **password_effect_time**. The default value of **password_effect_time** is **0**, indicating that the validity period is disabled. In this case, no value is returned by **rolpassworddeadline**.

Status code: 400

Table 4-357 Response body parameters

Parameter	Type	Description
error_code	String	Error code. Minimum length: 8 characters Maximum length: 36 characters
error_msg	String	Error message. Minimum length: 2 characters Maximum length: 512 characters

Example Request

```
https://{Endpoint}/rds/gaussdb/v3/instances/c95385c0b04f4e2cb2af3e5ae0dc5c7bin14/db-users
```

Example Response

Status code: 200

Database users queried.


```
{
  "users" : [ {
    "name" : "root",
    "attribute" : {
      "rolsuper" : false,
      "rolinherit" : true,
      "rolcreaterole" : true,
      "rolcreatedb" : true,
      "rolcanlogin" : true,
      "rolconnlimit" : -1,
      "rolreplication" : false,
      "rolbypassrls" : false,
      "rolpassworddeadline" : "88 days 01:01:05"
    },
    "memberof" :
"{gs_role_copy_files,gs_role_signal_backend,gs_role_tablespace,gs_role_replication,gs_role_account_lock}",
    "lock_status" : false
  } ],
  "total_count" : 1
}
```


Status Code

Status Code	Description
200	Normal
400	Abnormal

Error Code

For details, see **Error Codes**.

4.4.15 Querying Connection Users, Applications, Table Statistics, and Lock Information of a Database

Function

This API is used to query the connection users, applications, table statistics, and lock information of a database.

URI

GET /gaussdb/v3/instances/{instance_id}/database/relation-and-lock

Table 4-358 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Table 4-359 Query parameters

Parameter	Mandatory	Type	Description
name	Yes	String	Name of the database to be queried.
table_limit	No	String	Number of records to be queried. The default value is **10**. The value cannot be a negative number. The minimum value is **1** and the maximum value is **100**.

Request Parameters

Table 4-360 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Response Parameters

Status code: 200

Table 4-361 Response body parameters

Parameter	Type	Description
user_count	String	Number of users connected to the database.
app_count	String	Number of applications connected to the database.
relations_info	Array of **Table 4-362** objects	Table statistics.
locks_info	Array of **Table 4-363** objects	Database lock details.
total_lock_count	String	Total number of locks in the database.
rel_lock_count	String	Total number of table-level locks in the database.
row_lock_count	String	Total number of row-level locks in the database.
other_lock_count	String	Total number of other locks in the database.

Table 4-362 RelationInfo

Parameter	Type	Description
table_name	String	Table name.
data_skew	String	Skew rate of table data.
sequence_scan_percent	String	Percentage of sequential scanning in all scanning methods.
live_rows	String	Number of active rows in the table.
dead_rows	String	Number of dead rows in the table.

Table 4-363 LockInfo

Parameter	Type	Description
table_name	String	Table name corresponding to the lock.

Parameter	Type	Description
pid	String	Logical ID of the server thread holding or awaiting the lock.
lock_obj_id	String	OID of the object in its system catalog.
lock_mode	String	Lock mode held or desired by the thread.
query_start_time	String	Start time of the query corresponding to the lock.
query_id	String	ID of the query statement corresponding to the lock.
is_granted	Boolean	Whether the query is holding or waiting for a lock.
grant_lock_type	String	Type of the locked object. The value can be **relation**, **extend**, **page**, **tuple**, **transactionid**, **virtualxid**, **object**, **userlock**, or **advisory**.
app_name	String	Application name corresponding to the lock.
user_name	String	Username corresponding to the lock.

Status code: 400

Table 4-364 Response body parameters

Parameter	Type	Description
error_code	String	Error code. Minimum length: 8 characters Maximum length: 36 characters
error_msg	String	Error message. Minimum length: 2 characters Maximum length: 512 characters

Example Request

Querying connection users, applications, table statistics, and lock information of the **postgres** database on the instance (ID: 4bfc77f4ff6f4fedafaf33c53186de58in14)

```
https://{Endpoint}/rds/gaussdb/v3/instances/4bfc77f4ff6f4fedafaf33c53186de58in14/database/relation-and-lock?name=postgres
```

Example Response

Status code: 200

Database table and lock information


```
{
  "user_count" : "2",
  "app_count" : "2",
  "relations_info" : [ {
    "table_name" : "tb1",
    "data_skew" : "0.1",
    "sequence_scan_percent" : "0.2",
    "live_rows" : "1256",
    "dead_rows" : "89547"
  } ],
  "locks_info" : [ {
    "table_name" : "tb1",
    "pid" : "140471800882944",
    "lock_obj_id" : "",
    "lock_mode" : "AccessShareLock",
    "query_start_time" : "1695351552",
    "query_id" : "290763650943359936",
    "is_granted" : true,
    "grant_lock_type" : "Table",
    "app_name" : "gsql",
    "user_name" : "rdsAdmin"
  } ],
  "total_lock_count" : "1",
  "rel_lock_count" : "1",
  "row_lock_count" : "0",
  "other_lock_count" : "0"
}
```


Status Code

Status Code	Description
200	Normal
400	Abnormal

Error Code

For details, see **Error Codes**.

4.4.16 Flashing Back a Table

Function

This API is only suitable for centralized instances.

1. You need to set **enable_default_ustore_table** to **on** and **enable_recyclebin** to **on**.
2. **recyclebin_retention_time** indicates the retention period of objects in the recycle bin. Objects in the recycle bin that exceed the retention period are automatically deleted. **undo_retention_time** indicates the retention period of the undo logs and is used to retrieve the deleted data. You need to set **recyclebin_retention_time** and **undo_retention_time** to be both greater than 0.

URI

POST /gaussdb/v3/instances/{instance_id}/db-table/flashback

Table 4-365 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-366 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	Yes	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Table 4-367 Request body parameters

Parameter	Mandatory	Type	Description
db_name	Yes	String	Database name.
schema_name	Yes	String	Schema name, which is case-insensitive.
table_name	Yes	String	Table name, which is case-insensitive.
action	Yes	String	Action. The value can be **back_drop** or **back_time**. **back_drop**: Rolls back the latest DROP operation on the current table. **back_time**: Roll back the data in the table to the specified point in time.

Parameter	Mandatory	Type	Description
flashback_time	No	String	This parameter is invalid when **action** is set to **back_drop**. This parameter is mandatory when **action** is set to **back_time**. The format is YYYY-MM-DD HH24:MI:SS, for example, **2023-05-08 17:36:02**. When a time point is used for flashback, there may be a 3s error.

Response Parameters

Status code: default

Table 4-368 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

```
POST
https://{Endpoint}/rds/gaussdb/v3/instances/0611f1bd8b00d5d32f17c017f15b599f/db-table/flashback
{
  "db_name" : "core",
  "schema_name" : "core",
  "table_name" : "test",
  "action" : "back_drop",
  "flashback_time" : "2023-06-07 06:04:12"
}
```

Example Response

None

Status Code

Status Code	Description
200	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.4.17 Creating a Database Role

Function

This API is used to create a database role of a specified instance.

URI

POST /gaussdb/v3/instances/{instance_id}/db-role

Table 4-369 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-370 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Table 4-371 Request body parameters

<table>
<tr><th>Parameter</th><th>Mandatory</th><th>Type</th><th>Description</th></tr>
<tr><td>name</td><td>Yes</td><td>String</td><td>Database role name. The database role name can contain 1 to 63 characters. Only letters, digits, and underscores (_) are allowed. It cannot start with pg or a digit, and must be different from system user or role names, including rdsAdmin, rdsRepl, rdsBackup, rdsMetric, root, dbmind_monitor, dbmind_monitor_agent, and hadr_disaster.
Minimum length: 1 character
Maximum length: 63 characters</td></tr>
<tr><td>password</td><td>Yes</td><td>String</td><td>Password of the database role. Contain at least three types of the following four types of characters: uppercase characters, lowercase characters, digits, and special characters (~!@#$%^&*()-_=+\|[{}];:,<.>/?). If the password contains characters other than the preceding characters, an error will be reported during statement execution.
Minimum length: 8 characters
Maximum length: 32 characters</td></tr>
</table>

Response Parameters

Status code: 400

Table 4-372 Response body parameters

Parameter	Type	Description
errCode	String	Error code. Minimum length: 8 characters Maximum length: 36 characters

Parameter	Type	Description
externalMessage	String	Error message. Minimum length: 2 characters Maximum length: 512 characters

Example Request

```
post https://{Endpoint}/rds/gaussdb/v3/instances/{instance_id}/db-role

{
  "name" : "readwrite",
  "password" : "xxxxx"
}
```

Example Response

None

Status Code

Status Code	Description
200	Normal
400	Abnormal

Error Code

For details, see **Error Codes**.

4.4.18 Deleting a Database Role

Function

This API is used to delete a database role from a specified instance.

URI

DELETE /gaussdb/v3/instances/{instance_id}/db-role

Table 4-373 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-374 Request header parameters

Parameter	Mandatory	Type	Description
X-Language	No	String	User token.
X-Mini-Token	Yes	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Table 4-375 Request body parameters

Parameter	Mandatory	Type	Description
name	Yes	String	Database role name. The database role name can contain 1 to 63 characters. Only letters, digits, and underscores (_) are allowed. It cannot start with **pg** or a digit, and must be different from system user or role names, including **rdsAdmin**, **rdsRepl**, **rdsBackup**, **rdsMetric**, **root**, **dbmind_monitor**, **dbmind_monitor_agent**, and **hadr_disaster**. Minimum length: 1 character Maximum length: 63 characters

Response Parameters

Status code: 400

Table 4-376 Response body parameters

Parameter	Type	Description
errCode	String	Error code. Minimum length: 8 characters Maximum length: 36 characters

Parameter	Type	Description
externalMessage	String	Error message. Minimum length: 2 characters Maximum length: 512 characters

Example Request

```
DELETE https://{Endpoint}/rds/gaussdb/v3/instances/{instance_id}/db-role

{
  "name" : "readwrite"
}
```

Example Response

None

Status Code

Status Code	Description
200	Normal
400	Abnormal

Error Code

For details, see **Error Codes**.

4.4.19 Granting Permissions to and Revoking Permissions from a Database User or a Database Role

Function

This API is used to grant permissions to or revoke permissions from a customized user or role on a specified instance.

URI

POST /gaussdb/v3.1/instances/{instance_id}/db-user/privileges

Table 4-377 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-378 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Table 4-379 Request body parameters

Parameter	Mandatory	Type	Description
any_permissions	No	**Table 4-380** object	Any attributes assigned to the user or role.
system_permissions	No	**Table 4-383** object	System attributes assigned to the user or role.
schema_permissions	No	**Table 4-386** object	Schema access permissions assigned to the user or role.
table_permissions	No	**Table 4-389** object	Schema table access permissions assigned to the user or role.
sequence_permissions	No	**Table 4-391** object	Schema sequence access permissions assigned to the user or role.
newly_table_permissions	No	**Table 4-393** object	Operation permissions to new tables assigned to the user or role.
newly_sequence_permissions	No	**Table 4-396** object	Operation permissions to new sequences assigned to the user or role.
user_role_permissions	No	**Table 4-399** object	Operation permissions to roles or users.

Table 4-380 AnyPermissions

Parameter	Mandatory	Type	Description
operation_type	Yes	String	Operation type. The value can be **grant** or **revoke**. Enumerated values: • **grant** • **revoke**
operations	Yes	Array of **Table 4-381** objects	Operations for assigning any attributes to the user or role. Array length: 1 to 30 characters

Table 4-381 AnyPermissionsOperation

Parameter	Mandatory	Type	Description
name	Yes	String	Username or role name.
database_name	No	String	Database name.
attributes	Yes	**Table 4-382** object	Permission description for assigning any attributes to the user or role.

Table 4-382 AnyPermissionsEntity

Parameter	Mandatory	Type	Description
select_any_table_enabled	No	Boolean	select_any_table permissions.
create_any_table_enabled	No	Boolean	create_any_table permissions.
alter_any_table_enabled	No	Boolean	alter_any_table permissions.
drop_any_table_enabled	No	Boolean	drop_any_table permissions.
insert_any_table_enabled	No	Boolean	insert_any_table permissions.
update_any_table_enabled	No	Boolean	update_any_table permissions.

Parameter	Mandatory	Type	Description
delete_any_table_enabled	No	Boolean	delete_any_table permissions.
create_any_sequence_enabled	No	Boolean	create_any_sequence permission.
alter_any_sequence_enabled	No	Boolean	alter_any_sequence permissions.
drop_any_sequence_enabled	No	Boolean	drop_any_sequence permissions.
select_any_sequence_enabled	No	Boolean	select_any_sequence permission.
create_any_index_enabled	No	Boolean	create_any_index permissions.
alter_any_index_enabled	No	Boolean	alter_any_index permissions.
drop_any_index_enabled	No	Boolean	drop_any_index permissions.
create_any_function_enabled	No	Boolean	create_any_function permissions.
execute_any_function_enabled	No	Boolean	execute_any_function permissions.
create_any_package_enabled	No	Boolean	create_any_package permissions.
execute_any_package_enabled	No	Boolean	execute_any_package permissions.
create_any_type_enabled	No	Boolean	create_any_type permissions.
alter_any_type_enabled	No	Boolean	alter_any_type permissions.
drop_any_type_enabled	No	Boolean	drop_any_type permissions.
create_any_trigger_enabled	No	Boolean	create_any_trigger permissions.

Parameter	Mandatory	Type	Description
alter_any_trigger_enabled	No	Boolean	alter_any_trigger permissions.
drop_any_trigger_enabled	No	Boolean	drop_any_trigger permissions.
create_any_synonym_enabled	No	Boolean	create_any_synonym permissions.
drop_any_synonym_enabled	No	Boolean	drop_any_synonym permissions.

Table 4-383 SystemPermissions

Parameter	Mandatory	Type	Description
operation_type	Yes	String	Operation type. The value can be **grant** or **revoke**. Enumerated values: • **grant** • **revoke**
operations	Yes	Array of **Table 4-384** objects	Operations for assigning system attributes to the user or role. Array length: 1 to 30 characters

Table 4-384 SystemPermissionsOperation

Parameter	Mandatory	Type	Description
name	Yes	String	Username or role name.
database_name	No	String	Database name.
attributes	Yes	**Table 4-385** object	Permission description for assigning system attributes to the user or role.

Table 4-385 SystemPermissionsEntity

Parameter	Mandatory	Type	Description
sysadmin_enabled	No	Boolean	Whether the user or role is a database administrator.
createdb_enabled	No	Boolean	Whether the user or role has permissions to create databases.
createrole_enabled	No	Boolean	Whether the user or role has permissions to create users.
auditadmin_enabled	No	Boolean	Whether the user or role is an audit administrator.
monadmin_enabled	No	Boolean	Whether the user or role is a monitor administrator.
opradmin_enabled	No	Boolean	Whether the user or role is an operation administrator.
poladmin_enabled	No	Boolean	Whether the user or role is a security policy administrator.
replication_enabled	No	Boolean	Whether streaming replication is supported.
login_enabled	No	Boolean	Whether the login is allowed.

Table 4-386 SchemaPermissions

Parameter	Mandatory	Type	Description
operation_type	Yes	String	Operation type. The value can be **grant** or **revoke**. Enumerated values: • **grant** • **revoke**
operations	Yes	Array of **Table 4-387** objects	Operations for assigning schema access permissions to the user or role. Array length: 1 to 30 characters

Table 4-387 SchemaPermissionsOperation

Parameter	Mandatory	Type	Description
name	Yes	String	Username or role name.
database_name	No	String	Database name.
schema_name	Yes	String	Schema name.
attributes	Yes	Table 4-388 object	Attribute description for assigning schema access permissions to the user or role.

Table 4-388 SchemaPermissionsEntity

Parameter	Mandatory	Type	Description
all_enabled	No	Boolean	All schema permissions.
usage_enabled	No	Boolean	USAGE permission for schemas.
create_enabled	No	Boolean	CREATE permissions for schemas.
alter_enabled	No	Boolean	ALTER permissions for schemas.
drop_enabled	No	Boolean	DROP permissions for schemas.
comment_enabled	No	Boolean	COMMENT permission for schemas.

Table 4-389 TablePermissions

Parameter	Mandatory	Type	Description
operation_type	Yes	String	Operation type. The value can be **grant** or **revoke**. Enumerated values: ● **grant** ● **revoke**

Parameter	Mandatory	Type	Description
operations	Yes	Array of **Table 4-390** objects	Operations for assigning table access permissions to the user or role. Array length: 1 to 30 characters

Table 4-390 TablePermissionsOperation

Parameter	Mandatory	Type	Description
name	Yes	String	Username or role name.
database_na me	No	String	Database name.
schema_name	Yes	String	Schema name.
table_name	No	String	Table name.
attributes	Yes	**Table 4-395** object	Attribute description for assigning table access permissions to the user or role.

Table 4-391 SequencePermissions

Parameter	Mandatory	Type	Description
operation_typ e	Yes	String	Operation type. The value can be **grant** or **revoke**. Enumerated values: • **grant** • **revoke**
operations	Yes	Array of **Table 4-392** objects	Operations for assigning sequence access permissions to the user or role. Array length: 1 to 30 characters

Table 4-392 SequencePermissionsOperation

Parameter	Mandatory	Type	Description
name	Yes	String	Username or role name.

Parameter	Mandatory	Type	Description
database_name	No	String	Database name.
schema_name	Yes	String	Schema name.
sequence_name	No	String	Sequence name.
attributes	Yes	Table 4-398 object	Attribute description for assigning sequence access permissions to the user or role.

Table 4-393 NewlyTablePermissions

Parameter	Mandatory	Type	Description
operation_type	Yes	String	Operation type. The value can be **grant** or **revoke**. Enumerated values: • **grant** • **revoke**
operations	Yes	Array of Table 4-394 objects	Operation for assigning table access permission of a user or role to another user or role. Array length: 1 to 30 characters

Table 4-394 NewlyTablePermissionsOperation

Parameter	Mandatory	Type	Description
source_name	Yes	String	Name of the source user or role who grants permissions to another user or role.
target_name	Yes	String	Name of the target user or role that permissions are assigned to.
database_name	No	String	Database name.
schema_name	No	String	Schema name.

Parameter	Mandatory	Type	Description
attributes	Yes	**Table 4-395** object	Attribute description for assigning table access permission of a user or role to another user or role.

Table 4-395 TablePermissionsEntity

Parameter	Mandatory	Type	Description
all_enabled	No	Boolean	All table permissions.
select_enabled	No	Boolean	SELECT permissions for tables.
insert_enabled	No	Boolean	INSERT permissions for tables.
update_enabled	No	Boolean	UPDATE permissions for tables.
delete_enabled	No	Boolean	DELETE permissions for tables.
truncate_enabled	No	Boolean	TRUNCATE permissions for tables.
references_enabled	No	Boolean	REFERENCES permissions for tables.
trigger_enabled	No	Boolean	TRIGGER permissions for tables.
alter_enabled	No	Boolean	ALTER permissions for tables.
drop_enabled	No	Boolean	DROP permissions for tables.
comment_enabled	No	Boolean	COMMENT permissions for tables.
index_enabled	No	Boolean	INDEX permissions for tables.
vacuum_enabled	No	Boolean	VACUUM permissions for tables.

Table 4-396 NewlySequencePermissions

Parameter	Mandatory	Type	Description
operation_type	Yes	String	Operation type. The value can be **grant** or **revoke**. Enumerated values: • **grant** • **revoke**
operations	Yes	Array of **Table 4-397** objects	Operation for assigning sequence access permission of a user or role to another user or role. Array length: 1 to 30 characters

Table 4-397 NewlySequencePermissionsOperation

Parameter	Mandatory	Type	Description
source_name	Yes	String	Name of the source user or role who grants permissions to another user or role.
target_name	Yes	String	Name of the target user or role that permissions are assigned to.
database_name	No	String	Database name.
schema_name	No	String	Schema name.
attributes	Yes	**Table 4-398** object	Attribute description for assigning sequence access permission of a user or role to another user or role.

Table 4-398 SequencePermissionsEntity

Parameter	Mandatory	Type	Description
all_enabled	No	Boolean	All permissions for sequences.
select_enabled	No	Boolean	SELECT permissions for sequences.
usage_enabled	No	Boolean	USAGE permissions for sequences.

Parameter	Mandatory	Type	Description
update_enabled	No	Boolean	UPDATE permissions for sequences.
alter_enabled	No	Boolean	ALTER permissions for sequences.
drop_enabled	No	Boolean	DROP permissions for sequences.
comment_enabled	No	Boolean	COMMENT permissions for sequences.

Table 4-399 UserRolePermissions

Parameter	Mandatory	Type	Description
operation_type	Yes	String	Operation type. The value can be **grant** or **revoke**. Enumerated values: • **grant** • **revoke**
operations	Yes	Array of **Table 4-400** objects	Operations for assigning a role's or user's permissions to another user or role. Array length: 1 to 30 characters

Table 4-400 UserRolePermissionsOperation

Parameter	Mandatory	Type	Description
source_name	Yes	String	Name of the source user or role who grants permissions to another user or role.
target_name	Yes	String	Name of the target user or role that permissions are assigned to.
database_name	No	String	Database name.

Parameter	Mandatory	Type	Description
is_reuse	No	Boolean	If the **value** is true, the authorized user can grant the permission to others. Default value: **false** Enumerated values: • **true** • **false**

Response Parameters

Status code: 400

Table 4-401 Response body parameters

Parameter	Type	Description
errCode	String	Error code. Minimum length: 8 characters Maximum length: 36 characters
externalMessage	String	Error message. Minimum length: 2 characters Maximum length: 512 characters

Example Request

```
post
https://{Endpoint}/rds/gaussdb/v3.1/instances/eac0c5157f0b43ecae51a56047f72776in14/db-user/privileges
{
  "any_permissions" : {
    "operation_type" : "revoke",
    "operations" : [ {
      "database_name" : "testdb",
      "name" : "readonly",
      "attributes" : {
        "select_any_table_enabled" : true
      }
    } ]
  }
}
```

Example Response

None

Status Code

Status Code	Description
200	Normal
400	Abnormal

Error Code

For details, see **Error Codes**.

4.5 Database Diagnostics and Optimization

4.5.1 Querying Node and Component Information on Database Key Views

Function

This API is used to query information about nodes (non-log CN or DN nodes) and their CN and DN components. To call the APIs for querying and killing sessions, you must use the node ID and component ID returned by the API. Constraints:

- This API is supported by distributed GaussDB instances that are deployed in independent or combined mode and centralized GaussDB instances.
- For distributed instances, the first node group contains a normal non-log node that contains a CN.
- For centralized instances, there is at least an available non-log primary node.

URI

GET /gaussdb/v3/instances/{instance_id}/key-view-execute-node

Table 4-402 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID. Minimum length: 32 characters Maximum length: 36 characters

Request Parameters

Table 4-403 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	Request token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Response Parameters

Status code: 200

Table 4-404 Response body parameters

Parameter	Type	Description
nodes	Array of **Table 4-405** objects	Node and component information.

Table 4-405 OpsKeyViewExecuteNode

Parameter	Type	Description
node_id	String	Node ID.
node_name	String	Node name.
role	String	Node role.
type	String	Node type.
distributed_id	String	Distribution ID.
component_id	String	Component ID.
detail	String	Details.

Status code: 400

Table 4-406 Response body parameters

Parameter	Type	Description
error_code	String	Error code. Minimum length: 8 characters Maximum length: 36 characters
error_msg	String	Error message. Minimum length: 2 characters Maximum length: 512 characters

Example Request

```
GET https://{Endpoint}/rds/gaussdb/v3/instances/725398ddd0dd41d88a609acf137698e5in14/key-view-execute-node
```

Example Response

Status code: 200

Success.


```
{
  "nodes" : [ {
    "type" : "DN",
    "detail" : "Normal",
    "node_id" : "faed3373b25a46c99c5897681ffcb285no14",
    "node_name" : "gauss-cdd_root_0",
    "role" : "master",
    "distributed_id" : "60011",
    "component_id" : "dn_6001"
  } ]
}
```


Status Code

Status Code	Description
200	Normal
400	Abnormal

Error Code

For details, see **Error Codes**.

4.5.2 Querying Real-Time Sessions of a Component on a Node

Function

This API is used to query real-time sessions of a component on a node.

Constraints

- The **node_id** and **component_id** parameters are mandatory. To obtain their value, see the API for querying node and component information.
- This API can only return the sessions that are generated when non-system built-in users (**rdsAdmin**, **rdsMetric**, **rdsBackup**, and **rdsRepl**) connect to databases. If the client IP address is empty, no sessions are returned. When connecting to the database, you need to use **-h** to configure the **hostname** (set **hostname** to the client IP address), for example, **gsql -h hostname -q postgres -p 8000 -U username -W password**.

URI

POST /gaussdb/v3/instances/{instance_id}/real-time-session

Table 4-407 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID. Minimum length: 32 characters Maximum length: 36 characters

Request Parameters

Table 4-408 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	Request token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Table 4-409 Request body parameters

Parameter	Mandatory	Type	Description
node_id	Yes	String	Node ID.
component_id	Yes	String	Component ID.

Parameter	Mandatory	Type	Description
query_info	No	**Table 4-410** object	Session filter criteria.
is_statistic_open	No	String	Whether to enable the collection of statistics on the number of transactions, rollback times, and SQL statements. The statistics collection is enabled by default if this parameter is not delivered. The value of this parameter can be: ● true ● false
limit	No	Integer	Maximum number of records to be returned in the query result. The value ranges from **1** to **200**.

Parameter	Mandatory	Type	Description
order_by	No	String	Sorting field of the query result. The value can be one of the following options. However, if **is_statistic_open** is set to **false**, the value cannot be **trans_num**, **rollback_num**, or **sql_num**. ● sessionid ● unique_sql_id ● pid ● datname ● client_addr ● usename ● waiting ● node_name ● block_sessionid ● wait_event ● state ● query_runtime ● exec_time ● backend_start ● xact_start ● query_start ● application_name ● trans_num ● rollback_num ● sql_num
order	No	String	Sorting order. This parameter is effective only when **order_by** is delivered. The value of this parameter can be: ● desc ● asc

Table 4-410 SessionQueryInfo

Parameter	Mandatory	Type	Description
database_name	No	String	Database name, which can contain 1 to 63 characters.

Parameter	Mandatory	Type	Description
client_ip	No	String	Client IP address.
user_name	No	String	Username, which can contain 1 to 63 characters.
wait	No	String	Whether to query blocked events. The value can be: • f • t
session_id	No	String	Session ID.
state	No	String	Session status.
wait_event	No	String	Wait event.
exec_time	No	String	Session length.
query_runtime	No	String	Statement execution duration.
unique_sql_id	No	String	Normalized SQL ID.

Response Parameters

Status code: 200

Table 4-411 Response body parameters

Parameter	Type	Description
sessions	Array of **Table 4-412** objects	Real-time sessions.

Table 4-412 RealTimeSession

Parameter	Type	Description
session_id	String	Session ID.
pid	String	Thread ID.
unique_sql_id	String	SQL ID.
database_name	String	Database.
client_ip	String	Client IP address.

Parameter	Type	Description
user_name	String	Username.
wait	String	Whether to wait.
block_session	String	Blocked session ID.
wait_event	String	Wait event.
state	String	Status.
query_runtime	String	Execution time of the statement.
query	String	SQL text.
back_end_start	Integer	Start time of the session.
transaction_start	Integer	Start time of the transaction.
query_start	Integer	Start time of the statement.
application_name	String	Application name.
exec_time	String	Session duration, in seconds.
trans_num	String	Number of transactions.
rollback_num	String	Number of rollbacks.
sql_num	String	Number of SQL statements.

Status code: 400

Table 4-413 Response body parameters

Parameter	Type	Description
error_code	String	Error code. Minimum length: 8 characters Maximum length: 36 characters
error_msg	String	Error message. Minimum length: 2 characters Maximum length: 512 characters

Example Request

```
POST https://{Endpoint}/rds/gaussdb/v3/instances/725398ddd0dd41d88a609acf137698e5in14/real-time-session
```

```
{
    "node_id": "bbf551e2a8c840ec9f7087690c0e63d0no14",
    "component_id": "dn_6002",
    "query_info": {
        "database_name": "db1",
        "client_ip": "11.*.*.*",
        "user_name": "user1",
                "wait": "f",
                "state": "active",
                "session_id": "725",
                "exec_time": "2",
                "wait_event": "wait gs_sleep",
                "query_runtime": "3"
    }
}
```


Example Response

Status code: 200

Success.


```
{
  "sessions" : [ {
    "session_id" : "725",
    "pid" : "140623987341056",
    "unique_sql_id" : "3545025713",
    "database_name" : "postgres",
    "client_ip" : "23.*.*.*",
    "user_name" : "root",
    "wait" : "f",
    "block_session" : "",
    "wait_event" : "none",
    "state" : "active",
    "query_runtime" : "00:00:22.641774",
    "query" : "select pg_sleep(100);",
    "back_end_start" : 1690448618645,
    "transaction_start" : 1690448618689,
    "query_start" : 1690448618689,
    "application_name" : "gsql"
     "exec_time": "97",
     "trans_num": "0",
     "rollback_num": "0",
     "sql_num": "0"
  } ]
}
```


Status Code

Status Code	Description
200	Normal
400	Abnormal

Error Code

For details, see **Error Codes**.

4.5.3 Killing Specified Sessions

Function

This API is used to kill one or more sessions of a component on a node. To call this API, you must obtain the IDs of sessions that you want to kill. To obtain the session IDs, you need to call the API for querying real-time sessions of a component on a node.

URI

POST /gaussdb/v3/instances/{instance_id}/kill-session

Table 4-414 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID. Minimum length: 32 characters Maximum length: 36 characters

Request Parameters

Table 4-415 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	Request token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Table 4-416 Request body parameters

Parameter	Mandatory	Type	Description
node_id	Yes	String	Node ID.
component_id	Yes	String	Component ID.
session_ids	Yes	Array of strings	Session IDs to be killed.

Response Parameters

Status code: 200

Table 4-417 Response body parameters

Parameter	Type	Description
session_ids	Array of strings	Session IDs to be killed.
success	Boolean	Whether the sessions are killed.

Example Request

```
POST https://{Endpoint}/rds/gaussdb/v3/instances/725398ddd0dd41d88a609acf137698e5in14/kill-session

{
  "node_id": "bbf551e2a8c840ec9f7087690c0e63d0no14",
  "component_id": "dn_6002",
   session_ids: [ "281440561845168" ]
}
```

Example Response

Status code: 200

Success.

```
{
   "session_ids" : ["281440561845168"],
   "success" : true
 }
```

Status Code

Status Code	Description
200	Normal

Error Code

For details, see **Error Codes**.

4.5.4 Killing All Idle Sessions

Function

This API is used to kill all idle sessions in the background of a component on a node. Exercise caution when using this function.

URI

POST /gaussdb/v3/instances/{instance_id}/kill-free-session

Table 4-418 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID. Minimum length: 32 characters Maximum length: 36 characters

Request Parameters

Table 4-419 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	Request token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Table 4-420 Request body parameters

Parameter	Mandatory	Type	Description
node_id	Yes	String	Node ID.
component_id	Yes	String	Component ID.

Response Parameters

Status code: 200

Table 4-421 Response body parameters

Parameter	Type	Description
success	Boolean	Whether all idle sessions are killed.

Example Request

```
POST https://{Endpoint}/rds/gaussdb/v3/instances/725398ddd0dd41d88a609acf137698e5in14/kill-free-session

{
  "node_id": "bbf551e2a8c840ec9f7087690c0e63d0no14",
  "component_id": "dn_6002"
}
```

Example Response

Status code: 200

Success.

```
{
  "success" : true
}
```

Status Code

Status Code	Description
200	Normal

Error Code

For details, see **Error Codes**.

4.5.5 Querying Table Analysis Information

Function

This API is used to query table analysis information.

Constraints

Data analysis is implemented based on WDR snapshots. Ensure that the following operations do not happen during the analysis period:

- No nodes in the instance are being rebooted within the specified time range.
- No primary/standby switchover is being performed within the specified time range.
- Performance metrics are not reset within the specified time range.
- No DROP operation is performed on the database within the specified time range.
- At least two snapshots must be generated within the specified time range.

URI

POST /gaussdb/v3/instances/{instance_id}/database-table-analysis

Table 4-422 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-423 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-token	Yes	String	Authentication token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Table 4-424 Request body parameters

Parameter	Mandatory	Type	Description
object_name	Yes	String	Table name, which can contain 1 to 63 characters. Fuzzy match is supported.
offset	Yes	String	Index offset. If **offset** is set to *N*, the resource query starts from the *N*+1 piece of data. The value is **0** by default, indicating that the query starts from the first piece of data. The value must be a number but cannot be a negative number.
limit	Yes	String	Number of records to be queried. The default value is **10**. The value cannot be a negative number. The minimum value is **1** and the maximum value is **100**.
start_time	Yes	Long	Start time. The value is a 13-digit UNIX timestamp, in milliseconds. The time zone is UTC+8.

Parameter	Mandatory	Type	Description
end_time	Yes	Long	End time. The value is a 13-digit UNIX timestamp, in milliseconds. The time zone is UTC+8.
order_by	No	String	Sorting field. The value can be **db_name**, **schema**, **rel_name**, **seq_scan**, **seq_tup_read**, **index_scan**, **index_tup_fetch**, **tuple_insert**, **tuple_update** or **tuple_delete**. The data is sorted in ascending. Currently, the sorting is controlled by the frontend. The sorting specified by the backend API does not take effect.

Response Parameters

Status code: 200

Table 4-425 Response body parameters

Parameter	Type	Description
total_count	Integer	Total number of records.
table_infos	Array of **Table 4-426** objects	Database table analysis information.

Table 4-426 DataBaseTableInfo

Parameter	Type	Description
db_name	String	Database name.
schema	String	Schema name.
rel_name	String	Table name.
seq_scan	Long	Number of sequential scans initiated by the table.
seq_tup_read	Long	Number of active rows fetched by sequential scans.
index_scan	Long	Number of index scans initiated by the table.

Parameter	Type	Description
index_tup_fetch	Long	Number of active rows fetched by index scans.
tuple_insert	Long	Number of inserted rows.
tuple_update	Long	Number of updated rows.
tuple_delete	Long	Number of deleted rows.
tuple_hot_update	Long	Number of rows HOT updated (with no separate index update required).
last_vacuum	String	Last time the table was cleared.
last_autovacuum	String	Last time the table was cleared by autovacuum.
last_analyze	String	Last time the table was analyzed.
last_auto_analyze	String	Last time the table was analyzed by autovacuum.
vacuum_count	Long	Number of times that the table is cleared.
autovacuum_count	Long	Number of times that the table is cleared by autovacuum.
analyze_count	Long	Number of times the table is manually analyzed.
auto_analyze_count	Long	Number of times that the table is analyzed by autovacuum.

Example Request

```
https://{Endpoint}/rds/gaussdb/v3/instances/6617d9e3e8874631992bb450d23cd8c8in14/database-table-analysis

 {
   "object_name" : "tbl_test",
   "offset" : 0,
   "limit" : 10,
   "start_time" : 1695176518716,
   "end_time" : 1695189910603,
   "order_by" : ""
 }
```

Example Response

Status code: 200

Database table analysis results queried.

```
{
  "table_infos" : [ {
    "db_name" : "postgres",
```

```
  "schema" : "root",
  "rel_name" : "tbl_test",
  "seq_scan" : 0,
  "seq_tup_read" : 0,
  "index_scan" : 0,
  "index_tup_fetch" : 0,
  "tuple_insert" : 0,
  "tuple_update" : 0,
  "tuple_delete" : 0,
  "tuple_hot_update" : 0,
  "last_vacuum" : "",
  "last_autovacuum" : "",
  "last_analyze" : "2023-09-20 10:49:47",
  "last_auto_analyze" : "2023-09-20 10:49:47",
  "vacuum_count" : 0,
  "autovacuum_count" : 0,
  "analyze_count" : 0,
  "auto_analyze_count" : 0
 } ],
 "total_count" : 1
}
```


Status Code

Status Code	Description
200	Normal

Error Code

For details, see **Error Codes**.

4.5.6 Querying Index Analysis Information

Function

This API is used to query index analysis information.

Constraints

Data analysis is implemented based on WDR snapshots. Ensure that the following operations do not happen during the analysis period:

- No nodes in the instance are being rebooted within the specified time range.
- No primary/standby switchover is being performed within the specified time range.
- Performance metrics are not reset within the specified time range.
- No DROP operation is performed on the database within the specified time range.
- At least two snapshots must be generated within the specified time range.

URI

POST /gaussdb/v3/instances/{instance_id}/database-index-analysis

Table 4-427 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-428 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-token	Yes	String	Authentication token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Table 4-429 Request body parameters

Parameter	Mandatory	Type	Description
object_name	Yes	String	Index name. Fuzzy match is supported.
offset	Yes	String	Index offset. If **offset** is set to *N*, the resource query starts from the *N*+1 piece of data. The value is **0** by default, indicating that the query starts from the first piece of data. The value must be a number but cannot be a negative number.
limit	Yes	String	Number of records to be queried. The default value is **10**. The value cannot be a negative number. The minimum value is **1** and the maximum value is **100**.
start_time	Yes	String	Start time. The value is a 13-digit UNIX timestamp, in milliseconds. The time zone is UTC+8.

Parameter	Mandatory	Type	Description
end_time	Yes	String	End time. The value is a 13-digit UNIX timestamp, in milliseconds. The time zone is UTC+8.
order_by	Yes	String	Sorting field. The value can be **db_name**, **schema**, **rel_name**, **index_rel_name**, **index_scan**, **index_tup_read**, or **index_tup_fetch**. The data is sorted in ascending. Currently, the sorting is controlled by the frontend. The sorting specified by the backend API does not take effect.

Response Parameters

Status code: 200

Table 4-430 Response body parameters

Parameter	Type	Description
total_count	Integer	Total number of records.
index_infos	Array of **Table 4-431** objects	Database index analysis information.

Table 4-431 DataBaseIndexInfo

Parameter	Type	Description
db_name	String	Database name.
schema	String	Schema name.
rel_name	String	Table name.
index_rel_name	String	Index name.
index_scan	Long	Number of index scans initiated on the index.
index_tup_read	Long	Number of index entries returned by scans on the index.
index_tup_fetch	Long	Number of live table rows fetched by simple index scans using the index.

Example Request

```
https://{Endpoint}/rds/gaussdb/v3/instances/6617d9e3e8874631992bb450d23cd8c8in14/database-index-analysis

{
  "object_name" : "id_index",
  "offset" : 0,
  "limit" : 10,
  "start_time" : 1695176518716,
  "end_time" : 1695190026233,
  "order_by" : ""
}
```

Example Response

Status code: 200

Database index analysis results queried.

```
{
  "index_infos" : [ {
    "db_name" : "postgres",
    "schema" : "root",
    "rel_name" : "tbl_test",
    "index_rel_name" : "id_index",
    "index_scan" : 0,
    "index_tup_read" : 0,
    "index_tup_fetch" : 0
  } ],
  "total_count" : 1
}
```

Status Code

Status Code	Description
200	Normal

Error Code

For details, see **Error Codes**.

4.5.7 Querying WDR Snapshot Status

Function

This API is used to query WDR snapshot status of a GaussDB instance.

URI

GET /gaussdb/v3/instances/{instance_id}/wdr-snapshot/status

Table 4-432 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-433 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Response Parameters

Status code: 200

Table 4-434 Response body parameters

Parameter	Type	Description
status	String	Switch status. Enumerated values: ● **ON** ● **OFF**
start_time	String	Earliest time when a snapshot is generated. The format is yyyy-mm-dd hh:mm:ss timezone. **timezone** indicates the time zone offset. For example, in the Beijing time zone, the time zone offset is shown as **+08**. If no snapshots are generated, the value is left blank.
end_time	String	Latest time when a snapshot is generated. The format is yyyy-mm-dd hh:mm:ss timezone. **timezone** indicates the time zone offset. For example, in the Beijing time zone, the time zone offset is shown as **+08**. If no snapshots are generated, the value is left blank.

Status code: default

Table 4-435 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

```
GET https://{Endpoint}/rds/gaussdb/v3/instances/{instance_id}/wdr-snapshot/status
```

Example Response

Status code: 200

WDR snapshot status returned.

```
{
  "status" : "ON",
  "start_time" : "2023-07-24 18:38:10.119531+08",
  "end_time" : "2023-08-03 11:10:31.850324+08"
}
```

Status Code

Status Code	Description
200	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.5.8 Enabling or Disabling WDR Snapshots

Function

This API is used to enable or disable WDR snapshots for a GaussDB instance. If this function is enabled, the kernel will periodically generate snapshots.

URI

PUT /gaussdb/v3/instances/{instance_id}/wdr-snapshot/status

Table 4-436 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-437 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Table 4-438 Request body parameters

Parameter	Mandatory	Type	Description
status	Yes	String	WDR snapshot status. Its value is case-insensitive. Enumerated values: • **ON** • **OFF**

Response Parameters

Status code: default

Table 4-439 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

```
PUT https://{Endpoint}/rds/gaussdb/v3/instances/{instance_id}/wdr-snapshot/status

{
  "status" : "ON"
}
```

Example Response

Status code: 202

Success.

```
{ }
```

Status Code

Status Code	Description
202	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.5.9 Querying Available WDR Snapshot Groups

Function

This API is used to query available WDR snapshot groups of a GaussDB instance. The time range where kernel reset (node restarts, primary/standby switchovers, RESET operations on performance metrics, and DROP DATABASE operations) occurs is excluded, and the snapshot list is split into multiple optional snapshot segment groups.

URI

GET /gaussdb/v3/instances/{instance_id}/wdr-snapshots/available-groups

Table 4-440 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Table 4-441 Query parameters

Parameter	Mandatory	Type	Description
begin_time	Yes	String	Query start time. The format is yyyy-mm-ddThh:mm:ssZ. **T** is the separator between calendar and hourly notation of time. **Z** indicates the time zone offset. The plus sign (+) in the time zone must be URL-encoded to %2B, and the minus sign (-) in the time zone does not need to be encoded. For example, in the Beijing time zone, the time zone offset is shown as **+0800** and the value (**2024-03-15T17:20:33+0800**) of **begin_time** needs to be encoded as **2024-03-15T17:20:33%2B0800**.
end_time	Yes	String	Query end time. The format is yyyy-mm-ddThh:mm:ssZ. **T** is the separator between calendar and hourly notation of time. **Z** indicates the time zone offset. The plus sign (+) in the time zone must be URL-encoded to %2B, and the minus sign (-) in the time zone does not need to be encoded. For example, in the Beijing time zone, the time zone offset is shown as **+0800** and the value (**2024-03-16T17:20:33+0800**) of **end_time** needs to be encoded as **2024-03-16T17:20:33%2B0800**.

Request Parameters

Table 4-442 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Response Parameters

Status code: 200

Table 4-443 Response body parameters

Parameter	Type	Description
groups	Array of **Table 4-444** objects	WDR snapshot groups.
total_count	Long	Total number of groups.

Table 4-444 WdrSnapshotGruop

Parameter	Type	Description
total_count	Long	Total number of snapshots in a group.
begin_time	Long	Group start time. It is the start time of the first snapshot in the group. The value is a 13-digit standard timestamp, for example, **1735704123000**.
end_time	Long	Group end time. It is the end time of the last snapshot in the group. The value is a 13-digit standard timestamp, for example, **1735709403000**.
snapshots	Array of **Table 4-445** objects	Snapshot records in a group.

Table 4-445 WdrSnapshotRecord

Parameter	Type	Description
id	String	Snapshot ID.
start_time	Long	Snapshot start time. The value is a 13-digit standard timestamp, for example, **1735704123000**.
end_time	Long	Snapshot end time. The value is a 13-digit standard timestamp, for example, **1735704133000**.

Status code: 400

Table 4-446 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

```
GET https://{Endpoint}/rds/gaussdb/v3/instances/{instance_id}/wdr-snapshots/available-groups?
begin_time=2025-01-01T12:01:10%2B0800&end_time=2025-01-01T16:01:10%2B0800
```

Example Response

Status code: 200

Available WDR snapshot groups queried.


```
{
  "total_count" : 2,
  "groups" : [ {
    "snapshots" : [ {
      "id" : "1",
      "start_time" : 1735704123000,
      "end_time" : 1735704603000
    }, {
      "id" : "2",
      "start_time" : 1735708203000,
      "end_time" : 1735709403000
    } ],
    "total_count" : 2,
    "begin_time" : 1735704123000,
    "end_time" : 1735709403000
  }, {
    "snapshots" : [ {
      "id" : "3",
      "start_time" : 1735713003000,
      "end_time" : 1735713183000
    }, {
      "id" : "4",
      "start_time" : 1735716783000,
```

```
      "end_time" : 1735717203000
    }, {
      "id" : "5",
      "start_time" : 1735717803000,
      "end_time" : 1735718103000
    } ],
    "total_count" : 3,
    "begin_time" : 1735713003000,
    "end_time" : 1735718103000
  } ]
}
```


Status Codes

Status Code	Description
200	Normal
400	Abnormal

Error Code

For details, see **Error Codes**.

4.5.10 Querying Collection Results for WDR Snapshot Reports

Function

This API is used to query collection results for WDR snapshot reports of GaussDB instances.

URI

GET /gaussdb/v3/instances/{instance_id}/wdr-snapshots

Table 4-447 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Table 4-448 Query parameters

Parameter	Mandatory	Type	Description
offset	No	Integer	Index offset. If **offset** is set to *N*, the resource query starts from the *N*+1 piece of data. The value is **0** by default, indicating that the query starts from the first piece of data. The value must be a number but cannot be a negative number.
limit	No	Integer	Number of records to be queried. The default value is **10**. The value cannot be a negative number. The minimum value is **1** and the maximum value is **100**.
start_time	No	String	Query start time. The format is yyyy-mm-ddThh:mm:ssZ. **T** is the separator between calendar and hourly notation of time. **Z** indicates the time zone offset. The plus sign (+) in the time zone must be URL-encoded to %2B, and the minus sign (-) in the time zone does not need to be encoded. For example, in the Beijing time zone, the time zone offset is shown as **+0800** and the value (**2024-03-15T17:20:33+0800**) of **start_time** needs to be encoded as **2024-03-15T17:20:33%2B0800**.

Parameter	Mandatory	Type	Description
end_time	No	String	Query end time. The format is yyyy-mm-ddThh:mm:ssZ. **T** is the separator between calendar and hourly notation of time. **Z** indicates the time zone offset. The plus sign (+) in the time zone must be URL-encoded to %2B, and the minus sign (-) in the time zone does not need to be encoded. For example, in the Beijing time zone, the time zone offset is shown as **+0800** and the value (**2024-03-16T17:20:33+0800**) of **end_time** needs to be encoded as **2024-03-16T17:20:33%2B0800**.
job_id	No	String	Job ID. After setting this parameter, you can query the snapshot report collection result of the specified job. Fuzzy match is not supported. You need to enter the complete job ID.
status	No	String	Status of the collection job. After setting this parameter, you can query the status of the corresponding collection job. Fuzzy match is not supported. The value is case-sensitive. Enter a complete valid value. Enumerated values: • **EXPORTING**: The collection is in progress. • **SUCCESS**: The collection is successful. • **FAILED**: The collection fails.

Parameter	Mandatory	Type	Description
wdr_type	No	String	Collection type. After setting this parameter, you can query the type of the corresponding collection job. Fuzzy match is not supported. The value is case-sensitive. Enter a complete valid value. Enumerated values: ● **cluster**: instance-level. ● **component**: component-level. ● **pdb**: tenant-level.
job_start_time	No	String	Time when the creation of the collection job starts. Results of jobs created on or after the start time can be queried. The format is yyyy-mm-ddThh:mm:ssZ. **T** is the separator between calendar and hourly notation of time. **Z** indicates the time zone offset. The plus sign (+) in the time zone must be URL-encoded to %2B, and the minus sign (-) in the time zone does not need to be encoded. For example, in the Beijing time zone, the time zone offset is shown as **+0800** and the value (**2024-03-15T17:20:33+0800**) of **job_start_time** needs to be encoded as **2024-03-15T17:20:33%2B0800**.

Parameter	Mandatory	Type	Description
job_end_time	No	String	Time when the collection job ends. Results of jobs created on or before the end time can be queried. The format is yyyy-mm-ddThh:mm:ssZ. **T** is the separator between calendar and hourly notation of time. **Z** indicates the time zone offset. The plus sign (+) in the time zone must be URL-encoded to %2B, and the minus sign (-) in the time zone does not need to be encoded. For example, in the Beijing time zone, the time zone offset is shown as **+0800** and the value (**2024-03-16T17:20:33+0800**) of **job_end_time** needs to be encoded as **2024-03-16T17:20:33%2B0800**.

Request Parameters

Table 4-449 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Response Parameters

Status code: 200

Table 4-450 Response body parameters

Parameter	Type	Description
wdr_snapshots	Array of **Table 4-451** objects	WDR snapshots in the list.
total_count	Integer	Total number of records.

Table 4-451 ListWdrSnapshotsResult

Parameter	Type	Description
job_id	String	Job ID.
file_size	Integer	File size, in KB. When the value of **status** is **SUCCESS**, the value of this parameter cannot be left blank.
wdr_type	String	Collection type. The value can be **cluster** (instance level) or **component** (component level). Enumerated values: ● **cluster**: instance-level. ● **component**: component-level. ● **pdb**: tenant-level.
start_time	String	Start time for querying WDR snapshots. The format is yyyy-mm-ddThh:mm:ssZ. **T** is the separator between the calendar and the hourly notation of time. **Z** indicates the time zone offset. The current time is always in the +0 time zone. For example, **2025-07-05T14:39:43+0000**.
end_time	String	End time for querying WDR snapshots. The format is yyyy-mm-ddThh:mm:ssZ. **T** is the separator between the calendar and the hourly notation of time. **Z** indicates the time zone offset. The current time is always in the +0 time zone. For example, 2025-07-05T15:40:03+0000.
job_create_time	String	Time when a WDR generation job was created. The format is yyyy-mm-ddThh:mm:ssZ. **T** is the separator between the calendar and the hourly notation of time. **Z** indicates the time zone offset. The current time is always in the +0 time zone. For example, **2025-07-08T10:57:59+0000**.

Parameter	Type	Description
start_snapshot_id	String	ID of the first comparison snapshot used to generate a WDR, for example, **20024**. This parameter is only valid for the collection job whose report generation mode is **snapshot_id**. This parameter is left blank if the report generation mode is **time_range**.
end_snapshot_id	String	ID of the second comparison snapshot used to generate a WDR, for example, **20025**. This parameter is only valid for the collection job whose report generation mode is **snapshot_id**. This parameter is left blank if the report generation mode is **time_range**.
status	String	Collection status. Enumerated values: ● **SUCCESS**: The collection is successful. ● **FAILED**: The collection fails. ● **EXPORTING**: The collection is in progress.
error_msg	String	Error information, which is used for O&M personnel to locate and analyze the fault.
notes	String	Remarks. When the collection type is component-level, this parameter indicates the ID of the component from which data is collected.
file_name	String	Reserved field. This field is always left blank in lightweight scenarios.
file_path	String	Reserved field. This field is always left blank in lightweight scenarios.
obs_bucket	String	Reserved field. This field is always left blank in lightweight scenarios.

Status code: 400

Table 4-452 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

```
GET https://{Endpoint}/rds/gaussdb/v3/instances/{instance_id}/wdr-snapshots?
status=SUCCESS&wdr_type=component
```

Example Response

Status code: 200

Success.


```
{
  "wdr_snapshots" : [ {
    "job_id" : "56c196275a9a59f6c4c047414d64a5a7",
    "file_size" : 1024,
    "wdr_type" : "component",
    "start_time" : "2023-07-24T18:38:10.11+0000",
    "end_time" : "2023-08-01T15:32:11.13+0000",
    "job_create_time" : "2023-07-28T20:30:10.22+0000",
    "status" : "SUCCESS",
    "notes" : "dn_6001,dn_6002"
    "file_name" : null,
    "file_path" : null,
    "obs_bucket: : null
  } ],
  "total_count" : 1
}
```


Status Code

Status Code	Description
200	Normal
400	Abnormal

Error Code

For details, see **Error Codes**.

4.5.11 Generating a WDR Snapshot Immediately

Function

This API is used to enable kernel to generate a WDR snapshot immediately.

URI

POST /gaussdb/v3/instances/{instance_id}/wdr-snapshots

Table 4-453 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-454 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Response Parameters

Status code: 400

Table 4-455 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

```
POST https://{Endpoint}/rds/gaussdb/v3/instances/{instance_id}/wdr-snapshots
```

Example Response

Status code: 200

Request result of generating a WDR snapshot immediately

```
{ }
```

Status Code

Status Code	Description
200	Normal
400	Abnormal

Error Code

For details, see **Error Codes**.

4.5.12 Collecting a WDR Snapshot Report

Function

This API is used to collect a WDR snapshot report of a GaussDB instance.

URI

POST /gaussdb/v3/instances/{instance_id}/wdr-snapshots/collect

Table 4-456 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-457 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Table 4-458 Request body parameters

Parameter	Mandatory	Type	Description
start_time	No	String	Snapshot start time. This parameter is available only when the time range mode is selected for snapshot report generation.

Parameter	Mandatory	Type	Description
end_time	No	String	Snapshot end time. This parameter is available only when the time range mode is selected for snapshot report generation.
start_snapshot_id	No	String	ID of the first snapshot to be compared. The kernel generates a report based on two snapshots. This parameter specifies the first snapshot. This parameter is valid only when the snapshot report generation mode is snapshot ID comparison. Notes: ● The snapshot ID must be valid and exist. ● The time of the first snapshot must be earlier than that of the second snapshot. ● No node reboot, primary/standby switchover, or DROP DATABASE operations were performed between the period when the two snapshots were taken.

Parameter	Mandatory	Type	Description
end_snapshot_id	No	String	ID of the second snapshot to be compared. The kernel generates a report based on two snapshots. This parameter specifies the second snapshot. This parameter is valid only when the snapshot report generation mode is snapshot ID comparison. Notes: ● The snapshot ID must be valid and exist. ● The time of the first snapshot must be earlier than that of the second snapshot. ● No node reboot, primary/standby switchover, or DROP DATABASE operations were performed between the period when the two snapshots were taken.
wdr_type	Yes	Array of strings	Collection type, which can be instance-level or component-level. For the instance level, enter **cluster**. For the component level, enter component names. The instance level and component level cannot be entered at the same time.

Parameter	Mandatory	Type	Description
mode	No	String	Snapshot report generation mode. This parameter can be left blank. By default, the time range mode is used. Enumerated values: ● **time_range**: time range mode. The **start_time** and **end_time** parameters take effect and are mandatory. The **start_snapshot_id** and **end_snapshot_id** parameters do not take effect and cannot be set. ● **snapshot_id**: snapshot ID comparison mode. The **start_time** and **end_time** parameters do not take effect and cannot be set. The **start_snapshot_id** and **end_snapshot_id** parameters take effect and are mandatory.

Response Parameters

Status code: 202

Table 4-459 Response body parameters

Parameter	Type	Description
job_id	String	Job ID.

Status code: 400

Table 4-460 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

- Collecting a component-level WDR snapshot report by time range

```
POST https://{Endpoint}/rds/gaussdb/v3/instances/{instance_id}/wdr-snapshots/collect

 {
   "start_time" : "2023-08-01T11:10:31+0800",
   "end_time" : "2023-08-02T11:10:31+0800",
   "wdr_type" : [ "dn_6001", "dn_6002" ]
 }
```

- Collecting an instance-level WDR snapshot report by time range

```
POST https://{Endpoint}/rds/gaussdb/v3/instances/{instance_id}/wdr-snapshots/collect

{
   "start_time" : "2023-08-01T11:10:31+0800",
   "end_time" : "2023-08-02T11:10:31+0800",
   "wdr_type" : [ "cluster" ],
   "mode": "time_range"
 }
```

- Collecting a component-level WDR snapshot report by snapshot ID comparison

```
POST https://{Endpoint}/rds/gaussdb/v3/instances/{instance_id}/wdr-snapshots/collect

{
   "start_snapshot_id" : "2002",
   "end_snapshot_id" : "2005",
   "wdr_type" : [ "dn_6001", "dn_6002" ],
   "mode": "snapshot_id"
 }
```

- Collecting an instance-level WDR snapshot report by snapshot ID comparison

```
POST https://{Endpoint}/rds/gaussdb/v3/instances/{instance_id}/wdr-snapshots/collect

{
   "start_snapshot_id" : "2002",
   "end_snapshot_id" : "2005",
   "wdr_type" : [ "cluster" ],
   "mode": "snapshot_id"
 }
```

Example Response

Status code: 202

Request result of collecting a WDR snapshot

```
{
  "job_id" : "56c196275a9a59f6c4c047414d64a5a7"
}
```

Status Code

Status Code	Description
202	Normal
400	Abnormal

Error Code

For details, see **Error Codes**.

4.5.13 Downloading a WDR Snapshot

Function

This API is used to download a WDR snapshot of a GaussDB instance.

URI

POST /gaussdb/v3/instances/{instance_id}/wdr-snapshots/download

Table 4-461 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-462 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Table 4-463 Request body parameter

Parameter	Mandatory	Type	Description
job_id	Yes	String	Job ID.

Response Parameters

Status code: 400

Table 4-464 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

```
POST https://{Endpoint}/rds/gaussdb/v3/instances/{instance_id}/wdr-snapshots/download

{
  "job_id" : "56c196275a9a59f6c4c047414d64a5a7"
}
```

Example Response

Status code: 200

Downloading a WDR snapshot of a GaussDB instance

```
{ }
```

Status Code

Status Code	Description
200	Normal
400	Abnormal

Error Code

For details, see **Error Codes**.

4.5.14 Obtaining Top SQL Data

Function

This API is used to obtain top SQL data in a period of time.

URI

POST /v3/instances/{instance_id}/top-sqls

Table 4-465 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-466 Request body parameters

Parameter	Mandatory	Type	Description
start_time	Yes	String	Start date.
end_time	Yes	String	End date.
db_user_name	No	String	Username.
sql_template	No	String	SQL template.
offset	No	Integer	Offset. The default value is **1**.
limit	No	Integer	Number of returned records. The default value is **10**. The value ranges from **1** to **100**.

Response Parameters

Status code: 200

Table 4-467 Response body parameters

Parameter	Type	Description
limit	Integer	Offset.
offset	Integer	Current page.
total_page	Integer	Total number of pages.
total_count	Integer	Total number of records.
rows	Array of **Table 4-468** objects	Top SQL list.

Table 4-468 TopSqlList

Parameter	Type	Description
avg_elapse_time	Integer	Average duration.
calls	Integer	Number of executions.
component_name	String	Component name.

Parameter	Type	Description
cpu_percent	String	CPU time (%).
cpu_time	Integer	Average CPU time.
db_time	Integer	Total time.
db_user_name	String	Username.
hash_count	Integer	Number of hashing execution times.
hash_time	Integer	Average hashing time.
io_percent	String	I/O time (%).
io_time	Integer	Average I/O time.
max_elapse_time	Integer	Maximum elapsed time.
net_recv_time	Integer	Average network receive time.
net_send_time	Integer	Average network transmit time.
returned_rows	Integer	Returned rows.
sql_template	String	SQL template.
time	String	Time.
tuples_fetched	Integer	Scanned rows.
unique_sql_id	Integer	Normalized SQL ID.

Status code: 400

Table 4-469 Response body parameters

Parameter	Type	Description
error_code	String	Error code. Minimum length: 8 characters Maximum length: 36 characters
error_msg	String	Error message. Minimum length: 2 characters Maximum length: 512 characters

Example Request

```
https://{Endpoint}/rds/v3/instances/abecfb3cc0784214955f574b80e4e117in14/top-sqls

{
  "start_time" : "2023-09-07T09:11:39.86Z",
  "end_time" : "2023-09-08T11:11:39.866Z",
  "limit" : 10,
  "offset" : 1
}
```

Example Response

Status code: 200

TOP SQL response body


```
{
  "limit" : 10,
  "offset" : 1,
  "total_page" : 1,
  "total_count" : 1,
  "rows" : [ {
    "cpu_time": 0.1,
    "max_elapse_time": 0.42,
    "component_name": "dn_6001_6002_6003",
    "tuples_fetched": 0.0,
    "hash_count": 0.0,
    "hash_time": 0.0,
    "returned_rows": 0,
    "net_recv_time": 0.03,
    "io_time": 0.0,
    "io_percent": "0.00%",
    "sql_template": "show synchronous_commit;",
    "calls": 3608,
    "db_time": 360.06,
    "db_user_name": "rdsAdmin",
    "avg_elapse_time": 0.1,
    "time": "2023-11-21T05:55:23.71Z",
    "net_send_time": 0.02,
"unique_sql_id": "1252930335",
"cpu_percent": "0.03%"
  } ]
}
```


Status Code

Status Code	Description
200	Normal
400	Abnormal

Error Code

For details, see **Error Codes**.

4.5.15 Querying Nodes Where Slow SQL Statements Are Executed

Function

This API is used to query nodes where slow SQL statements are executed.

URI

POST /gaussdb/v3/instances/{instance_id}/slow-sql-execute-node

Table 4-470 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-471 Request body parameter

Parameter	Mandatory	Type	Description
action	No	String	Action. For slow SQL statements, the value is **slow**. Minimum length: 2 characters Maximum length: 512 characters

Response Parameters

Status code: 200

Table 4-472 Response body parameters

Parameter	Type	Description
nodes	Array of **Table 4-473** objects	Node information.

Table 4-473 SlowSqlNodeList

Parameter	Type	Description
node_id	String	Node ID.

Parameter	Type	Description
node_name	String	Node name.
role	String	Node role.
instance_id	String	Instance ID.
component_type	String	Component type.

Status code: 400

Table 4-474 Response body parameters

Parameter	Type	Description
error_code	String	Error code. Minimum length: 8 characters Maximum length: 36 characters
error_msg	String	Error message. Minimum length: 2 characters Maximum length: 512 characters

Example Request

```
https://{Endpoint}/rds/gaussdb/v3/instances/d04686c6baae4f65a742771186f47b7ain14/slow-sql-execute-node

{
  "action" : "slow"
}
```

Example Response

Status code: 200

Node information is queried.

```
{
  "nodes" : [ {
    "node_id" : "0d9ca6ce53ef4c158aab7084e9885108no14",
    "node_name" : "gauss-mcs-xhy_gaussdbv5_hcs2-0_0",
    "role" : "master",
    "instance_id" : "d04686c6baae4f65a742771186f47b7ain14",
    "component_type" : "CN"
  } ]
}
```

Status Code

Status Code	Description
200	Normal
400	Abnormal

Error Code

For details, see **Error Codes**.

4.5.16 Querying Slow SQL Statements

Function

This API is used to query slow SQL statements.

URI

POST /gaussdb/v3/instances/{instance_id}/slow-sql-list

Table 4-475 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-476 Request body parameters

Parameter	Mandatory	Type	Description
start_time	Yes	Long	Start time, in milliseconds. The value is a 13-digit UNIX timestamp. The time zone is UTC.
end_time	Yes	Long	End time, in milliseconds. The value is a 13-digit UNIX timestamp. The time zone is UTC.
instance_id	Yes	String	Instance ID, which must be the same as that in the path.

Parameter	Mandatory	Type	Description
threshold	Yes	Integer	Slow SQL threshold, in seconds. The value must range from **1** to **99**.
node_ids	Yes	Array of strings	Node IDs. The array cannot be null or empty.

Response Parameters

Status code: 200

Table 4-477 Response body parameters

Parameter	Type	Description
total	Integer	Total number of records.
slow_sql_infos	Array of **Table 4-478** objects	Slow SQL statements.

Table 4-478 SlowSqlListRes

Parameter	Type	Description
schema_name	String	Schema name.
db_name	String	Database name.
sql_id	String	SQL ID.
user_name	String	Username.
node_id	String	Node ID.
node_name	String	Node name.
sql_text	String	SQL template.
query_plan	String	Execution plan.
query_plan_optimize	String	Execution plan (optimized).
calls	Integer	Number of executions.
avg_exec_time	String	Average time (ms).
avg_cpu_time	String	CPU time (ms).
avg_io_time	String	I/O time (ms).

Parameter	Type	Description
avg_returned_rows	Integer	Returned rows.
avg_fetched_rows	Integer	Scanned rows.
buffer_hit_ratio	String	Hit rate.

Status code: 400

Table 4-479 Response body parameters

Parameter	Type	Description
error_code	String	Error code. Minimum length: 8 characters Maximum length: 36 characters
error_msg	String	Error message. Minimum length: 2 characters Maximum length: 512 characters

Example Request

```
https://{Endpoint}/rds/gaussdb/v3/instances/d04686c6baae4f65a742771186f47b7ain14/slow-sql-list

{
  "instance_id" : "d04686c6baae4f65a742771186f47b7ain14",
  "node_ids" : [ "88391813e1cb4e93899b210e880fdb1bno20" ],
  "start_time" : 1686903754933,
  "end_time" : 1686990154933,
  "threshold" : 1
}
```

Example Response

Status code: 200

Response body

```
{
  "slow_sql_infos" : [ {
    "schema_name": "root,public",
    "db_name": "postgres",
    "sql_id" : "6789",
    "user_name" : "john_doe",
    "sql_text" : "SELECT * FROM large_table WHERE column1 = 'value';",
    "query_plan" : "Seq Scan on large_table ...",
    "query_plan_optimize": "Datanode Name: dn_6001\n id | operation |    cost    | rows | width \n----
+-----------+------------+------+-------\n  1 | Result    | 0.00..0.01 |    1 |     0 \n",
    "calls" : 5,
    "avg_exec_time" : "5000000",
```

```
    "avg_cpu_time" : "1000000",
    "avg_io_time" : "2000000",
    "avg_returned_rows" : 1000,
    "avg_fetched_rows" : 5000,
    "buffer_hit_ratio" : "0.8",
    "node_id" : "88391813e1cb4e93899b210e880fdb1bno20",
    "node_name" : "Node1"
  } ],
"total":1
}
```


Status Code

Status Code	Description
200	Normal
400	Abnormal

Error Code

For details, see **Error Codes**.

4.5.17 Querying Details About a Slow SQL Statement

Function

This API is used to query details about a slow SQL statement.

URI

POST /gaussdb/v3/instances/{instance_id}/slow-sql-detail

Table 4-480 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-481 Request body parameters

Parameter	Mandatory	Type	Description
start_time	Yes	Long	Start time, in milliseconds. The value is a 13-digit UNIX timestamp. The time zone is UTC.

Parameter	Mandatory	Type	Description
end_time	Yes	Long	End time, in milliseconds. The value is a 13-digit UNIX timestamp. The time zone is UTC.
instance_id	Yes	String	Instance ID.
sql_id	Yes	String	ID of the slow SQL statement.
node_ids	Yes	Array of strings	Node IDs. The array cannot be null or empty.

Response Parameters

Status code: 200

Table 4-482 Response body parameters

Parameter	Type	Description
slow_sql_details	Array of **Table 4-483** objects	Details of slow SQL statements.
total	Integer	Total number of records.

Table 4-483 SlowSqlDetails

Parameter	Type	Description
schema_name	String	Schema name.
db_name	String	Database name.
sql_id	String	SQL ID.
user_name	String	Username.
client_ip	String	Client IP address.
client_port	String	Client port.
node_id	String	Node ID.
node_name	String	Node name.
sql_text	String	SQL template.
query_plan	String	Execution plan.
query_plan_optimize	String	Execution plan (optimized).

Parameter	Type	Description
start_time	Integer	Start time (ms).
finish_time	Integer	End time (ms).
returned_rows	Integer	Returned rows.
fetched_rows	Integer	Scanned rows.
fetched_pages	Integer	Scanned pages.
hit_pages	Integer	Hit pages.
total_time	String	Total valid DB time (unit: ms), which is accumulated if multiple threads are involved.
cpu_time	String	CPU time (ms).
plan_time	String	Planned time.
io_time	String	I/O time.
lock_count	Integer	Number of locks.

Status code: 400

Table 4-484 Response body parameters

Parameter	Type	Description
error_code	String	Error code. Minimum length: 8 characters Maximum length: 36 characters
error_msg	String	Error message. Minimum length: 2 characters Maximum length: 512 characters

Example Request

```
https://{Endpoint}/rds/gaussdb/v3/instances/d04686c6baae4f65a742771186f47b7ain14/slow-sql-detail

{
  "instance_id" : "d04686c6baae4f65a742771186f47b7ain14",
  "node_ids" : [ "88391813e1cb4e93899b210e880fdb1bno20" ],
  "start_time" : 1686903754933,
  "end_time" : 1686990154933,
  "sql_id" : 123
}
```

Example Response

Status code: 200

Response body of SQL details


```
{
  "slow_sql_details" : [ {
    "schema_name": "root,public",
    "db_name": "postgres",
    "sql_id" : "12345",
    "user_name" : "john_doe",
    "client_ip" : "192.168.*.*",
    "client_port" : "5432",
    "sql_text" : "SELECT * FROM large_table WHERE column1 = 'value';",
    "query_plan" : "Seq Scan on large_table ...",
    "query_plan_optimize": "Datanode Name: dn_6001\n id | operation |    cost    | rows | width \n----
+-----------+------------+------+-------\n  1 | Result    | 0.00..0.01 |    1 |     0 \n",
    "start_time" : 1619452000,
    "finish_time" : 1619453000,
    "returned_rows" : 1000,
    "fetched_rows" : 5000,
    "fetched_pages" : 50,
    "hit_pages" : 25,
    "total_time" : "1000000",
    "cpu_time" : "500000",
    "plan_time" : "100000",
    "io_time" : "400000",
    "lock_count" : 1,
    "lock_time" : "1000",
    "node_id" : "88391813e1cb4e93899b210e880fdb1bno20",
    "node_name" : "Node1"
  } ],
"total": 1
}
```


Status Code

Status Code	Description
200	Normal
400	Abnormal

Error Code

For details, see **Error Codes**.

4.5.18 Creating a SQL Patch

Function

This API is used to create an SQL patch for a slow SQL statement. The value of **sql_id** is obtained from the **Slow SQL** page.

URI

POST /gaussdb/v3/instances/{instance_id}/sql-patch

Table 4-485 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-486 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Table 4-487 Request body parameters

Parameter	Mandatory	Type	Description
node_id	Yes	String	Node ID. Only CN/DN node IDs are supported.
sql_id	Yes	String	ID of the slow SQL statement, which is obtained from the **Slow SQL** page.
patch_name	Yes	String	Patch name. Minimum length: 1 character Maximum length: 63 characters
hint	Yes	String	Hint content. The hint content cannot contain special characters: |;<>`\'{}[]~?!\n Minimum length: 1 character Maximum length: 1024 characters

Response Parameters

Status code: 400

Table 4-488 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

```
POST https://{Endpoint}/rds/gaussdb/v3/instances/d8e6ca5a624745bcb546a227aa3ae1cfin14/sql-patch

{
  "node_id" : "d8e6ca5a624745bcb546a227aa3ae1cfno14",
  "sql_id" : "28f61af50fc9452aa0ed5ea25c3cc3d3",
  "patch_name" : "patch1",
  "hint" : "indexscan(t1)",
}
```

Example Response

Status code: 200

Success

```
success
```

Status code: 400

Error

```
{
  "error_code" : "DBS.2x",
  "error_msg" : "Cannot accept sql patches with duplicate unique sql id."
}
```

Status Code

Status Code	Description
200	Normal
400	Abnormal

Error Code

For details, see **Error Codes**.

4.5.19 Querying SQL Patch Information

Function

This API is used to query the SQL patch information of a slow SQL statement based on the SQL ID. The value of **sql_id** is obtained from the **Slow SQL** page.

URI

GET /gaussdb/v3/instances/{instance_id}/sql-patch

Table 4-489 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-490 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Table 4-491 Request query parameters

Parameter	Mandatory	Type	Description
node_id	Yes	String	Node ID. Only CN/DN node IDs are supported.
sql_id	Yes	String	ID of the slow SQL statement, which is obtained from the **Slow SQL** page.

Response Parameters

Status code: 200

Table 4-492 Response body parameters

Parameter	Type	Description
patch_name	String	Patch name.
hint	String	Patch content (hint text).

Parameter	Type	Description
patch_status	String	Patch status.

Status code: 400

Table 4-493 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

```
GET https://{Endpoint}/rds/gaussdb/v3/instances/pd7e6ca5a624745bcb546a227aa3ae1cfin14/sql-patch

{
  "node_id" : "d8e6ca5a624745bcb546a227aa3ae1cfno14",
  "sql_id" : "28f61af50fc9452aa0ed5ea25c3cc3d3"
}
```

Example Response

Status code: 200

Success

```
{
  "patch_name" : "patch1",
  "hint" : "indexscan(t1)",
  "patch_status" : "enabled/disabled"
}
```

Status code: 400

Error

```
{
  "error_code" : "DBS.2x",
  "error_msg" : "Cannot accept sql patches with duplicate unique sql id."
}
```

Status Code

Status Code	Description
200	Normal
400	Abnormal

Error Code

For details, see **Error Codes**.

4.5.20 Enabling or Disabling SQL Patch or Deleting a SQL Patch

Function

This API is used to enable SQL Patch, disable SQL Patch, or Delete a SQL patch.

URI

POST /gaussdb/v3/instances/{instance_id}/handle-sql-patch

Table 4-494 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-495 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Table 4-496 Request body parameters

Parameter	Mandatory	Type	Description
node_id	Yes	String	Node ID. Only CN/DN node IDs are supported.
sql_id	Yes	String	ID of the slow SQL statement, which is obtained from the **Slow SQL** page.

Parameter	Mandatory	Type	Description
patch_name	Yes	String	Patch name. Minimum length: 1 character Maximum length: 63 characters
action	Yes	String	Operations. The value can be **open** (enabling SQL Patch), **close** (disabling SQL Patch), or **drop** (deleting a SQL patch). Enumerated values: • **open** • **close** • **drop**

Response Parameters

Status code: 400

Table 4-497 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

```
POST https://{Endpoint}/rds/gaussdb/v3/instances/ds26ca5a624745bcb546a227aa3ae1cfin14/handle-sql-patch

{
  "node_id" : "d8e6ca5a624745bcb546a227aa3ae1cfno14",
  "sql_id" : "28f61af50fc9452aa0ed5ea25c3cc3d3",
  "patch_name" : "patch1",
  "action" : "open/close/drop"
}
```

Example Response

Status code: 200

Success

```
success
```

Status code: 400

Error

```
{
  "error_code" : "DBS.2x",
  "error_msg" : "Cannot accept sql patches with duplicate unique sql id."
}
```

Status Code

Status Code	Description
200	Normal
400	Abnormal

Error Code

For details, see **Error Codes**.

4.6 Intelligent O&M

4.6.1 Managing an Instance or Canceling Management for an Instance

Function

This API is used to manage an instance or cancel the management for an instance.

URI

POST /v3/db-assist/manage-instance

Request Parameters

Table 4-498 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Table 4-499 Request body parameters

Parameter	Mandatory	Type	Description
db_assist_id	Yes	String	ID of the DBMind instance.
instance_ids	Yes	Array of strings	IDs of the instances to be managed or unmanaged.
manage_type	Yes	String	Management status. **ENABLED** indicates that the instances are managed, and **DISABLED** indicates that the instances are not managed. Enumerated values: • **ENABLED** • **DISABLED**

Response Parameters

Status code: 202

Table 4-500 Response body parameters

Parameter	Type	Description
job_id	String	Job ID.

Status code: default

Table 4-501 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

Managing instances

```
POST https://{Endpoint}/rds/v3/db-assist/manage-instance

{
  "db_assist_id" : "6997795f659840d595232225d9a70d2din19",
  "instance_ids" : [ "ad40c3ff2603429c9eb5511361047933in14",
"ad40c3ff2603429c9eb5511361047933in15" ],
  "manage_type" : "ENABLED"
}
```

Example Response

Status code: 202

Success.


```
{
  "job_id" : "92e9d5907fedef045b5f93e338730caf"
}
```


Status Code

Status Code	Description
202	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.6.2 Querying Instances That Have Been Managed or Can Be Managed by a DBMind Instance

Function

This API is used to query instances that have been managed or can be managed by a DBMind Instance.

Constraints

This API is used to query instances that have been managed or can be managed by a DBMind Instance.

URI

GET /v3/db-assist/manageable-instances

Table 4-502 Query parameters

Parameter	Mandatory	Type	Description
db_assist_id	Yes	String	ID of the DBMind instance. When you query instances that can be managed, only the instances whose architecture is the same as that of the DBMind instance are displayed.

Parameter	Mandatory	Type	Description
instance_id	No	String	Instance ID.
instance_name	No	String	Instance name.
manage_type	Yes	String	Management type. The value can be **ENABLED** (managed instances) or **DISABLED** (unmanaged instances). Enumerated values: • **ENABLED** • **DISABLED**
offset	No	Integer	Index offset. If **offset** is set to *N*, the resource query starts from the *N*+1 piece of data. The value is **0** by default, indicating that the query starts from the first piece of data. The value must be a number but cannot be a negative number. Minimum value: **0** Default value: **0**
limit	No	Integer	Number of records to be queried. The default value is **100**. The value cannot be a negative number. The minimum value is **1** and the maximum value is **100**. Minimum value: **1** Maximum value: **100** Default value: **100**

Request Parameters

Table 4-503 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.

Parameter	Mandatory	Type	Description
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Response Parameters

Status code: 200

Table 4-504 Response body parameters

Parameter	Type	Description
managed_instances	Array of **Table 4-505** objects	Managed instances.
unmanaged_instances	Array of **Table 4-505** objects	Instances that can be managed.
total_count	Integer	Total number of instances.

Table 4-505 DBAssistManageableInstances

Parameter	Type	Description
id	String	Instance ID.
name	String	Instance name.
status	String	Instance status.
manage_status	String	Management status.
engine_name	Object	DB engine name.
engine_version	String	DB engine version.
ha_mode	String	HA mode.
volume_type	String	Storage type.
virtualization_type	String	Virtual server type.
server_type	String	Server type.

Parameter	Type	Description
vpc_id	String	Cloud ID.
vpc_ids	Array of strings	Cloud IDs.
actions	Array of strings	Operations.

Status code: default

Table 4-506 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

```
GET https://{Endpoint}/rds/v3/db-assist/manageable-instances?
db_assist_id={db_assist_id}&manage_type={manage_type}
```

Example Response

Status code: 200

Success.


```
{
  "managed_instances" : [ ],
  "unmanaged_instances" : [ {
    "id" : "30ea03e211d047d09d8ca42d6176e3d3in14",
    "name" : "gauss-for-dbmind",
    "status" : "SHUTDOWN",
    "manage_status" : null,
    "upgrade_status" : null,
    "engine_name" : "gaussdbv5",
    "engine_version" : "8.0.1",
    "ha_mode" : "Ha",
    "volume_type" : "LOCALSSD",
    "virtualization_type" : "bms",
    "server_type" : "BMS",
    "vpc_id" : null,
    "vpc_ids" : [ "d3ab50fc-3d09-4821-90d5-f398035b2cef" ],
    "actions" : [ "START_DB" ]
  } ],
  "total" : 1
}
```

Status Code

Status Code	Description
200	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.6.3 (Recommended) Querying Instances That Have Been Managed or Can Be Managed by a DBMind Instance

Function

This API is used to query instances that have been managed or can be managed by a DBMind Instance.

Constraints

This API is used to query instances that have been managed or can be managed by a DBMind Instance.

URI

GET /v3.1/db-assist/manageable-instances

Table 4-507 Query parameters

Parameter	Mandatory	Type	Description
db_assist_id	Yes	String	ID of the DBMind instance. When you query instances that can be managed, only the instances whose architecture is the same as that of the DBMind instance are displayed.
instance_id	No	String	Instance ID.
instance_name	No	String	Instance name.

Parameter	Mandatory	Type	Description
manage_type	Yes	String	Management type. The value can be **ENABLED** (managed instances) or **DISABLED** (unmanaged instances). Enumerated values: ● **ENABLED** ● **DISABLED**
offset	No	Integer	Index offset. If **offset** is set to *N*, the resource query starts from the *N*+1 piece of data. The value is **0** by default, indicating that the query starts from the first piece of data. The value must be a number but cannot be a negative number. Minimum value: **0** Default value: **0**
limit	No	Integer	Number of records to be queried. The default value is **100**. The value cannot be a negative number. The minimum value is **1** and the maximum value is **100**. Minimum value: **1** Maximum value: **100** Default value: **100**

Request Parameters

Table 4-508 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Response Parameters

Status code: 200

Table 4-509 Response body parameters

Parameter	Type	Description
managed_instances	Array of **Table 4-510** objects	Managed instances.
unmanaged_instances	Array of **Table 4-510** objects	Instances that can be managed.
total_count	Integer	Total number of instances.

Table 4-510 DBAssistManageableInstances

Parameter	Type	Description
id	String	Instance ID.
name	String	Instance name.
status	String	Instance status.
manage_status	String	Management status.
engine_name	Object	DB engine name.
engine_version	String	DB engine version.
ha_mode	String	HA mode.
volume_type	String	Storage type.
virtualization_type	String	Virtual server type.
server_type	String	Server type.
vpc_id	String	Cloud ID.
vpc_ids	Array of strings	Cloud IDs.
actions	Array of strings	Operations.

Status code: default

Table 4-511 Response body parameters

Parameter	Type	Description
error_code	String	Error Code
error_msg	String	Error message.

Example Request

```
GET https://{Endpoint}/rds/v3.1/db-assist/manageable-instances?
db_assist_id={db_assist_id}&manage_type={manage_type}
```

Example Response

Status code: 200

Success.


```
{
  "managed_instances" : [ ],
  "unmanaged_instances" : [ {
    "id" : "30ea03e211d047d09d8ca42d6176e3d3in14",
    "name" : "gauss-for-dbmind",
    "status" : "SHUTDOWN",
    "manage_status" : null,
    "upgrade_status" : null,
    "engine_name" : "gaussdbv5",
    "engine_version" : "V2.0-8.0.1",
    "ha_mode" : "Ha",
    "volume_type" : "LOCALSSD",
    "virtualization_type" : "bms",
    "server_type" : "BMS",
    "vpc_id" : null,
    "vpc_ids" : [ "d3ab50fc-3d09-4821-90d5-f398035b2cef" ],
    "actions" : [ "START_DB" ]
  } ],
  "total" : 1
}
```


Status Code

Status Code	Description
200	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.6.4 Querying DBMind Instances

Function

This API is used to query DBMind instances.

Constraints

This API is used to only query DBMind instances.

URI

GET /v3/db-assist/instances

Table 4-512 Query parameters

Parameter	Mandatory	Type	Description
instance_id	No	String	Instance ID.
instance_name	No	String	Instance name.
offset	No	Integer	Index offset. If **offset** is set to *N*, the resource query starts from the *N*+1 piece of data. The value is **0** by default, indicating that the query starts from the first piece of data. The value must be a number but cannot be a negative number. Minimum value: **0** Default value: **0**
limit	No	Integer	Number of records to be queried. The default value is **100**. The value cannot be a negative number. The minimum value is **1** and the maximum value is **100**. Minimum value: **1** Maximum value: **100** Default value: **100**

Request Parameters

Table 4-513 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	No	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Response Parameters

Status code: 200

Table 4-514 Response body parameters

Parameter	Type	Description
instances	Array of **Table 4-515** objects	Instances.
total_count	Integer	Total number of instances.

Table 4-515 DBAssistInstanceInfo

Parameter	Type	Description
id	String	Instance ID.
name	String	Instance name.
port	Integer	Instance port.
cpu	String	CPU size.
arch	String	Architecture type of the DBMind instance.
mem	String	Memory size.
actions	Array of strings	Operations.
engine_name	String	Engine name.
engine_version	String	DB engine version.
project_id	String	Project ID.

Parameter	Type	Description
instance_status	String	Instance status. Value: ● **normal** ● **abnormal** ● **creating** ● **createfail** ● **managing** ● **manage_failed** ● **data_disk_full** ● **shutdown**
create_at	Long	Creation time.
ha_mode	String	Instance type.
volume_type	String	Storage type.
virtualization_type	String	Virtual server type. Enumerated values: ● **XEN** ● **KVM** ● **MCS** ● **ARM** ● **BMS**
server_type	String	Server type.
spec_code	String	Instance specifications code.
managed_num	Integer	Number of managed instances.
manageable_limit	Integer	Maximum number of managed instances.
metadata_id	String	Metadata instance ID.
metadata_name	String	Metadata instance name.
vpc_id	Object	Cloud ID.
vpc_ids	Array of strings	Cloud IDs.

Status code: default

Table 4-516 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

```
GET https://{Endpoint}/rds/v3/db-assist/instances
```

Example Response

Status code: 200

Success.


```
{
  "instances" : [
  {
    "id" : "2653666aced0487ba749611e6863e467in20",
    "name" : "liujian-456322-12-22-02",
    "port" : 8000,
    "cpu" : "8",
    "arch" : "ARM",
    "mem" : "64",
    "actions" : null,
    "engine_name" : "gaussdbv5-dbmind",
    "engine_version" : "8.1.0",
    "project_id" : "179a755507c841b2bef7906f50d001ee",
    "instance_status" : "FAILED",
    "create_at" : 1703222887678,
    "ha_mode" : "Ha",
    "volume_type" : "LOCALSSD",
    "virtualization_type" : "bms",
    "server_type" : "BMS",
    "spec_code" : "physical.gaussdb.opengauss.8u.64g.arm",
    "managed_num" : 0,
    "manageable_limit" : 8,
    "metadata_id" : "2653666aced0487ba749611e6863e467in20",
    "metadata_name" : "liujian-456322-12-22-02",
    "vpc_id" : null,
    "vpc_ids" : [ "85354f1d-1a81-4633-b522-9ed1f0fa45ff" ]
  }
  ],
  "total_count" : 1
}
```


Status Code

Status Code	Description
200	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.6.5 (Recommended) Querying DBMind Instances

Function

This API is used to query DBMind instances.

Constraints

This API is used to only query DBMind instances.

URI

GET /v3.1/db-assist/instances

Table 4-517 Query parameters

Parameter	Mandatory	Type	Description
instance_id	No	String	Instance ID.
instance_name	No	String	Instance name.
offset	No	Integer	Index offset. If **offset** is set to *N*, the resource query starts from the *N*+1 piece of data. The value is **0** by default, indicating that the query starts from the first piece of data. The value must be a number but cannot be a negative number. Minimum value: **0** Default value: **0**
limit	No	Integer	Number of records to be queried. The default value is **100**. The value cannot be a negative number. The minimum value is **1** and the maximum value is **100**. Minimum value: **1** Maximum value: **100** Default value: **100**

Request Parameters

Table 4-518 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	No	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Response Parameters

Status code: 200

Table 4-519 Response body parameters

Parameter	Type	Description
instances	Array of **Table 4-520** objects	Instances.
total_count	Integer	Total number of instances.

Table 4-520 DBAssistInstanceInfo

Parameter	Type	Description
id	String	Instance ID.
name	String	Instance name.
port	Integer	Instance port.
cpu	String	CPU size.
arch	String	Architecture type of the DBMind instance.
mem	String	Memory size.
actions	Array of strings	Operations.
engine_name	String	Engine name.
engine_version	String	DB engine version.
project_id	String	Project ID.

Parameter	Type	Description
instance_status	String	Instance status. Value: • **normal** • **abnormal** • **creating** • **createfail** • **managing** • **manage_failed** • **data_disk_full** • **shutdown**
create_at	Long	Creation time.
ha_mode	String	Instance type.
volume_type	String	Storage type.
virtualization_type	String	Virtual server type. Enumerated values: • **XEN** • **KVM** • **MCS** • **ARM** • **BMS**
server_type	String	Server type.
spec_code	String	Instance specifications code.
managed_num	Integer	Number of managed instances.
manageable_limit	Integer	Maximum number of managed instances.
metadata_id	String	Metadata instance ID.
metadata_name	String	Metadata instance name.
vpc_id	Object	Cloud ID.
vpc_ids	Array of strings	Cloud IDs.

Status code: default

Table 4-521 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

```
GET https://{Endpoint}/rds/v3.1/db-assist/instances
```

Example Response

Status code: 200

Success.


```
{
  "instances" : [
  {
    "id" : "2653666aced0487ba749611e6863e467in20",
    "name" : "liujian-456322-12-22-02",
    "port" : 8000,
    "cpu" : "8",
    "arch" : "ARM",
    "mem" : "64",
    "actions" : null,
    "engine_name" : "gaussdbv5-dbmind",
    "engine_version" : "V2.0-8.1.0",
    "project_id" : "179a755507c841b2bef7906f50d001ee",
    "instance_status" : "FAILED",
    "create_at" : 1703222887678,
    "ha_mode" : "Ha",
    "volume_type" : "LOCALSSD",
    "virtualization_type" : "bms",
    "server_type" : "BMS",
    "spec_code" : "physical.gaussdb.opengauss.8u.64g.arm",
    "managed_num" : 0,
    "manageable_limit" : 8,
    "metadata_id" : "2653666aced0487ba749611e6863e467in20",
    "metadata_name" : "liujian-456322-12-22-02",
    "vpc_id" : null,
    "vpc_ids" : [ "85354f1d-1a81-4633-b522-9ed1f0fa45ff" ]
   }
  ],
  "total_count" : 1
}
{
    "total_count": 1,
    "data": [
        {
            "id": "a60b49af-1d66-415a-b930-9fc4cbd3c264",
            "cmd_name": "query_specified_thread_memory",
            "cmd_type": "SQL",
            "cmd": "select * from gs_get_thread_memctx_detail({tid},{contextname});",
            "call_type": "sync",
            "constraints": [
                {
                    "name": "db_version",
                    "min_value": "3.100.0",
                    "max_value": null,
                    "enum_value": null
                },
                {
```

```
                "name": "kernel_version",
                "min_value": "503",
                "max_value": null,
                "enum_value": null
              }
            ],
            "param": [
              {
                "param_id": "e7523d6f-7ab0-4312-bb34-4b8070b11615",
                "param_name": "tid",
                "param_type": "single",
                "data_type": "string",
                "required": true,
                "default_value": null,
                "pattern": "[0-9]{1,256}",
                "description": "Thread ID"
              },
              {
                "param_id": "880084d8-44b9-42fe-adf7-b37032580566",
                "param_name": "contextname",
                "param_type": "single",
                "data_type": "string",
                "required": true,
                "default_value": null,
                "pattern": "[A-Za-z0-9_\\-\\s]{1,256}",
                "description": "Memory context name (contextname column in the result returned by
gs_session_memory_context)"
              }
            ],
            "is_support": true,
            "description": "Memory usage of the thread with the ID"
            "remark": "",
            "node_type": null,
            "cluster_mode": "All",
            "category": "memory"
          }
        ]
}
```


Status Code

Status Code	Description
200	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.6.6 Querying Common O&M Commands (SQL and Python Commands)

Function

This API is used to query preset common O&M commands of TPOPS. You can query SQL and Python commands supported by a specified instance. The command information includes the command name, command ID, command content, function description, supported kernel version, command type, and command parameter list.

URI

GET /gaussdb/v3/instances/{instance_id}/common-operations

Table 4-522 Query parameters

Parameter	Mandatory	Type	Description
cmd_type	Yes	String	Command type. Enumerated values: • **SQL** or **sql**: SQL command • **PYTHON** or **python**: Python command
is_show_detail	No	Boolean	Whether the command text is displayed in the returned result list. Enumerated values: • **true**: The command details are displayed. • **false**: The command details are not displayed. The default value is **false**.
cmd_name	No	String	Query by command name. Fuzzy query is supported.
cmd_id	No	String	Specified command ID used to query the corresponding command. Fuzzy query is not supported. If the deployment model (centralized or distributed) of the instance does not meet the requirements of the specified command, the returned query result is empty.
offset	No	Integer	Index offset. The query starts from the next piece of data indexed by this parameter. The default value is **0**, indicating that the query starts from the first data record. The value cannot be a negative number.

Parameter	Mandatory	Type	Description
limit	No	Integer	Number of records to be queried. The default value is **10**. The value cannot be a negative number. The minimum value is **1**, and the maximum value is **1000**.

Request Parameters

Table 4-523 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Response Parameters

Status code: 200

Table 4-524 Response body parameters

Parameter	Type	Description
data	Array of **CommonOmOperationDetail** objects	Command details. For details, see **Table 4-525**.
total_count	Integer	Total number of preset O&M commands supported by an instance.

Table 4-525 CommonOmOperationDetail

Parameter	Type	Description
id	String	O&M command ID.

Parameter	Type	Description
cmd_name	String	O&M command name.
cmd_type	String	O&M command type. Enumerated values: ● **SQL**: SQL command. ● **PYTHON**: Python command.
cmd	String	O&M command content.
call_type	String	Call type of the command delivered to the Agent. Enumerated values: ● **sync**: synchronous command delivery ● **async**: asynchronous command delivery Note that SQL commands are synchronous, while Python commands are usually asynchronous because they take longer to execute. After an asynchronous command is delivered, you need to call the API for querying the asynchronous command execution result and poll the API repeatedly until the final execution result is returned.
constraints	Array of **ConstraintItem** objects	Constraint items. For details, see **Table 4-526**.
param	Array of **OperationParam** objects	Command parameters. For details, see **Table 4-527**.
is_support	Boolean	Whether an instance supports the command. The constraints (such as the engine version requirements of the database instance) in the command configuration are used for verification. If the verification is successful, the instance supports the command.
node_type	String	Type of the node where the command is executed. Enumerated values: ● **dn**: DN ● **cn**: CN ● Other: reserved scenario, to be determined If this parameter is left empty, there is no constraint.

Parameter	Type	Description
description	String	Description of the command function.
remark	String	Command-related remarks.
cluster_mode	String	Deployment model of the instance on which the command is restricted. Enumerated values: ● **Ha**: The command can be executed only on centralized instances. ● **Distributed**: The command can be executed only on distributed instances. ● **All**: The command can be executed on all instances.
category	String	Category of the command by service domain.

Table 4-526 ConstraintItem

Parameter	Type	Description
name	String	Constraint item name.
min_value	String	Minimum value.
max_value	String	Maximum value.
enum_value	Array of strings	Enumerated values.

Table 4-527 OperationParam

Parameter	Type	Description
param_id	String	Parameter ID. Each parameter of a preset command in TPOPS is mapped to a unique ID and its configuration. The ID uniquely identifies the parameter configuration preset in the system.
param_name	String	Parameter name. When a parameter is delivered, the TPOPS platform verifies the parameter validity based on the actual service scenario (associated with related verification and matching rules) of the parameter.

Parameter	Type	Description
param_type	String	Parameter type. It specifies whether a parameter is of single-value or multi-value (array) type. The value can be **single** or **array**. The default value is **single**.
data_type	String	Data type. It specifies the data type of a parameter field. Enumerated values: • **string**: character string • **integer**: integer • **Bool**: Boolean • **enum**: enumeration • **float**: floating-point number • **time_str**: formatted time, for example, **2023-08-01T11:10:31+0800**. • **time_stamp**: timestamp, for example, **1754586061000** The value range of a specific data type is associated with a configuration item on TPOPS to store the corresponding verification matching rule. The rule is used to verify the value of this parameter during command delivery. For details about the constraints on each parameter, see the **pattern** field, or contact technical support or the administrator.
required	Boolean	Whether it is mandatory. It specifies whether a parameter is mandatory when a command is delivered. Enumerated values: • **false**: The parameter is optional. In this case, if the parameter has a default value (not empty), the default value is used when the command is delivered. • **true**: The parameter is mandatory. Currently, this parameter is valid only for a single-value parameter (that is, **param_type** is set to **single**).

Parameter	Type	Description
default_value	String	Default value of a parameter, which is stored as a character string. If this value is not empty and the corresponding parameter is optional (that is, **required** is set to **false**), this default value is used to fill in the parameter when the command is delivered. Currently, this parameter is valid only for a single-value parameter (that is, **param_type** is set to **single**).
pattern	String	Pattern matching of parameter values. When a command is delivered, this matching mode is used to perform regular expression verification on parameter validity. If a parameter is of array or multi-value type (that is, **param_type** is set to **array**), regular expression verification is performed on each value of the parameter in the array.
description	String	Description of a parameter. It can include possible precautions.

Status code: 400

Table 4-528 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

- Querying the details of common O&M SQL commands by page
```
GET https://{Endpoint}/rds/gaussdb/v3/instances/e8a3df12517e45e49d6eca167e72b934in14/common-operations?cmd_type=SQL&is_show_detail=true&offset=0&limit=2
```
- Querying details of common O&M Python commands by page and filtering the commands by command name
```
GET https://{Endpoint}/rds/gaussdb/v3/instances/e8a3df12517e45e49d6eca167e72b934in14/common-operations?cmd_type=PYTHON&is_show_detail=false&cmd_name=auto_emergency_scenario_data_collect_ha&offset=0&limit=100
```

Example Response

- **Status code: 200**

Querying the details of common O&M SQL commands by page with the command content displayed


```
{
  "total_count": 64,
  "data": [
    {
      "id": "a60b49af-1d66-415a-b930-9fc4cbd3c264",
      "cmd_name": "query_specified_thread_memory",
      "cmd_type": "SQL",
      "cmd": "select * from gs_get_thread_memctx_detail({tid},{contextname});",
      "call_type": "sync",
      "constraints": [
        {
          "name": "db_version",
          "min_value": "3.100.0",
          "max_value": null,
          "enum_value": null
        },
        {
          "name": "kernel_version",
          "min_value": "503",
          "max_value": null,
          "enum_value": null
        }
      ],
      "param": [
        {
          "param_id": "e7523d6f-7ab0-4312-bb34-4b8070b11615",
          "param_name": "tid",
          "param_type": "single",
          "data_type": "string",
          "required": true,
          "default_value": null,
          "pattern": "[0-9]{1,256}",
          "description": "Thread ID"
        },
        {
          "param_id": "880084d8-44b9-42fe-adf7-b37032580566",
          "param_name": "contextname",
          "param_type": "single",
          "data_type": "string",
          "required": true,
          "default_value": null,
          "pattern": "[A-Za-z0-9_\\-\\s]{1,256}",
          "description": "Memory context name (contextname column in the result returned by 
gs_session_memory_context)"
        }
      ],
      "is_support": true,
      "description": "Memory usage of the thread with the ID"
      "remark": "",
      "node_type": null,
      "cluster_mode": "All",
      "category": "memory"
    },{
      "id": "7dce7897-9254-4aa7-9a36-7234715117e7",
      "cmd_name": "query_flow_control",
      "cmd_type": "SQL",
      "cmd": "select * from dbe_perf.global_recovery_status order by 1,2 limit 100;",
      "call_type": "sync",
      "constraints": [
        {
          "name": "db_version",
          "min_value": "3.100.0",
          "max_value": null,
          "enum_value": null
        },
        {
          "name": "kernel_version",
```

```
        "min_value": "503",
        "max_value": null,
        "enum_value": null
      }
    ],
    "param": [],
    "is_support": true,
    "description": "Flow control",
    "remark": "",
    "node_type": null,
    "cluster_mode": "All",
    "category": "flow_limit"
  }
 ]
}
```

- **Status code: 200**

  Querying details of common O&M Python commands by page and filtering the commands by command name without displaying the command content

```
{
 "total_count": 1,
 "data": [
  {
   "id": "965e830e-8dcf-4bfd-92f1-943e6bf9fd32",
   "cmd_name": "auto_emergency_scenario_data_collect_ha",
   "cmd_type": "PYTHON",
   "cmd": null,
   "call_type": "async",
   "constraints": [
    {
     "name": "db_version",
     "min_value": "3.100.0",
     "max_value": null,
     "enum_value": null
    },
    {
     "name": "kernel_version",
     "min_value": "503",
     "max_value": null,
     "enum_value": null
    }
   ],
   "param": [
    {
     "param_id": "14d28a98-c76e-411b-87ff-4e8ccfe32df0",
     "param_name": "database",
     "param_type": "single",
     "data_type": "string",
     "required": false,
     "default_value": "postgres",
     "pattern": "^[a-zA-Z0-9_\\\\-\\\\.]{1,256}$",
     "description": "Database name. This parameter is optional. The default value is postgres."
    },
    {
     "param_id": "1a1d59f7-b4c1-412c-90da-cd2dbf316564",
     "param_name": "username",
     "param_type": "single",
     "data_type": "string",
     "required": false,
     "default_value": "${rdsAdminUser}",
     "pattern": "^[a-zA-Z0-9_\\\\-\\\\.]{1,256}$",
     "description": "Login username, which is optional. The default username is rdsAdmin."
    },
    {
     "param_id": "a1f2da0e-2fda-4f1c-9ecc-dc71403ab1a7",
     "param_name": "password",
     "param_type": "single",
     "data_type": "string",
```

```
        "required": false,
        "default_value": "${rdsAdminPwd}",
        "pattern": "^(.){1,512}$",
        "description": "Login password, which is optional. The default value is the password of user rdsAdmin."
      },
      {
        "param_id": "cd0de0b6-cf95-4c0a-8ba2-d05e56b6c737",
        "param_name": "type",
        "param_type": "single",
        "data_type": "enum",
        "required": true,
        "default_value": null,
        "pattern": "^slot_slow_collector|slot_memory_collector|memory_collector|cpu_collector|wait_event_collector$",
        "description": "Collection/diagnosis type, which is mandatory. The options are as follows: slot_slow_collector: Information collection for slow logical replication progress. slot_memory_collector: Information collection for high memory usage in logical decoding. memory_collector: Information collection for high dynamic memory usage. cpu_collector: Information collection for CPU spikes. wait_event_collector: Information collection for wait events."
      },
      {
        "param_id": "a3fa5a3e-5c40-4691-9c6c-c5a35647319f",
        "param_name": "start-time",
        "param_type": "single",
        "data_type": "string",
        "required": false,
        "default_value": "${last_one_hour_timestamp}",
        "pattern": "^[\\d]{13}$",
        "description": "Start timestamp, which is optional. The value is a 13-digit timestamp. The default value is the timestamp one hour before the current time. This parameter is required only when type is set to wait_event_collector. In other scenarios, this parameter will be ignored during execution."
      },
      {
        "param_id": "e63de6f4-2c3c-4083-b109-caab6c5fb8af",
        "param_name": "end-time",
        "param_type": "single",
        "data_type": "string",
        "required": false,
        "default_value": "${current_timestamp}",
        "pattern": "^[\\d]{13}$",
        "description": "End timestamp, which is optional. The value is a 13-digit timestamp. The default value is the current timestamp. This parameter is required only when type is set to wait_event_collector. In other scenarios, this parameter will be ignored during execution."
      }
    ],
    "is_support": true,
    "description": "Emergency O&M operations for centralized instances: Collect fault locating information in a specified scenario."
    "remark": "",
    "node_type": null,
    "cluster_mode": "Ha",
    "category": "emergency"
  }
 ]
}
```


Status Codes

Status Code	Description
200	Normal
400	Abnormal

Error Codes

For details, see **Error Codes**.

4.6.7 Delivering Common O&M Commands (Synchronous and Asynchronous Commands)

Function

This API is used to deliver common O&M commands.

- For a synchronous command, the command execution result is returned after the command is delivered.
- For an asynchronous command, the command execution task ID is returned after the command is delivered. You can call the query API in **Querying the Execution Results of Asynchronous O&M Commands** to obtain the command status. You can call the download API in **Downloading the Execution Result File of an Asynchronous O&M Command** to obtain the command execution result.

The Agent uses **rdsAdmin** to execute SQL commands and **Ruby** to execute Python commands when the commands are delivered.

Prerequisites

- The Adapter needs to connect to port 8000 of the corresponding database component on the target node. Therefore, port 8000 must be enabled on the node. Otherwise, the command execution will fail.

  Before delivering a command, you are advised to verify that the corresponding database component on the node is normal and the ports are enabled.
- It is recommended that commands be delivered to the following target nodes to avoid call failures:
  - Distributed edition: normal CN
  - Centralized edition: normal primary and standby DNs
- Before delivering Python commands, ensure that the SFTP server on TPOPS is running properly and the storage space of TPOPS is sufficient.

URI

POST /gaussdb/v3/instances/{instance_id}/common-operations/invoke

Request Parameters

Table 4-529 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.

Parameter	Mandatory	Type	Description
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Table 4-530 Request body parameters

Parameter	Mandatory	Type	Description
cmd_type	Yes	String	O&M command type. Enumerated values: ● **SQL** or **sql**: SQL command ● **PYTHON** or **python**: Python command The Agent uses **rdsAdmin** to execute SQL commands and **Ruby** to execute Python commands when the commands are delivered.
id	No	String	O&M command ID. When searching for a command, the system prioritizes **id** over **cmd_name**. That is, if **id** is empty, **cmd_name** is used to search for a predefined command. **id** and **cmd_name** cannot both be empty.
cmd_name	No	String	O&M command name. When searching for a command, the system prioritizes **id** over **cmd_name**. If **id** is empty, **cmd_name** is used to search for a predefined command. **id** and **cmd_name** cannot both be empty.

Parameter	Mandatory	Type	Description
invoke_type	No	String	Command delivery mode, which is reserved and does not need to be set. The value is **default**, indicating that only commands in the predefined whitelist can be delivered. The predefined commands to be executed are retrieved in the priority order of **id** over **cmd_name**.
param	No	Array of **OperationValue** objects	List of parameter values for the command being delivered. For details, see **Table 4-531**.
node_id	Yes	String	ID of the target node.

Table 4-531 OperationValue

Parameter	Mandatory	Type	Description
param_name	Yes	String	Parameter name. You must enter the parameter name in the command parameter list. Otherwise, a failure will occur.

Parameter	Mandatory	Type	Description
param_value	Yes	String	Parameter value. When setting this parameter, refer to the type returned by the query command. When delivering this parameter, the TPOPS platform verifies the parameter validity based on the actual service scenario (associated with related verification and matching rules). The parameter value is transferred as a character string. (If an array needs to be transferred, serialize the array and convert it to a character string.) After the parameter is delivered, the system parses, converts, and verifies the parameter based on the parameter and data type. ● Example 1: When a single integer or floating-point parameter is transferred, the value can be **1**, **-2**, **100**, **0.1**, or **3.14235**. ● Example 2: When a single string or Boolean parameter is transferred, the value can be **value**, **zhangsan**, **true**, or **false**. ● Example 3: When a single formatted time parameter is transferred, the value can be **2024-10-19 11:10:25**. ● Example 4: When a single timestamp parameter is transferred, the value can be **1695026670878** or **1577836800**. ● Example 5: When an integer array is transferred, the value can be **[12,101]**. ● Example 6: When a string array is transferred, the value can be **[\"value_1\",\"value_2\"]**.

Response Parameters

Status code: 202

Table 4-532 Response body parameters

Parameter	Type	Description
call_type	String	Call type of the command delivered to the Agent. Enumerated values: ● **sync**: synchronous command delivery ● **async**: asynchronous command delivery Note that SQL commands are currently synchronous, while Python commands are usually asynchronous because they take longer to execute.
task_id	String	ID of the internal task for calling the API. The value is also the request ID when the Agent delivers the calling request. If the calling fails, you can quickly locate the corresponding log information on the Agent based on the value. For asynchronous command delivery, you can call the API for querying the asynchronous O&M command execution result based on this ID to obtain the command execution status and result data.
status	String	Status of the command delivery result. Enumerated values: ● **running**: The inspection task is being executed. ● **finished**: The task is complete. ● **failed**: The task fails. For an asynchronous command, the value is **running**, indicating that the command is running in the Agent backend.
result	Object of **CommonCmdOperationResult**	Command delivery result. For an asynchronous command, the value is **null**. For details, see **Table 4-533**.

Table 4-533 CommonCmdOperationResult

Parameter	Type	Description
type	String	Data type. It specifies the data type of the returned result. Enumerated values: ● **text**: plain text or a JSON string. The data is a plain text string, which is mainly used in the result data returned after a Python synchronous command is executed. ● **hex**: hexadecimal string. This value is reserved. ● **array**: object array. The data is an array, which is mainly used in the result list returned after a SQL command is executed. ● **url**: redirection to a file on the SFTP server. The data is resource data (including basic information about the SFTP server and the storage information and path of the result file), which is mainly used in the result polling of Python asynchronous command execution. **CAUTION** ● In this scenario, the command is an asynchronous command, and the execution result is redirected to the SFTP server of TPOPS for temporary storage. The result file is retained for three days by default and is automatically deleted by TPOPS upon expiration. ● Ensure that the SFTP server on TPOPS is running properly and the storage space of TPOPS is sufficient.
data	Object of **URIDataResult**	Data content object. Enumerated values: ● **text**, **hex**, or **array**: execution result of synchronous commands. ● **url**: resource data for asynchronous commands. The data includes the storage information about the result data redirection, such as the basic information about the SFTP server and the path where the result file is saved. For details about the object structure, see **Table 4-534**.

Table 4-534 URIDataResult

Parameter	Type	Description
url	String	Full path of the result data redirected to the SFTP server, including the default SFTP root directory. Example: **/sftpservice/om_cmd/e8a3df12517e45e49d6eca167e72b934in14/output_auto_emergency_scenario_data_collect_ha_ComOmCmd-e2c361d3-b000-4fa4-8414-2a0f2b76c1b9.log**
file_name	String	Name of the file where the command output is stored on the SFTP server. Example: **output_auto_emergency_scenario_data_collect_ha_ComOmCmd-e2c361d3-b000-4fa4-8414-2a0f2b76c1b9.log**
file_path	String	File path, relative to the default root directory (**base_dir**), of the command output on the SFTP server. Example: **om_cmd/e8a3df12517e45e49d6eca167e72b934in14/output_auto_emergency_scenario_data_collect_ha_ComOmCmd-e2c361d3-b000-4fa4-8414-2a0f2b76c1b9.log**
file_save_dir	String	File directory, relative to the default root directory (**base_dir**), of the command output on the SFTP server. The value is in the format of **om_cmd/***{Instance ID}*, for example, **om_cmd/e8a3df12517e45e49d6eca167e72b934in14/**.
storage_mode	String	Storage type. In lightweight deployment, the default value is **sftp**, indicating that the SFTP server is used for storage.
sftp_server	Object of **SftpServer**	Information about the SFTP server on TPOPS. For details, see **Table 4-535**.

Table 4-535 SftpServer

Parameter	Type	Description
addr	String	IP address of the SFTP server.
port	Integer	Port number of the SFTP server

Parameter	Type	Description
base_dir	String	Default root directory for storing asynchronous command result files.

Status code: 400

Table 4-536 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

- Delivering general O&M commands on a centralized instance

```
POST https://{Endpoint}/rds/gaussdb/v3/instances/e8a3df12517e45e49d6eca167e72b934in14/
common-operations/invoke
{
  "id": "965e830e-8dcf-4bfd-92f1-943e6bf9fd32",
  "cmd_type": "python",
  "node_id": "2702e57f535b4a6f8cc82d9ba52f0464no14",
  "param": [
    {
      "param_name": "type",
      "param_value": "cpu_collector"
    },
    {
      "param_name": "start-time",
      "param_value": "1754586061000"
    },
    {
      "param_name": "end-time",
      "param_value": "1757264461000"
    }
  ]
}
```

- Delivering general O&M commands on a distributed instance

```
POST https://{Endpoint}/rds/gaussdb/v3/instances/e8a3df12517e45e49d6eca167e72b934in14/
common-operations/invoke
{
  "id": "2b066bbf-c957-4c89-b61d-f215d7320c51",
  "cmd_type": "python",
  "node_id": "dc96ae0de3b04d7a8679406c3a21fa90no14",
  "param": [
      {
      "param_name": "type",
      "param_value": "cpu_collector"
      },
      {
      "param_name": "start-time",
      "param_value": "1754586061000"
      },
      {
      "param_name": "end-time",
      "param_value": "1757264461000"
      },
```

```
    {
    "param_name": "component-name",
    "param_value": "cn_5001"
    }
  ]
}
```

Example Response

Status code: 202

Delivering a common O&M Python asynchronous command predefined to run the emergency O&M command for collecting high CPU usage information

```
{
  "call_type": "async",
  "task_id": "ComOmCmd-e2c361d3-b000-4fa4-8414-2a0f2b76c1b9",
  "status": "running",
  "result": null
}
```

Status Codes

Status Code	Description
202	Normal
400	Abnormal

Error Codes

For details, see **Error Codes**.

4.6.8 Querying the Execution Results of Asynchronous O&M Commands

Function

This API is used to query the execution status and result of an asynchronous O&M command.

Currently, all SQL commands are synchronous commands. Do not use this API for SQL commands. This API is used to query the execution results of asynchronous Python commands.

Prerequisites

- Verify that the commands whose execution results are to be queried are Python commands instead of SQL commands.
- After an asynchronous O&M command is executed, this API needs to be called to query the execution status (in progress, completed, or failed) and result of the command.

- The instance node ID and task ID transferred in the request body for querying the execution results of asynchronous commands must be the same as those in the corresponding response body. Otherwise, the API will fail to be called.
- Python O&M commands return a large amount of data. When the commands are executed on the Agent, the final results are dumped to the SFTP server of TPOPS. This query API returns the basic login information of the SFTP service, including the name and directory of the asynchronous command execution result file.
  - In this scenario, the execution result is redirected to the SFTP server of TPOPS and temporarily stored as a text file. The file is retained for three days by default and is automatically deleted upon expiration.
  - If the caller has connected to the SFTP server, the caller can read the command execution result based on the saved directory and file name.
- Ensure that the SFTP server on TPOPS is running properly and the storage space of TPOPS is sufficient. The command execution result file is usually less than or equal to 5 MB.

URI

POST /gaussdb/v3/instances/{instance_id}/common-operations/async-result

Request Parameters

Table 4-537 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Table 4-538 Request body parameters

Parameter	Mandatory	Type	Description
task_id	Yes	String	Asynchronous request task ID. When calling the API, set this parameter to the value of **task_id** in the response body of the asynchronous command delivery API. Otherwise, the call will fail.

Parameter	Mandatory	Type	Description
node_id	Yes	String	ID of the target node. The value must be the same as that of **node_id** in the request body for delivering the asynchronous O&M command. Otherwise, the API call will fail.

Response Parameters

Status code: 200

Table 4-539 Response body parameters

Parameter	Type	Description
status	String	Asynchronous command execution status. Enumerated values: ● **running**: execution in progress ● **finished**: execution completed ● **failed**: execution failed If the value is **running**, the command is being executed in the backend. You need to wait for a period of time before calling the API. The API can be periodically called to poll the execution progress and result data of an asynchronous command.
result	Object of **CommonCmdOperationResult**	Asynchronous command execution result data. For details, see **Table 4-533**.
error_code	String	Error code. Exception thrown during the execution process.
error_msg	String	Error details. Information about the error that occurred during the execution process.

Status code: 400

Table 4-540 Response body parameters

Parameter	Type	Description
error_code	String	Error code.

Parameter	Type	Description
error_msg	String	Error message.

Example Request

```
POST https://{Endpoint}/rds/gaussdb/v3/instances/e8a3df12517e45e49d6eca167e72b934in14/common-
operations/async-result
{
  "task_id": "ComOmCmd-e2c361d3-b000-4fa4-8414-2a0f2b76c1b9",
  "node_id": "2702e57f535b4a6f8cc82d9ba52f0464no14"
}
```

Example Response

Status code: 200

Querying the execution status and result data of an asynchronous Python command (emergency O&M command – collecting high CPU usage information)

```
{
  "status": "finished",
  "result": {
    "type": "url",
    "data": {
      "url": "/sftpservice/om_cmd/e8a3df12517e45e49d6eca167e72b934in14/
output_auto_emergency_scenario_data_collect_ha_ComOmCmd-e2c361d3-
b000-4fa4-8414-2a0f2b76c1b9.log",
      "file_name": "output_auto_emergency_scenario_data_collect_ha_ComOmCmd-e2c361d3-
b000-4fa4-8414-2a0f2b76c1b9.log",
      "file_path": "om_cmd/e8a3df12517e45e49d6eca167e72b934in14/
output_auto_emergency_scenario_data_collect_ha_ComOmCmd-e2c361d3-
b000-4fa4-8414-2a0f2b76c1b9.log",
      "file_save_dir": "om_cmd/e8a3df12517e45e49d6eca167e72b934in14/",
      "storage_mode": "sftp",
      "sftp_server": {
        "sftp_url": "192.168.122.100",
        "addr": "192.168.122.100",
        "port": "10022",
        "base_dir": "/sftpservice/"
      }
    }
  },
  "error_code": null,
  "error_msg": null
}
```

Status Codes

Status Code	Description
200	Normal
400	Abnormal

Error Codes

For details, see **Error Codes**.

4.6.9 Downloading the Execution Result File of an Asynchronous O&M Command

Function

This API is used to download the execution result file of an asynchronous O&M command whose output is redirected to the SFTP server.

Prerequisites

- This API is valid only for asynchronous commands. The execution results must be redirected to a file and dumped in the SFTP server of TPOPS.
- The asynchronous command has been successfully executed when this API is called. If the command is still being executed or fails to be executed, this API cannot be called.
- The temporary result file stored in the SFTP server is still valid. This API cannot be called if the result file, which was redirected to the SFTP server, has expired for more than three days.
- The instance node ID and task ID transferred in the request body for querying the execution results of asynchronous commands must be the same as those in the corresponding response body. Otherwise, the API will fail to be called.
- Ensure that the SFTP server on TPOPS is running properly and the storage space of TPOPS is sufficient. The command execution result file is usually less than or equal to 5 MB.

URI

POST /gaussdb/v3/instances/{instance_id}/common-operations/download-result

Request Parameters

Table 4-541 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Table 4-542 Request body parameters

Parameter	Mandatory	Type	Description
task_id	Yes	String	Asynchronous request task ID. When calling the API, set this parameter to the value of **task_id** in the response body of the asynchronous command delivery API. Otherwise, the call will fail.
node_id	Yes	String	ID of the target node. The value must be the same as that of **node_id** in the request body for delivering the asynchronous O&M command. Otherwise, the API call will fail.

Response Parameters

Status code: 200

None

The API returns a file download stream. The file name format is **output_***{cmd_name}***-***{task_id}***.log**. *{cmd_name}* indicates the command name, and *{task_id}* indicates the task ID returned after the asynchronous command is delivered.

Status code: 400

Table 4-543 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

```
POST https://{Endpoint}/rds/gaussdb/v3/instances/e8a3df12517e45e49d6eca167e72b934in14/common-operations/download-result
{
  "task_id": "ComOmCmd-e2c361d3-b000-4fa4-8414-2a0f2b76c1b9",
  "node_id": "2702e57f535b4a6f8cc82d9ba52f0464no14"
}
```

Example Response

Status code: 200

None

Status Codes

Status Code	Description
200	Normal
400	Abnormal

Error Codes

For details, see **Error Codes**.

4.7 SQL Throttling

4.7.1 Creating a Throttling Task

Function

This API is used to create a throttling task based on the specific scope and type.

URI

POST /gaussdb/v3/instances/{instance_id}/limit-task

Table 4-544 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-545 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.

Table 4-546 Request body parameters

Parameter	Mandatory	Type	Description
task_scope	Yes	String	Throttling task scope. Currently, **SQL** and **SESSION** are supported.
start_time	No	String	Task start time. The value must be later than or equal to the two minutes before the current time (UTC time). The format must be yyyy-mm-ddThh:mm:ssZ. When you configure this parameter, **task_scope** must be configured.
end_time	No	String	Task end time. The value must be later than the task start time. The format must be yyyy-mm-ddThh:mm:ssZ. The current time is the UTC time. When you configure this parameter, **task_scope** must be configured.
limit_type	Yes	String	Throttling type. When **task_scope** is set to **SQL**, the value can be **SQL_ID** or **SQL_TYPE**. When **task_scope** is set to **SESSION**, the value can be **SESSION_ACTIVE_MAX_COUNT**.
limit_type_value	Yes	String	Throttling type value. When **limit_type** is set to **SQL_ID**, the value is the SQL ID of the selected template. When **limit_type** is set to **SQL_TYPE**, the value is **select**, **update**, **insert**, **delete**, or **merge**. When **limit_type** is set to **SESSION_ACTIVE_MAX_COUNT**, the value can only be **CPU_OR_MEMORY**.

Parameter	Mandatory	Type	Description
key_words	No	String	Keyword. This parameter is returned only when **limit_type** is set to **SQL_TYPE**. You can enter 2 to 100 keywords and separate multiple keywords by commas (,). Each keyword can contain 2 to 64 characters and cannot start and end with a space. The specifical characters ("\{}) and **null** are not allowed.
task_name	Yes	String	Throttling task name. This parameter is mandatory. The value can contain only uppercase letters, lowercase letters, underscores (_), digits, and dollar signs ($).
sql_model	No	String	SQL template of CNs. If **limit_type** is set to **SQL_ID**, this parameter is mandatory.
parallel_size	Yes	Integer	Maximum concurrency. The value can be 0 or a positive integer. Value range: **0** to **2147483647**.
cpu_utilization	No	Integer	CPU usage threshold. The value is a positive integer ranging from [0, 100). If **limit_type** is **SESSION_ACTIVE_MAX_COUNT**, this parameter is mandatory. The parameter and **memory_utilization** cannot be both set to **0**. If you only need one of them for throttling, set the other threshold to **0**.

Parameter	Mandatory	Type	Description
memory_utilization	No	Integer	Memory usage threshold. The value is a positive integer ranging from [0, 100). If **limit_type** is **SESSION_ACTIVE_MAX_COUNT**, this parameter is mandatory. The parameter and **cpu_utilization** cannot be both set to **0**. If you only need one of them for throttling, set the other threshold to **0**.
databases	No	String	Databases. Databases are separated by commas (,). If **limit_type** is set to **SQL_TYPE**, this parameter is mandatory.
node_infos	No	Array of Table 4-547 objects	CN node information list. If **limit_type** is set to **SQL_ID**, this parameter is mandatory.

Table 4-547 CreateLimitTaskNodeOption

Parameter	Mandatory	Type	Description
node_id	Yes	String	Node ID.
sql_id	Yes	String	ID of the SQL statement executed on the node. If the **limit_type** is set to **SQL_ID**, the value must be the same as that of **limit_type_value**.

Response Parameters

Status code: 200

Table 4-548 Response body parameters

Parameter	Type	Description
task_id	String	Throttling task ID.
task_scope	String	Throttling scope, which is the same as the value of request parameter **task_scope**.
limit_type	String	Throttling type, which is the same as the value of request parameter **limit_type**.

Parameter	Type	Description
limit_type_value	String	Throttling type value, which is the same as the value of request parameter **limit_type_value**.
databases	String	Databases. Databases are separated by commas (,).
task_name	String	Throttling task name.
rule_name	String	Rule name.
sql_model	String	SQL template. This parameter is returned only when **limit_type** is set to **SQL_ID**. Its value is the same as the request parameter **sql_model**.
key_words	Array of strings	Keyword. This parameter is returned only when **limit_type** is set to **SQL_TYPE**. Its value is the same as the request parameter **key_words**.
status	String	Throttling task status. The value can be **CREATING**, **UPDATING**, **DELETING**, **WAIT_EXECUTE**, **EXECUTING**, **TIME_OVER**, **DELETED**, **CREATE_FAILED**, **UPDATE_FAILED**, **DELETE_FAILED**, **EXCEPTION** or **NODE_SHUT_DOWN**.
instance_id	String	Instance ID.
parallel_size	Integer	Maximum concurrency. Its value is the same as the request parameter **parallel_size**.
cpu_utilization	Integer	CPU usage threshold. This parameter is returned only when **limit_type** is set to **SESSION_ACTIVE_MAX_COUNT**. Only a positive integer is returned.
memory_utilization	Integer	Memory usage threshold. This parameter is returned only when **limit_type** is set to **SESSION_ACTIVE_MAX_COUNT**. Only a positive integer is returned.
start_time	String	Start time of the throttling task in the format of yyyy-mm-ddThh:mm:ssZ (UTC time). Its value is the same as the request parameter **start_time**.
end_time	String	End time of the throttling task in the format of yyyy-mm-ddThh:mm:ssZ (UTC time). Its value is the same as the request parameter **end_time**.
created	String	Creation time in the "yyyy-mm-ddThh:mm:ssZ" format.
updated	String	Update time in the "yyyy-mm-ddThh:mm:ssZ" format.

Parameter	Type	Description
creator	String	Creator.
modifier	String	Modifier.
node_infos	Array of Table 4-549 objects	CN node information list. If the **limit_type** is **SQL_ID**, this value is returned and is the same as the value of the request parameter **node_infos**.

Table 4-549 CreateLimitTaskNodeResult

Parameter	Type	Description
node_id	String	Node ID.
sql_id	String	ID of the SQL statement executed on the node.

Status code: default

Table 4-550 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

Creating a throttling task

```
POST https://{Endpoint}/rds/gaussdb/v3/instances/3d39c18788b54a919bab633874c159dfin14/limit-task

{
  "task_scope" : "SQL",
  "start_time" : "2023-06-06T01:47:20+0800",
  "end_time" : "2023-06-07T01:47:20+0800",
  "limit_type" : "SQL_ID",
  "limit_type_value" : "39b6a1a",
  "task_name" : "test1",
  "sql_model" : "select * from a where b = ?",
  "parallel_size" : 100,
  "databases" : "test1",
  "node_infos" : [ {
    "node_id" : "46d996fdda594f58b17fe509061e0893no14",
    "sql_id" : "39b6a1a"
  } ]
}
POST https://{Endpoint}/rds/gaussdb/v3/instances/3d39c18788b54a919bab633874c159dfin14/limit-task

{
  "task_scope" : "SQL",
  "start_time" : "2023-06-06T01:47:20+0800",
  "end_time" : "2023-06-07T01:47:20+0800",
```

```
  "limit_type" : "SQL_TYPE",
  "limit_type_value" : "select",
  "task_name" : "test1",
  "key_words" : "table1,id",
  "parallel_size" : 100,
  "databases" : "test1"
 }
POST https://{Endpoint}/rds/gaussdb/v3/instances/3d39c18788b54a919bab633874c159dfin14/limit-task

 {
  "task_name" : "test",
  "task_scope" : "SESSION",
  "start_time" : "2023-06-06T01:47:20+0800",
  "end_time" : "2023-06-07T01:47:20+0800",
  "limit_type" : "SESSION_ACTIVE_MAX_COUNT",
  "limit_type_value" : "CPU_OR_MEMORY",
  "cpu_utilization" : 80,
  "memory_utilization" : 80,
  "parallel_size" : 100
 }
```

Example Response

Status code: 200

Success.

```
{
  "task_id" : "59b6a1a278844ac48119d86512e0000",
  "task_scope" : "SQL",
  "task_name" : "test1",
  "limit_type" : "SQL_TYPE",
  "limit_type_value" : "select",
  "instance_id" : "cb651ac71c5a447685ef981e44a0422fin14",
  "key_words" : "test",
  "databases" : "test1",
  "status" : "creating",
  "parallel_size" : 100,
  "rule_name" : "dsa48119d86512e0000bin066a1a27",
  "start_time" : "2023-12-30T02:00:00Z",
  "end_time" : "2023-12-30T02:00:00Z",
  "created" : "2023-12-30T02:00:00Z",
  "updated" : "2023-12-30T02:00:00Z",
  "creator" : "dbs_rds_guangzhou_l00417929_01"
}
```

Status Code

Status Code	Description
200	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.7.2 Deleting a Throttling Task

Function

This API is used to delete a throttling task based on the task ID.

URI

DELETE /gaussdb/v3/instances/{instance_id}/limit-task/{task_id}

Table 4-551 URI parameter

Parameter	Mandatory	Type	Description
task_id	Yes	String	Throttling task ID.
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-552 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.

Response Parameters

Status code: 500

Table 4-553 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

Deleting the throttling task whose **task_id** is **96854ba7-8d50-4a3c-8fd8-210a7390e9d1**

```
DELETE https://{Endpoint}/rds/gaussdb/v3/instances/3d39c18788b54a919bab633874c159dfin14/limit-task/
96854ba7-8d50-4a3c-8fd8-210a7390e9d1
```

Example Response

None

Status Code

Status Code	Description
200	Normal
500	Abnormal

Error Code

For details, see **Error Codes**.

4.7.3 Querying Throttling Tasks Based on Search Criteria

Function

This API is used to query throttling tasks based on search criteria.

URI

GET /gaussdb/v3/instances/{instance_id}/limit-task-list

Table 4-554 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Table 4-555 Query parameters

Parameter	Mandatory	Type	Description
task_scope	No	String	Throttling task scope. Currently, SQL and SESSION are supported.
limit_type	No	String	Throttling type. The value can be **SQL_ID**, **SQL_TYPE**, or **SESSION_ACTIVE_MAX_COUNT**.
limit_type_value	No	String	Throttling type value. Fuzzy match is supported.
task_name	No	String	Throttling task name. Fuzzy match is supported.
sql_model	No	String	SQL template. Fuzzy match is supported.

Parameter	Mandatory	Type	Description
rule_name	No	String	Rule name.
offset	No	Integer	Index offset. If **offset** is set to *N*, the resource query starts from the *N*+1 piece of data. The value is **0** by default, indicating that the query starts from the first piece of data. The value must be a number but cannot be a negative number. Value: **0** to **10000**
limit	No	Integer	Number of records to be queried. The default value is **10**. The value cannot be a negative number. The minimum value is **1** and the maximum value is **100**.

Request Parameters

Table 4-556 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.

Response Parameters

Status code: 200

Table 4-557 Response body parameters

Parameter	Type	Description
limit_task_list	Array of **Table 4-558** objects	Throttling task information in the list.
limit	Integer	Number of records to be queried.
offset	Integer	Index offset.
total_count	Integer	Total number of throttling tasks.

Table 4-558 ListLimitTaskResponseResult

Parameter	Type	Description
task_id	String	Throttling task ID.
task_scope	String	Throttling task scope.
limit_type	String	Throttling task type.
limit_type_value	String	Throttling task type value.
task_name	String	Throttling task name.
databases	String	Databases. Databases are separated by commas (,).
sql_model	String	SQL template. This parameter is returned only when the task type is **SQL_ID**.
key_words	String	Keyword. This parameter is returned only when the task type is **SQL_TYPE**.
status	String	Throttling task status. The value can be **CREATING**, **UPDATING**, **DELETING**, **WAIT_EXECUTE**, **EXECUTING**, **TIME_OVER**, **DELETED**, **CREATE_FAILED**, **UPDATE_FAILED**, **DELETE_FAILED**, **EXCEPTION** or **NODE_SHUT_DOWN**.
instance_id	String	Instance ID.
rule_name	String	Rule name.
parallel_size	Integer	Maximum concurrency.
start_time	String	Start time of the throttling task in the format of yyyy-mm-ddThh:mm:ssZ (UTC time). Replace **+** in the time zone with **%2B**.
end_time	String	End time of the throttling task in the format of yyyy-mm-ddThh:mm:ssZ (UTC time). Replace **+** in the time zone with **%2B**.
cpu_utilization	Integer	CPU usage. This parameter is returned only when the task type is **SESSION_ACTIVE_MAX_COUNT**.
memory_utilization	Integer	Memory usage. This parameter is returned only when the task type is **SESSION_ACTIVE_MAX_COUNT**.
created	String	Creation time in the format of yyyy-mm-ddThh:mm:ssZ (UTC time).
updated	String	Update time in the format of yyyy-mm-ddThh:mm:ssZ (UTC time).

Parameter	Type	Description
creator	String	Creator.
modifier	String	Modifier.
node_infos	Array of **Table 4-559** objects	CN node information list.
node_id	String	Node ID.

Table 4-559 ShowLimitTaskNodeOption

Parameter	Type	Description
node_id	String	Node ID.
sql_id	String	ID of the SQL statement executed on the node.

Status code: default

Table 4-560 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

- Querying throttling tasks based on search criteria
```
GET https://{Endpoint}/rds/gaussdb/v3/instances/3d39c18788b54a919bab633874c159dfin14/limit-task-list?offset=1&limit=10&limit_type=SQL_ID
```
- Querying throttling tasks based on search criteria
```
GET https://{Endpoint}/rds/gaussdb/v3/instances/3d39c18788b54a919bab633874c159dfin14/limit-task-list?offset=1&limit=10&limit_type=SQL_ID&task_name=test
```
- Querying throttling tasks based on the specified time range
```
GET https://{Endpoint}/rds/gaussdb/v3/instances/3d39c18788b54a919bab633874c159dfin14/limit-task-list?offset=0&limit=10&start_time=2024-01-11T08:41:45%2B0000&end_time=2024-01-18T08:41:51%2B0000
```

Example Response

Status code: 200

Success.

```
{
  "limit_task_list" : [ {
```

```
    "task_id" : "59b6a1a278844ac48119d86512e0000",
    "task_scope" : "SQL",
    "limit_type" : "SQL_ID",
    "limit_type_value" : 2348983611,
    "sql_model" : "select * from table where id = {id}",
    "status" : "creating",
    "key_words" : null,
    "instance_id" : "39b6a1a278844ac48119d86512e0000bin06",
    "parallel_size" : 100,
    "rule_name" : "dsa48119d86512e0000bin066a1a27",
    "databases" : "test1",
    "task_name" : "test1",
    "start_time" : "2023-12-30T02:00:00Z",
    "end_time" : "2023-12-30T02:00:00Z",
    "created" : "2023-12-28T01:55:08Z,",
    "updated" : "2023-12-28T01:55:08Z,",
    "creator" : "test,",
    "modifier" : "null,",
    "cpu_utilization" : null,
    "memory_utilization" : null,
    "node_infos" : [ {
      "node_id" : "cf521ccca2d74c9e89569a9828b6adcbno14",
      "sql_id" : "44ac48119d8"
    } ] ,
    "node_id": "cf521ccca2d74c9e89569a9828b6adcbno14",
  } ],
  "total_count" : 1,
  "limit" : 10,
  "offset" : 1
}
```

Status Code

Status Code	Description
200	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.7.4 Querying SQL Templates of a Specified Node

Function

This API is used to query SQL templates of a specified node.

URI

GET /gaussdb/v3/instances/{instance_id}/list-node-limit-sql-model

Table 4-561 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Table 4-562 Query parameters

Parameter	Mandatory	Type	Description
node_id	Yes	String	Node ID.
sql_model	No	String	SQL template. The value can contain only uppercase letters, lowercase letters, underscores (_), digits, spaces, and special characters ($*?=+;()><,.").
offset	No	Integer	Index offset. If **offset** is set to *N*, the resource query starts from the *N*+1 piece of data. The value is **0** by default, indicating that the query starts from the first piece of data. The value must be a number but cannot be a negative number. Value: **0** to **10000**
limit	No	Integer	Number of records to be queried. The default value is **10**. The value cannot be a negative number. The minimum value is **1** and the maximum value is **100**.

Request Parameters

Table 4-563 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.

Response Parameters

Status code: 200

Table 4-564 Response body parameters

Parameter	Type	Description
node_limit_sql_model_list	Array of **Table 4-565** objects	Information about the SQL template for SQL throttling.
limit	Integer	Number of records to be queried.
offset	Integer	Index offset.
total_count	Integer	Total number of SQL templates.

Table 4-565 ListNodeLimitSqlModelResponseResult

Parameter	Type	Description
sql_id	String	SQL ID of the throttling task.
sql_model	String	SQL template of the throttling task.

Status code: default

Table 4-566 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

Querying SQL templates of a specified node

```
GET https://{Endpoint}/rds/gaussdb/v3/instances/3d39c18788b54a919bab633874c159dfin14/list-node-limit-sql-model?node_id=46d996fdda594f58b17fe509061e0893no14
```

Example Response

Status code: 200

Success.


```
{
  "list" : [ {
    "sql_id" : "59b6a1a278844ac48119d86512e0000",
    "sql_model" : "select * from table where id= ?",
  } ],
  "total_count" : 1,
  "limit" : 10,
  "offset" : 1
}
```

Status Code

Status Code	Description
200	Success.
default	Client or server error.

Error Code

For details, see **Error Codes**.

4.8 Host Key Management

4.8.1 Generating a Key Pair for Host Connection

Function

This API is used to generate a key pair for host connection.

Constraints

- A maximum of 20 key pairs can be created.
- If a key pair is being used by a host, you can forcibly delete it as needed.
- To use a key pair to connect to a host without entering a username and password, you need to download the public key in the key pair and copy the key content to the **~/.ssh/authorized_keys** file on the host.
- Currently, the key encryption algorithm can only be RSA.
- You are advised to delete and update your key pair periodically.

URI

POST /gaussdb/v3/hosts/key-pair

Request Parameters

Table 4-567 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.

Parameter	Mandatory	Type	Description
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Table 4-568 Request body parameters

Parameter	Mandatory	Type	Description
name	No	String	SSH private key name, which is randomly generated by default. It cannot exceed 128 bytes.
description	No	String	SSH private key description. It cannot exceed 256 bytes.
key_algorithm	No	String	Key algorithm. Default value: **RSA_3072** Enumerated values: ● **RSA_3072** ● **RSA_4096**

Response Parameters

Status code: 200

Table 4-569 Response body parameters

Parameter	Type	Description
id	String	Private key ID.

Example Request

```
POST https://{Endpoint}/rds/gaussdb/v3/hosts/key-pair

{
  "name" : "ssh-123456",
  "description": " It is used to connect hosts in data center 1.",
  "key_algorithm" : "RSA_3072"
}
```

Example Response

Status code: 200

Success


```
{
  "id" : "5d001c5c-ca85-4e4f-a891-debf125ed1ef"
}
```


Status Code

Status Code	Description
200	Normal

Error Code

For details, see **Error Codes**.

4.8.2 Querying Key Pair Information

Function

This API is used to query key pair information.

URI

GET /gaussdb/v3/hosts/key-pair

Table 4-570 Query parameters

Parameter	Mandatory	Type	Description
id	No	String	Private key ID.
name	No	String	Private key name.
description	No	String	Private key description.
offset	No	Integer	Index offset. If **offset** is set to *N*, the resource query starts from the *N*+1 piece of data. The value is **0** by default, indicating that the query starts from the first piece of data. The value must be a number but cannot be a negative number.

Parameter	Mandatory	Type	Description
limit	No	Integer	Number of queried private key pairs. The default value is **10**. The value cannot be a negative number. The minimum value is **1** and the maximum value is **20**.

Request Parameters

Table 4-571 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Response Parameters

Status code: 200

Table 4-572 Response body parameters

Parameter	Type	Description
total_count	Integer	Number of private key pairs.
key_pairs	**Table 4-573** object	Private key pairs.

Table 4-573 QueryKeyPiarResult

Parameter	Type	Description
id	String	Private key ID.
name	String	Private key name. It can contain up to 128 bytes.
created_at	String	Creation time.

Parameter	Type	Description
description	String	Private key description. It can contain up to 256 bytes.
host_nums	Integer	Number of bound hosts.

Example Request

```
GET https://{Endpoint}/rds/gaussdb/v3/hosts/key-pair
```

Example Response

Status code: 200

Success


```
{
  "total_count" : 1,
  "key_pairs" : [ {
    "id" : "ff2e6f95-a93c-4c10-b020-d5b0244d94ef",
    "name" : "SSH-123456",
    "description": "It is used by data center 1",
    "created_at" : "2023-12-18 14:47:32",
    "host_nums" : 1
  } ]
}
```


Status Code

Status Code	Description
200	Normal

Error Code

For details, see **Error Codes**.

4.8.3 Modifying Key Information

Function

This API is used to modify key information.

Constraints

The private key name must be unique.

URI

PUT /gaussdb/v3/hosts/key-pair

Request Parameters

Table 4-574 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Table 4-575 Request body parameters

Parameter	Mandatory	Type	Description
id	Yes	String	Private key ID.
name	No	String	Modified private key name. The private key name must be unique and cannot exceed 128 bytes.
description	No	String	Modified private key description. It cannot exceed 256 bytes.

Response Parameters

None

Example Request

```
PUT https://{Endpoint}/rds/gaussdb/v3/hosts/key-pair

{
  "id" : "ff2e6f95-a93c-4c10-b020-d5b0244d94ef",
  "name" : "ssh_key",
  "description" : "Key pair for data center 2"
}
```

Example Response

None

Status Code

Status Code	Description
200	Normal

Error Code

For details, see **Error Codes**.

4.8.4 Deleting a Key Pair

Function

This API is used to delete a key pair.

Constraints

- A key pair cannot be deleted when it is bound to a host by default, but you can forcibly delete it as needed.
- A deleted key pair cannot be retrieved. If you want to use it again, create a key pair again.

URI

DELETE /gaussdb/v3/hosts/key-pair

Request Parameters

Table 4-576 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Table 4-577 Request body parameters

Parameter	Mandatory	Type	Description
id	Yes	String	Private key ID.

Parameter	Mandatory	Type	Description
forced	No	Boolean	Whether to forcibly delete the key pair. Default value: **false** Enumerated values: ● **true** ● **false**

Response Parameters

None

Example Request

```
DELETE https://{Endpoint}/rds/gaussdb/v3/hosts/key-pair

{
  "id" : "59ec7944-757a-4429-b448-bad7207ce7c6",
  "forced" : true
}
```

Example Response

None

Status Code

Status Code	Description
200	Normal

Error Code

For details, see **Error Codes**.

4.8.5 Downloading a Public Key

Function

This API is used to download a public key.

Constraints

You need to copy the public key content to **~/.ssh/authorized_keys** on a host.

URI

GET /gaussdb/v3/hosts/public-key

Table 4-578 Query parameters

Parameter	Mandatory	Type	Description
id	Yes	String	Private key ID.

Request Parameters

Table 4-579 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Response Parameters

Status code: 200

Table 4-580 Response body parameters

Parameter	Type	Description
-	File	Success

Example Request

```
GET https://{Endpoint}/rds/gaussdb/v3/hosts/public-key?id=aa6d92e2-79a6-4a71-843f-b810b69b23a4
```

Example Response

None

Status Code

Status Code	Description
200	Normal

Error Code

For details, see **Error Codes**.

4.9 Data Center Management

4.9.1 Querying Hosts

Function

This API is used to query hosts.

URI

GET /gaussdb/v3/hosts

Table 4-581 Query parameters

Parameter	Mandatory	Type	Description
id	No	String	Host ID.
name	No	String	Host alias.
type	No	String	Host type.
status	No	String	Host status. Enumerated values: • **initializing** • **initialized** • **initialize_failed** • **unmanaged** • **in_use** • **deleting** • **delete_failed** • **modifying**
data_center_id	No	String	Data center ID.
data_center	No	String	Data center alias.
instance_id	No	String	Instance ID.
instance_name	No	String	Instance name.
manage_ip	No	String	Host management IP address.

Parameter	Mandatory	Type	Description
offset	No	Integer	Index offset. If **offset** is set to *N*, the resource query starts from the *N*+1 piece of data. The value is **0** by default, indicating that the query starts from the first piece of data. The value must be a number but cannot be a negative number.
limit	No	Integer	Number of hosts. The default value is **10**. The value cannot be a negative number. The minimum value is **1** and the maximum value is **100**.
sort_key	No	String	Sorting key. Enumerated values: ● **name** ● **type** ● **status** ● **created_at**
sort_order	No	String	Sorting order. It determines whether data is sorted in ascending or descending order. Enumerated values: ● **asc** ● **desc**

Request Parameters

Table 4-582 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	Yes	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Response Parameters

Status code: 200

Table 4-583 Response body parameters

Parameter	Type	Description
total_count	Integer	Number of hosts.
hosts	Array of **Table 4-584** objects	Hosts.

Table 4-584 HostResult

Parameter	Type	Description
id	String	Host ID.
name	String	Host alias.
description	String	Host description.
type	String	Host type.
ssh_port	Integer	SSH service port of the host.
ssh_user_name	String	SSH username of the host.
ssh_key_id	String	SSH private key ID of the host.
data_center	String	Data center where the host is installed.
data_center_id	String	ID of the data center where the host is installed.
os	**Table 4-585** object	OS information about the host.
spec	**Table 4-586** object	Host specifications.
ip	**Table 4-587** object	IP address of the host.
instance_id	String	ID of the instance to which the host belongs.
instance_name	String	Name of the instance to which the host belongs.
node_id	String	ID of the node to which the host belongs.
node_status	String	Status of the node to which the host belongs.

Parameter	Type	Description
status	String	Host status. Enumerated values: • **initializing** • **initialized** • **initialize_failed** • **unmanaged** • **in_use** • **deleting** • **delete_failed** • **modifying**
created_at	String	Creation time in the "yyyy-mm-ddThh:mm:ssZ" format. **T** is the separator between calendar and hourly notation of time. **Z** indicates the time zone offset. For example, in the Beijing time zone, the offset is **+0800**.
updated_at	String	Update time in the "yyyy-mm-ddThh:mm:ssZ" format. **T** is the separator between calendar and hourly notation of time. **Z** indicates the time zone offset. For example, in the Beijing time zone, the offset is **+0800**.
job_id	String	Job ID.
async_task_error	**Table 4-588** object	Host task error information.
agent_version	String	Host agent version.
engine	String	Name of the engine to which the host belongs.
dir_settings	Arrays of **Table 4-589**	Host directory settings.

Table 4-585 OsOption

Parameter	Type	Description
os_type	String	OS of the host.
os_version	String	OS version of the host.

Parameter	Type	Description
os_arch	String	Architecture type of the host. It can be x86 or Arm. Enumerated values: ● **x86** ● **arm**
small_os_version	string	Minor OS version of the host. Enumerated values: ● **kylin V10 SP1** ● **kylin V10 SP2** ● **kylin V10 SP3** ● **hce 2.0** ● **sles 12.5** ● **uos 20** ● **bclinux 21.10** ● **openeuler 22.03 SP4** ● **nfs 4.0**

Table 4-586 SpecOption

Parameter	Type	Description
vcpus	String	Number of vCPUs on the host.
cpu_spec	String	Host CPU vendor.
mem	String	Total memory of the host.
root_disk_size	String	System disk size of the host.
data_disk_size	String	Data disk size of the host.
storage_type	String	Storage type of the host.

Table 4-587 IpOption

Parameter	Type	Description
manage_ip	String	Management IP address of the host.
data_ip	String	Data IP address of the host.
virtual_ip	String	Service IP address of the host.
float_ip	String	Floating IP address of the host.

Parameter	Type	Description
storage_ip	String	Storage IP address of a host.
storage_data_ip	String	Storage data IP address of a host.
manage_ipv6	String	Management IPv6 address of the host.
data_ipv6	String	Data IPv6 address of the host.
virtual_ipv6	String	Service IPv6 address of the host.
float_ipv6	String	Floating IPv6 address of the host.

Table 4-588 AsyncTaskErrorOption

Parameter	Type	Description
error_code	String	Task error code.
error_msg	String	Task error message.

Table 4-589 DirSettingOption

Parameter	Type	Description
dir_name	String	Directory name.
dir_path	String	Directory path.

Status code: default

Table 4-590 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

```
GET  https://{Endpoint}/rds/gaussdb/v3/hosts
```

Example Response

Status code: 200

Success.

```
{
  "total_count" : 1,
  "hosts" : [ {
    "id" : "5d001c5c-ca85-4e4f-a891-debf125ed1ef",
    "name" : "test_hosts",
    "description": " The host is used to build a GaussDB database.",
    "type" : "BMS",
    "ssh_port" : 22,
    "ssh_user_name" : "root",
    "ssh_key_id" : null,
    "data_center" : "AZ1",
    "data_center_id" : "59ec7944-757a-4429-b448-bad7207ce7c6",
    "os" : {
      "os_type" : "kylin",
      "os_version" : "Kylin Linux Advanced Server V10 (Sword)",
      "os_arch" : "x86" ,
      "small_os_version": "kylin V10 SP1"
    },
    "spec" : {
      "vcpus" : 8,
      "cpu_spec" : "intel",
      "mem" : 64,
      "root_disk_size" : 40,
      "data_disk_size" : 300,
      "storage_type" : "localssd"
    },
    "ip" : {
      "manage_ip" : "192.168.0.1",
      "data_ip" : "192.168.0.2",
      "virtual_ip" : "192.168.0.3",
      "float_ip" : null ,
      "storage_ip" : null ,
      "storage_data_ip" : null ,
      "manage_ipv6" : "fd00:aaaa:20:cb:7a2e:5ff6:831b:8b6c",
      "data_ipv6" : "2001:0db8:85a3:0000:0000:8a2e:0370:7334",
      "virtual_ipv6" : "2001:0db8:0a0b:12f0:0000:0000:0000:1a2b",
      "float_ipv6" : null
    },
    "instance_id" : "367d35dd71f64ffc8eb95074ddf5fdb6in14",
    "instance_name" : "Test_instance",
    "node_id" : "6e7766ce192c4606b9e799efc41befe3no14",
    "node_status": "normal",
    "status" : "in_use",
    "created_at" : "2023-05-27T03:38:51+0800",
    "updated_at" : "2023-06-02T04:38:51+0800",
    "job_id" : "7ec75921f0fdc-4a64-9b97-0b52fe666437",
    "async_task_error" : null,
    "agent_version": "gaussdbv5_agent_24.7.32.0",
    "engine": "gaussdbv5",
    "dir_settings" : [ {
       "dir_name" : "software_dir",
       "dir_path" : "/dbs"
     }, {
       "dir_name" : "log_dir",
       "dir_path" : "/home/Ruby/log"
     } ]
  } ]
}
```

Status Code

Status Code	Description
200	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.9.2 Querying Hosts (Recommended)

Function

This API is used to query hosts.

URI

GET /gaussdb/v3.1/hosts

Table 4-591 Query parameters

Parameter	Mandatory	Type	Description
id	No	String	Host ID.
name	No	String	Host alias.
type	No	String	Host type.
status	No	String	Host status. Enumerated values: ● **initializing** ● **initialized** ● **initialize_failed** ● **unmanaged** ● **in_use** ● **deleting** ● **delete_failed** ● **modifying**
data_center_id	No	String	Data center ID.
data_center	No	String	Data center alias.
instance_id	No	String	Instance ID.
instance_name	No	String	Instance name.
manage_ip	No	String	Host management IP address.

Parameter	Mandatory	Type	Description
offset	No	Integer	Index offset. If **offset** is set to *N*, the resource query starts from the *N*+1 data entry. The default value is **0**, indicating that the query starts from the first data entry. The value must be a number but cannot be a negative number.
limit	No	Integer	Number of hosts. The default value is **10**. The value cannot be a negative number. The minimum value is **1**, and the maximum value is **100**.
sort_key	No	String	Sorting key. Enumerated values: ● **name** ● **type** ● **status** ● **created_at**
sort_order	No	String	Sorting order. It determines whether data is sorted in ascending or descending order. Enumerated values: ● **asc** ● **desc**

Request Parameters

Table 4-592 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	Yes	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Response Parameters

Status code: 200

Table 4-593 Response body parameters

Parameter	Type	Description
total_count	Integer	Number of hosts.
hosts	Array of **Table 4-594** objects	Hosts.

Table 4-594 HostResultV2

Parameter	Type	Description
id	String	Host ID.
name	String	Host alias.
description	String	Host description.
type	String	Host type.
ssh_port	Integer	SSH service port of the host.
ssh_user_name	String	SSH username of the host.
ssh_key_id	String	SSH private key ID of the host.
data_center	String	Data center where the host is installed.
data_center_id	String	ID of the data center where the host is installed.
os	**Table 4-595** object	OS information about the host.
spec	**Table 4-596** object	Host specifications.
ip	**Table 4-597** object	Host IP address.
associate_nodes	**Table 4-598** object	Information about the nodes associated with a host.

Parameter	Type	Description
status	String	Host status. Enumerated values: • **initializing** • **initialized** • **initialize_failed** • **unmanaged** • **in_use** • **deleting** • **delete_failed** • **modifying**
created_at	String	Creation time in the "yyyy-mm-ddThh:mm:ssZ" format. **T** is the separator between calendar and hourly notation of time. **Z** indicates the time zone offset. For example, in the Beijing time zone, the offset is **+0800**.
updated_at	String	Update time in the "yyyy-mm-ddThh:mm:ssZ" format. **T** is the separator between calendar and hourly notation of time. **Z** indicates the time zone offset. For example, in the Beijing time zone, the offset is **+0800**.
job_id	String	Job ID.
async_task_error	**Table 4-599** object	Host task error information.
agent_version	String	Host agent version.
engine_name	String	Name of the engine to which the host belongs.
dir_settings	Arrays of **Table 4-600**	Host directory settings.

Table 4-595 OsOption

Parameter	Type	Description
os_type	String	OS of the host.
os_version	String	OS version of the host.

Parameter	Type	Description
os_arch	String	Architecture type of the host. It can be x86 or Arm. Enumerated values: • **x86** • **arm**
small_os_version	string	Minor OS version of the host. Enumerated values: • **kylin V10 SP1** • **kylin V10 SP2** • **kylin V10 SP3** • **hce 2.0** • **sles 12.5** • **uos 20** • **bclinux 21.10** • **openeuler 22.03 SP4** • **nfs 4.0**

Table 4-596 SpecOption

Parameter	Type	Description
vcpus	String	Number of vCPUs on the host.
cpu_spec	String	Host CPU vendor.
mem	String	Total memory of the host.
root_disk_size	String	System disk size of the host.
data_disk_size	String	Data disk size of the host.
storage_type	String	Host storage type.

Table 4-597 IpOption

Parameter	Type	Description
manage_ip	String	Management IP address of the host.
data_ip	String	Data IP address of the host.
virtual_ip	String	Service IP address of the host.
float_ip	String	Floating IP address of the host.

Parameter	Type	Description
storage_ip	String	Storage IP address of a host.
storage_data_ip	String	Storage data IP address of a host.
manage_ipv6	String	Management IPv6 address of the host.
data_ipv6	String	Data IPv6 address of the host.
virtual_ipv6	String	Service IPv6 address of the host.
float_ipv6	String	Floating IPv6 address of the host.

Table 4-598 AssociateNodeOption

Parameter	Type	Description
node_id	String	ID of the associated node.
node_status	String	Status of the associated node.
instance_id	String	ID of the instance to which the node belongs.
instance_name	String	Name of the instance to which the node belongs.

Table 4-599 AsyncTaskErrorOption

Parameter	Type	Description
error_code	String	Task error code.
error_msg	String	Task error message.

Table 4-600 DirSettingOption

Parameter	Type	Description
dir_name	String	Directory name.
dir_path	String	Directory path.

Status code: default

Table 4-601 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

```
GET  https://{Endpoint}/rds/gaussdb/v3.1/hosts
```

Example Response

Status code: 200

Success.


```
{
  "total_count" : 1,
  "hosts" : [ {
    "id" : "5d001c5c-ca85-4e4f-a891-debf125ed1ef",
    "name" : "test_hosts",
    "type" : "BMS",
    "status" : "in_use",
    "os" : {
      "os_type" : "kylin",
      "os_arch" : "x86" ,
      "os_version" : "Kylin Linux Advanced Server V10 (Sword)",
      "small_os_version": "kylin V10 SP1"
    },
    "spec" : {
      "vcpus" : 8,
      "cpu_spec" : "intel",
      "mem" : 64,
      "root_disk_size" : 40,
      "data_disk_size" : 300,
      "storage_type" : "localssd"
    },
    "ip" : {
      "manage_ip" : "192.168.0.1",
      "data_ip" : "192.168.0.2",
      "virtual_ip" : "192.168.0.3",
      "float_ip" : null ,
      "storage_ip" : null ,
      "storage_data_ip" : null ,
      "manage_ipv6" : "fd00:aaaa:20:cb:7a2e:5ff6:831b:8b6c",
      "data_ipv6" : "2001:0db8:85a3:0000:0000:8a2e:0370:7334",
      "virtual_ipv6" : "2001:0db8:0a0b:12f0:0000:0000:0000:1a2b",
      "float_ipv6" : null
    },
    "associate_nodes" : [ {
      "node_id" : "6e7766ce192c4606b9e799efc41befe3no14",
      "node_status" : "normal",
      "instance_id" : "367d35dd71f64ffc8eb95074ddf5fdb6in14",
      "instance_name" : "Test_instance" ,
    } ],
    "ssh_port" : 22,
    "ssh_user_name" : "root",
    "ssh_key_id" : null,
    "data_center" : "AZ1",
    "data_center_id" : "59ec7944-757a-4429-b448-bad7207ce7c6",
    "created_at" : "2023-05-27T03:38:51+0800",
    "updated_at" : "2023-06-02T04:38:51+0800",
    "engine_name": "gaussdbv5",
```

```
  "agent_version": "gaussdbv5_agent_24.7.32.0",
  "dir_settings" : [ {
    "dir_name" : "software_dir",
    "dir_path" : "/dbs"
   }, {
    "dir_name" : "log_dir",
    "dir_path" : "/home/Ruby/log"
   } ],
  "description": " The host is used to build a GaussDB database.",
  "job_id" : "7ec75921f0fdc-4a64-9b97-0b52fe666437",
  "async_task_error" : null
 } ]
}
```


Status Codes

Status Code	Description
200	Normal
default	Abnormal

Error Codes

For details, see **Error Codes**.

4.9.3 Adding Hosts

Function

This API is used to add hosts.

A maximum of 50 hosts can be added in batches.

Constraints

- For details about the supported host specifications, see the API for querying supported host specifications.
- To add a host over SSH, you need to use a username and password or use an SSH private key.

URI

POST /gaussdb/v3/hosts

Request Parameters

Table 4-602 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.

Parameter	Mandatory	Type	Description
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Table 4-603 Request body parameter

Parameter	Mandatory	Type	Description
hosts	Yes	Array of **Table 4-604** objects	Hosts to be added.

Table 4-604 HostOption

Parameter	Mandatory	Type	Description
name	No	String	Host alias. It contains up to 64 bytes.
type	Yes	String	Host type. The value can be **BMS** (a host, which can be a physical machine or VM).
os_type	No	String	● If the host type is **BMS**, the value can be **kylin**, **uniontech**, **hce**, **suse**, **bclinux**, **openeuler**, or **nfs**. ● If the host type is **CONTAINER**, the value can be **kylin**, **uniontech**, or **hce**.
cpu_spec	No	String	Host CPU vendor. The value can be **intel**, **kunpeng**, or **hygon**.
storage_type	Yes	String	Host storage type. The value can be **localssd** or **dorado**.
ip	Yes	**Table 4-605** object	IP address of the host.

Parameter	Mandatory	Type	Description
is_use_ssh	No	Boolean	Whether to use SSH to add hosts. Set this parameter to **true**. The default value is **true**. If the host type is **CONTAINER** and the host is brought online in Agent mode, set this parameter to **false**. Otherwise, set it to **true**.
ssh_port	No	Integer	SSH service port of the host. This parameter is mandatory when an SSH private key or username/password is used. The value ranges from 1 to 65535. The following ports are restricted: 8000, 8635, 9000, 9200, 9300, 12017, and 20050.
ssh_user_name	No	String	SSH username of the host. The default value is **root**.
ssh_user_pwd	No	String	Password of the SSH user of the host. This parameter needs to be set when a username and password are used. If the SSH user of the host is a non-root user, this parameter needs to be set.
os_root_pwd	No	String	Password of the **root** user of the host. This parameter needs to be set when a username and password are used. If the SSH user of the host is **root**, set either **ssh_user_pwd** (preferred) or **os_root_pwd**.
ssh_key_id	No	String	SSH private key ID, which is required when an SSH private key is used.
data_center_id	Yes	String	ID of the data center where the host is installed.
description	No	String	Host description. It can contain up to 100 characters.
is_force_rm_chroot_dir	No	Boolean	Whether to forcibly clear the sandbox directory.

Parameter	Mandatory	Type	Description
installation_mode	No	String	Installation mode. The value can be **dm** (DM mode) or **chroot** (sandbox mode). The default value is **chroot**. If a host is used to install a Dorado instance or DBMind instance, this parameter must be set to **chroot** (sandbox mode).
dir_settings	No	Arrays of **Table 4-606**	Host directory settings.

Table 4-605 AddIpOption

Parameter	Mandatory	Type	Description
manage_ip	Yes	String	Management IP address of the host.
data_ip	Yes	String	Data IP address of the host.
virtual_ip	Yes	String	Service IP address of the host.
storage_ip	No	String	Storage IP address of a host.
storage_data_ip	No	String	Storage data IP address of a host.
manage_ipv6	No	String	Management IPv6 address of the host. This parameter is mandatory when the host network protocol is IPv6.
data_ipv6	No	String	Data IPv6 address of the host. This parameter is mandatory when the host network protocol is IPv6.
virtual_ipv6	No	String	Service IPv6 address of the host. This parameter is mandatory when the host network protocol is IPv6.

Table 4-606 DirSettingOption

Parameter	Type	Description
dir_name	String	Directory name.

Parameter	Type	Description
dir_path	String	Directory path.

Response Parameters

Status code: 202

Table 4-607 Response body parameters

Parameter	Type	Description
job_id	String	Job ID.
hosts	Array of **Table 4-608** objects	Hosts.

Table 4-608 HostResponseOption

Parameter	Type	Description
id	String	Host ID.
manage_ip	String	Management IP address of the host.

Status code: default

Table 4-609 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

- Adding a single host using a username and password (as the **root** user)

```
POST https://{Endpoint}/rds/gaussdb/v3/hosts
{
  "hosts" : [ {
    "name" : "test_hosts",
    "type" : "BMS",
    "os_type" : "kylin",
    "cpu_spec" : "intel",
    "storage_type" : "localssd",
    "ip" : {
      "manage_ip" : "192.168.0.1",
      "data_ip" : "192.168.0.2",
      "virtual_ip" : "192.168.0.3"
    },
    "ssh_port" : 22,
```

```
    "ssh_user_name" : "root",
    "ssh_user_pwd" : "Test_345612",
    "os_root_pwd" : "Test_345612",
    "data_center_id" : "59ec7944-757a-4429-b448-bad7207ce7c6",
    "description": " The host is used to build a GaussDB database.",
    "is_force_rm_chroot_dir" : false,
    "installation_mode" : "dm",
    "dir_settings" : [ {
      "dir_name" : "software_dir",
      "dir_path" : "/dbs"
     }, {
      "dir_name" : "log_dir",
      "dir_path" : "/home/Ruby/log"
     } ]
  } ]
}
```

- Adding a single host using a username and password (as a non-**root** user)

```
POST https://{Endpoint}/rds/gaussdb/v3/hosts
{
  "hosts" : [ {
    "name" : "test_hosts",
    "type" : "BMS",
    "os_type" : "kylin",
    "cpu_spec" : "intel",
    "storage_type" : "localssd",
    "ip" : {
      "manage_ip" : "192.168.0.1",
      "data_ip" : "192.168.0.2",
      "virtual_ip" : "192.168.0.3"
    },
    "ssh_port" : 22,
    "ssh_user_name" : "testuser",
    "ssh_user_pwd" : "Test_345612",
    "data_center_id" : "59ec7944-757a-4429-b448-bad7207ce7c6",
    "description": " The host is used to build a GaussDB database.",
    "is_force_rm_chroot_dir" : false,
    "installation_mode" : "dm"
  } ]
}
```

- Adding a single host using SSH private key authorization

```
POST https://{Endpoint}/rds/gaussdb/v3/hosts
{
  "hosts" : [ {
    "name" : "test_hosts",
    "type" : "BMS",
    "os_type" : "kylin",
    "cpu_spec" : "intel",
    "storage_type" : "localssd",
    "ip" : {
      "manage_ip" : "192.168.0.1",
      "data_ip" : "192.168.0.2",
      "virtual_ip" : "192.168.0.3"
    },
    "ssh_port" : 22,
    "ssh_user_name" : "root",
    "ssh_key_id" : "c8ec7944-757a-4429-b448-bad7207ce7c6",
    "data_center_id" : "59ec7944-757a-4429-b448-bad7207ce7c6",
    "description": " The host test_hosts is used to build a GaussDB database.",
    "is_force_rm_chroot_dir" : false
  } ]
}
```

- Adding IPv4 and IPv6 hosts at a time

```
POST https://{Endpoint}/rds/gaussdb/v3/hosts
{
  "hosts" : [ {
    "name" : "test_hosts",
    "type" : "BMS",
    "os_type" : "kylin",
```

```
  "cpu_spec" : "intel",
  "storage_type" : "localssd",
  "ip" : {
    "manage_ip" : "192.168.0.1",
    "data_ip" : "192.168.0.2",
    "virtual_ip" : "192.168.0.3"
  },
  "ssh_port" : 22,
  "data_center_id" : "59ec7944-757a-4429-b448-bad7207ce7c6",
  "ssh_user_name" : "root",
  "ssh_user_pwd" : "Test_345612",
  "os_root_pwd" : "Test_345612",
  "description": "The IPv4 host test_hosts is used to build a GaussDB database.",
  "is_force_rm_chroot_dir" : false
 }, {
  "name" : "test_hosts2",
  "type" : "BMS",
  "os_type" : "kylin",
  "cpu_spec" : "intel",
  "storage_type" : "localssd",
  "ip" : {
    "manage_ip" : "192.168.0.4",
    "data_ip" : "192.168.0.5",
    "virtual_ip" : "192.168.0.6",
    "manage_ipv6" : "fd00:aaaa:20:cb:7a2e:5ff6:831b:8b6c",
    "data_ipv6" : "2001:0db8:85a3:0000:0000:8a2e:0370:7334",
    "virtual_ipv6": "2001:0db8:0a0b:12f0:0000:0000:0000:1a2b"
  },
  "ssh_port" : 22,
  "ssh_user_name" : "testuser",
  "ssh_user_pwd" : "Test_345612",
  "data_center_id" : "59ec7944-757a-4429-b448-bad7207ce7c6",
  "description": "The IPv6 host test_hosts2 is used to build a GaussDB database.",
  "is_force_rm_chroot_dir" : false
 } ]
}
```

Example Response

Status code: 202

Success.

- Adding a host

```
{
 "job_id" : "ff2e6f95-a93c-4c10-b020-d5b0244d94ef",
 "hosts" : [ {
  "id" : "5d001c5c-ca85-4e4f-a891-debf125ed1ef",
  "manage_ip" : "192.168.0.1"
 } ]
}
```

- Adding hosts in batches

```
{
 "job_id" : "ff2e6f95-a93c-4c10-b020-d5b0244d94ef",
 "hosts" : [ {
  "id" : "5d001c5c-ca85-4e4f-a891-debf125ed1ef",
  "manage_ip" : "192.168.0.1"
 }, {
  "id" : "995ea66a-c3f7-43f3-8eac-1393ba0c59ce",
  "manage_ip" : "192.168.0.4"
 } ]
}
```

Status Code

Status Code	Description
202	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.9.4 Modifying Static Host Information

Function

This API is used to modify the static information about hosts.

Constraints

- The static information about a host cannot be modified when the host is in the **Initializing**, **Modifying**, or **Deleting** state.
- The IP address and data center of a host can only be changed when the host is in the **Initialization completed** or **Deletion failed** state.
- Hosts support IPv4-to-IPv6 conversion, but do not support IPv6-to-IPv4 conversion.
- You can modify static information about up to 50 hosts in batches.

URI

PUT /gaussdb/v3/hosts

Request Parameters

Table 4-610 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Table 4-611 Request body parameters

Parameter	Mandatory	Type	Description
hosts	Yes	Array of **Table 4-612** objects	Hosts to be modified.

Table 4-612 ModifyHostOption

Parameter	Mandatory	Type	Description
manage_ip	Yes	String	Management IP address of the host, which cannot be changed.
name	No	String	Host alias. It contains up to 64 bytes.
data_ip	No	String	Data IP address of the host.
virtual_ip	No	String	Service IP address of the host.
storage_ip	No	String	Storage IP address of a host.
storage_data_ip	No	String	Storage data IP address of a host.
manage_ipv6	No	String	Management IPv6 address of the host.
data_ipv6	No	String	Data IPv6 address of the host.
virtual_ipv6	No	String	Service IPv6 address of the host.
data_center_id	No	String	ID of the data center where the host is installed.
description	No	String	Host description. It can contain up to 100 characters.
ssh_key_id	No	String	SSH private key ID.

Response Parameters

Status code: 200

Table 4-613 Response body parameters

Parameter	Type	Description
job_id	String	Job ID.

Parameter	Type	Description
hosts	Array of **Table 4-614** objects	Hosts.

Table 4-614 HostResponseOption

Parameter	Type	Description
id	String	Host ID.
manage_ip	String	Management IP address of the host.

Status code: default

Table 4-615 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

- Changing the data IP address of the host

```
PUT  https://{Endpoint}/rds/gaussdb/v3/hosts

{
  "hosts" : [ {
    "manage_ip" : "192.168.0.1",
    "data_ip" : "192.168.0.2"
  } ]
}
```

- IPv4-to-IPv6 conversion of the host

```
PUT  https://{Endpoint}/rds/gaussdb/v3/hosts

{
  "hosts" : [ {
    "manage_ip" : "192.168.0.1",
    "manage_ipv6" : "fd00:aaaa:20:cb:7a2e:5ff6:831b:8b6c",
    "data_ipv6" : "2001:0db8:85a3:0000:0000:8a2e:0370:7334",
    "virtual_ipv6": "2001:0db8:0a0b:12f0:0000:0000:0000:1a2b"
  } ]
}
```

Example Response

Status code: 202

Success.

```
{
  "job_id" : "ff2e6f95-a93c-4c10-b020-d5b0244d94ef",
```

```
"hosts" : [ {
  "id" : "5d001c5c-ca85-4e4f-a891-debf125ed1ef",
  "manage_ip" : "192.168.0.1"
 } ]
}
```


Status Code

Status Code	Description
202	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.9.5 Deleting a Host

Function

This API is used to delete a host.

A maximum of 50 hosts can be deleted in batches.

Constraints

- Hosts cannot be deleted when they are in the **Initializing**, **Modifying**, or **Deleting** status.
- A host in the **In use** state can be deleted when **Forced Deletion** is selected, HA is enabled for the host, and there are abnormal nodes on the host.
- If a host is added using an SSH private key and there is a public key on the host, you do not need to configure the **ssh_key_id** field when deleting the host using an SSH private key.
- If a host is deleted using an SSH private key, you need to ensure that there is a public key on the host.

URI

DELETE /gaussdb/v3/hosts

Request Parameters

Table 4-616 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.

Parameter	Mandatory	Type	Description
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Table 4-617 Request body parameters

Parameter	Mandatory	Type	Description
hosts	Yes	Array of **Table 4-618** objects	Hosts to be deleted.

Table 4-618 RemoveHostOption

Parameter	Mandatory	Type	Description
manage_ip	Yes	String	Management IP address of the host.
ssh_port	No	Integer	SSH service port of the host. This parameter is mandatory when an SSH private key or username/password is used. The value ranges from 1 to 65535. The following ports are restricted: 8000, 8635, 9000, 9200, 9300, 12017, and 20050.
ssh_user_name	No	String	SSH username of the host. The default value is **root**.
ssh_user_pwd	No	String	Password of the SSH user of the host. This parameter needs to be set when a username and password are used. If the SSH user of the host is a non-root user, this parameter needs to be set.

Parameter	Mandatory	Type	Description
os_root_pwd	No	String	Password of the **root** user of the host. This parameter needs to be set when a username and password are used. If the SSH user of the host is **root**, set either **ssh_user_pwd** (preferred) or **os_root_pwd**.
ssh_key_id	No	String	SSH private key ID for logging in to the host. This parameter needs to be set when an SSH private key is used.
is_force_rm_host	No	Boolean	Whether to forcibly delete a host. The default value is **false**.

Response Parameters

Status code: 202

Table 4-619 Response body parameters

Parameter	Type	Description
job_id	String	Job ID.
hosts	Array of **Table 4-620** objects	Hosts.

Table 4-620 HostResponseOption

Parameter	Type	Description
id	String	Host ID.
manage_ip	String	Management IP address of the host.

Status code: default

Table 4-621 Response body parameters

Parameter	Type	Description
error_code	String	Error code.

Parameter	Type	Description
error_msg	String	Error message.

Example Request

- Deleting a single host using a username and password (as the **root** user)

```
DELETE  https://{Endpoint}/rds/gaussdb/v3/hosts
{
  "hosts" : [ {
    "manage_ip" : "192.168.0.1",
    "ssh_port" : 22,
    "ssh_user_name" : "root",
    "ssh_user_pwd" : "Test_345612",
    "os_root_pwd" : "Test_345612"
  } ]
}
```

- Deleting a single host using a username and password (as a non-**root** user)

```
DELETE  https://{Endpoint}/rds/gaussdb/v3/hosts
{
  "hosts" : [ {
    "manage_ip" : "192.168.0.1",
    "ssh_port" : 22,
    "ssh_user_name" : "testuser",
    "ssh_user_pwd" : "Test_345612"
  } ]
}
```

- Forcibly deleting a single host using an SSH private key

```
DELETE  https://{Endpoint}/rds/gaussdb/v3/hosts
{
  "hosts" : [ {
    "manage_ip" : "192.168.0.1",
    "ssh_port" : 22,
    "ssh_user_name" : "root",
    "ssh_key_id" : "53001c5c-ca85-4e4f-a891-debf125ed1ef"
    "is_force_rm_host": true
  } ]
}
```

- Deleting multiple hosts at a time

```
DELETE  https://{Endpoint}/rds/gaussdb/v3/hosts
{
   "hosts" : [ {
     "manage_ip" : "192.168.0.1",
     "ssh_port" : 22,
     "ssh_user_name" : "testuser",
     "ssh_user_pwd" : "Test_345612"
   }, {
     "manage_ip" : "192.168.0.2",
     "ssh_port" : 22,
     "ssh_user_name" : "root",
     "ssh_key_id" : "53001c5c-ca85-4e4f-a891-debf125ed1ef"
   } ]
}
```

Example Response

Status code: 202

Success.

- Host deleted.

```
{
  "job_id" : "ff2e6f95-a93c-4c10-b020-d5b0244d94ef",
  "hosts" : [ {
    "id" : "5d001c5c-ca85-4e4f-a891-debf125ed1ef",
    "manage_ip" : "192.168.0.1"
  } ]
}
```


- Hosts deleted.


```
{
  "job_id" : "ff2e6f95-a93c-4c10-b020-d5b0244d94ef",
  "hosts" : [ {
    "id" : "5d001c5c-ca85-4e4f-a891-debf125ed1ef",
    "manage_ip" : "192.168.0.1"
  }, {
    "id" : "995ea66a-c3f7-43f3-8eac-1393ba0c59ce",
    "manage_ip" : "192.168.0.4"
  } ]
}
```


Status Code

Status Code	Description
202	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.9.6 Querying the Standardization Check Result of a Host

Function

This API is used to query the standardization check result of a host.

URI

GET /gaussdb/v3/hosts/{host_id}/detection-results

Table 4-622 URI parameter

Parameter	Mandatory	Type	Description
host_id	Yes	String	Host ID.

Table 4-623 Query parameters

Parameter	Mandatory	Type	Description
offset	No	Integer	Index offset. If **offset** is set to *N*, the resource query starts from the *N*+1 piece of data. The value is **0** by default, indicating that the query starts from the first piece of data. The value must be a number but cannot be a negative number.
limit	No	Integer	Number of host check items. The default value is **50**. The value cannot be a negative number. The minimum value is **1** and the maximum value is **100**.
sort_key	No	String	Sorting key. Enumerated value: **result**
sort_order	No	String	Sorting order. It determines whether data is sorted in ascending or descending order. Enumerated values: ● **asc** ● **desc**

Request Parameters

Table 4-624 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	Yes	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Response Parameters

Status code: 200

Table 4-625 Response body parameters

Parameter	Type	Description
total_count	Integer	Number of check items.
detection_results	Array of **Table 4-626** objects	Check result details.

Table 4-626 DetectionResult

Parameter	Type	Description
id	String	Check Item ID.
name	String	Check item name.
description	String	Check item description.
expect_result	String	Expected value.
actual_result	String	Current value.
result	String	Check result. Enumerated values: ● **passed** ● **warning** ● **unpassed**

Status code: default

Table 4-627 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

```
GET  https://{Endpoint}/rds/gaussdb/v3/hosts/5d001c5c-ca85-4e4f-a891-debf125ed1ef/detection-results
```

Example Response

Status code: 200

Success.

```
{
  "total_count" : 1,
```

```
  "detection_results" : [ {
    "id" : "b1398c19-1ef8-4eff-a5bc-0cc2658cd6c8",
    "name" : "CPU",
    "description": "vCPUs",
    "expect_result" : "64",
    "actual_result" : "64",
     "result" : "passed"
  } ]
}
```


Status Code

Status Code	Description
200	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.9.7 Downloading a Host Import Template

Function

This API is used to download a host import template.

URI

GET /gaussdb/v3/hosts/template-file

Request Parameters

Table 4-628 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Response Parameters

Status code: 200

Table 4-629 Response body parameters

Parameter	Type	Description
-	File	Success.

Status code: default

Table 4-630 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

```
GET  https://{Endpoint}/rds/gaussdb/v3/hosts/template-file
```

Example Response

None

Status Code

Status Code	Description
200	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.9.8 Batch Importing Host Files for Check

Function

- This API is used to import host files for check in batches.
- A maximum of 50 hosts can be imported in batches. The file for importing hosts in batches must be in .xls or .xlsx format. The file size cannot exceed 500 KB.

URI

POST /gaussdb/v3/hosts/file-verification

Request Parameters

Table 4-631 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Table 4-632 Request body parameter

Parameter	Mandatory	Type	Description
file	Yes	String	File content, which is a hexadecimal byte array.

Response Parameters

Status code: 200

Table 4-633 Response body parameters

Parameter	Type	Description
total_count	Integer	Number of hosts in the file.
fail_count	Integer	Number of hosts that fail to be checked.
hosts	Array of **Table 4-634** objects	Hosts that are displayed after the check.
fail_hosts	Array of strings	Information about the host that fails to be checked. The format is *key(value:) in xy*, where *key* indicates the name of the field to be checked, *value* indicates the current value, and *xy* indicates that the column *x* and row *y* where the field is located.

Table 4-634 HostOption

Parameter	Type	Description
name	String	Host alias.
type	String	Host type.
os_type	String	Host OS.
cpu_spec	String	Host CPU vendor.
storage_type	String	Host storage type.
ip	**Table 4-635** object	Host IP address.
is_use_ssh	Boolean	Whether to use SSH to add hosts.
ssh_port	Integer	SSH service port of the host.
ssh_user_name	String	SSH username of the host.
ssh_user_pwd	String	Password of the SSH user for logging in to the host.
os_root_pwd	String	Password of the **root** user for logging in to the host.
ssh_key_id	String	SSH private key ID of the host.
data_center_id	String	ID of the data center where the host is installed.
description	String	Host description.
is_force_rm_chroot_dir	Boolean	Whether to forcibly clear the sandbox directory.
installation_mode	String	Installation mode. The value can be **dm** (DM mode) or **chroot** (sandbox mode).
dir_settings	Arrays of **Table 4-636**	Host directory settings.

Table 4-635 AddIpOption

Parameter	Type	Description
manage_ip	String	Management IP address of the host.
data_ip	String	Data IP address of the host.
virtual_ip	String	Service IP address of the host.
storage_ip	String	Storage IP address of a host.

Parameter	Type	Description
storage_data_ip	String	Storage data IP address of a host.
manage_ipv6	String	Management IPv6 address of the host.
data_ipv6	String	Data IPv6 address of the host.
virtual_ipv6	String	Service IPv6 address of the host.

Table 4-636 DirSettingOption

Parameter	Type	Description
dir_name	String	Directory name.
dir_path	String	Directory path.

Example Request

```
POST  https://{Endpoint}/rds/gaussdb/v3/hosts/file-verification

{
  "file" : "\"Content-Disposition: form-data; name=\"****\"; filename=\"****.xlsx\" Content-Type: application/
vnd.ms-excel\"****"
}
```

Example Response

Status code: 200

Success.

- Batch importing of host files passes the check.

```
{
  "total_count" : 1,
  "fail_count" : null,
  "hosts" : [ {
    "name" : "test_hosts",
    "type" : "BMS",
    "os_type" : "kylin",
    "cpu_spec" : "intel",
    "storage_type" : "localssd",
    "ip" : {
      "manage_ip" : "192.168.0.1",
      "data_ip" : "192.168.0.2",
      "virtual_ip" : "192.168.0.3",
      "manage_ipv6" : "fd00:aaaa:20:cb:7a2e:5ff6:831b:8b6c",
      "data_ipv6" : "2001:0db8:85a3:0000:0000:8a2e:0370:7334",
      "virtual_ipv6" : "2001:0db8:0a0b:12f0:0000:0000:0000:1a2b"
    },
    "ssh_port" : 22,
    "ssh_user_name" : "root",
    "ssh_user_pwd" : "Test_345612",
    "os_root_pwd" : "Test_345612",
    "data_center_id" : "59ec7944-757a-4429-b448-bad7207ce7c6",
    "description": " The host is used to build a GaussDB database.",
    "is_force_rm_chroot_dir" : false,
```

```
  "installation_mode" : "chroot",
  "dir_settings" : [ {
    "dir_name" : "software_dir",
    "dir_path" : "/dbs"
   }, {
    "dir_name" : "log_dir",
    "dir_path" : "/home/Ruby/log"
   } ]
 } ],
 "fail_hosts" : null
}
```

- Batch importing of host files does not pass the check.

```
{
  "total_count" : null,
  "fail_count" : 1,
  "hosts" : null,
  "fail_hosts" : [ "please check error in [\"ip(value:100.1) in E2\"]" ]
}
```

Status Code

Status Code	Description
200	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.9.9 Querying Available Host Specifications

Function

This API is used to query available host specifications.

Constraints

Hosts using HCE (**hce**) or BC-Linux (**bclinux**) and flash storage (**dorado**) cannot be added.

URI

GET /gaussdb/v3/hosts/specifications

Request Parameters

Table 4-637 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.

Parameter	Mandatory	Type	Description
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Response Parameters

Status code: 200

Table 4-638 Response body parameters

Parameter	Type	Description
types	Array of strings	Available host types. The value can be: ● **BMS**: a host, which can be a physical machine or a VM Value: ● **BMS**
os_types	Array of strings	Available host OSs. The value can be: ● Kylin: **kylin** ● UnionTech: **uniontech** ● Huawei Cloud EulerOS (HCE): **hce** ● SUSE: **suse** ● BC-Linux: **bclinux** ● openEuler: openeuler ● **nfs**: NFS China Enumerated values: ● **kylin** ● **uniontech** ● **hce** ● **suse** ● **bclinux** ● **openeuler** ● **nfs**

Parameter	Type	Description
storage_types	Array of strings	Available host storage types. The value can be: • Local SSD: **localssd** • Flash storage: **dorado** Enumerated values: • **localssd** • **dorado**
cpu_specs	Array of strings	Available host CPU vendors. The value can be: • Intel: **intel** • Kunpeng: **kunpeng** • Hygon: **hygon** Enumerated values: • **intel** • **kunpeng** • **hygon**
key_algorithm	Array of strings	Supported key algorithms. Value range: • RSA_3072 • RSA_4096 Enumerated values: • **RSA_3072** • **RSA_4096**

Status code: default

Table 4-639 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

```
GET  https://{Endpoint}/rds/gaussdb/v3/hosts/specifications
```

Example Response

Status code: 200

Success.

```
{
  "types" : [ "BMS" ],
```

```
 "os_types" : [ "kylin", "uniontech", "hce", "suse", "bclinux", "openeuler" , "nfs" ],
 "storage_types" : [ "localssd", "dorado" ],
 "cpu_specs" : [ "intel", "kunpeng", "hygon" ],
 "key_algorithm": ["RSA_3072", "RSA_4096"]
}
```

Status Code

Status Code	Description
200	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.9.10 Querying Data Centers

Function

This API is used to query data centers.

URI

GET /gaussdb/v3/data-centers

Table 4-640 Query parameters

Parameter	Mandatory	Type	Description
id	No	String	Data center ID.
name	No	String	Data center alias.
offset	No	Integer	Index offset. If **offset** is set to *N*, the resource query starts from the *N*+1 piece of data. The value is **0** by default, indicating that the query starts from the first piece of data. The value must be a number but cannot be a negative number.

Parameter	Mandatory	Type	Description
limit	No	Integer	Number of queried data centers. The default value is **10**. The value cannot be a negative number. The minimum value is **1** and the maximum value is **100**.
sort_key	No	String	Sorting key. Enumerated values: • **name** • **hosts_num**
sort_order	No	String	Sorting order. It determines whether data is sorted in ascending or descending order. Enumerated values: • **asc** • **desc**

Request Parameters

Table 4-641 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Response Parameters

Status code: 200

Table 4-642 Response body parameters

Parameter	Type	Description
total_count	Integer	Total number of data centers.

Parameter	Type	Description
data_centers	Array of **Table 4-643** objects	Data centers.

Table 4-643 DataCenterResult

Parameter	Type	Description
id	String	Data center ID.
name	String	Data center alias.
description	String	Data center description.
hosts_num	Integer	Number of hosts in the data center.

Status code: default

Table 4-644 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

```
GET  https://{Endpoint}/rds/gaussdb/v3/data-centers
```

Example Response

Status code: 200

Success.

```
{
  "total_count" : 1,
  "data_centers" : [ {
    "id" : "59ec7944-757a-4429-b448-bad7207ce7c6",
    "name" : "AZ1",
    "description": " data center 1",
    "hosts_num" : 10
  } ]
}
```

Status Code

Status Code	Description
200	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.9.11 Adding a Data Center

Function

This API is used to add a data center.

Constraints

- The data center alias must be unique.
- A maximum of 50 data centers can be added in batches.

URI

POST /gaussdb/v3/data-centers

Request Parameters

Table 4-645 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Table 4-646 Request body parameters

Parameter	Mandatory	Type	Description
data_centers	Yes	Array of **Table 4-647** objects	Data centers.

Table 4-647 AddDataCenterOption

Parameter	Mandatory	Type	Description
name	Yes	String	Data center alias. It can contain up to 20 characters. Only letters, numbers, hyphens (-), and underscores (_) are allowed.
description	No	String	Data center description. It can contain up to 100 characters.

Response Parameters

Status code: default

Table 4-648 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

```
POST  https://{Endpoint}/rds/gaussdb/v3/data-centers

{
  "data_centers" : [ {
    "name" : "AZ1"
  } ]
}
```

Example Response

None

Status Code

Status Code	Description
200	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.9.12 Editing Data Center Information

Function

This API is used to edit data center information.

Constraints

- The data center alias must be unique.
- If the data center is associated with a host that is in the **Unmanaged** or **In use** state, the data center alias cannot be changed.
- A maximum of 50 data centers can be edited in batches.

URI

PUT /gaussdb/v3/data-centers

Request Parameters

Table 4-649 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Table 4-650 Request body parameter

Parameter	Mandatory	Type	Description
data_centers	Yes	Array of **Table 4-651** objects	Data centers to be edited.

Table 4-651 ModifyDataCenterOption

Parameter	Mandatory	Type	Description
id	Yes	String	Data center ID.
name	No	String	Data center alias. It can contain up to 20 characters. Only letters, numbers, hyphens (-), and underscores (_) are allowed. (Select at least one optional parameter.)
description	No	String	Data center description. It can contain up to 100 characters. (Select at least one optional parameter.)

Response Parameters

Status code: default

Table 4-652 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

```
PUT  https://{Endpoint}/rds/gaussdb/v3/data-centers

{
  "data_centers" : [ {
    "id" : "59ec7944-757a-4429-b448-bad7207ce7c6",
    "name" : "AZ2"
  } ]
}
```

Example Response

None

Status Code

Status Code	Description
200	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.9.13 Deleting a Data Center

Function

This API is used to delete a data center.

Constraints

- If a host is associated with a data center, the data center cannot be deleted.
- A maximum of 50 data centers can be deleted in batches.

URI

DELETE /gaussdb/v3/data-centers

Request Parameters

Table 4-653 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Table 4-654 Request body parameter

Parameter	Mandatory	Type	Description
data_center_ids	Yes	Array of strings	IDs of data centers to be deleted.

Response Parameters

Status code: default

Table 4-655 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

```
DELETE  https://{Endpoint}/rds/gaussdb/v3/data-centers

{
  "data_center_ids" : [ "59ec7944-757a-4429-b448-bad7207ce7c6", "0904510d-52bf-4e40-8a39-f2b6403d85c8" ]
}
```

Example Response

None

Status Code

Status Code	Description
200	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.10 Routine Inspection

4.10.1 Querying Inspection Tasks

Function

This API is used to query inspection tasks by page and sort inspection tasks by field.

URI

GET /gaussdb/v3/inspections

Table 4-656 Query parameters

Parameter	Mandatory	Type	Description
limit	No	Integer	Number of returned records. Minimum value: **1** Maximum value: **100** Default value: **10**
offset	No	Integer	Query offset. Minimum value: **0** Maximum value: **2147483647** Default value: **0**
inspection_name	No	String	Inspection task name. Minimum length: 1 character Maximum length: 64 characters
instance_name	No	String	Instance name. Minimum length: 1 character Maximum length: 128 characters
create_time_start	No	String	Start time of the creation, for example, **2024/01/11 15:47:53**. If this parameter is set separately, it does not take effect. You need to configure both **create_time_end** and **create_time_start**.
create_time_end	No	String	End time of the creation, for example, **2024/01/12 15:47:53**. If this parameter is set separately, it does not take effect. You need to configure both **create_time_end** and **create_time_start**.
last_execute_time_start	No	String	Start time of the last execution, for example, **2024/01/11 15:47:53**. If this parameter is set separately, it does not take effect. You need to configure both **last_execute_time_start** and **last_execute_time_end**.

Parameter	Mandatory	Type	Description
last_execute_time_end	No	String	End time of the last execution, for example, 2024/01/12 15:47:53. If this parameter is set separately, it does not take effect. You need to configure both **last_execute_time_start** and **last_execute_time_end**.
next_execute_time_start	No	String	Start time of the next execution, for example, 2024/01/11 15:47:53. If this parameter is set separately, it does not take effect. You need to configure both **next_execute_time_start** and **next_execute_time_end**.
next_execute_time_end	No	String	End time of the next execution, for example, 2024/01/12 15:47:53. If this parameter is set separately, it does not take effect. You need to configure both **next_execute_time_start** and **next_execute_time_end**.
sort	No	String	Sorting field. Default value: **create_at** Enumerated values: ● **inspection_name** ● **create_at** ● **last_execute_time** ● **next_execute_time**
order	No	String	Sorting order. Default value: **desc** Enumerated values: ● **desc** ● **asc**

Request Parameters

Table 4-657 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Response Parameters

Status code: 201

Table 4-658 Response header parameter

Parameter	Type	Description
X-TRACE-ID	String	Request ID used for task tracing.

Table 4-659 Response body parameters

Parameter	Type	Description
total_count	Integer	Total number of inspection items. Minimum value: **0** Maximum value: **20000**
inspections	Array of **Table 4-660** objects	Inspection items. Array length: 0 to 2000 characters

Table 4-660 InspectionInfo

Parameter	Type	Description
id	String	Inspection task ID. Minimum length: 1 character Maximum length: 64 characters
inspection_name	String	Inspection task name.

Parameter	Type	Description
inspection_scope	String	Inspection type.
instance_names	String	Character string concatenated by the instance name.
inspection_policy	String	Inspection policy.
scene	String	Inspection scenario.
status	String	Inspection status.
last_execute_time	String	Last execution time.
next_execute_time	String	Next execution time.
create_at	String	Creation time.

Status code: 400

Table 4-661 Response header parameter

Parameter	Type	Description
X-TRACE-ID	String	Request ID used for task tracing.

Table 4-662 Response body parameters

Parameter	Type	Description
errCode	String	Error code. Minimum length: 8 characters Maximum length: 36 characters
externalMessage	String	Error message. Minimum length: 2 characters Maximum length: 512 characters

Example Request

Querying inspection tasks

```
GET https://{Endpoint}/rds/gaussdb/v3/inspections?limit=10&offset=0&inspection_name=inspection_test
```

Example Response

Status code: 201

Success


```
{
  "total_count" : 9,
  "inspections" : [ {
    "id" : "e130d0d7-fd96-442e-a2de-c3b9cdac9f50",
    "scene" : null,
    "status" : "completed",
    "inspection_name" : "inspection_test",
    "inspection_scope" : "real_time",
    "instance_names" : "gauss-for-dbmind",
    "inspection_policy" : "single",
    "last_execute_time" : 1700724936556,
    "next_execute_time" : 1700623847872,
    "create_at" : 1700623847872
  } ]
}
```


Status Code

Status Code	Description
201	Normal
400	Abnormal

Error Code

For details, see **Error Codes**.

4.10.2 Creating an Inspection Task

Function

This API is used to create an inspection task. You can select inspection items and instance as needed.

URI

POST /gaussdb/v3/inspections

Request Parameters

Table 4-663 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.

Parameter	Mandatory	Type	Description
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Table 4-664 Request body parameters

Parameter	Mandatory	Type	Description
inspection_name	Yes	String	Inspection task name. The task name must start with a letter and consist of 4 to 64 characters. It can contain only letters, digits, hyphens (-), and underscores (_). Minimum length: 4 characters Maximum length: 64 characters
instance_ids	Yes	Array of strings	Instance IDs. Array length: 1 to 200 characters
inspection_scope	Yes	String	Inspection type. Enumerated values: ● **real_time** ● **daily** ● **weekly** ● **monthly**

Parameter	Mandatory	Type	Description
inspection_policy	Yes	String	Inspection policy. If the value is **single** (one-time inspection), the value of **inspection_period** can be **later** (executed later) or **soon** (executed immediately). If the value is **periodic** (periodic inspection), the value of **inspection_period** can be **fixed** (executed at a specified interval), **daily** (executed by day), **weekly** (executed by week), and **monthly** (executed by month). Enumerated values: • **single** • **periodic**
inspection_period	Yes	String	Inspection period. You need to set this parameter based on the inspection policy. Enumerated values: • **fixed** • **daily** • **weekly** • **monthly** • **later** • **soon**
inspection_time	No	String	Inspection time, for example, **2024-01-22T11:00:59+08:00**.
item_ids	Yes	Array of strings	Inspection item ID. Array length: 1 to 100 characters
task_start_time	No	String	Task start time, for example, **2024-01-22T11:00:59+08:00**.

Parameter	Mandatory	Type	Description
inspection_day	No	String	Inspection time. This parameter is specified when the inspection policy is periodic inspection. If the inspection is performed by day, the value is fixed to **0**. If the inspection is performed by week, the value ranges from **1** to **7**, indicating that the inspection is performed from Monday to Sunday. If the inspection is performed by month, the value ranges from **1** to **31**, indicating that the inspection is performed from the first day to the 31st day within one month.

Response Parameters

Status code: 201

Table 4-665 Response header parameter

Parameter	Type	Description
X-TRACE-ID	String	Request ID used for task tracing.

Status code: 400

Table 4-666 Response header parameter

Parameter	Type	Description
X-TRACE-ID	String	Request ID used for task tracing.

Table 4-667 Response body parameters

Parameter	Type	Description
errCode	String	Error code. Minimum length: 8 characters Maximum length: 36 characters

Parameter	Type	Description
externalMessage	String	Error message. Minimum length: 2 characters Maximum length: 512 characters

Example Request

Creating an inspection task

```
POST https://{Endpoint}/rds/gaussdb/v3/inspections

{
  "inspection_name" : "inspection_test",
  "instance_ids" : [ "619d3e78f61b4be68bc5aa0b59edcf7b", "219d3e78f61b4be68bc5aa0b59edcf7b" ],
  "inspection_scope" : "daily",
  "inspection_policy" : "periodic",
  "inspection_period" : "daily",
  "inspection_time" : "01:00:00",
  "item_ids" : [ "619d3e78f61b4be68bc5aa0b59edcf7b", "619d3e78f61b4be68bc5aa0b59edhf7b" ]
}
```

Example Response

Status code: 201

Success

```
success
```

Status Code

Status Code	Description
201	Normal
400	Abnormal

Error Code

For details, see **Error Codes**.

4.10.3 Deleting Inspection Tasks in Batches

Function

This API is used to delete inspection tasks in batches.

URI

POST /gaussdb/v3/inspections/batch-delete

Request Parameters

Table 4-668 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Table 4-669 Request body parameter

Parameter	Mandatory	Type	Description
inspection_ids	Yes	Array of strings	IDs of the selected inspection tasks to be deleted.

Response Parameters

Status code: 201

Table 4-670 Response header parameter

Parameter	Type	Description
X-TRACE-ID	String	Request ID used for task tracing.

Table 4-671 Response body parameters

Parameter	Type	Description
results	Array of **Table 4-672** objects	Deletion result.

Table 4-672 DeleteInspectionResult

Parameter	Type	Description
inspction_id	String	Inspection task ID.

Parameter	Type	Description
result	String	Deletion result. Enumerated values: ● **success** ● **failure**

Status code: 400

Table 4-673 Response header parameter

Parameter	Type	Description
X-TRACE-ID	String	Request ID used for task tracing.

Table 4-674 Response body parameters

Parameter	Type	Description
errCode	String	Error code. Minimum length: 8 characters Maximum length: 36 characters
externalMessage	String	Error message. Minimum length: 2 characters Maximum length: 512 characters

Example Request

Deleting inspection tasks in batches.

```
POST https://{Endpoint}/rds/gaussdb/v3/inspections/batch-delete

{
  "inspection_ids" : [ "6b13d4ba7268438dbb4fe6869ab5fcf0", "d101c49ab9f64b0ca93e289d9b401d7f" ]
}
```

Example Response

Status code: 201

Success

```
{
  "results" : [ {
    "inspction_id" : "d101c49ab9f64b0ca93e289d9b401d7f",
    "result" : "success"
  }, {
    "inspction_id" : "405e791367f3419fa262e75ccb202e5c",
    "result" : "failure"
```

```
 }]
}
```

Status Code

Status Code	Description
201	Normal
400	Abnormal

Error Code

For details, see **Error Codes**.

4.10.4 Executing an Inspection Task

Function

This API is used to execute an inspection task.

URI

POST /gaussdb/v3/inspections/{inspection_id}/start

Table 4-675 URI parameter

Parameter	Mandatory	Type	Description
inspection_id	Yes	String	Inspection task ID. Minimum length: 1 character Maximum length: 64 characters

Request Parameters

Table 4-676 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Response Parameters

Status code: 201

Table 4-677 Response header parameter

Parameter	Type	Description
X-TRACE-ID	String	Request ID used for task tracing.

Status code: 400

Table 4-678 Response header parameter

Parameter	Type	Description
X-TRACE-ID	String	Request ID used for task tracing.

Table 4-679 Response body parameters

Parameter	Type	Description
errCode	String	Error code. Minimum length: 8 characters Maximum length: 36 characters
externalMessage	String	Error message. Minimum length: 2 characters Maximum length: 512 characters

Example Request

Executing an inspection task again

```
POST https://{Endpoint}/rds/gaussdb/v3/inspections/{inspection_id}/start

https://{Endpoint}/rds/gaussdb/v3/inspections/b5df9c14-cacc-4489-aedd-629d7b20c616/start
```

Example Response

Status code: 201

Success

```
success
```

Status Code

Status Code	Description
201	Normal
400	Abnormal

Error Code

For details, see **Error Codes**.

4.10.5 Querying Details of an Inspection Task

Function

This API is used to query the details of an inspection task based on the inspection task ID.

URI

GET /gaussdb/v3/inspections/{inspection_id}

Table 4-680 URI parameter

Parameter	Mandatory	Type	Description
inspection_id	Yes	String	Inspection task ID. Minimum length: 1 character Maximum length: 64 characters

Request Parameters

Table 4-681 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Response Parameters

Status code: 201

Table 4-682 Response header parameter

Parameter	Type	Description
X-TRACE-ID	String	Request ID used for task tracing.

Table 4-683 Response body parameters

Parameter	Type	Description
inspection_name	String	Inspection task name. Minimum length: 1 character Maximum length: 64 characters
inspection_id	String	Inspection ID.
instance_ids	Array of strings	Instance IDs. Array length: 1 to 200 characters
inspection_scope	String	Inspection type. Enumerated values: ● **real_time** ● **daily** ● **weekly** ● **monthly**
inspection_policy	String	Inspection policy. If the value is **single** (one-time inspection), the value of **inspection_period** can be **later** (executed later) or **soon** (executed immediately). If the value is **periodic** (periodic inspection), the value of **inspection_period** can be **fixed** (executed at a specified interval), **daily** (executed by day), **weekly** (executed by week), and **monthly** (executed by month). Enumerated values: ● **single** ● **periodic**

Parameter	Type	Description
inspection_period	String	Inspection period. You need to set this parameter based on the inspection policy. Enumerated values: • **fixed** • **daily** • **weekly** • **monthly** • **later** • **soon**
inspection_time	String	Inspection time.
item_ids	Array of strings	Inspection item ID. Array length: 1 to 100 characters
task_start_time	String	Task start time.
inspection_day	String	Inspection time, by day.

Status code: 400

Table 4-684 Response header parameter

Parameter	Type	Description
X-TRACE-ID	String	Request ID used for task tracing.

Table 4-685 Response body parameters

Parameter	Type	Description
errCode	String	Error code. Minimum length: 8 characters Maximum length: 36 characters
externalMessage	String	Error message. Minimum length: 2 characters Maximum length: 512 characters

Example Request

Querying details of an inspection task

```
GET https://{Endpoint}/rds/gaussdb/v3/inspections/{inspection_id}
```

Example Response

Status code: 201

Success


```
{
  "inspection_id" : "60505768-89b0-4435-b170-0285d213cdb4",
  "inspection_name" : "test8",
  "instance_ids" : null,
  "inspection_scope" : "real_time",
  "inspection_policy" : "single",
  "inspection_period" : "soon",
  "inspection_day" : null,
  "inspection_time" : "2024-01-11T15:50:12+08:00",
  "item_ids" : [ "ae348ee4-5139-4777-8808-d68791549de3", "ad1b4bf5-7be8-4708-a10e-bee6a89f35fe",
"c4473978-73b0-43b8-b233-7aceca98507b", "12526548-a4ea-478b-b1bd-1dc2ff83cb5a", "b3dab237-
c5e4-40da-8174-71182898a359", "1613f70e-27e3-426c-96f6-4834c883ab97", "5ba48f64-1792-4c77-802e-
c8fdd6aa38fa" ],
  "task_start_time" : "2024-01-11T15:50:12+08:00"
}
```


Status Code

Status Code	Description
201	Normal
400	Abnormal

Error Code

For details, see **Error Codes**.

4.10.6 Modifying an Inspection Task

Function

This API is used to modify an inspection task.

URI

PUT /gaussdb/v3/inspections/{inspection_id}

Table 4-686 URI parameter

Parameter	Mandatory	Type	Description
inspection_id	Yes	String	Inspection task ID. Minimum length: 1 character Maximum length: 64 characters

Request Parameters

Table 4-687 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Table 4-688 Request body parameters

Parameter	Mandatory	Type	Description
instance_ids	Yes	Array of strings	Name. Array length: 1 to 200 characters
inspection_scope	Yes	String	Inspection type. Enumerated values: ● **real_time** ● **daily** ● **weekly** ● **monthly**

Parameter	Mandatory	Type	Description
inspection_policy	Yes	String	Inspection policy. ● If the value is **single** (one-time inspection), the value of **inspection_period** can be **later** (executed later) or **soon** (executed immediately). ● If the value is **periodic** (periodic inspection), the value of **inspection_period** can be **fixed** (executed at a specified interval), **daily** (executed by day), **weekly** (executed by week), and **monthly** (executed by month). Enumerated values: ● **single** ● **periodic**
inspection_period	Yes	String	Inspection period. You need to set this parameter based on the inspection policy. Enumerated values: ● **fixed** ● **daily** ● **weekly** ● **monthly** ● **later** ● **soon**
inspection_time	No	String	Inspection time, for example, **2024-01-22T11:00:59+08:00**.
item_ids	Yes	Array of strings	Inspection item ID. Array length: 1 to 100 characters
task_start_time	No	String	Task start time, for example, **2024-01-22T11:00:59+08:00**.

Response Parameters

Status code: 201

Table 4-689 Response header parameter

Parameter	Type	Description
X-TRACE-ID	String	Request ID used for task tracing.

Status code: 400

Table 4-690 Response header parameter

Parameter	Type	Description
X-TRACE-ID	String	Request ID used for task tracing.

Table 4-691 Response body parameters

Parameter	Type	Description
errCode	String	Error code. Minimum length: 8 characters Maximum length: 36 characters
externalMessage	String	Error message. Minimum length: 2 characters Maximum length: 512 characters

Example Request

Modifying an inspection task

```
PUT https://{Endpoint}/rds/gaussdb/v3/inspections/{inspection_id}

{
  "inspection_name" : "test_3",
  "instance_ids" : [ "9a60f8d8fb7942df89aa0f1d92e03a6ein14" ],
  "inspection_scope" : "real_time",
  "inspection_policy" : "single",
  "item_ids" : [ "ae348ee4-5139-4777-8808-d68791549de3", "ad1b4bf5-7be8-4708-a10e-bee6a89f35fe",
"c4473978-73b0-43b8-b233-7aceca98507b", "12526548-a4ea-478b-b1bd-1dc2ff83cb5a", "b3dab237-
c5e4-40da-8174-71182898a359", "1613f70e-27e3-426c-96f6-4834c883ab97" ],
  "inspection_period" : "soon",
  "task_start_time" : "2024-01-04T19:53:33+08:00",
  "inspection_time" : "2024-01-04T19:53:33+08:00"
}
```

Example Response

Status code: 201

Success

```
success
```

Status Code

Status Code	Description
201	Normal
400	Abnormal

Error Code

For details, see **Error Codes**.

4.10.7 Querying Inspection Result Statistics

Function

This API is used to query inspection result statistics.

URI

GET /gaussdb/v3/inspections/records/{record_id}/statistic

Table 4-692 URI parameter

Parameter	Mandatory	Type	Description
record_id	Yes	String	Inspection record ID.

Request Parameters

Table 4-693 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Response Parameters

Status code: 200

Table 4-694 Response header parameter

Parameter	Type	Description
X-TRACE-ID	String	Request ID used for task tracing.

Table 4-695 Response body parameters

Parameter	Type	Description
global_statistics	**Table 4-696** object	Status statistics.
category_statistics	Array of **Table 4-697** objects	Category statistics. Array length: 0 to 200 characters

Table 4-696 global_statistics

Parameter	Type	Description
fail	Integer	Failed inspection items. Minimum value: **0** Maximum value: **200**
abnormal	Integer	Abnormal inspection items. Minimum value: **0** Maximum value: **200**
normal	Integer	Normal inspection items. Minimum value: **0** Maximum value: **200**
total	String	Total number of inspection items.

Table 4-697 InspectionsStatisticCategory

Parameter	Type	Description
fail	Integer	Failed inspection items. Minimum value: **0** Maximum value: **200**
abnormal	Integer	Abnormal inspection items. Minimum value: **0** Maximum value: **200**

Parameter	Type	Description
normal	Integer	Normal inspection items. Minimum value: **0** Maximum value: **200**
name	String	Inspection category. Minimum length: 0 character Maximum length: 200 characters
total	Integer	Total number of inspection items.

Status code: 400

Table 4-698 Response header parameter

Parameter	Type	Description
X-TRACE-ID	String	Request ID used for task tracing.

Table 4-699 Response body parameters

Parameter	Type	Description
errCode	String	Error code. Minimum length: 8 characters Maximum length: 36 characters
externalMessage	String	Error message. Minimum length: 2 characters Maximum length: 512 characters

Example Request

```
GET https://{Endpoint}/rds/gaussdb/v3/inspections/records/{record_id}/statistic
```

Example Response

Status code: 200

Success

```
{
  "global_statistics" : {
    "total" : 11,
    "fail" : 11,
    "abnormal" : 0,
    "normal" : 0
  },
```



```
"category_statistics" : [ {
  "total" : 1,
  "fail" : 1,
  "abnormal" : 0,
  "normal" : 0,
  "name": "database resources"
 }, {
  "total" : 4,
  "fail" : 4,
  "abnormal" : 0,
  "normal" : 0,
  "name": " database performance"
 }, {
  "total" : 1,
  "fail" : 1,
  "abnormal" : 0,
  "normal" : 0,
  "name": "instance status"
 }, {
  "total" : 5,
  "fail" : 5,
  "abnormal" : 0,
  "normal" : 0,
  "name": "system resource"
 } ]
}
```

Status Code

Status Code	Description
200	Normal
400	Abnormal

Error Code

For details, see **Error Codes**.

4.10.8 Querying Execution Details of Inspection Items

Function

This API is used to query execution details of inspection items in batches.

URI

GET /gaussdb/v3/inspections/records/{record_id}

Table 4-700 URI parameter

Parameter	Mandatory	Type	Description
record_id	Yes	String	Inspection record ID.

Request Parameters

Table 4-701 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Response Parameters

Status code: 200

Table 4-702 Response header parameter

Parameter	Type	Description
X-TRACE-ID	String	Request ID used for task tracing.

Table 4-703 Response body parameters

Parameter	Type	Description
total_count	Integer	Total number of inspection items. Minimum value: **0** Maximum value: **2000**
instance_id	String	Instance ID.
start_time	String	Start timestamp.
end_time	String	End timestamp.
inspection_results	Array of **Table 4-704** objects	Inspection item results Array length: 0 to 200 characters

Table 4-704 inspection_results

Parameter	Type	Description
name	String	Inspection item name. Minimum length: 0 character Maximum length: 200 characters
description	String	Inspection item description. Minimum length: 0 character Maximum length: 1024 characters
inspection_category	String	Inspection category. Minimum length: 0 character Maximum length: 200 characters
status	String	Inspection status. Minimum length: 0 character Maximum length: 64 characters
inspection_result	String	Inspection result. Minimum length: 0 character Maximum length: 1024 characters
possible_causes	String	Possible cause. Minimum length: 0 character Maximum length: 1024 characters
suggestions	String	Optimization suggestion. Minimum length: 0 character Maximum length: 1024 characters
item	String	Inspection item.
indicators	Array of **Table 4-705** objects	Key metrics. Array length: 0 to 200 characters

Table 4-705 indicators

Parameter	Type	Description
name	String	Metric name. Minimum length: 0 character Maximum length: 200 characters
unit	String	Unit. Minimum length: 0 character Maximum length: 64 characters

Parameter	Type	Description
metrics	Array of **Table 4-706** objects	Node metric data. Array length: 0 to 200 characters

Table 4-706 metrics

Parameter	Type	Description
ip	String	Node IP address. Minimum length: 0 character Maximum length: 64 characters
name	String	Node name. Minimum length: 0 character Maximum length: 200 characters
max	Number	Maximum value. Minimum value: **0** Maximum value: **10000**
min	Number	Minimum value. Minimum value: **0** Maximum value: **10000**
average	Number	Average value. Minimum value: **0** Maximum value: **10000**
p95	String	95th percentile.
data	**Table 4-707** object	Metric data.

Table 4-707 data

Parameter	Type	Description
time	Array of strings	Time data. Minimum length: 0 character Maximum length: 200 characters Array length: 0 to 200 characters

Parameter	Type	Description
metrics	Array of numbers	Time data. Minimum value: **0** Maximum value: **10000** Array length: 0 to 200 characters

Status code: 400

Table 4-708 Response header parameter

Parameter	Type	Description
X-TRACE-ID	String	Request ID used for task tracing.

Table 4-709 Response body parameters

Parameter	Type	Description
errCode	String	Error code. Minimum length: 8 characters Maximum length: 36 characters
externalMessage	String	Error message. Minimum length: 2 characters Maximum length: 512 characters

Example Request

```
GET https://{Endpoint}/rds/gaussdb/v3/inspections/records/{record_id}
```

Example Response

Status code: 200

Success


```
{
  "total_count" : 2,
  "instance_id" : "b7dbb6b9987448d7a0498860f170842fin14",
  "start_time" : 1704368860019,
  "end_time" : 1704369907550,
  "inspection_results" : [ {
    "item" : "db_size",
    "name": " database size",
    "description": " check and record the size of each database.",
    "status" : "fail",
    "suggestions": " check whether the content and format of the inspection data returned by DBMind are correct in the database.",
    "indicators" : null,
```

```
    "inspection_category": " database resources",
    "inspection_result": " inspection failed",
    "possible_causes": " inspection data format is incorrect."
  }, {
    "item" : "long_transaction",
    "name": " long transaction",
    "description": " check whether there are active transactions that have not ended for more than 12 hours.
If yes, the inspection fails.",
    "status" : "fail",
    "suggestions": " check whether the content and format of the inspection data returned by DBMind are
correct in the database.",
    "indicators" : null,
    "inspection_category": " database performance",
    "inspection_result": " inspection failed",
    "possible_causes": " inspection data format is incorrect."
  } ]
}
```

Status Code

Status Code	Description
200	Normal
400	Abnormal

Error Code

For details, see **Error Codes**.

4.10.9 Querying Inspection Items

Function

This API is used to query inspection items. The inspection category and inspection items are returned.

URI

GET /gaussdb/v3/inspections/items

Request Parameters

Table 4-710 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.

Parameter	Mandatory	Type	Description
X-Language	Yes	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Response Parameters

Status code: 200

Table 4-711 Response header parameter

Parameter	Type	Description
X-TRACE-ID	String	Request ID used for task tracing.

Table 4-712 Response body parameters

Parameter	Type	Description
total_count	String	Total number of inspection items.
inspections	Array of **Table 4-713** objects	Inspection items.

Table 4-713 InspectionCategory

Parameter	Type	Description
name	String	Inspection type name. Minimum length: 1 character Maximum length: 32 characters The inspection categories include diagnosis and optimization, instance status, database resources, database performance, and system resources.
items	Array of **Table 4-714** objects	Inspection subitems.

Table 4-714 InspectionItems

Parameter	Type	Description
id	String	Inspection item ID.
name	String	Inspection item name.
description	String	Inspection item description.

Status code: 400

Table 4-715 Response header parameter

Parameter	Type	Description
X-TRACE-ID	String	Request ID used for task tracing.

Table 4-716 Response body parameters

Parameter	Type	Description
errCode	String	Error code. Minimum length: 8 characters Maximum length: 36 characters
externalMessage	String	Error message. Minimum length: 2 characters Maximum length: 512 characters

Example Request

Querying inspection categories and inspection items

```
GET https://{Endpoint}/rds/gaussdb/v3/inspections/items
```

Example Response

Status code: 200

Success

```
{
  "total_count" : 1,
  "inspections" : [ {
    "name": "database resources"
    "items" : [ {
      "id" : "405e791367f3419fa262e75ccb202e5c",
      "name" : "CPU",
      "description": " check the CPU usage of each node."
    }, {
      "id" : "405e791367f3419fa262e75ccb202e5c",
```

```
      "name": " memory",
      "description": " check the memory usage of each node."
    } ]
  } ]
}
```


Status Code

Status Code	Description
200	Normal
400	Abnormal

Error Code

For details, see **Error Codes**.

4.10.10 Querying Inspection Records

Function

This API is used to query inspection records in the inspection results. The backend supports pagination and sorting.

URI

POST /gaussdb/v3/inspections/records

Request Parameters

Table 4-717 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Table 4-718 Request body parameters

Parameter	Mandatory	Type	Description
instance_name	No	Array of strings	Instance name.
status	No	String	Inspection execution status. Enumerated values: **running**, **completed**, and **failed**
task_name	No	String	Inspection task name.
inspection_start_time_range	No	Array of Table 4-719 objects	Execution time range.
limit	Yes	Integer	The value cannot be a negative number. The minimum value is **1** and the maximum value is **100**.
offset	Yes	Integer	Index offset. The query starts from the first piece of data. The value must be a number and cannot be a negative number.
sort	No	String	Sorting key. Default value: **start_time** Enumerated values: ● **instance_name** ● **status** ● **task_name** ● **start_time** ● **end_time**
order	No	String	Sorting order. Default value: **desc** Enumerated values: ● **desc** ● **asc**

Table 4-719 TimeRange

Parameter	Mandatory	Type	Description
start_time	No	String	Execution start time, for example, **1705892651000**

Parameter	Mandatory	Type	Description
end_time	No	String	Execution end time, for example, **1705892651000**

Response Parameters

Status code: 200

Table 4-720 Response header parameter

Parameter	Type	Description
X-TRACE-ID	String	Request ID used for task tracing.

Table 4-721 Response body parameters

Parameter	Type	Description
records	Array of **Table 4-722** objects	Inspection records.
total_count	Integer	Total number of inspection records.

Table 4-722 InspectionsRecords

Parameter	Type	Description
record_id	String	Inspection record ID.
instance_name	String	Instance name.
inspection_policy	String	Inspection policy.
inspection_scope	String	Inspection period.
task_name	String	Inspection task name.
start_time	Integer	Inspection start time.
end_time	Integer	Inspection end time.
progress	String	Inspection progress.
instance_id	String	Instance ID.

Parameter	Type	Description
status	String	Inspection result status. Enumerated values: executing and completed
total_count	Integer	Total number of inspection items.
failed_count	Integer	Number of failed inspection items.
abnormal_count	Integer	Number of abnormal inspection items.
normal_count	Integer	Number of normal inspection items.

Status code: 400

Table 4-723 Response header parameter

Parameter	Type	Description
X-TRACE-ID	String	Request ID used for task tracing.

Table 4-724 Response body parameters

Parameter	Type	Description
errCode	String	Error code. Minimum length: 8 characters Maximum length: 36 characters
externalMessage	String	Error message. Minimum length: 2 characters Maximum length: 512 characters

Example Request

```
POST https://{Endpoint}/rds/gaussdb/v3/inspections/records

{
  "limit" : 10,
  "offset" : 0
}
```

Example Response

Status code: 200

Success

```
{
  "records" : [ {
```

```
    "status" : "completed",
    "progress" : "100",
    "record_id" : "835059db-4b9d-4f6a-8ffb-00f78e5f4c03",
    "instance_id" : "26.84.49.99",
    "instance_name" : null,
    "inspection_policy" : "single",
    "inspection_scope" : "real_time",
    "task_name" : "jWelpAFJCdQqQgGOjtBtXgWh",
    "start_time" : 1705043968671,
    "end_time" : 1705043969141,
    "total_count" : 6,
    "failed_count" : 6,
    "abnormal_count" : 0,
    "normal_count" : 0
  } ],
  "total_count" : 42
}
```


Status Code

Status Code	Description
200	Normal
400	Abnormal

Error Code

For details, see **Error Codes**.

4.10.11 Deleting Inspection Records in Batches

Function

This API is used to delete inspection records.

URI

POST /gaussdb/v3/inspections/records/batch-delete

Request Parameters

Table 4-725 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.

Parameter	Mandatory	Type	Description
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Table 4-726 Request body parameter

Parameter	Mandatory	Type	Description
[record_id]	No	Array of strings	IDs of inspection records to be deleted.

Response Parameters

Status code: 200

Table 4-727 Response header parameter

Parameter	Type	Description
X-TRACE-ID	String	Request ID used for task tracing.

Table 4-728 Response body parameters

Parameter	Type	Description
records	**Table 4-729** object	Execution records to be deleted.
total_count	Integer	Number of records to be deleted.

Table 4-729 DeleteRecordsResult

Parameter	Type	Description
record_id	String	ID of the record to be deleted.
result	String	Deletion result. Enumerated values: **success** or **failure**

Status code: 400

Table 4-730 Response header parameter

Parameter	Type	Description
X-TRACE-ID	String	Request ID used for task tracing.

Table 4-731 Response body parameters

Parameter	Type	Description
errCode	String	Error code. Minimum length: 8 characters Maximum length: 36 characters
externalMessage	String	Error message. Minimum length: 2 characters Maximum length: 512 characters

Example Request

Deleting inspection records

```
POST https://{Endpoint}/rds/gaussdb/v3/inspections/records/batch-delete

[
"09b3eb561c5a491f931d6de5e4fe8ffcin14", "09b3eb561c5a491f931d6de5e4adf23413"
]
```

Example Response

Status code: 200

Success

```
{
  "records" : [ {
    "result" : "success",
    "record_id" : "27e4a583-8738-42f7-8114-9e2869ac5393"
  } ],
  "total_count" : 1
}
```

Status Code

Status Code	Description
200	Normal
400	Abnormal

Error Code

For details, see **Error Codes**.

4.10.12 Querying Inspection Scenarios

Function

This API is used to query inspection scenarios. Inspection items in the scenarios are automatically selected during front-end interaction.

URI

GET /gaussdb/v3/inspections/scenes

Request Parameters

Table 4-732 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	Yes	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Response Parameters

Status code: 200

Table 4-733 Response header parameter

Parameter	Type	Description
X-TRACE-ID	String	Request ID used for task tracing.

Table 4-734 Response body parameters

Parameter	Type	Description
total_count	Integer	Total number of inspection scenarios.
scenes	Array of **Table 4-735** objects	Inspection scenarios.

Table 4-735 QueryInspectionSceneResponseBody

Parameter	Type	Description
scene_id	String	Scenario ID. Minimum length: 1 character Maximum length: 64 characters
name	String	Scenario name.
is_default	Boolean	Whether the template is the default one.
item_ids	Array of integers	Inspection item IDs.

Status code: 400

Table 4-736 Response header parameter

Parameter	Type	Description
X-TRACE-ID	String	Request ID used for task tracing.

Table 4-737 Response body parameters

Parameter	Type	Description
errCode	String	Error code. Minimum length: 8 characters Maximum length: 36 characters
externalMessage	String	Error message. Minimum length: 2 characters Maximum length: 512 characters

Example Request

Querying inspection scenarios

```
GET https://{Endpoint}/rds/gaussdb/v3/inspections/scenes
```

Example Response

Status code: 200

Success

```
{
  "total_count" : 1,
  "scenes" : [ {
```

```
    "name" : "Health Check",
    "scene_id" : null,
    "is_default" : null,
    "item_ids" : [ "c4473978-73b0-43b8-b233-7aceca98507b", "12526548-a4ea-478b-b1bd-1dc2ff83cb5a",
"b3dab237-c5e4-40da-8174-71182898a359", "1613f70e-27e3-426c-96f6-4834c883ab97",
"ae348ee4-5139-4777-8808-d68791549de3", "ad1b4bf5-7be8-4708-a10e-bee6a89f35fe",
"5ba48f64-1792-4c77-802e-c8fdd6aa38fa", "1778ce4e-82af-426f-af90-f55ae062387e",
"37b40377-47e6-4392-9748-a567c5d068aa", "b2b89b48-bbd1-46e8-ad7d-012247a23d2f",
"bb09847e-905d-4d65-8d81-4a19458dea1e" ]
  } ]
}
```

Status Code

Status Code	Description
200	Normal
400	Abnormal

Error Code

For details, see **Error Codes**.

4.10.13 Adding an Inspection Scenario

Function

This API is used to add an inspection scenario. The inspection scenario name and associated inspection items are returned.

URI

POST /gaussdb/v3/inspections/scenes

Request Parameters

Table 4-738 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Table 4-739 Request body parameters

Parameter	Mandatory	Type	Description
name	Yes	String	Scenario name. The name must start with a letter and consist of 4 to 64 characters. It can contain only letters, digits, hyphens (-), and underscores (_). Minimum length: 4 characters Maximum length: 64 characters
item_ids	Yes	Array of strings	Inspection item IDs. Array length: 1 to 100 characters

Response Parameters

Status code: 200

Table 4-740 Response header parameter

Parameter	Type	Description
X-TRACE-ID	String	Request ID used for task tracing.

Status code: 400

Table 4-741 Response header parameter

Parameter	Type	Description
X-TRACE-ID	String	Request ID used for task tracing.

Table 4-742 Response body parameters

Parameter	Type	Description
errCode	String	Error code. Minimum length: 8 characters Maximum length: 36 characters

Parameter	Type	Description
externalMessage	String	Error message. Minimum length: 2 characters Maximum length: 512 characters

Example Request

Creating an inspection scenario

```
POST https://{Endpoint}/rds/gaussdb/v3/inspections/scenes

{
  "name" : "test_scenes",
  "item_ids" : [ "405e791367f3419fa262e75ccb202e5c", "405e791367f3419fa262e75ccb202e5c" ]
}
```

Example Response

Status code: 200

Success

```
success
```

Status Code

Status Code	Description
200	Normal
400	Abnormal

Error Code

For details, see **Error Codes**.

4.10.14 Deleting an Inspection Scenario

Function

This API is used to delete an inspection scenario.

URI

DELETE /gaussdb/v3/inspections/scenes

Request Parameters

Table 4-743 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	Yes	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Table 4-744 Request body parameter

Parameter	Mandatory	Type	Description
names	Yes	Array of strings	Scenario name array.

Response Parameters

Status code: 200

Table 4-745 Response header parameter

Parameter	Type	Description
X-TRACE-ID	String	Request ID used for task tracing.

Table 4-746 Response body parameters

Parameter	Type	Description
results	Array of **Table 4-747** objects	Inspection items. Array length: 0 to 2000 characters

Table 4-747 DeleteInspectionInfo

Parameter	Type	Description
inspection_id	String	Inspection task name.

Parameter	Type	Description
result	String	Number of results.

Status code: 400

Table 4-748 Response header parameter

Parameter	Type	Description
X-TRACE-ID	String	Request ID used for task tracing.

Table 4-749 Response body parameters

Parameter	Type	Description
errCode	String	Error code. Minimum length: 8 characters Maximum length: 36 characters
externalMessage	String	Error message. Minimum length: 2 characters Maximum length: 512 characters

Example Request

Deleting an inspection scenario

```
DELETE https://{Endpoint}/rds/gaussdb/v3/inspections/scenes

{
  "names" : [ "test" ]
}
```

Example Response

Status code: 200

Success

```
{
  "results" : [ {
    "result" : "6",
    "inspection_id" : "c4473978-73b0-43b8-b233-7aceca98507b"
  } ]
}
```

Status Code

Status Code	Description
200	Normal
400	Abnormal

Error Code

For details, see **Error Codes**.

4.10.15 Modifying an Inspection Scenario

Function

This API is used to modify an inspection scenario. The inspection items in the inspection scenario will be modified.

URI

PUT /gaussdb/v3/inspections/scenes

Request Parameters

Table 4-750 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Table 4-751 Request body parameters

Parameter	Mandatory	Type	Description
name	Yes	String	Name of the original scenario. The name must start with a letter and consist of 4 to 64 characters. It can contain only letters, digits, hyphens (-), and underscores (_). Minimum length: 4 characters Maximum length: 64 characters
item_ids	Yes	Array of strings	Inspection item IDs.
modify_name	Yes	String	Name of the new scenario. The name must start with a letter and consist of 4 to 64 characters. It can contain only letters, digits, hyphens (-), and underscores (_). Minimum length: 4 characters Maximum length: 64 characters

Response Parameters

Status code: 200

Table 4-752 Response header parameter

Parameter	Type	Description
X-TRACE-ID	String	Request ID used for task tracing.

Status code: 400

Table 4-753 Response header parameter

Parameter	Type	Description
X-TRACE-ID	String	Request ID used for task tracing.

Table 4-754 Response body parameters

Parameter	Type	Description
errCode	String	Error code. Minimum length: 8 characters Maximum length: 36 characters
externalMessage	String	Error message. Minimum length: 2 characters Maximum length: 512 characters

Example Request

Modifying an inspection scenario

```
PUT https://{Endpoint}/rds/gaussdb/v3/inspections/scenes

{
  "name" : "test",
  "item_ids" : [ "ae348ee4-5139-4777-8808-d68791549de3", "ad1b4bf5-7be8-4708-a10e-bee6a89f35fe" ],
  "modify_name" : "test"
}
```

Example Response

Status code: 200

Modification results for inspection scenarios.

```
success
```

Status Code

Status Code	Description
200	Normal
400	Abnormal

Error Code

For details, see **Error Codes**.

4.11 Instance Management

4.11.1 Configuring Extended Information for an Instance

Function

This API is used to configure the extended information for an instance.

URI

PUT /gaussdb/v3/instances/{instance_id}/extend-info

Table 4-755 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-756 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token. Minimum length: 0 character Maximum length: 32768 characters
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Table 4-757 Request body parameters

Parameter	Mandatory	Type	Description
key	Yes	String	Key of the extended information. Value: • BackupSet • XbsaRoute
value	Yes	String	Value of extended Information.

Response Parameters

None

Example Request

Configuring the **BackupSet** extended information for an instance

```
PUT https://{Endpoint}/rds/gaussdb/v3/instances/dsfae23fsfdsae3435in14/extend-info

{
  "key" : "BackupSet",
  "value" : "/etc/xbsa.conf"
}
```

Configuring the **XbsaRoute** extended information for an instance

```
PUT https://{Endpoint}/rds/gaussdb/v3/instances/dsfae23fsfdsae3435in14/extend-info

{
    "key": "XbsaRoute",
    "value": "{\"ip\":\"***.***.***.***\",\"port\":\"***\"}"
}
```

Example Response

None

Status Code

Status Code	Description
200	Normal

Error Code

For details, see **Error Codes**.

4.11.2 Querying Extended Information of an Instance

Function

This API is used to query the extended information of an instance.

URI

GET /gaussdb/v3/instances/{instance_id}/extend-info

Table 4-758 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-759 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token. Minimum length: 0 character Maximum length: 32768 characters
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Table 4-760 Query parameters

Parameter	Mandatory	Type	Description
key	Yes	String	Key of the extended information.

Response Parameters

Status code: 200

Table 4-761 Response body parameters

Parameter	Type	Description
key	String	Key of the extended information.
value	String	Value of extended Information.

Example Request

Querying the extended information of an instance

```
GET https://{Endpoint}/rds/gaussdb/v3/instances/dsfae23fsfdsae3435in14/extend-info?key=BackupSet
```

Example Response

Status code: 200

Success.

```
{
  "key" : "BackupSet",
  "value" : "/etc/xbsa.conf"
}
```


Status Code

Status Code	Description
200	Normal

Error Code

For details, see **Error Codes**.

4.11.3 Querying Components of an Instance

Function

This API is used to query components of an instance.

URI

GET /gaussdb/v3/instances/{instance_id}/components

Table 4-762 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Table 4-763 Query parameters

Parameter	Mandatory	Type	Description
limit	No	Integer	Number of records to be queried. The default value is **100**. The value cannot be a negative number. The minimum value is **1** and the maximum value is **100**. Minimum value: **1** Maximum value: **100** Default value: **100**

Parameter	Mandatory	Type	Description
offset	No	Integer	Index offset. If **offset** is set to *N*, the resource query starts from the *N*+1 piece of data. The value is **0** by default, indicating that the query starts from the first piece of data. The value must be a number but cannot be a negative number. Minimum value: **0** Default value: **0**
component_type	No	String	Component type. Default value: **ALL** Enumerated values: • **ALL** • **CN** • **DN** • **CM** • **GTM** • **ETCD**
availability_zone_id	No	String	ID of the AZ where the primary component is located. The default value is **ALL**, indicating that component information of nodes in all AZs of the instance is queried. When you query the AZ where a primary DN is located, the information of all DNs in the same shard as the primary DN is displayed. When you query the AZ where a CN is located, only the CN information in the AZ is displayed. When you query the AZ where a component (except CNs or DNs) is located, information about all components of the same type is returned. If there is no such a component, no information is returned. Default value: **ALL**
node_name	No	String	Node name.
node_id	No	String	Node ID.

Parameter	Mandatory	Type	Description
hostname	No	String	Host name.

Request Parameters

Table 4-764 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Response Parameters

Status code: 200

Table 4-765 Response body parameters

Parameter	Type	Description
[*Array elements*]	Array of **Table 4-766** objects	Successful response.

Table 4-766 GetComponentsResponse

Parameter	Type	Description
nodes	Array of **Table 4-767** objects	Node information.
total_count	Integer	Number of components.

Table 4-767 nodes

Parameter	Type	Description
components	Array of **Table 4-768** objects	Component information.

Parameter	Type	Description
id	String	Node ID.
name	String	Node name.
hostname	String	Host name.
availability_zone_id	String	AZ ID.
description	String	Node description.
status	String	Node status.

Table 4-768 components

Parameter	Type	Description
id	String	Component ID, which can be: ● **cm_***: database management module, for example, **cm_3**. ● **gtm_***: global transaction manager, for example, **gtm_1003**. ● **cn_***: coordinator node, for example, **cn_6001**. ● **dn_***: data node, for example, **dn_6002**. ● **etcd_***: editable text configuration daemon, for example, **etcd_7003**.
role	String	Role, which can be: ● **master**: primary node ● **slave**: standby node ● **readreplica**: read replica
status	String	Node status, which can be: ● **Primary**: primary CM, DN, CN, or GTM ● **Standby**: standby CM, DN, CN, or GTM ● **StateLeader**: primary etcd ● **StateFollower**: standby etcd
distributed_id	String	Distribution ID (for example, **60011**).

Parameter	Type	Description
type	String	Component type, which can be: ● **CM**: database management mode. It provides the ability to manage and monitor the status of each function unit and physical resources in a system. ● **GTM**: global transaction manager. It generates and maintains the global transaction IDs, transaction snapshots, timestamps, and sequences that must be unique globally. ● **CN**: coordinator node, which schedules tasks to DNs. ● **DN**: data node, which stores data, executes data query tasks, and returns execution results to CNs. ● **ETCD**: editable text configuration daemon. It is a storage service in key-value format in a distributed environment, which is used for configuration sharing and service discovery (registration and search).
detail	String	Status details, which can be: ● **Normal**: The component is normal. ● **Abnormal**: The component is abnormal.

Example Request

Querying components

```
GET https://{Endpoint}/rds/gaussdb/v3/instances/3d39c18788b54a919bab633874c159dfin14/components?
offset=0&limit=30&component_type=DN
```

Example Response

Status code: 200

Success.

```
{
  "nodes": [
    {
      "components": [
        {
          "id": "cm_3",
          "role": "master",
          "status": "Standby",
          "distributed_id": null,
          "type": "CM",
          "detail": null
        },
        {
          "id": "gtm_1003",
```

```
        "role": "master",
        "status": "Standby",
        "distributed_id": null,
        "type": "GTM",
        "detail": "Connection ok"
      },
      {
        "id": "cn_5003",
        "role": "master",
        "status": "Normal",
        "distributed_id": null,
        "type": "CN",
        "detail": null
      },
      {
        "id": "etcd_7003",
        "role": "master",
        "status": "StateLeader",
        "distributed_id": null,
        "type": "ETCD",
        "detail": null
      },
      {
        "id": "dn_6003",
        "role": "master",
        "status": "Standby",
        "distributed_id": 60011,
        "type": "DN",
        "detail": "Normal"
      },
      {
        "id": "dn_6006",
        "role": "master",
        "status": "Standby",
        "distributed_id": 60042,
        "type": "DN",
        "detail": "Normal"
      },
      {
        "id": "dn_6007",
        "role": "master",
        "status": "Primary",
        "distributed_id": 60073,
        "type": "DN",
        "detail": "Normal"
      }
    ],
    "id": "085b6f5e8e3243a09ee1ecef6968abc8no14",
    "name": "gauss-lizhonggui_gaussdbv5_hcs2-0_2",
    "hostname": "lzk-at-47Jftu-0002",
    "availability_zone_- id": "az_03",
    "description": "az_03",
    "status": "normal"
  },
  {
    "components": [
      {
        "id": "cm_2",
        "role": "master",
        "status": "Standby",
        "distributed_id": null,
        "type": "CM",
        "detail": null
      },
      {
        "id": "gtm_1002",
        "role": "master",
        "status": "Standby",
        "distributed_id": null,
```

```
      "type": "GTM",
      "detail": "Connection ok"
    },
    {
      "id": "cn_5002",
      "role": "master",
      "status": "Normal",
      "distributed_id": null,
      "type": "CN",
      "detail": null
    },
    {
      "id": "etcd_7002",
      "role": "master",
      "status": "StateFollower",
      "distributed_id": null,
      "type": "ETCD",
      "detail": null
    },
    {
      "id": "dn_6002",
      "role": "master",
      "status": "Standby",
      "distributed_id": 60011,
      "type": "DN",
      "detail": "Normal"
    },
    {
      "id": "dn_6004",
      "role": "master",
      "status": "Primary",
      "distributed_id": 60042,
      "type": "DN",
      "detail": "Normal"
    },
    {
      "id": "dn_6008",
      "role": "master",
      "status": "Standby",
      "distributed_id": 60073,
      "type": "DN",
      "detail": "Normal"
    }
  ],
  "id": "9440ea1b42da492b9d78ee6a35fe26e2no14",
  "name": "gauss-lizhonggui_gaussdbv5_hcs2-0_1",
  "hostname": "lzk-at-47Jftu-0001",
  "availability_zone_- id": "az_03",
  "description": "az_03",
  "status": "normal"
},
{
  "components": [
    {
      "id": "cm_1",
      "role": "master",
      "status": "Primary",
      "distributed_id": null,
      "type": "CM",
      "detail": null
    },
    {
      "id": "gtm_1001",
      "role": "master",
      "status": "Primary",
      "distributed_id": null,
      "type": "GTM",
      "detail": "Connection ok"
    },
```

```
      {
        "id": "cn_5001",
        "role": "master",
        "status": "Normal",
        "distributed_id": null,
        "type": "CN",
        "detail": null
      },
      {
        "id": "etcd_7001",
        "role": "master",
        "status": "StateFollower",
        "distributed_id": null,
        "type": "ETCD",
        "detail": null
      },
      {
        "id": "dn_6001",
        "role": "master",
        "status": "Primary",
        "distributed_id": 60011,
        "type": "DN",
        "detail": "Normal"
      },
      {
        "id": "dn_6005",
        "role": "master",
        "status": "Standby",
        "distributed_id": 60042,
        "type": "DN",
        "detail": "Normal"
      },
      {
        "id": "dn_6009",
        "role": "master",
        "status": "Standby",
        "distributed_id": 60073,
        "type": "DN",
        "detail": "Normal"
      }
    ],
    "id": "a0db0718462045e28fcb52c2afb0e1e0no14",
    "name": "gauss-lizhonggui_gaussdbv5_hcs2-0_0",
    "hostname": "lzk-cdd-6PE05Z-0002",
    "availability_zone_- id": "az_03",
    "description": "az_03",
    "status": "normal"
  }
 ],
 "total_count": 3
}
```


Status Code

Status Code	Description
200	Normal

Error Code

For details, see **Error Codes**.

4.11.4 Performing a Primary/Standby DN Switchover

Function

This API is used to perform a primary/standby DN switchover.

URI

POST /gaussdb/v3/instances/{instance_id}/switch-shard

Table 4-769 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-770 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Table 4-771 Request body parameter

Parameter	Mandatory	Type	Description
shards	Yes	Array of **Table 4-772** objects	Shards.

Table 4-772 shards

Parameter	Mandatory	Type	Description
node_id	Yes	String	ID of the node where the standby DN to be promoted to primary is deployed.

Parameter	Mandatory	Type	Description
component_id	Yes	String	ID of the standby DN to be promoted to primary. It can contain up to 7 characters. It cannot be null, an empty string or spaces. Before verifying and using it, spaces are automatically filtered out. To obtain the value, call the API for querying components of an instance.

Response Parameters

Status code: 200

Table 4-773 Response body parameters

Parameter	Type	Description
job_id	String	Job ID.

Example Request

- Switching primary and standby DNs in multiple shards

```
POST https://{Endpoint}/rds/gaussdb/v3/instances/f54427e5af55418b89e304babf47e52ain14/switch-shard

{
  "shards" : [ {
    "node_id" : "07bea8968a084a9191d85f5d9ba21d5cno14",
    "component_id" : "dn_6002"
  }, {
    "node_id" : "07bea8968a084a9191d85f5d9ba21d5cno14",
    "component_id" : "dn_6004"
  }, {
    "node_id" : "39e2b4c4d8104cd6b780457b5d6537efno14",
    "component_id" : "dn_6009"
  } ]
}
```

- Switching primary and standby DNs in a shard

```
POST https://{Endpoint}/rds/gaussdb/v3/instances/f54427e5af55418b89e304babf47e52ain14/switch-shard

{
  "shards" : [ {
    "node_id" : "07bea8968a084a9191d85f5d9ba21d5cno14",
    "component_id" : "dn_6002"
  } ]
}
```

Example Response

Status code: 200

Success.


```
{
  "job_id" : "975d1377bc8b70ebf3c41d3a1fdbc434"
}
```


Status Code

Status Code	Description
200	Normal

Error Code

For details, see **Error Codes**.

4.11.5 Deleting an Instance

Function

This API is used to uninstall an instance or cancel the management for an instance.

Constraints

You can simultaneously send a maximum of 60 requests of deleting instances per minute by calling this API.

URI

DELETE /gaussdb/v3/instances/{instance_id}/management

Table 4-774 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-775 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.

Parameter	Mandatory	Type	Description
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Table 4-776 Request body parameters

Parameter	Mandatory	Type	Description
clear_data	No	Boolean	Whether to clear data. **true** indicates that the instance is deleted, and **false** indicates that the instance management is canceled.
keep_db_backup	No	Integer	Number of days for retaining automated full backups. This parameter takes effect only when an instance is deleted. If it is set to **0**, the instance is deleted immediately. Value range: **0-2562** Minimum value: **0** Maximum value: **2562**
keep_difference_backup	No	Integer	Number of days for retaining differential backups. This parameter takes effect only when an instance is deleted. If it is set to **0**, the instance is deleted immediately. Value range: **0-2562** Minimum value: **0** Maximum value: **2562**
rds_admin_password	No	String	Password of user **rdsAdmin**. If **clear_data** and **is_force_remove_instance** are not specified, **rds_admin_password** must be specified.
is_force_remove_instance	No	Boolean	Whether to forcibly remove the instance.

Response Parameters

Status code: 202

Table 4-777 Response body parameters

Parameter	Type	Description
instance_id	String	Instance ID.
job_id	String	Job ID.

Status code: default

Table 4-778 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

- Uninstalling an instance

```
DELETE https://{Endpoint}/rds/gaussdb/v3/instances/cc6fd964d93f4003851dfc29d57d30a5in14/
management

{
  "rds_admin_password" : null,
  "is_force_remove_instance" : false,
  "clear_data" : true,
  "keep_db_backup" : 0,
  "keep_difference_backup" : 0
}
```

- Canceling management for an instance

```
DELETE https://{Endpoint}/rds/gaussdb/v3/instances/cc6fd964d93f4003851dfc29d57d30a5in14/
management

{
  "rds_admin_password" : "***",
  "is_force_remove_instance" : false,
  "clear_data" : false,
  "keep_db_backup" : 0,
  "keep_difference_backup" : 0
}
```

Example Response

Status code: 202

Success.

```
{
  "instance_id" : "cc6fd964d93f4003851dfc29d57d30a5in14",
  "job_id" : "3bcaff3eb858e9b94ef5a3010e7f65be"
}
```

Status Code

Status Code	Description
202	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.11.6 Creating an Instance or Restoring Data to a New Instance

Function

This API is used to create an instance or restore data to a new instance.

URI

POST /gaussdb/v3/instances

Request Parameters

Table 4-779 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Table 4-780 Request body parameters

Parameter	Mandatory	Type	Description
name	Yes	String	Instance name. The name must consist of 4 to 64 characters and start with a letter. It can contain only letters (case-sensitive), digits, hyphens (-), and underscores (_).
os_type	Yes	String	Operating system, which must be the same as the selected host operating system. The value can be: • **Kylin** • **UnionTech** • **bclinux**: BC-Linux • **HCE**: Huawei Cloud EulerOS • **SUSE**: SUSE Linux Enterprise Server • openEuler: openeuler • **nfs**: NFS China The value is case-insensitive. Enumerated values: • **Kylin** • **UnionTech** • **bclinux** • **Hce** • **Suse** • **openeuler** • **nfs**
datastore	No	**Table 4-781** object	DB engine version. This parameter is mandatory only when an instance is created and is invalid when data is restored to a new instance.
host_type	Yes	String	Host type: Currently, only hosts are supported. The value is case-insensitive. Enumerated value: **BMS**

Parameter	Mandatory	Type	Description
cpu_spec	Yes	String	CPU vendor, which must be the same as the CPU vendor of the selected host. The value can be: • **Intel** • **Kunpeng** • **Hygon** The value is case-insensitive.
arch	Yes	String	Architecture type. The value is case-insensitive. Enumerated values: • **X86** • **ARM**
ha	No	**Table 4-782** object	Database type. This parameter is mandatory only when an instance is created and is invalid when data is restored to a new instance.

Parameter	Mandatory	Type	Description
solution	Yes	String	Instance type. The value can be: Centralized: ● **single**: single-replica ● **double**: 2 nodes (1 primary + 1 standby + 1 log) ● **logger**: 3 nodes (1 primary + 1 standby + 1 log) ● **triset**: 3 nodes (1 primary + 2 standby) ● **quadruset**: 5 nodes (1 primary + 3 standby, 1 arbitration node) ● **pentaset**: 5 nodes (1 primary + 4 standby) ● **doublenoetcd**: 2 nodes (1 primary + 1 standby + 1 log) (without ETCD) ● **loggernoetcd**: 3 nodes (1 primary + 1 standby + 1 log) (without ETCD) ● **trisetnoetcd**: 3 nodes (1 primary + 2 standby, without ETCD) ● **quadrusetnoetcd**: 5 nodes (1 primary + 3 standby, 1 arbitration node) (without ETCD) ● **pentasetnoetcd**: 5 nodes (1 primary + 4 standby, without ETCD) To use mode without ETCD, you need to enable the whitelist feature **gaussdb_feature_supportDccCluster**. After the whitelist is enabled, **double**, **triset**, **quadruset**, and **pentaset** are automatically converted to the mode without ETCD. Distributed deployment: ● **hcs1**: 9 nodes. There are 4 shards and each shard contains one primary DN and three standby DNs.

Parameter	Mandatory	Type	Description
			• **hcs2**: 3 nodes. There are 3 shards and each shard contains one primary DN and two standby DNs. • **hcs3**: 4 nodes. There are 4 shards and each shard contains one primary DN and three standby DNs. • **hcs4**: 5 nodes. There are 4 shards and each shard contains one primary DN and three standby DNs. • **hcs6**: 1 node. There are 3 shards and each shard contains one primary DN. • **hcs9**: 1 node. There are 4 shards and each shard contains one primary DN. Enumerated values: • **single** • **double** • **logger** • **triset** • **quadruset** • **pentaset** • **doublenoetcd** • **loggernoetcd** • **trisetnoetcd** • **quadrusetnoetcd** • **pentasetnoetcd** • **hcs1** • **hcs2** • **hcs3** • **hcs4** • **hcs6** • **hcs9**

Parameter	Mandatory	Type	Description
port	No	String	Instance port. The default value is **8000**. Port range: **1024** to **39989**. The following ports are reserved for system use and cannot be used: **2378** to **2380**, **2400**, **4999** to **5001**, **5100**, **5500**, **5999** to **6001**, **6009** to **6010**, **6500**, **8015**, **8097**, **8098**, **8181**, **9090**, **9100**, **9180**, **9187**, **9200**, **12016**, **12017**, **20049**, **20050**, **21731**, **21732**, **32122** to **32126**, or **39001**.
password	Yes	String	Password of user **root**. The following requirements must be met: ● The value cannot be empty. ● The value should consist of 8 to 32 characters, including at least three of the following: uppercase letters, lowercase letters, digits, and special characters ~!@#%^*-_=+?, Enter a strong password to improve security, preventing security risks such as brute force cracking.
manage_ip	No	Array of strings	Hosts.
master_az_ip	No	Array of strings	Hosts in the primary AZ. This parameter is available only when **solution** is set to **hcs1**, **triset**, **quadruset**, **trisetnoetcd**, or **quadrusetnoetcd**.
slave_az_ip	No	Array of strings	Hosts in the standby AZ. This parameter is available only when **solution** is set to **hcs1**, **quadruset**, or **quadrusetnoetcd**.
arbiter_az_ip	No	Array of strings	Hosts in the arbitration AZ. This parameter is available only when **solution** is set to **hcs1**, **hcs4**, **quadruset**, or **quadrusetnoetcd**.

Parameter	Mandatory	Type	Description
logger_az_ip	No	Array of strings	Hosts in the log AZ. This parameter is available only when **solution** is set to **logger** or **loggernoetcd**.
float_ip	No	String	Floating IP address. This parameter is only available when you create a non-single-node instance with a floating IP address in the centralized deployment.
configuration_id	No	String	Parameter template ID. If this parameter is not specified, the default parameter template is used.
enable_force_switch	No	Boolean	Failover policy. This parameter is available only when 2-node centralized instances are created. • **true**: indicates the failover priority is set to availability, log nodes are deployed on standby nodes. • **false**: indicates the failover priority is set to reliability, log nodes are deployed on standby nodes.
time_zone	No	String	Time zone.
diy_install_settings	No	**Table 4-783** object	Installation policy that you configure. This parameter is mandatory only when a GaussDB instance with the DM installation mode is created. It is also mandatory when data is restored to a new instance.
restore_point	No	**Table 4-786** object	ID of the backup that will be used to restore data or timestamp that data will be restored to. This parameter is available only when data is restored to a new instance and is invalid when an instance is created.

Parameter	Mandatory	Type	Description
xbsa_ssl_certs	No	**Table 4-789** object	XBSA certificate. This parameter is available only when data is restored to a new instance and is invalid when an instance is created.
xbsa_extension_infos	No	Array of **Table 4-790** objects	XBSA extended attribute. This parameter is mandatory when cross-cloud backup restoration is performed or the original instance has been deleted. Otherwise, the target instance cannot communicate with XBSA devices.
storage_device	No	**Table 4-791** objects	NAS storage device information. If this parameter is not specified, no NAS device is mounted to the instance, and the backup function is unavailable. This parameter is invalid when a DBMind instance is created.
is_cdb	No	Boolean	Whether to enable a multitenant container. This parameter is supported only when there are 16 vCPUs and 128 GB memory or more available. It is mandatory when the original instance is a multitenant instance and its data is restored to a new instance.

Parameter	Mandatory	Type	Description
small_os_version	No	String	OS version, which must be the same as the selected host operating system. The value can be: • Kylin: **kylin V10 SP1**, **kylin V10 SP2**, or **kylin V10 SP3** • UnionTech OS: **uos 20** • HCE: **hce 2.0** • SLES: **sles 12.5** • BC-Linux: **bclinux 21.10** • openEuler: **22.03 SP4** • **nfs 4.0**: NFS China The value is case-insensitive.
lower_case_table_names	No	Integer	Case sensitivity of M-compatible table names. Value range: • **0** (default value): The names are case-sensitive. • **1**: The names are case-insensitive. **NOTICE** • This parameter is only available for centralized instances of version V2.0-8.203 or later. • When data is restored to a new instance, the setting of this parameter for the new instance must be the same as that for the original instance.
dorado_shared_volume	No	**Table 4-793** object	Dorado shared volume information, which is mandatory for creating a Dorado instance.

Table 4-781 DatastoreInfo

Parameter	Mandatory	Type	Description
type	Yes	String	Engine type. The value can be: • **GaussDB**: A GaussDB instance is to be created. The value is case insensitive. • **gaussdbv5-dbmind**: A DBMind instance is to be created. The value is case insensitive. Enumerated values: • **GaussDB** • **gaussdbv5-dbmind**
version	No	String	Engine version. For details, see Querying DB Engine Versions. If this parameter is not specified during GaussDB instance creation, the latest version is used by default. This parameter is mandatory when a DBMind instance is created.

Table 4-782 CustomerHaOption

Parameter	Mandatory	Type	Description
mode	Yes	String	Instance type. The value can be **centralization_standard** for centralized instances or **combined** for distributed instances. Enumerated values: • **centralization_standard** • **combined**

Parameter	Mandatory	Type	Description
consistency	Yes	String	Transaction consistency. This parameter is valid only for distributed instances. The value can be: ● **strong**: strong consistency ● **eventual**: eventual consistency Strong consistency parameter by default. The value is case-insensitive. For a centralized instance, the value can only be **strong**. Enumerated values: ● **strong** ● **eventual**
replication_mode	Yes	String	Replication mode for the standby node. The value can only be **sync**. Enumerated value: **sync**
consistency_protocol	No	String	Replica consistency protocol. Only Paxos is supported when there is no ETCD in an instance. Enumerated values: ● **quorum** ● **paxos**

Table 4-783 DIYInstallSettings

Parameter	Mandatory	Type	Description
dir_settings	No	Array of **Table 4-784** objects	Installation directory settings.
component_distributions	No	Array of **Table 4-785** objects	IP address distribution of installation components.

Table 4-784 DirInstallSettings

Parameter	Mandatory	Type	Description
dir_name	Yes	String	Directory name. Currently, the following directories are supported: • **install_dir**: software directory • **log_dir**: log directory • **dn_dir**: data directory for DNs • **etcd_dir**: data directory for ETCDs Enumerated values: • **install_dir** • **log_dir** • **dn_dir** • **etcd_dir**
dir	Yes	String	Directory.

Table 4-785 ComponentDistributions

Parameter	Mandatory	Type	Description
component	Yes	String	Component name. The components depend on the instance type. Currently, the following components are supported: • cn • cm • gtm • etcd • dn1_group • dn2_group • dn3_group • dn4_group Enumerated values: • **cn** • **cm** • **gtm** • **etcd** • **dn1_group** • **dn2_group** • **dn3_group** • **dn4_group**
ip	Yes	String	Host IP address.

Table 4-786 RestorePoint

Parameter	Mandatory	Type	Description
restore_time	No	String	Specified point in time that data will be restored to, in timestamp.
instance_id	Yes	String	Original instance ID.
backup_id	No	String	ID of the backup that will be used to restore data.

Parameter	Mandatory	Type	Description
type	No	String	Restoration type. Value: ● **backup**: indicates data is restored using backups. In this mode, **type** is optional and **backup_id** is mandatory. ● **timestamp**: indicates data is restored using point-in-time recovery (PITR). In this mode, **type** and **restore_time** are both mandatory. Enumerated values: ● **backup** ● **timestamp**
table_list	No	Array of **Table 4-787** objects	Table-level backup information. This parameter is required only when databases and tables are restored. **Constraints**: ● Only instances later than V2.0-3.200 support NAS-based table-level backups. Only instances of the V2.0-8.200 version and later support XBSA-based table-level backups. ● A maximum of 100 databases or tables can be restored at the same time. If there are more than 100 databases or tables, you are advised to use instance-level restoration.

Parameter	Mandatory	Type	Description
schema_type	No	String	**Explanation**: Source backup type. This parameter is mandatory for PITR restoration and determines whether to use a full backup or a table-level backup for restoration. During restoration using a backup file, the backup type is automatically obtained based on the backup ID, and you do not need to set this parameter. **Constraints**: N/A **Value range**: ● **INSTANCE**: instance-level backup ● **DATABASE_TABLE**: table-level backup **Default value**: INSTANCE
restore_config	No	**Table 4-792** object	Restoration configurations.

Table 4-787 TableListInfo

Parameter	Mandatory	Type	Description
table_list	Yes	Array of **Table 4-788** objects	New database tables that data will be restored to.

Table 4-788 TableInfo

Parameter	Mandatory	Type	Description
db_name	Yes	String	**Explanation**: Name of the database to be restored. **Constraints**: • M-compatible databases do not support database- and table-level backups and restorations. • The following template databases and system users cannot be restored: – Template databases: **postgres**, **template0**, **templatem**, and **template1**. – System users: **rdsAdmin**, **rdsMetric**, **rdsBackup**, and **rdsRepl**. **Value range**: N/A **Default value**: N/A
schema_name	No	String	**Explanation**: Name of the schema to be restored. **Constraints**: System-level schemas, such as **dbe_application_info** and **rdsRepl**, cannot be restored. **Value range**: N/A **Default value**: N/A

<table>
<tr><th>Parameter</th><th>Mandatory</th><th>Type</th><th>Description</th></tr>
<tr><td>table_name</td><td>No</td><td>String</td><td>Explanation:
Name of the table to be restored.
Constraints:
N/A
Value range:
N/A
Default value:
N/A</td></tr>
<tr><td>new_db_name</td><td>No</td><td>String</td><td>Explanation:
Name of the new database.
Constraints:
● The name cannot be the same as that of existing databases in the original or existing instance.
● It must be different from template databases. Template databases: postgres, template0, templatem, template1, and templatea. The database name cannot be the system users rdsAdmin, rdsMetric, rdsBackup, and rdsRepl.
Value range:
A database name can contain 1 to 63 characters. Only letters, digits, and underscores (_) are allowed. It cannot start with pg or a digit.
Default Value:
N/A</td></tr>
</table>

Parameter	Mandatory	Type	Description
new_schema_name	No	String	**Explanation**: Name of the new schema. **Constraints**: ● The name cannot be the same as that of existing schemas in the original or existing instance. ● The name cannot be the same as that of a template database or an existing schema. Template databases include **postgres**, **template0**, **template1**, **templatem**, and **templatea**. Existing schemas include **public** and **information_schema**. In addition, system-level schemas such as **dbe_application_info** and **rdsRepl** cannot be restored. **Value range**: A schema name can contain 1 to 63 characters. Only letters, digits, and underscores (_) are allowed. It cannot start with **pg** or a digit. **Default value**: N/A
new_table_name	No	String	**Explanation**: Name of the new table. **Constraints**: The name cannot be the same as that of existing tables in the original or existing instance. **Value range**: A schema name can contain 1 to 63 characters. Only letters, digits, and underscores (_) are allowed. It cannot start with **pg** or a digit. **Default value**: N/A

Table 4-789 XbsaSslCerts

Parameter	Mandatory	Type	Description
ca_cert_pem	Yes	String	Root certificate (Base64-encoded).
client_crt	Yes	String	User certificate (Base64-encoded).
client_key	Yes	String	User private key (Base64-encoded).
rand_pass	Yes	String	Password of the private key file.

Table 4-790 XbsaExtensionReq

Parameter	Mandatory	Type	Description
key	Yes	String	Key of the extended attribute.
value	Yes	String	Value of the extended attribute.

Table 4-791 StorageDevice

Parameter	Mandatory	Type	Description
device_id	Yes	String	Storage device ID. This parameter is mandatory when a storage device is specified.
device_directory	Yes	String	User-defined storage path, which starts with a slash (/). The storage device path consists of the default directory and custom directory.

Table 4-792 RestoreConfig

Parameter	Mandatory	Type	Description
restore_parallel_degree	No	String	Number of concurrent restoration tasks. A restoration task is split into multiple subtasks to improve restoration efficiency. The default value is **1**. Value: **1**, **2**, **4**, or **8** **NOTE** • This parameter is only available for instances whose DB engine version is V2.0-9.0.0.SPC0100 or later. • This parameter is only available for an instance having at least 4 vCPUs. • This parameter is only available when an instance-level restoration is performed.

Table 4-793 DoradoSharedVolume

Parameter	Mandatory	Type	Description
cms_shared_volume	Yes	**Table 4-794** object	CMS shared volume information.
dn_shared_volume	Yes	**Table 4-795** object	DN shared volume information.

Table 4-794 CmsSharedVolume

Parameter	Mandatory	Type	Description
wwn	Yes	String	WWN of a Dorado storage device. The value is the NGUID for a NoF device or WWN for an FC device.

Table 4-795 DnSharedVolume

Parameter	Mandatory	Type	Description
wwn	Yes	String	WWN of a Dorado storage device. The value is the NGUID for a NoF device or WWN for an FC device.

Response Parameters

Status code: 202

Table 4-796 Response body parameters

Parameter	Type	Description
job_id	String	Job ID.
instance_id	String	Instance ID.
name	String	Instance name.
os_type	String	Operating system. Enumerated values: ● **Kylin** ● **UnionTech** ● **bclinux** ● **Hce** ● **Suse** ● **openEuler**
datastore	**Table 4-797** object	DB engine version.
ha	**Table 4-798** object	Database type.

Parameter	Type	Description
solution	String	Instance type. Centralized: ● **single**: single-replica ● **double**: 2 nodes (1 primary + 1 standby + 1 log) ● **logger**: 3 nodes (1 primary + 1 standby + 1 log) ● **triset**: 3 nodes (1 primary + 2 standby) ● **quadruset**: 5 nodes (1 primary + 3 standby, 1 arbitration node) ● **pentaset**: 5 nodes (1 primary + 4 standby) ● **doublenoetcd**: 2 nodes (1 primary + 1 standby + 1 log) (without ETCD) ● **loggernoetcd**: 3 nodes (1 primary + 1 standby + 1 log) (without ETCD) ● **trisetnoetcd**: 3 nodes (1 primary + 2 standby, without ETCD) ● **quadrusetnoetcd**: 5 nodes (1 primary + 3 standby, 1 arbitration node) (without ETCD) ● **pentasetnoetcd**: 5 nodes (1 primary + 4 standby, without ETCD) Distributed deployment: ● **hcs1**: 9 nodes. There are 4 shards and each shard contains one primary DN and three standby DNs. ● **hcs2**: 3 nodes. There are 3 shards and each shard contains one primary DN and two standby DNs. ● **hcs3**: 4 nodes. There are 4 shards and each shard contains one primary DN and three standby DNs. ● **hcs4**: 5 nodes. There are 4 shards and each shard contains one primary DN and three standby DNs. ● **hcs6**: 1 node. There are 3 shards and each shard contains one primary DN. ● **hcs9**: 1 node. There are 4 shards and each shard contains one primary DN. Enumerated values: ● **single** ● **double** ● **logger** ● **triset**

Parameter	Type	Description
		• **quadruset** • **pentaset** • **doublenoetcd** • **loggernoetcd** • **trisetnoetcd** • **quadrusetnoetcd** • **pentasetnoetcd** • **hcs1** • **hcs2** • **hcs3** • **hcs4** • **hcs6** • **hcs9**
port	String	Port.
manage_ip	Array of strings	Host IP addresses in the list.
float_ip	String	Floating IP address.
configuration_id	String	Parameter template ID.
time_zone	String	Time zone.
backup_strategy	**Table 4-799** object	Automated backup policy. If the **storage_device** parameter is left blank, the returned value is empty.

Table 4-797 DatastoreInfoResponse

Parameter	Type	Description
type	String	Engine type
version	String	DB engine version.

Table 4-798 CustomerHaOptionResponse

Parameter	Type	Description
mode	String	Instance model.
consistency	String	Transaction consistency

Parameter	Type	Description
replication_mode	String	Replication mode for the standby node. This parameter is fixed to **sync**, which indicates synchronous replication.
consistency_protocol	String	Replica consistency protocol.

Table 4-799 CustomerBackupStrategy

Parameter	Type	Description
start_time	String	Backup time window. The creation of an automated backup will be triggered during the backup time window. The value cannot be empty. It must be a valid value in the "hh:mm-HH:MM" format. The current time is in the UTC format. ● The **HH** value must be 1 greater than the **hh** value. ● The values of **mm** and **MM** must be the same and must be set to **00**. Example value: ● 08:00-09:00 ● 23:00-00:00
keep_days	Integer	Retention days for specific backups. Value: **0** to **732**. The default value **7**. If this parameter is set to **0**, the default value **7** is used.

Example Request

- Creating a centralized instance (**solution** is set to **single**)

```
POST https://{Endpoint}/rds/gaussdb/v3/instances

{
  "name" : "gaussdb-s10d",
  "os_type" : "kylin",
  "datastore" : {
    "type" : "GaussDB",
    "version" : "1.1"
  },
  "host_type" : "BMS",
  "cpu_spec" : "intel",
  "arch" : "X86",
  "ha" : {
    "mode" : "centralization_standard",
    "consistency" : "strong",
    "replication_mode" : "sync",
    "consistency_protocol" : "quorum"
  },
```

```
  "solution" : "single",
  "port" : "8000",
  "password" : "***",
  "manage_ip" : [ "127.*.*.1" ]
}
```

- Creating a centralized instance with high service availability (**solution** is set to **double**)

```
POST https://{Endpoint}/rds/gaussdb/v3/instances

{
  "name": "gaussdb-s10d",
  "os_type": "kylin",
  "datastore": {
    "type": "GaussDB",
    "version": "8.200"
  },
  "host_type": "BMS",
  "cpu_spec": "intel",
  "arch": "X86",
  "ha": {
    "mode": "centralization_standard",
    "consistency": "strong",
    "replication_mode": "sync",
    "consistency_protocol": "paxos"
  },
  "solution": "double",
  "port": "8000",
  "password": "Ga***",
  "manage_ip": [
    "127.*.*.1",
    "127.*.*.2"
  ],
  "enable_force_switch": true
}
```

- Creating a centralized logger instance

```
POST https://{Endpoint}/rds/gaussdb/v3/instances
{
  "name" : "gaussdb-s10d",
  "os_type" : "kylin",
  "datastore" : {
    "type" : "GaussDB",
    "version" : "1.1"
  },
  "host_type" : "BMS",
  "cpu_spec" : "intel",
  "arch" : "X86",
  "ha" : {
    "mode" : "centralization_standard",
    "consistency" : "strong",
    "replication_mode" : "sync",
    "consistency_protocol" : "paxos"
  },
  "solution" : "logger",
  "port" : "8000",
  "password" : "***",
  "manage_ip" : [ "127.*.*.1", "127.*.*.2", "127.*.*.3" ]
}
```

- Creating a centralized instance (**solution** is set to **triset**)

```
POST https://{Endpoint}/rds/gaussdb/v3/instances

{
  "name" : "gaussdb-s10d",
  "os_type" : "kylin",
  "datastore" : {
    "type" : "GaussDB",
    "version" : "1.1"
  },
```

```
  "host_type" : "BMS",
  "cpu_spec" : "intel",
  "arch" : "X86",
  "ha" : {
    "mode" : "centralization_standard",
    "consistency" : "strong",
    "replication_mode" : "sync",
    "consistency_protocol" : "quorum"
  },
  "solution" : "triset",
  "port" : "8000",
  "password" : "***",
  "manage_ip" : [ "127.*.*.1", "127.*.*.2", "127.*.*.3" ],
  "master_az_ip" : [ "127.*.*.1" ]
}
```

- Creating a centralized instance (**solution** is set to **quadruset**)

```
POST https://{Endpoint}/rds/gaussdb/v3/instances

{
  "name" : "gaussdb-s10d",
  "os_type" : "kylin",
  "datastore" : {
    "type" : "GaussDB",
    "version" : "1.1"
  },
  "host_type" : "BMS",
  "cpu_spec" : "intel",
  "arch" : "X86",
  "ha" : {
    "mode" : "centralization_standard",
    "consistency" : "strong",
    "replication_mode" : "sync",
    "consistency_protocol" : "quorum"
  },
  "solution" : "quadruset",
  "port" : "8000",
  "password" : "***",
  "manage_ip" : [ "127.*.*.1", "127.*.*.2", "127.*.*.3", "127.*.*.4", "127.*.*.5" ],
  "master_az_ip" : [ "127.*.*.1", "127.*.*.2" ],
  "slave_az_ip" : [ "127.*.*.3", "127.*.*.4" ],
  "arbiter_az_ip" : [ "127.*.*.5" ]
}
```

- Creating a centralized instance (**solution** is set to **pentaset**)

```
POST https://{Endpoint}/rds/gaussdb/v3/instances

{
  "name" : "gaussdb-s10d",
  "os_type" : "kylin",
  "datastore" : {
    "type" : "GaussDB",
    "version" : "1.1"
  },
  "host_type" : "BMS",
  "cpu_spec" : "intel",
  "arch" : "X86",
  "ha" : {
    "mode" : "centralization_standard",
    "consistency" : "strong",
    "replication_mode" : "sync",
    "consistency_protocol" : "quorum"
  },
  "solution" : "pentaset",
  "port" : "8000",
  "password" : "***",
  "manage_ip" : [ "127.*.*.1", "127.*.*.2", "127.*.*.3", "127.*.*.4", "127.*.*.5" ]
}
```

- Creating a distributed instance (**solution** is set to **hcs1**)

```
POST https://{Endpoint}/rds/gaussdb/v3/instances

{
  "name" : "gaussdb-s10d",
  "os_type" : "kylin",
  "datastore" : {
    "type" : "GaussDB",
    "version" : "1.1"
  },
  "host_type" : "BMS",
  "cpu_spec" : "intel",
  "arch" : "X86",
  "ha" : {
    "mode" : "combined",
    "consistency" : "strong",
    "replication_mode" : "sync",
    "consistency_protocol" : "quorum"
  },
  "solution" : "hcs1",
  "port" : "8000",
  "password" : "***",
  "manage_ip" : [ "127.*.*.1", "127.*.*.2", "127.*.*.3", "127.*.*.4", "127.*.*.5", "127.*.*.6", "127.*.*.7", 
"127.*.*.8", "127.*.*.9" ],
  "master_az_ip" : [ "127.*.*.1", "127.*.*.2", "127.*.*.3", "127.*.*.4" ],
  "slave_az_ip" : [ "127.*.*.5", "127.*.*.6", "127.*.*.7", "127.*.*.8" ],
  "arbiter_az_ip" : [ "127.*.*.9" ]
}
```

- Creating a distributed instance (**solution** is set to **hcs2**)

```
POST https://{Endpoint}/rds/gaussdb/v3/instances

{
  "name" : "gaussdb-s10d",
  "os_type" : "kylin",
  "datastore" : {
    "type" : "GaussDB",
    "version" : "1.1"
  },
  "host_type" : "BMS",
  "cpu_spec" : "intel",
  "arch" : "X86",
  "ha" : {
    "mode" : "combined",
    "consistency" : "strong",
    "replication_mode" : "sync",
    "consistency_protocol" : "quorum"
  },
  "solution" : "hcs2",
  "port" : "8000",
  "password" : "***",
  "manage_ip" : [ "127.*.*.1", "127.*.*.2", "127.*.*.3" ]
}
```

- Creating a distributed instance (**solution** is set to **hcs3**)

```
POST https://{Endpoint}/rds/gaussdb/v3/instances

{
  "name" : "gaussdb-s10d",
  "os_type" : "kylin",
  "datastore" : {
    "type" : "GaussDB",
    "version" : "1.1"
  },
  "host_type" : "BMS",
  "cpu_spec" : "intel",
  "arch" : "X86",
  "ha" : {
    "mode" : "combined",
    "consistency" : "strong",
    "replication_mode" : "sync",
```

```
    "consistency_protocol" : "quorum"
  },
  "solution" : "hcs3",
  "port" : "8000",
  "password" : "***",
  "manage_ip" : [ "127.*.*.1", "127.*.*.2", "127.*.*.3", "127.*.*.4" ]
}
```

- Creating a distributed instance (**solution** is set to **hcs4**)

```
POST https://{Endpoint}/rds/gaussdb/v3/instances

{
  "name" : "gaussdb-s10d",
  "os_type" : "kylin",
  "datastore" : {
    "type" : "GaussDB",
    "version" : "1.1"
  },
  "host_type" : "BMS",
  "cpu_spec" : "intel",
  "arch" : "X86",
  "ha" : {
    "mode" : "combined",
    "consistency" : "strong",
    "replication_mode" : "sync",
    "consistency_protocol" : "quorum"
  },
  "solution" : "hcs4",
  "port" : "8000",
  "password" : "***",
  "manage_ip" : [ "127.*.*.1", "127.*.*.2", "127.*.*.3", "127.*.*.4", "127.*.*.5" ],
  "arbiter_az_ip" : [ "127.*.*.5" ]
}
```

- Creating a distributed instance (**solution** is set to **hcs6**)

```
POST https://{Endpoint}/rds/gaussdb/v3/instances

{
  "name" : "gaussdb-s10d",
  "os_type" : "kylin",
  "datastore" : {
    "type" : "GaussDB",
    "version" : "1.1"
  },
  "host_type" : "BMS",
  "cpu_spec" : "intel",
  "arch" : "X86",
  "ha" : {
    "mode" : "combined",
    "consistency" : "strong",
    "replication_mode" : "sync",
    "consistency_protocol" : "quorum"
  },
  "solution" : "hcs6",
  "port" : "8000",
  "password" : "***",
  "manage_ip" : [ "127.*.*.1" ]
}
```

- Creating a centralized instance with a floating IP address (**solution** is set to **triset**)

```
POST https://{Endpoint}/rds/gaussdb/v3/instances

{
  "name" : "gaussdb-s10d",
  "os_type" : "kylin",
  "datastore" : {
    "type" : "GaussDB",
    "version" : "1.1"
  },
```

```
  "host_type" : "BMS",
  "cpu_spec" : "intel",
  "arch" : "X86",
  "ha" : {
    "mode" : "centralization_standard",
    "consistency" : "strong",
    "replication_mode" : "sync",
    "consistency_protocol" : "quorum"
  },
  "solution" : "triset",
  "port" : "8000",
  "password" : "***",
  "manage_ip" : [ "127.*.*.1", "127.*.*.2", "127.*.*.3" ],
  "master_az_ip" : [ "127.*.*.1" ],
  "float_ip" : "127.*.*.2"
}
```

- Restoring a centralized instance with 5 nodes

```
POST https://{Endpoint}/rds/gaussdb/v3/instances

{
  "name" : "gaussdb-s10d",
  "os_type" : "Kylin",
  "host_type" : "BMS",
  "cpu_spec" : "Intel",
  "arch" : "X86",
  "solution" : "quadruset",
  "diy_install_settings": {
    "dir_settings": [
     {
      "dir_name": "dn_dir",
      "dir": "/opt/cluster/var/lib/engine/data1/data"
      },
      {
       "dir_name": "etcd_dir",
       "dir": "/opt/cluster/usr/local/etcd"
       },
       {
        "dir_name": "install_dir",
        "dir": "/opt/cluster"
       },
       {
        "dir_name": "log_dir",
        "dir": "/opt/cluster/var/lib/log"
       }
    ]
  },
  "port" : "8000",
  "password" : "***",
  "manage_ip" : [ "127.*.*.1", "127.*.*.2", "127.*.*.3", "127.*.*.4", "127.*.*.5" ],
  "master_az_ip" : [ "127.*.*.1", "127.*.*.2" ],
  "slave_az_ip" : [ "127.*.*.3", "127.*.*.4" ],
  "arbiter_az_ip" : [ "127.*.*.5" ],
  "restore_point" : {
    "backup_id" : "bde1999aa29a4aa5a990dabcac9d6210br14",
    "instance_id" : "bde1999aa29a4aa5a990dabcac9d6210in14"
  }
}
```

- Restoring a centralized instance with 5 nodes to a specified point in time

```
POST https://{Endpoint}/rds/gaussdb/v3/instances

{
  "name" : "gaussdb-s10d",
  "os_type" : "Kylin",
  "host_type" : "BMS",
  "cpu_spec" : "Intel",
  "arch" : "X86",
  "solution" : "quadruset",
  "diy_install_settings": {
```

```
    "dir_settings": [
     {
      "dir_name": "dn_dir",
      "dir": "/opt/cluster/var/lib/engine/data1/data"
      },
      {
       "dir_name": "etcd_dir",
       "dir": "/opt/cluster/usr/local/etcd"
       },
       {
        "dir_name": "install_dir",
        "dir": "/opt/cluster"
       },
       {
        "dir_name": "log_dir",
        "dir": "/opt/cluster/var/lib/log"
       }
    ]
  },
  "port" : "8000",
  "password" : "***",
  "manage_ip" : [ "127.*.*.1", "127.*.*.2", "127.*.*.3", "127.*.*.4", "127.*.*.5" ],
  "master_az_ip" : [ "127.*.*.1", "127.*.*.2" ],
  "slave_az_ip" : [ "127.*.*.3", "127.*.*.4" ],
  "arbiter_az_ip" : [ "127.*.*.5" ],
  "restore_point" : {
    "restore_time" : "1234564789",
    "instance_id" : "bde1999aa29a4aa5a990dabcac9d6210in14"
  }
}
```

- Creating a BMS-based centralized DB instance managed by DBMind

```
POST https://{Endpoint}/rds/gaussdb/v3/instances

{
  "region": "mini_01",
  "name": "gauss-dbmind-f7d0",
  "host_type": "bms",
  "os_type": "Kylin",
  "cpu_spec": "intel",
  "arch": "x86",
  "datastore": {
    "type": "gaussdbv5-dbmind",
    "version": "8.201"
  },
  "ha": {
    "mode": "centralization_standard",
    "consistency": "strong",
    "replication_mode": "sync"
  },
  "manage_ip": [
    "192.168.122.105"
  ],
  "solution": "single",
  "seccompDisabled": false,
  "enable_force_switch": false
}
```

- Restoring a centralized instance with 5 nodes and a floating IP address

```
POST https://{Endpoint}/rds/gaussdb/v3/instances

{
  "name" : "gaussdb-s10d",
  "os_type" : "Kylin",
  "host_type" : "BMS",
  "cpu_spec" : "Intel",
  "arch" : "X86",
  "solution" : "quadruset",
  "diy_install_settings": {
    "dir_settings": [
```

```
    {
     "dir_name": "dn_dir",
     "dir": "/opt/cluster/var/lib/engine/data1/data"
     },
     {
      "dir_name": "etcd_dir",
      "dir": "/opt/cluster/usr/local/etcd"
      },
      {
       "dir_name": "install_dir",
       "dir": "/opt/cluster"
      },
      {
       "dir_name": "log_dir",
       "dir": "/opt/cluster/var/lib/log"
      }
    ]
  },
  "port" : "8000",
  "password" : "***",
  "manage_ip" : [ "127.*.*.1", "127.*.*.2", "127.*.*.3", "127.*.*.4", "127.*.*.5" ],
  "master_az_ip" : [ "127.*.*.1", "127.*.*.2" ],
  "slave_az_ip" : [ "127.*.*.3", "127.*.*.4" ],
  "arbiter_az_ip" : [ "127.*.*.5" ],
  "float_ip" : "127.*.*.2",
  "restore_point" : {
    "backup_id" : "bde1999aa29a4aa5a990dabcac9d6210br14",
    "instance_id" : "bde1999aa29a4aa5a990dabcac9d6210in14"
  }
}
```

- Restoring data to a new centralized instance with 3 nodes (1 primary + 2 standby) using an XBSA full backup

```
POST https://{Endpoint}/rds/gaussdb/v3/instances

{
  "name": "gauss-restore-xbsa",
  "host_type": "bms",
  "os_type": "Kylin",
  "cpu_spec": "intel",
  "arch": "x86",
  "solution": "triset",
  "diy_install_settings": {
    "dir_settings": [
     {
      "dir_name": "dn_dir",
      "dir": "/opt/cluster/var/lib/engine/data1/data"
     },
     {
      "dir_name": "etcd_dir",
      "dir": "/opt/cluster/usr/local/etcd"
      },
      {
       "dir_name": "install_dir",
       "dir": "/opt/cluster"
      },
      {
       "dir_name": "log_dir",
       "dir": "/opt/cluster/var/lib/log"
      }
    ]
  },
  "port": "8000",
  "password": "***",
  "manage_ip": [
    "100.107.*.*",
    "100.99.222.*",
    "100.112.42.*"
  ],
```

```
  "master_az_ip": [
    "100.107.*.*"
  ],
  "restore_point": {
    "backup_id": "bde1999aa29a4aa5a990dabcac9d6210br14",
    "instance_id": "bde1999aa29a4aa5a990dabcac9d6210in14",
    "type": "backup"
  },
  "xbsa_extension_infos": [
    {
      "key": "BackupSet",
      "value": "/opt/xbsa/conf/xbsa_gaussdb_restore.conf"
    },
    {
      "key": "XbsaRoute",
      "value": "{\"port\":\"5000\",\"ip\":\"172.168.1.68\"}"
    }
  ],
  "xbsa_ssl_certs": {
    "ca_cert_pem": "Q2VydGlma******tCg==",
    "client_crt": "Q2VydG*****tLS0K",
    "client_key": "LS0t****LS0tLQo=",
    "rand_pass": "U*****8"
  }
}
```

- Restoring data to a specified point in time of a new distributed instance with 5 nodes using an XBSA backup

```
POST https://{Endpoint}/rds/gaussdb/v3/instances

{
  "name": "gauss-restore-pitr-xbsa",
  "host_type": "bms",
  "os_type": "Kylin",
  "cpu_spec": "intel",
  "arch": "x86",
  "solution": "hcs4",
  "diy_install_settings": {
    "dir_settings": [
     {
      "dir_name": "dn_dir",
      "dir": "/opt/cluster/var/lib/engine/data1/data"
      },
      {
       "dir_name": "etcd_dir",
       "dir": "/opt/cluster/usr/local/etcd"
       },
       {
        "dir_name": "install_dir",
        "dir": "/opt/cluster"
       },
       {
        "dir_name": "log_dir",
        "dir": "/opt/cluster/var/lib/log"
       }
    ]
  },
  "port": "8000",
  "password": "***",
  "manage_ip": [
    "127.0.0.*",
    "127.0.0.*",
    "127.0.0.*",
    "127.0.0.*",
    "127.0.0.*"
  ],
  "arbiter_az_ip": [
    "127.0.0.*"
  ],
  "float_ip":
```

```
    "127.0.2.*",
  "restore_point": {
    "restore_point": "1710138043000",
    "instance_id": "bde1999aa29a4aa5a990dabcac9d6210in14",
    "type": "timestamp"
  },
  "xbsa_extension_infos": [
    {
      "key": "BackupSet",
      "value": "/opt/xbsa/conf/xbsa_gaussdb_restore.conf"
    },
    {
      "key": "XbsaRoute",
      "value": "{\"port\":\"5000\",\"ip\":\"172.168.1.68\"}"
    }
  ],
  "xbsa_ssl_certs": {
    "ca_cert_pem": "Q2VydGlma******tCg==",
    "client_crt": "Q2VydG*****tLS0K",
    "client_key": "LS0t****LS0tLQo=",
    "rand_pass": "U*****8"
  }
}
```

- Manually creating a distributed instance (**solution** is set to **hcs2**)

```
POST https://{Endpoint}/rds/gaussdb/v3/instances

{
  "name": "gaussdb-s10d",
  "os_type": "kylin",
  "datastore": {
    "type": "GaussDB",
    "version": "8.200"
  },
  "host_type": "BMS",
  "cpu_spec": "intel",
  "arch": "X86",
  "ha": {
    "mode": "combined",
    "consistency": "strong",
    "replication_mode": "sync",
    "consistency_protocol": "quorum"
  },
  "solution": "hcs2",
  "port": "8000",
  "password": "***",
  "manage_ip": [
    "127.*.*.1",
    "127.*.*.2",
    "127.*.*.3"
  ],
  "diy_install_settings": {
    "dir_settings": [
      {
        "dir_name": "dn_dir",
        "dir": "/opt/cluster/var/lib/engine/data1/data"
      },
      {
        "dir_name": "etcd_dir",
        "dir": "/opt/cluster/usr/local/etcd"
      },
      {
        "dir_name": "install_dir",
        "dir": "/opt/cluster"
      },
      {
        "dir_name": "log_dir",
        "dir": "/opt/cluster/var/lib/log"
      }
    ],
```

```
"component_distributions": [
  {
    "component": "cm",
    "ip": "127.*.*.1"
  },
  {
    "component": "cn",
    "ip": "127.*.*.1"
  },
  {
    "component": "dn1_group",
    "ip": "127.*.*.1"
  },
  {
    "component": "dn2_group",
    "ip": "127.*.*.1"
  },
  {
    "component": "dn3_group",
    "ip": "127.*.*.1"
  },
  {
    "component": "etcd",
    "ip": "127.*.*.1"
  },
  {
    "component": "gtm",
    "ip": "127.*.*.1"
  },
  {
    "component": "cm",
    "ip": "127.*.*.2"
  },
  {
    "component": "cn",
    "ip": "127.*.*.2"
  },
  {
    "component": "dn1_group",
    "ip": "127.*.*.2"
  },
  {
    "component": "dn2_group",
    "ip": "127.*.*.2"
  },
  {
    "component": "dn3_group",
    "ip": "127.*.*.2"
  },
  {
    "component": "etcd",
    "ip": "127.*.*.2"
  },
  {
    "component": "gtm",
    "ip": "127.*.*.2"
  },
  {
    "component": "cm",
    "ip": "127.*.*.3"
  },
  {
    "component": "cn",
    "ip": "127.*.*.3"
  },
  {
    "component": "dn1_group",
    "ip": "127.*.*.3"
  },
```

```
      {
        "component": "dn2_group",
        "ip": "127.*.*.3"
      },
      {
        "component": "dn3_group",
        "ip": "127.*.*.3"
      },
      {
        "component": "etcd",
        "ip": "127.*.*.3"
      },
      {
        "component": "gtm",
        "ip": "127.*.*.3"
      }
    ]
  }
}
```


Example Response

Status code: 202

Normal


```
{
  "instance_id" : "63be94303826495ca95373660396a0d2in14",
  "job_id" : "f0b5e495-7ad5-3536-a820-2f176d3b8f47",
  "name" : "gaussdb-s10d",
  "os_type" : "Kylin",
  "datastore" : {
    "type" : "GaussDB",
    "version" : "1.1"
  },
  "ha" : {
    "mode" : "centralization_standard",
    "consistency" : "strong",
    "replication_mode" : "sync",
    "consistency_protocol" : "quorum"
  },
  "solution" : "triset",
  "port" : "8000",
  "manage_ip" : [ "127.*.*.1", "127.*.*.2", "127.*.*.3" ],
  "float_ip" : "127.0.2.2",
  "configuration_id" : "b000d7c91f1749da87315700793a11d4pr14",
  "time_zone" : "UTC+08:00",
  "backup_strategy" : {
    "start_time" : "17:00-18:00",
    "keep_days" : 7
  }
}
```


Status Code

Status Code	Description
202	Normal

Error Code

For details, see **Error Codes**.

4.11.7 (Recommended) Creating an Instance or Restoring Data to a New Instance

Function

This API is used to create an instance or restore data to a new instance.

URI

POST /gaussdb/v3.1/instances

Request Parameters

Table 4-800 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Table 4-801 Request body parameters

Parameter	Mandatory	Type	Description
name	Yes	String	Instance name. The name must consist of 4 to 64 characters and start with a letter. It can contain only letters (case-sensitive), digits, hyphens (-), and underscores (_).

Parameter	Mandatory	Type	Description
os_type	Yes	String	Operating system, which must be the same as the selected host operating system. The value can be: • **Kylin** • **UnionTech** • **bclinux**: BC-Linux • **HCE**: Huawei Cloud EulerOS • **SUSE**: SUSE Linux Enterprise Server • **openeuler**: openEuler • **nfs**: NFS China The value is case-insensitive. Enumerated values: • **Kylin** • **UnionTech** • **bclinux** • **Hce** • **Suse** • **openeuler** • **nfs**
datastore	No	**Table 4-802** object	DB engine version. This parameter is mandatory only when an instance is created and is invalid when data is restored to a new instance.
host_type	Yes	String	Host type: Currently, only hosts are supported. The value is case-insensitive. Enumerated value: **BMS**
cpu_spec	Yes	String	CPU vendor, which must be the same as the CPU vendor of the selected host. The value can be: • **Intel** • **Kunpeng** • **Hygon** The value is case-insensitive.

Parameter	Mandatory	Type	Description
arch	Yes	String	Architecture type. The value is case-insensitive. Enumerated values: • **X86** • **ARM**
ha	No	**Table 4-803** object	Database type. This parameter is mandatory only when an instance is created and is invalid when data is restored to a new instance.

Parameter	Mandatory	Type	Description
solution	Yes	String	Instance type. Value range: Centralized: • **single**: single-replica • **double**: 2 nodes (1 primary + 1 standby + 1 log) • **logger**: 3 nodes (1 primary + 1 standby + 1 log) • **triset**: 3 nodes (1 primary + 2 standby) • **quadruset**: 5 nodes (1 primary + 3 standby, 1 arbitration node) • **pentaset**: 5 nodes (1 primary + 4 standby) • **doublenoetcd**: 2 nodes (1 primary + 1 standby + 1 log) (without ETCD) • **loggernoetcd**: 3 nodes (1 primary + 1 standby + 1 log) (without ETCD) • **trisetnoetcd**: 3 nodes (1 primary + 2 standby, without ETCD) • **quadrusetnoetcd**: 5 nodes (1 primary + 3 standby, 1 arbitration node) (without ETCD) • **pentasetnoetcd**: 5 nodes (1 primary + 4 standby, without ETCD) To use mode without ETCD, you need to enable the whitelist feature **gaussdb_feature_supportDccCluster**. After the whitelist is enabled, **double**, **triset**, **quadruset**, and **pentaset** are automatically converted to the mode without ETCD. Distributed deployment: • **hcs1**: 9 nodes. There are 4 shards and each shard contains one primary DN and three standby DNs. • **hcs2**: 3 nodes. There are 3 shards and each shard

Parameter	Mandatory	Type	Description
			contains one primary DN and two standby DNs. • **hcs3**: 4 nodes. There are 4 shards and each shard contains one primary DN and three standby DNs. • **hcs4**: 5 nodes. There are 4 shards and each shard contains one primary DN and three standby DNs. • **hcs6**: 1 node. There are 3 shards and each shard contains one primary DN. • **hcs9**: 1 node. There are 4 shards and each shard contains one primary DN. Enumerated values: • **single** • **double** • **logger** • **triset** • **quadruset** • **pentaset** • **doublenoetcd** • **loggernoetcd** • **trisetnoetcd** • **quadrusetnoetcd** • **pentasetnoetcd** • **hcs1** • **hcs2** • **hcs3** • **hcs4** • **hcs6** • **hcs9**

Parameter	Mandatory	Type	Description
port	No	String	Instance port. Port range: **1024** to **39989**. The following ports are reserved for system use and cannot be used: **2378** to **2380**, **2400**, **4999** to **5001**, **5100**, **5500**, **5999** to **6001**, **6009** to **6010**, **6500**, **8015**, **8097**, **8098**, **8181**, **9090**, **9100**, **9180**, **9187**, **9200**, **12016**, **12017**, **20049**, **20050**, **21731**, **21732**, **32122** to **32126**, or **39001**.
password	Yes	String	Password of user **root**. The following requirements must be met: ● The value cannot be empty. ● The value should consist of 8 to 32 characters, including at least three of the following: uppercase letters, lowercase letters, digits, and special characters ~!@#%^*-_=+?, Enter a strong password to improve security, preventing security risks such as brute force cracking.
manage_ip	No	Array of strings	Hosts.
master_az_ip	No	Array of strings	Hosts in the primary AZ. This parameter is available only when **solution** is set to **hcs1**, **triset**, **quadruset**, **trisetnoetcd**, or **quadrusetnoetcd**.
slave_az_ip	No	Array of strings	Hosts in the standby AZ. This parameter is available only when **solution** is set to **hcs1**, **quadruset**, or **quadrusetnoetcd**.
arbiter_az_ip	No	Array of strings	Hosts in the arbitration AZ. This parameter is available only when **solution** is set to **hcs1**, **hcs4**, **quadruset**, or **quadrusetnoetcd**.

Parameter	Mandatory	Type	Description
logger_az_ip	No	Array of strings	Hosts in the log AZ. This parameter is available only when **solution** is set to **logger** or **loggernoetcd**.
float_ip	No	String	Floating IP address. This parameter is only available when you create a non-single-node instance with a floating IP address in the centralized deployment.
configuration_id	No	String	Parameter template ID. If this parameter is not specified, the default parameter template is used.
enable_force_switch	No	Boolean	Failover policy. This parameter is available only when 2-node centralized instances are created. • **true**: indicates the failover priority is set to availability, log nodes are deployed on standby nodes. • **false**: indicates the failover priority is set to reliability, log nodes are deployed on standby nodes.
time_zone	No	String	Time zone.
diy_install_settings	No	**Table 4-804** object	Installation policy that you configure. This parameter is mandatory only when a GaussDB instance with the DM installation mode is created. It is also mandatory when data is restored to a new instance.
restore_point	No	**Table 4-807** object	ID of the backup that will be used to restore data or timestamp that data will be restored to. This parameter is available only when data is restored to a new instance and is invalid when an instance is created.

Parameter	Mandatory	Type	Description
xbsa_ssl_certs	No	**Table 4-810** object	XBSA certificate. This parameter is available only when data is restored to a new instance and is invalid when an instance is created.
xbsa_extension_infos	No	Array of **Table 4-811** objects	XBSA extended attribute. This parameter is mandatory when cross-cloud backup restoration is performed or the original instance has been deleted. Otherwise, the target instance cannot communicate with XBSA devices.
storage_device	No	**Table 4-812** objects	NAS storage device information. If this parameter is not specified, no NAS device is mounted to the instance, and the backup function is unavailable. This parameter is invalid when a DBMind instance is created.
is_cdb	No	Boolean	Whether to enable a multitenant container. This parameter is supported only when there are 16 vCPUs and 128 GB memory or more available. It is mandatory when the original instance is a multitenant instance and its data is restored to a new instance.

Parameter	Mandatory	Type	Description
small_os_version	No	String	OS version, which must be the same as the selected host operating system. The value can be: • Kylin: **kylin V10 SP1**, **kylin V10 SP2**, or **kylin V10 SP3** • UnionTech OS: **uos 20** • HCE: **hce 2.0** • SLES: **sles 12.5** • BC-Linux: **bclinux 21.10** • openEuler: 22.03 SP4 • **nfs 4.0**: NFS China The value is case-insensitive.
lower_case_table_names	No	Integer	Case sensitivity of M-compatible table names. Value range: • **0** (default value): The names are case-sensitive. • **1**: The names are case-insensitive. **NOTICE** • This parameter is only available for centralized instances of version V2.0-8.203 or later. • When data is restored to a new instance, the setting of this parameter for the new instance must be the same as that for the original instance.
dorado_shared_volume	No	**Table 4-818** object	Dorado shared volume information, which is mandatory for creating a Dorado instance.

Table 4-802 DatastoreInfo

Parameter	Mandatory	Type	Description
type	Yes	String	Engine type. The value can be: ● **GaussDB**: A GaussDB instance is to be created. The value is case insensitive. ● **gaussdbv5-dbmind**: A DBMind instance is to be created. The value is case insensitive. Enumerated values: ● **GaussDB** ● **gaussdbv5-dbmind**
version	No	String	Engine version. For details, see Querying DB Engine Versions. If this parameter is not specified during GaussDB instance creation, the latest version is used by default. This parameter is mandatory when a DBMind instance is created.

Table 4-803 CustomerHaOption

Parameter	Mandatory	Type	Description
mode	Yes	String	Instance type. The value can be **centralization_standard** for centralized instances or **combined** for distributed instances. Enumerated values: ● **centralization_standard** ● **combined**

Parameter	Mandatory	Type	Description
consistency	Yes	String	Transaction consistency. This parameter is valid only for distributed instances. Value range: • **strong**: strong consistency • **eventual**: eventual consistency Strong consistency parameter by default. The value is case-insensitive. For a centralized instance, the value can only be **strong**. Enumerated values: • **strong** • **eventual**
replication_mode	Yes	String	Replication mode for the standby node. The value can only be **sync**. Enumerated value: **sync**
consistency_protocol	No	String	Replica consistency protocol. Only Paxos is supported when there is no ETCD in an instance. Enumerated values: • **quorum** • **paxos**

Table 4-804 DIYInstallSettings

Parameter	Mandatory	Type	Description
dir_settings	No	Array of **Table 4-805** objects	Installation directory settings.
component_distributions	No	Array of **Table 4-806** objects	IP address distribution of installation components.

Table 4-805 DirInstallSettings

Parameter	Mandatory	Type	Description
dir_name	Yes	String	Directory name. Currently, the following directories are supported: • **install_dir**: software directory • **log_dir**: log directory • **dn_dir**: data directory for DNs • **etcd_dir**: data directory for ETCDs Enumerated values: • **install_dir** • **log_dir** • **dn_dir** • **etcd_dir**
dir	Yes	String	Directory.

Table 4-806 ComponentDistributions

Parameter	Mandatory	Type	Description
component	Yes	String	Component name. The components depend on the instance type. Currently, the following components are supported: • cn • cm • gtm • etcd • dn1_group • dn2_group • dn3_group • dn4_group Enumerated values: • **cn** • **cm** • **gtm** • **etcd** • **dn1_group** • **dn2_group** • **dn3_group** • **dn4_group**
ip	Yes	String	Host IP address.

Table 4-807 RestorePoint

Parameter	Mandatory	Type	Description
restore_time	No	String	Specified point in time that data will be restored to, in timestamp.
instance_id	Yes	String	Original instance ID.
backup_id	No	String	ID of the backup that will be used to restore data.

Parameter	Mandatory	Type	Description
type	No	String	Restoration type. Value: • **backup**: indicates data is restored using backups. In this mode, **type** is optional and **backup_id** is mandatory. • **timestamp**: indicates data is restored using point-in-time recovery (PITR). In this mode, **type** and **restore_time** are both mandatory. Enumerated values: • **backup** • **timestamp**
table_list	No	Array of **Table 4-808** objects	Table-level backup information. This parameter is required only when databases and tables are restored. **Constraints**: • Only instances later than V2.0-3.200 support NAS-based table-level backups. Only instances of the V2.0-8.200 version and later support XBSA-based table-level backups. • A maximum of 100 databases or tables can be restored at the same time. If there are more than 100 databases or tables, you are advised to use instance-level restoration.

Parameter	Mandatory	Type	Description
schema_type	No	String	**Explanation**: Source backup type. This parameter is mandatory for PITR restoration and determines whether to use a full backup or a table-level backup for restoration. During restoration using a backup file, the backup type is automatically obtained based on the backup ID, and you do not need to set this parameter. **Constraints**: N/A **Value range**: ● **INSTANCE**: instance-level backup ● **DATABASE_TABLE**: table-level backup **Default value**: INSTANCE
restore_config	No	**Table 4-813** object	Restoration configurations.

Table 4-808 TableListInfo

Parameter	Mandatory	Type	Description
table_list	Yes	Array of **Table 4-809** objects	New database tables that data will be restored to.

Table 4-809 TableInfo

Parameter	Mandatory	Type	Description
db_name	Yes	String	**Explanation**: Name of the database to be restored. **Constraints**: ● M-compatible databases do not support database- and table-level backups and restorations. ● The following template databases and system users cannot be restored: – Template databases: **postgres**, **template0**, **templatem**, and **template1**. – System users: **rdsAdmin**, **rdsMetric**, **rdsBackup**, and **rdsRepl**. **Value range**: N/A **Default value**: N/A
schema_name	No	String	**Explanation**: Name of the schema to be restored. **Constraints**: System-level schemas, such as **dbe_application_info** and **rdsRepl**, cannot be restored. **Value range**: N/A **Default value**: N/A

Parameter	Mandatory	Type	Description
table_name	No	String	**Explanation**: Name of the table to be restored. **Constraints**: N/A **Value range**: N/A **Default value**: N/A
new_db_name	No	String	**Explanation**: Name of the new database. **Constraints**: ● The name cannot be the same as that of existing databases in the original or existing instance. ● It must be different from template databases. Template databases: **postgres**, **template0**, **templatem**, **template1**, and **templatea**. The database name cannot be the system users **rdsAdmin**, **rdsMetric**, **rdsBackup**, and **rdsRepl**. **Value range**: A database name can contain 1 to 63 characters. Only letters, digits, and underscores (_) are allowed. It cannot start with **pg** or a digit. **Default value**: N/A

Parameter	Mandatory	Type	Description
new_schema_name	No	String	**Explanation**: Name of the new schema. **Constraints**: ● The name cannot be the same as that of existing schemas in the original or existing instance. ● The name cannot be the same as that of a template database or an existing schema. Template databases include **postgres**, **template0**, **template1**, **templatem**, and **templatea**. Existing schemas include **public** and **information_schema**. In addition, system-level schemas such as **dbe_application_info** and **rdsRepl** cannot be restored. **Value range**: A schema name can contain 1 to 63 characters. Only letters, digits, and underscores (_) are allowed. It cannot start with **pg** or a digit. **Default value**: N/A
new_table_name	No	String	**Explanation**: Name of the new table. **Constraints**: The name cannot be the same as that of existing tables in the original or existing instance. **Value range**: A schema name can contain 1 to 63 characters. Only letters, digits, and underscores (_) are allowed. It cannot start with **pg** or a digit. **Default value**: N/A

Table 4-810 XbsaSslCerts

Parameter	Mandatory	Type	Description
ca_cert_pem	Yes	String	Root certificate (Base64-encoded).
client_crt	Yes	String	User certificate (Base64-encoded).
client_key	Yes	String	User private key (Base64-encoded).
rand_pass	Yes	String	Password of the private key file.

Table 4-811 XbsaExtensionReq

Parameter	Mandatory	Type	Description
key	Yes	String	Key of the extended attribute.
value	Yes	String	Value of the extended attribute.

Table 4-812 StorageDevice

Parameter	Mandatory	Type	Description
device_id	Yes	String	Storage device ID. This parameter is mandatory when a storage device is specified.
device_directory	Yes	String	User-defined storage path, which starts with a slash (/). The storage device path consists of the default directory and custom directory.

Table 4-813 RestoreConfig

Parameter	Mandatory	Type	Description
restore_parallel_degree	No	String	Number of concurrent restoration tasks. A restoration task is split into multiple subtasks to improve restoration efficiency. The default value is **1**. Value: **1**, **2**, **4**, or **8** **NOTE** • This parameter is only available for instances whose DB engine version is V2.0-9.0.0.SPC0100 or later. • This parameter is only available for an instance having at least 4 vCPUs. • This parameter is only available when an instance-level restoration is performed.

Response Parameters

Status code: 202

Table 4-814 Response body parameters

Parameter	Type	Description
job_id	String	Job ID.
instance_id	String	Instance ID.
name	String	Instance name.
os_type	String	Operating system. Enumerated values: • **Kylin** • **UnionTech** • **bclinux** • **Hce** • **Suse** • **openEuler**
datastore	**Table 4-815** object	DB engine version.
ha	**Table 4-816** object	Database type.

Parameter	Type	Description
solution	String	Instance type. Centralized: ● **single**: single-replica ● **double**: 2 nodes (1 primary + 1 standby + 1 log) ● **logger**: 3 nodes (1 primary + 1 standby + 1 log) ● **triset**: 3 nodes (1 primary + 2 standby) ● **quadruset**: 5 nodes (1 primary + 3 standby, 1 arbitration node) ● **pentaset**: 5 nodes (1 primary + 4 standby) ● **doublenoetcd**: 2 nodes (1 primary + 1 standby + 1 log) (without ETCD) ● **loggernoetcd**: 3 nodes (1 primary + 1 standby + 1 log) (without ETCD) ● **trisetnoetcd**: 3 nodes (1 primary + 2 standby, without ETCD) ● **quadrusetnoetcd**: 5 nodes (1 primary + 3 standby, 1 arbitration node) (without ETCD) ● **pentasetnoetcd**: 5 nodes (1 primary + 4 standby, without ETCD) Distributed deployment: ● **hcs1**: 9 nodes. There are 4 shards and each shard contains one primary DN and three standby DNs. ● **hcs2**: 3 nodes. There are 3 shards and each shard contains one primary DN and two standby DNs. ● **hcs3**: 4 nodes. There are 4 shards and each shard contains one primary DN and three standby DNs. ● **hcs4**: 5 nodes. There are 4 shards and each shard contains one primary DN and three standby DNs. ● **hcs6**: 1 node. There are 3 shards and each shard contains one primary DN. ● **hcs9**: 1 node. There are 4 shards and each shard contains one primary DN. Enumerated values: ● **single** ● **double** ● **logger** ● **triset**

Parameter	Type	Description
		• **quadruset** • **pentaset** • **doublenoetcd** • **loggernoetcd** • **trisetnoetcd** • **quadrusetnoetcd** • **pentasetnoetcd** • **hcs1** • **hcs2** • **hcs3** • **hcs4** • **hcs6** • **hcs9**
port	String	Port.
manage_ip	Array of strings	Host IP addresses in the list.
float_ip	String	Floating IP address.
configuration_id	String	Parameter template ID.
time_zone	String	Time zone.
backup_strategy	**Table 4-817** object	Automated backup policy. If the **storage_device** parameter is left blank, the returned value is empty.

Table 4-815 DatastoreInfoResponse

Parameter	Type	Description
type	String	Engine type
version	String	DB engine version.

Table 4-816 CustomerHaOptionResponse

Parameter	Type	Description
mode	String	Instance type.
consistency	String	Transaction consistency

Parameter	Type	Description
replication_mode	String	Replication mode for the standby node. This parameter is fixed to **sync**, which indicates synchronous replication.
consistency_protocol	String	Replica consistency protocol.

Table 4-817 CustomerBackupStrategy

Parameter	Type	Description
start_time	String	Backup time window. The creation of an automated backup will be triggered during the backup time window. The value cannot be empty. It must be a valid value in the "hh:mm-HH:MM" format. The current time is in the UTC format. ● The **HH** value must be 1 greater than the **hh** value. ● The values of **mm** and **MM** must be the same and must be set to **00**. Example value: ● 08:00-09:00 ● 23:00-00:00
keep_days	Integer	Retention days for specific backups. Value: **0** to **732**. The default value **7**. If this parameter is set to **0**, the default value **7** is used.

Table 4-818 DoradoSharedVolume

Parameter	Mandatory	Type	Description
cms_shared_volume	Yes	**Table 4-819** object	CMS shared volume information.
dn_shared_volume	Yes	**Table 4-820** object	DN shared volume information.

Table 4-819 CmsSharedVolume

Parameter	Mandatory	Type	Description
wwn	Yes	String	WWN of a Dorado storage device. The value is the NGUID for a NoF device or WWN for an FC device.

Table 4-820 DnSharedVolume

Parameter	Mandatory	Type	Description
wwn	Yes	String	WWN of a Dorado storage device. The value is the NGUID for a NoF device or WWN for an FC device.

Example Request

- Creating a centralized instance (**solution** is set to **single**)

```
POST https://{Endpoint}/rds/gaussdb/v3.1/instances

{
  "name" : "gaussdb-s10d",
  "os_type" : "kylin",
  "datastore" : {
    "type" : "GaussDB",
    "version" : "V2.0-1.1"
  },
  "host_type" : "BMS",
  "cpu_spec" : "intel",
  "arch" : "X86",
  "ha" : {
    "mode" : "centralization_standard",
    "consistency" : "strong",
    "replication_mode" : "sync",
    "consistency_protocol" : "quorum"
  },
  "solution" : "single",
  "port" : "8000",
  "password" : "***",
  "manage_ip" : [ "127.*.*.1" ]
}
```

- Creating a centralized instance with high service availability (**solution** is set to **double**)

```
POST https://{Endpoint}/rds/gaussdb/v3.1/instances

{
  "name": "gaussdb-s10d",
  "os_type": "kylin",
  "datastore": {
    "type": "GaussDB",
    "version": "V2.0-8.200"
  },
  "host_type": "BMS",
  "cpu_spec": "intel",
  "arch": "X86",
  "ha": {
```

```
    "mode": "centralization_standard",
    "consistency": "strong",
    "replication_mode": "sync",
    "consistency_protocol": "paxos"
  },
  "solution": "double",
  "port": "8000",
  "password": "Ga***",
  "manage_ip": [
    "127.*.*.1",
    "127.*.*.2"
  ],
  "enable_force_switch": true
}
```

- Creating a centralized logger instance

```
POST https://{Endpoint}/rds/gaussdb/v3/instances
{
  "name" : "gaussdb-s10d",
  "os_type" : "kylin",
  "datastore" : {
    "type" : "GaussDB",
    "version" : "1.1"
  },
  "host_type" : "BMS",
  "cpu_spec" : "intel",
  "arch" : "X86",
  "ha" : {
    "mode" : "centralization_standard",
    "consistency" : "strong",
    "replication_mode" : "sync",
    "consistency_protocol" : "paxos"
  },
  "solution" : "logger",
  "port" : "8000",
  "password" : "***",
  "manage_ip" : [ "127.*.*.1", "127.*.*.2", "127.*.*.3" ]
}
```

- Creating a centralized instance (**solution** is set to **triset**)

```
POST https://{Endpoint}/rds/gaussdb/v3.1/instances

{
  "name" : "gaussdb-s10d",
  "os_type" : "kylin",
  "datastore" : {
    "type" : "GaussDB",
    "version" : "V2.0-1.1"
  },
  "host_type" : "BMS",
  "cpu_spec" : "intel",
  "arch" : "X86",
  "ha" : {
    "mode" : "centralization_standard",
    "consistency" : "strong",
    "replication_mode" : "sync",
    "consistency_protocol" : "quorum"
  },
  "solution" : "triset",
  "port" : "8000",
  "password" : "***",
  "manage_ip" : [ "127.*.*.1", "127.*.*.2", "127.*.*.3" ],
  "master_az_ip" : [ "127.*.*.1" ]
}
```

- Creating a centralized instance (**solution** is set to **quadruset**)

```
POST https://{Endpoint}/rds/gaussdb/v3.1/instances

{
  "name" : "gaussdb-s10d",
  "os_type" : "kylin",
```

```
  "datastore" : {
    "type" : "GaussDB",
    "version" : "V2.0-1.1"
  },
  "host_type" : "BMS",
  "cpu_spec" : "intel",
  "arch" : "X86",
  "ha" : {
    "mode" : "centralization_standard",
    "consistency" : "strong",
    "replication_mode" : "sync",
    "consistency_protocol" : "quorum"
  },
  "solution" : "quadruset",
  "port" : "8000",
  "password" : "***",
  "manage_ip" : [ "127.*.*.1", "127.*.*.2", "127.*.*.3", "127.*.*.4", "127.*.*.5" ],
  "master_az_ip" : [ "127.*.*.1", "127.*.*.2" ],
  "slave_az_ip" : [ "127.*.*.3", "127.*.*.4" ],
  "arbiter_az_ip" : [ "127.*.*.5" ]
}
```

- Creating a centralized instance (**solution** is set to **pentaset**)

```
POST https://{Endpoint}/rds/gaussdb/v3.1/instances

{
  "name" : "gaussdb-s10d",
  "os_type" : "kylin",
  "datastore" : {
    "type" : "GaussDB",
    "version" : "V2.0-1.1"
  },
  "host_type" : "BMS",
  "cpu_spec" : "intel",
  "arch" : "X86",
  "ha" : {
    "mode" : "centralization_standard",
    "consistency" : "strong",
    "replication_mode" : "sync",
    "consistency_protocol" : "quorum"
  },
  "solution" : "pentaset",
  "port" : "8000",
  "password" : "***",
  "manage_ip" : [ "127.*.*.1", "127.*.*.2", "127.*.*.3", "127.*.*.4", "127.*.*.5" ]
}
```

- Creating a distributed instance (**solution** is set to **hcs1**)

```
POST https://{Endpoint}/rds/gaussdb/v3.1/instances

{
  "name" : "gaussdb-s10d",
  "os_type" : "kylin",
  "datastore" : {
    "type" : "GaussDB",
    "version" : "V2.0-1.1"
  },
  "host_type" : "BMS",
  "cpu_spec" : "intel",
  "arch" : "X86",
  "ha" : {
    "mode" : "combined",
    "consistency" : "strong",
    "replication_mode" : "sync",
    "consistency_protocol" : "quorum"
  },
  "solution" : "hcs1",
  "port" : "8000",
  "password" : "***",
  "manage_ip" : [ "127.*.*.1", "127.*.*.2", "127.*.*.3", "127.*.*.4", "127.*.*.5", "127.*.*.6", "127.*.*.7",
```

```
"127.*.*.8", "127.*.*.9" ],
  "master_az_ip" : [ "127.*.*.1", "127.*.*.2", "127.*.*.3", "127.*.*.4" ],
  "slave_az_ip" : [ "127.*.*.5", "127.*.*.6", "127.*.*.7", "127.*.*.8" ],
  "arbiter_az_ip" : [ "127.*.*.9" ]
}
```

- Creating a distributed instance (**solution** is set to **hcs2**)

```
POST https://{Endpoint}/rds/gaussdb/v3.1/instances

{
  "name" : "gaussdb-s10d",
  "os_type" : "kylin",
  "datastore" : {
    "type" : "GaussDB",
    "version" : "V2.0-1.1"
  },
  "host_type" : "BMS",
  "cpu_spec" : "intel",
  "arch" : "X86",
  "ha" : {
    "mode" : "combined",
    "consistency" : "strong",
    "replication_mode" : "sync",
    "consistency_protocol" : "quorum"
  },
  "solution" : "hcs2",
  "port" : "8000",
  "password" : "***",
  "manage_ip" : [ "127.*.*.1", "127.*.*.2", "127.*.*.3" ]
}
```

- Creating a distributed instance (**solution** is set to **hcs3**)

```
POST https://{Endpoint}/rds/gaussdb/v3.1/instances

{
  "name" : "gaussdb-s10d",
  "os_type" : "kylin",
  "datastore" : {
    "type" : "GaussDB",
    "version" : "V2.0-1.1"
  },
  "host_type" : "BMS",
  "cpu_spec" : "intel",
  "arch" : "X86",
  "ha" : {
    "mode" : "combined",
    "consistency" : "strong",
    "replication_mode" : "sync",
    "consistency_protocol" : "quorum"
  },
  "solution" : "hcs3",
  "port" : "8000",
  "password" : "***",
  "manage_ip" : [ "127.*.*.1", "127.*.*.2", "127.*.*.3", "127.*.*.4" ]
}
```

- Creating a distributed instance (**solution** is set to **hcs4**)

```
POST https://{Endpoint}/rds/gaussdb/v3.1/instances

{
  "name" : "gaussdb-s10d",
  "os_type" : "kylin",
  "datastore" : {
    "type" : "GaussDB",
    "version" : "V2.0-1.1"
  },
  "host_type" : "BMS",
  "cpu_spec" : "intel",
  "arch" : "X86",
  "ha" : {
    "mode" : "combined",
```

```
    "consistency" : "strong",
    "replication_mode" : "sync",
    "consistency_protocol" : "quorum"
  },
  "solution" : "hcs4",
  "port" : "8000",
  "password" : "***",
  "manage_ip" : [ "127.*.*.1", "127.*.*.2", "127.*.*.3", "127.*.*.4", "127.*.*.5" ],
  "arbiter_az_ip" : [ "127.*.*.5" ]
}
```

- Creating a distributed instance (**solution** is set to **hcs6**)

```
POST https://{Endpoint}/rds/gaussdb/v3.1/instances

{
  "name" : "gaussdb-s10d",
  "os_type" : "kylin",
  "datastore" : {
    "type" : "GaussDB",
    "version" : "V2.0-1.1"
  },
  "host_type" : "BMS",
  "cpu_spec" : "intel",
  "arch" : "X86",
  "ha" : {
    "mode" : "combined",
    "consistency" : "strong",
    "replication_mode" : "sync",
    "consistency_protocol" : "quorum"
  },
  "solution" : "hcs6",
  "port" : "8000",
  "password" : "***",
  "manage_ip" : [ "127.*.*.1" ]
}
```

- Creating a centralized instance with a floating IP address (**solution** is set to **triset**)

```
POST https://{Endpoint}/rds/gaussdb/v3.1/instances

{
  "name" : "gaussdb-s10d",
  "os_type" : "kylin",
  "datastore" : {
    "type" : "GaussDB",
    "version" : "V2.0-1.1"
  },
  "host_type" : "BMS",
  "cpu_spec" : "intel",
  "arch" : "X86",
  "ha" : {
    "mode" : "centralization_standard",
    "consistency" : "strong",
    "replication_mode" : "sync",
    "consistency_protocol" : "quorum"
  },
  "solution" : "triset",
  "port" : "8000",
  "password" : "***",
  "manage_ip" : [ "127.*.*.1", "127.*.*.2", "127.*.*.3" ],
  "master_az_ip" : [ "127.*.*.1" ],
  "float_ip" : "127.*.*.2"
}
```

- Restoring a centralized instance with 5 nodes

```
POST https://{Endpoint}/rds/gaussdb/v3.1/instances

{
  "name" : "gaussdb-s10d",
  "os_type" : "Kylin",
```

```
  "host_type" : "BMS",
  "cpu_spec" : "Intel",
  "arch" : "X86",
  "solution" : "quadruset",
  "port" : "8000",
  "password" : "***",
  "manage_ip" : [ "127.*.*.1", "127.*.*.2", "127.*.*.3", "127.*.*.4", "127.*.*.5" ],
  "master_az_ip" : [ "127.*.*.1", "127.*.*.2" ],
  "slave_az_ip" : [ "127.*.*.3", "127.*.*.4" ],
  "arbiter_az_ip" : [ "127.*.*.5" ],
  "restore_point" : {
    "backup_id" : "bde1999aa29a4aa5a990dabcac9d6210br14",
    "instance_id" : "bde1999aa29a4aa5a990dabcac9d6210in14"
  }
}
```

- Restoring a centralized instance with 5 nodes to a specified point in time

```
POST https://{Endpoint}/rds/gaussdb/v3.1/instances

{
  "name" : "gaussdb-s10d",
  "os_type" : "Kylin",
  "host_type" : "BMS",
  "cpu_spec" : "Intel",
  "arch" : "X86",
  "solution" : "quadruset",
  "port" : "8000",
  "password" : "***",
  "manage_ip" : [ "127.*.*.1", "127.*.*.2", "127.*.*.3", "127.*.*.4", "127.*.*.5" ],
  "master_az_ip" : [ "127.*.*.1", "127.*.*.2" ],
  "slave_az_ip" : [ "127.*.*.3", "127.*.*.4" ],
  "arbiter_az_ip" : [ "127.*.*.5" ],
  "restore_point" : {
    "restore_time" : "1234564789",
    "instance_id" : "bde1999aa29a4aa5a990dabcac9d6210in14"
  }
}
```

- Creating a BMS-based centralized DB instance managed by DBMind

```
POST https://{Endpoint}/rds/gaussdb/v3.1/instances

{
  "region": "mini_01",
  "name": "gauss-dbmind-f7d0",
  "host_type": "bms",
  "os_type": "Kylin",
  "cpu_spec": "intel",
  "arch": "x86",
  "datastore": {
    "type": "gaussdbv5-dbmind",
    "version": "V2.0-8.201"
  },
  "ha": {
    "mode": "centralization_standard",
    "consistency": "strong",
    "replication_mode": "sync"
  },
  "manage_ip": [
    "192.168.122.105"
  ],
  "solution": "single",
  "seccompDisabled": false,
  "enable_force_switch": false
}
```

- Restoring database tables to a new centralized instance with 5 nodes

```
POST https://{Endpoint}/rds/gaussdb/v3.1/instances

{
  "name" : "gaussdb-s10d",
  "os_type" : "Kylin",
```

```
  "host_type" : "BMS",
  "cpu_spec" : "Intel",
  "arch" : "X86",
  "solution" : "quadruset",
  "port" : "8000",
  "password" : "***",
  "manage_ip" : [ "127.*.*.1", "127.*.*.2", "127.*.*.3", "127.*.*.4", "127.*.*.5" ],
  "master_az_ip" : [ "127.*.*.1", "127.*.*.2" ],
  "slave_az_ip" : [ "127.*.*.3", "127.*.*.4" ],
  "arbiter_az_ip" : [ "127.*.*.5" ],
  "restore_point" : {
    "backup_id" : "bde1999aa29a4aa5a990dabcac9d6210br14",
    "instance_id" : "bde1999aa29a4aa5a990dabcac9d6210in14",
    "schema_type" : "DATABASE_TABLE",
    "table_list" : [ {
      "db_name" : "test_1",
      "table_name" : "table_1",
      "schema_name" : "schema_1",
      "new_db_name" : "test_2",
      "new_table_name" : "table_2",
      "new_schema_name" : "schema_2"
    }
 ]
  }
}
```

- Restoring a centralized instance with 5 nodes and a floating IP address

```
POST https://{Endpoint}/rds/gaussdb/v3.1/instances

{
  "name" : "gaussdb-s10d",
  "os_type" : "Kylin",
  "host_type" : "BMS",
  "cpu_spec" : "Intel",
  "arch" : "X86",
  "solution" : "quadruset",
  "port" : "8000",
  "password" : "***",
  "manage_ip" : [ "127.*.*.1", "127.*.*.2", "127.*.*.3", "127.*.*.4", "127.*.*.5" ],
  "master_az_ip" : [ "127.*.*.1", "127.*.*.2" ],
  "slave_az_ip" : [ "127.*.*.3", "127.*.*.4" ],
  "arbiter_az_ip" : [ "127.*.*.5" ],
  "float_ip" : "127.*.*.2",
  "restore_point" : {
    "backup_id" : "bde1999aa29a4aa5a990dabcac9d6210br14",
    "instance_id" : "bde1999aa29a4aa5a990dabcac9d6210in14"
  }
}
```

- Restoring data to a new centralized instance with 3 nodes (1 primary + 2 standby) using an XBSA full backup

```
POST https://{Endpoint}/rds/gaussdb/v3.1/instances

{
  "name": "gauss-restore-xbsa",
  "host_type": "bms",
  "os_type": "Kylin",
  "cpu_spec": "intel",
  "arch": "x86",
  "solution": "triset",
  "port": "8000",
  "password": "***",
  "manage_ip": [
    "100.107.*.*",
    "100.99.222.*",
    "100.112.42.*"
  ],
  "master_az_ip": [
    "100.107.*.*"
  ],
```

```
  "restore_point": {
    "backup_id": "bde1999aa29a4aa5a990dabcac9d6210br14",
    "instance_id": "bde1999aa29a4aa5a990dabcac9d6210in14",
    "type": "backup"
  },
  "xbsa_extension_infos": [
    {
      "key": "BackupSet",
      "value": "/opt/xbsa/conf/xbsa_gaussdb_restore.conf"
    },
    {
      "key": "XbsaRoute",
      "value": "{\"port\":\"5000\",\"ip\":\"172.168.1.68\"}"
    }
  ],
  "xbsa_ssl_certs": {
    "ca_cert_pem": "Q2VydGlma******tCg==",
    "client_crt": "Q2VydG*****tLS0K",
    "client_key": "LS0t****LS0tLQo=",
    "rand_pass": "U*****8"
  }
}
```

- Restoring data to a specified point in time of a new distributed instance with 5 nodes using an XBSA backup

```
POST https://{Endpoint}/rds/gaussdb/v3.1/instances

{
  "name": "gauss-restore-pitr-xbsa",
  "host_type": "bms",
  "os_type": "Kylin",
  "cpu_spec": "intel",
  "arch": "x86",
  "solution": "hcs4",
  "port": "8000",
  "password": "***",
  "manage_ip": [
    "127.0.0.*",
    "127.0.0.*",
    "127.0.0.*",
    "127.0.0.*",
    "127.0.0.*"
  ],
  "arbiter_az_ip": [
    "127.0.0.*"
  ],
  "float_ip":
    "127.0.2.*",
  "restore_point": {
    "restore_point": "1710138043000",
    "instance_id": "bde1999aa29a4aa5a990dabcac9d6210in14",
    "type": "timestamp"
  },
  "xbsa_extension_infos": [
    {
      "key": "BackupSet",
      "value": "/opt/xbsa/conf/xbsa_gaussdb_restore.conf"
    },
    {
      "key": "XbsaRoute",
      "value": "{\"port\":\"5000\",\"ip\":\"172.168.1.68\"}"
    }
  ],
  "xbsa_ssl_certs": {
    "ca_cert_pem": "Q2VydGlma******tCg==",
    "client_crt": "Q2VydG*****tLS0K",
    "client_key": "LS0t****LS0tLQo=",
    "rand_pass": "U*****8"
  }
}
```

- Manually creating a distributed instance (**solution** is set to **hcs2**)

```
POST https://{Endpoint}/rds/gaussdb/v3.1/instances

{
  "name": "gaussdb-s10d",
  "os_type": "kylin",
  "datastore": {
    "type": "GaussDB",
    "version": "V2.0-8.200"
  },
  "host_type": "BMS",
  "cpu_spec": "intel",
  "arch": "X86",
  "ha": {
    "mode": "combined",
    "consistency": "strong",
    "replication_mode": "sync",
    "consistency_protocol": "quorum"
  },
  "solution": "hcs2",
  "port": "8000",
  "password": "***",
  "manage_ip": [
    "127.*.*.1",
    "127.*.*.2",
    "127.*.*.3"
  ],
  "diy_install_settings": {
    "dir_settings": [
      {
        "dir_name": "dn_dir",
        "dir": "/opt/cluster/var/lib/engine/data1/data"
      },
      {
        "dir_name": "etcd_dir",
        "dir": "/opt/cluster/usr/local/etcd"
      },
      {
        "dir_name": "install_dir",
        "dir": "/opt/cluster"
      },
      {
        "dir_name": "log_dir",
        "dir": "/opt/cluster/var/lib/log"
      }
    ],
    "component_distributions": [
      {
        "component": "cm",
        "ip": "127.*.*.1"
      },
      {
        "component": "cn",
        "ip": "127.*.*.1"
      },
      {
        "component": "dn1_group",
        "ip": "127.*.*.1"
      },
      {
        "component": "dn2_group",
        "ip": "127.*.*.1"
      },
      {
        "component": "dn3_group",
        "ip": "127.*.*.1"
      },
      {
        "component": "etcd",
```

```
        "ip": "127.*.*.1"
      },
      {
        "component": "gtm",
        "ip": "127.*.*.1"
      },
      {
        "component": "cm",
        "ip": "127.*.*.2"
      },
      {
        "component": "cn",
        "ip": "127.*.*.2"
      },
      {
        "component": "dn1_group",
        "ip": "127.*.*.2"
      },
      {
        "component": "dn2_group",
        "ip": "127.*.*.2"
      },
      {
        "component": "dn3_group",
        "ip": "127.*.*.2"
      },
      {
        "component": "etcd",
        "ip": "127.*.*.2"
      },
      {
        "component": "gtm",
        "ip": "127.*.*.2"
      },
      {
        "component": "cm",
        "ip": "127.*.*.3"
      },
      {
        "component": "cn",
        "ip": "127.*.*.3"
      },
      {
        "component": "dn1_group",
        "ip": "127.*.*.3"
      },
      {
        "component": "dn2_group",
        "ip": "127.*.*.3"
      },
      {
        "component": "dn3_group",
        "ip": "127.*.*.3"
      },
      {
        "component": "etcd",
        "ip": "127.*.*.3"
      },
      {
        "component": "gtm",
        "ip": "127.*.*.3"
      }
    ]
  }
}
```

Example Response

Status code: 202

Normal


```
{
  "instance_id" : "63be94303826495ca95373660396a0d2in14",
  "job_id" : "f0b5e495-7ad5-3536-a820-2f176d3b8f47",
  "name" : "gaussdb-s10d",
  "os_type" : "Kylin",
  "datastore" : {
    "type" : "GaussDB",
    "version" : "V2.0-1.1"
  },
  "ha" : {
    "mode" : "centralization_standard",
    "consistency" : "strong",
    "replication_mode" : "sync",
    "consistency_protocol" : "quorum"
  },
  "solution" : "triset",
  "port" : "8000",
  "manage_ip" : [ "127.*.*.1", "127.*.*.2", "127.*.*.3" ],
  "float_ip" : "127.0.2.2",
  "configuration_id" : "b000d7c91f1749da87315700793a11d4pr14",
  "time_zone" : "UTC+08:00",
  "backup_strategy" : {
    "start_time" : "17:00-18:00",
    "keep_days" : 7
  }
}
```


Status Code

Status Code	Description
202	Normal

Error Code

For details, see **Error Codes**.

4.11.8 Querying Instances or Instance Details

Function

This API is used to query instance or instance details.

URI

GET /gaussdb/v3.1/instances

Table 4-821 Query parameters

Parameter	Mandatory	Type	Description
id	No	String	Instance ID. The asterisk (*) is reserved for the system. If the instance ID starts with *, it indicates that fuzzy match is performed based on the value following *. Otherwise, the exact match is performed based on the instance ID. The value cannot contain only asterisks (*).
name	No	String	Instance name. The asterisk (*) is reserved for the system. If the instance name starts with *, it indicates that fuzzy match is performed based on the value following *. Otherwise, the exact match is performed based on the instance name. The value cannot contain only asterisks (*).
type	No	String	Instance type to be queried. Currently, the value is only **Combined** (case-sensitive), indicating the distributed instance. Currently, the value can be "Ha" (case sensitive), which corresponds to centralized instances. Enumerated values: • Combined • **Ha**
datastore_type	No	String	Database type. Its value is case-insensitive. • **GaussDB**
offset	No	Integer	Index offset. If **offset** is set to *N*, the resource query starts from the *N*+1 piece of data. The value is **0** by default, indicating that the query starts from the first piece of data. The value must be a number but cannot be a negative number.

Parameter	Mandatory	Type	Description
limit	No	Integer	Number of records to be queried. The default value is **100**. The value cannot be a negative number. The minimum value is **1** and the maximum value is **100**.

Request Parameters

Table 4-822 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Response Parameters

Status code: 200

Table 4-823 Response body parameters

Parameter	Type	Description
instances	Array of **Table 4-824** objects	Instance information.
total_count	Integer	Total number of records.

Table 4-824 ListInstanceResult

Parameter	Type	Description
id	String	Instance ID.
name	String	Instance name.

Parameter	Type	Description
status	String	Instance status. Values: ● **BUILD**: The instance is being created. ● **BUILD_FAILED**: The instance failed to be created. ● **ACTIVE**: The instance is normal. ● **FAILED**: The instance is abnormal. ● **EXPANDING**: Read replicas or DNs are being added to the instance. ● If the value is **REBOOTING**, the instance is being rebooted. ● **REDUCING**: Read replicas are being deleted. ● **UPGRADING**: The instance is being upgraded. ● **RESTORING**: The instance is being restored. ● **SWITCHOVER**: A primary/standby switchover is being performed. ● **MIGRATING**: The instance is being migrated. ● **BACKING UP**: The instance is being backed up. ● **CHANGING DEPLOYMENT SOLUTION**: The deployment model of the instance is being changed (its replicas are being added or reduced.). ● **STORAGE FULL**: The instance storage is full. ● **REPAIRING**: The instance is being repaired. ● **SHUTDOWN**: The instance is stopped.
private_ips	Array of strings	Data IP addresses.
private_ipv6s	Array of strings	Service IPv6 addresses.
port	Integer	Database port.
type	String	Instance type. Value: **Combined** (distributed) or **centralization_standard** (centralized)
ha	**Table 4-825** object	This parameter is returned for distributed and centralized instances.
replica_num	Integer	Number of replicas.
datastore	**Table 4-826** object	Database information.
created	String	Creation time.

Parameter	Type	Description
updated	String	Update time.
db_user_name	String	Default username.
flavor_ref	String	Specification code.
flavor_info	**Table 4-827** object	Flavor information.
volume	**Table 4-828** object	Volume information.
backup_strategy	**Table 4-829** object	Backup policy.
nodes	Array of **Table 4-830** objects	Instance node information.
instance_mode	String	**enterprise**: enterprise edition
time_zone	String	Time zone.

Table 4-825 ListHaResult

Parameter	Type	Description
consistency	String	Database consistency type. This parameter is available only for distributed instances. The value can be **strong** or **eventual**, indicating strong consistency and eventual consistency, respectively. Enumerated values: • **strong** • **eventual**
replication_mode	String	Replication mode for the standby node. The value cannot be empty. Value: **sync** (synchronization mode)
consistency_protocol	String	Replica consistency protocol. Enumerated value: • **quorum** • paxos

Table 4-826 ListDatastore

Parameter	Type	Description
type	String	DB engine.
version	String	Major DB engine version.
complete_version	String	Minor DB engine version.
complete_kernel_version	String	Kernel engine version.

Table 4-827 ListFlavorInfo

Parameter	Type	Description
vcpu	Integer	Number of vCPUs.
mem	Integer	Memory size.

Table 4-828 ListVolume

Parameter	Type	Description
type	String	Storage type.
size	Integer	Storage.

Table 4-829 OpenGaussBackupStrategyForListResponse

Parameter	Type	Description
start_time	String	Backup time window. The creation of an automated backup will be triggered during the backup time window. The current time is the UTC time.
keep_days	Integer	Number of days to retain the generated backup files. Value: **1** to **732**

Table 4-830 NodeResult

Parameter	Type	Description
id	String	Node ID.

Parameter	Type	Description
name	String	Node name.
hostname	String	Host name.
role	String	Node type. The value can be **master** or **slave**, indicating the primary or standby node. The parameter can also be set to other values.
status	String	Node status.
availability_zone	String	AZ.
private_ip	String	Service IP address, which is used for service access.
data_ip	String	Data IP address, which is used for communication between DB instance components.
management_ip	String	Management IP address, which is used for communication between GaussDB Management Platform (TPOPS) and database nodes.
private_ipv6	String	Service IPv6 address, which is used for service access. Dual-stack access is supported.
data_ipv6	String	Data IPv6 address, which is used for communication between DB instance components.
management_ipv6	String	Management IPv6 address used for communication between TPOPS and database nodes.
component_names	String	Instance component information, which is not updated in real time.

Example Request

Querying instances based on search criteria

```
GET https://{Endpoint}/rds/gaussdb/v3.1/instances?
id=ed7cc6166ec24360a5ed5c5c9c2ed726in14&name=hy&type=Enterprise&datastore_type=GaussDB&offset=
0&limit=10
```

Example Response

Status code: 200

Success.


```
{
  "instances" : [ {
    "id" : "b331ed66cc3249f78bc20737308c01f4in14",
```

```
"status" : "ACTIVE",
"name" : "gauss-9e88",
"port" : 8000,
"type" : "centralization_standard",
"ha" : {
  "consistency" : "strong",
  "replication_mode" : "sync"
},
"region" : "cn-xianhz-1",
"datastore" : {
  "type" : "GaussDB",
  "version" : "1.3",
  "complete_version" : "1.3.0",
  "complete_kernel_version" : "505.2.1"
},
"created" : "2021-01-15 01:46:40 UTC",
"updated" : "2021-01-15 02:05:03 UTC",
"volume" : {
  "type" : "ULTRAHIGH",
  "size" : 120
},
"nodes" : [ {
  "id" : "02ebf757aaf94074855f49cc6e0e4712no14",
  "name" : "gauss-9e88_root_0",
  "hostname" : "gauss-9e88_root_0",
  "role" : "master",
  "status" : "ACTIVE",
  "availability_zone" : "az2xahz",
  "private_ip" : "192.168.0.*",
  "data_ip" : "192.168.0.*",
  "management_ip" : "192.168.0.*",
  "component_names" : "cm_1:Primary, dn_6001:6001:Primary"
}, {
  "id" : "0a87b8ecbfeb46aba1409cfc0f0d5c34no14",
  "name" : "gauss-9e88_root_1",
  "hostname" : "gauss-9e88_root_1",
  "role" : "slave",
  "status" : "ACTIVE",
  "availability_zone" : "az2xahz",
  "private_ip" : "192.168.0.*",
  "data_ip" : "192.168.0.*",
  "management_ip" : "192.168.0.*",
  "component_names" : "cm_1:Primary, dn_6001:6001:Primary"
}, {
  "id" : "2d9fec1ab3834936b074d63acf48b1f2no14",
  "name" : "gauss-9e88_root_2",
  "hostname" : "gauss-9e88_root_2",
  "role" : "slave",
  "status" : "ACTIVE",
  "availability_zone" : "az2xahz",
  "private_ip" : "192.168.0.*",
  "data_ip" : "192.168.0.*",
  "management_ip" : "192.168.0.*",
  "component_names" : "cm_1:Primary, dn_6001:6001:Primary"
} ],
"private_ips" : [ "192.168.0.* / 192.168.0.* / 192.168.0.*" ],
"replica_num" : 3,
"db_user_name" : "root",
"flavor_ref" : "gaussdb.ee.bms.2xlarge.x868.ha",
"flavor_info" : {
  "vcpu" : 8,
  "mem" : 64
},
"backup_strategy" : {
  "start_time" : "19:00-20:00",
  "keep_days" : 7
},
"instance_mode" : "enterprise",
"time_zone" : "UTC+08:00"
```

```
  } ],
  "total_count" : 1
}
```

Status Code

Status Code	Description
200	Normal

Error Code

For details, see **Error Codes**.

4.11.9 (Recommended) Querying Instances or Instance Details

Function

This API is used to query instance or instance details.

Constraints

You can query the instance list by calling the API. The number of concurrent requests cannot exceed 10 QPS.

URI

GET /gaussdb/v3.2/instances

Table 4-831 Query parameters

Parameter	Mandatory	Type	Description
id	No	String	Instance ID. The asterisk (*) is reserved for the system. If the instance ID starts with *, it indicates that fuzzy match is performed based on the value following *. Otherwise, the exact match is performed based on the instance ID. The value cannot contain only asterisks (*).

Parameter	Mandatory	Type	Description
name	No	String	Instance name. The asterisk (*) is reserved for the system. If the instance name starts with *, it indicates that fuzzy match is performed based on the value following *. Otherwise, the exact match is performed based on the instance name. The value cannot contain only asterisks (*).
type	No	String	Instance type to be queried. Currently, the value is only **Combined** (case-sensitive), indicating the distributed instance. Currently, the value can be "Ha" (case sensitive), which corresponds to centralized instances. Enumerated values: ● Combined ● Ha
datastore_type	No	String	Database type. Its value is case-insensitive. ● GaussDB
offset	No	Integer	Index offset. If **offset** is set to *N*, the resource query starts from the *N*+1 piece of data. The value is **0** by default, indicating that the query starts from the first piece of data. The value must be a number but cannot be a negative number.
limit	No	Integer	Number of records to be queried. The default value is **100**. The value cannot be a negative number. The minimum value is **1** and the maximum value is **100**.

Request Parameters

Table 4-832 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Languages Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Response Parameters

Status code: 200

Table 4-833 Response body parameters

Parameter	Type	Description
instances	Array of **Table 4-834** objects	Instance information.
total_count	Integer	Total number of records.

Table 4-834 ListInstanceResult

Parameter	Type	Description
id	String	Instance ID.
name	String	Instance name.

Parameter	Type	Description
status	String	Instance status. Values: ● **BUILD**: The instance is being created. ● **BUILD_FAILED**: The instance failed to be created. ● **ACTIVE**: The instance is normal. ● **FAILED**: The instance is abnormal. ● **EXPANDING**: Read replicas or DNs are being added to the instance. ● If the value is **REBOOTING**, the instance is being rebooted. ● **REDUCING**: Read replicas are being deleted. ● **UPGRADING**: The instance is being upgraded. ● **RESTORING**: The instance is being restored. ● **SWITCHOVER**: A primary/standby switchover is being performed. ● **MIGRATING**: The instance is being migrated. ● **BACKING UP**: The instance is being backed up. ● **CHANGING DEPLOYMENT SOLUTION**: The deployment model of the instance is being changed (its replicas are being added or reduced.). ● **STORAGE FULL**: The instance storage is full. ● **REPAIRING**: The instance is being repaired. ● **SHUTDOWN**: The instance is stopped.
private_ips	Array of strings	Data IP addresses.
private_ipv6s	Array of strings	Service IPv6 addresses.
port	Integer	Database port.
type	String	Instance type. The value can be **Combined** (a distributed instance) or **centralization_standard** (a centralized instance).
ha	**Table 4-835** object	This parameter is returned for distributed and centralized instances.
replica_num	Integer	Number of replicas.

Parameter	Type	Description
datastore	**Table 4-836** object	Database information.
created	String	Creation time.
updated	String	Update time.
db_user_name	String	Default username.
flavor_ref	String	Specification code.
flavor_info	**Table 4-837** object	Flavor information.
volume	**Table 4-838** object	Volume information.
backup_strategy	**Table 4-839** object	Backup policy.
nodes	Array of **Table 4-840** objects	Instance node information.
instance_mode	String	**enterprise**: enterprise edition
time_zone	String	Time zone.

Table 4-835 ListHaResult

Parameter	Type	Description
consistency	String	Database consistency type. This parameter is available only for distributed instances. The value can be **strong** or **eventual**, indicating strong consistency and eventual consistency, respectively. Enumerated values: ● strong ● eventual
replication_mode	String	Replication mode for the standby node. The value cannot be empty. Value: **sync** (synchronization mode)
consistency_protocol	String	Replica consistency protocol. Enumerated values: ● quorum ● paxos

Table 4-836 ListDatastore

Parameter	Type	Description
type	String	DB engine.
version	String	Major DB engine version.
complete_version	String	Minor DB engine version.
complete_kernel_version	String	Kernel engine version.

Table 4-837 ListFlavorInfo

Parameter	Type	Description
vcpu	Integer	Number of vCPUs.
mem	Integer	Memory size.

Table 4-838 ListVolume

Parameter	Type	Description
type	String	Storage type.
size	Integer	Storage.

Table 4-839 OpenGaussBackupStrategyForListResponse

Parameter	Type	Description
start_time	String	Backup time window. The creation of an automated backup will be triggered during the backup time window. The current time is the UTC time.
keep_days	Integer	Number of days to retain the generated backup files. Value: **1** to **732**

Table 4-840 NodeResult

Parameter	Type	Description
id	String	Node ID.
name	String	Node name.
hostname	String	Host name.
role	String	Node type. The value can be **master** or **slave**, indicating the primary or standby node. The parameter can also be set to other values.
status	String	Node status.
availability_zone	String	AZ.
private_ip	String	Service IP address, which is used for service access.
data_ip	String	Data IP address, which is used for communication between DB instance components.
management_ip	String	Management IP address, which is used for communication between GaussDB Management Platform (TPOPS) and database nodes.
private_ipv6	String	Service IPv6 address, which is used for service access. Dual-stack access is supported.
data_ipv6	String	Data IPv6 address, which is used for communication between DB instance components.
management_ipv6	String	Management IPv6 address used for communication between TPOPS and database nodes.
components_names	String	Instance component information, which is not updated in real time.

Example Request

Querying instances based on search criteria

```
GET https://{Endpoint}/rds/gaussdb/v3.2/instances?
id=ed7cc6166ec24360a5ed5c5c9c2ed726in14&name=hy&type=Enterprise&datastore_type=GaussDB&offset=
0&limit=10
```

Example Response

Status code: 200

Success.


```
{
  "instances" : [ {
    "id" : "b331ed66cc3249f78bc20737308c01f4in14",
    "status" : "ACTIVE",
    "name" : "gauss-9e88",
    "port" : 8000,
    "type" : "centralization_standard",
    "ha" : {
      "consistency" : "strong",
      "replication_mode" : "sync"
    },
    "region" : "cn-xianhz-1",
    "datastore" : {
      "type" : "GaussDB",
      "version" : "V2.0-1.3",
      "complete_version" : "1.3.0",
      "complete_kernel_version" : "505.2.1"
    },
    "created" : "2021-01-15 01:46:40 UTC",
    "updated" : "2021-01-15 02:05:03 UTC",
    "volume" : {
      "type" : "ULTRAHIGH",
      "size" : 120
    },
    "nodes" : [ {
      "id" : "02ebf757aaf94074855f49cc6e0e4712no14",
      "name" : "gauss-9e88_root_0",
      "hostname" : "gauss-9e88_root_0",
      "role" : "master",
      "status" : "ACTIVE",
      "availability_zone" : "az2xahz",
      "private_ip" : "192.168.0.*",
      "data_ip" : "192.168.0.*",
      "management_ip" : "192.168.0.*",
      "component_names" : "cm_1:Primary, dn_6001:6001:Primary"
    }, {
      "id" : "0a87b8ecbfeb46aba1409cfc0f0d5c34no14",
      "name" : "gauss-9e88_root_1",
      "hostname" : "gauss-9e88_root_1",
      "role" : "slave",
      "status" : "ACTIVE",
      "availability_zone" : "az2xahz",
      "private_ip" : "192.168.0.*",
      "data_ip" : "192.168.0.*",
      "management_ip" : "192.168.0.*",
      "component_names" : "cm_1:Primary, dn_6001:6001:Primary"
    }, {
      "id" : "2d9fec1ab3834936b074d63acf48b1f2no14",
      "name" : "gauss-9e88_root_2",
      "hostname" : "gauss-9e88_root_2",
      "role" : "slave",
      "status" : "ACTIVE",
      "availability_zone" : "az2xahz",
      "private_ip" : "192.168.0.*",
      "data_ip" : "192.168.0.*",
      "management_ip" : "192.168.0.*",
      "component_names" : "cm_1:Primary, dn_6001:6001:Primary"
    } ],
    "private_ips" : [ "192.168.0.* / 192.168.0.* / 192.168.0.*" ],
    "replica_num" : 3,
    "db_user_name" : "root",
    "flavor_ref" : "gaussdb.ee.bms.2xlarge.x868.ha",
    "flavor_info" : {
      "vcpu" : 8,
      "mem" : 64
    },
    "backup_strategy" : {
      "start_time" : "19:00-20:00",
```

```
    "keep_days" : 7
   },
   "instance_mode" : "enterprise",
   "time_zone" : "UTC+08:00"
 } ],
 "total_count" : 1
}
```


Status Code

Status Code	Description
200	Normal

Error Code

For details, see **Error Codes**.

4.11.10 Stopping an Instance or Node

Function

This API is used to stop an instance or node.

URI

POST /gaussdb/v3/instances/{instance_id}/db-stop

Table 4-841 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-842 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Table 4-843 Request body parameters

Parameter	Mandatory	Type	Description
node_ids	Yes	String	ID of the node to be stopped. The value cannot be **null**. If this parameter is left blank, the instance is stopped.

Response Parameters

Status code: 202

Table 4-844 Response body parameters

Parameter	Type	Description
job_id	String	Job ID.

Status code: default

Table 4-845 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

- Stopping a node (**node_ids** is configured)
```
https://{Endpoint}/rds/gaussdb/v3/instances/d8e6ca5a624745bcb546a227aa3ae1cfin14/db-stop

 {
   "node_ids" : [ "a4e6ca5a624745bcb546a227aa373kcfin14" ]
 }
```
- Stoping the instance (**node_ids** is left blank)
```
https://{Endpoint}/rds/gaussdb/v3/instances/d8e6ca5a624745bcb546a227aa3ae1cfin14/db-stop

 {
   "node_ids": []
```

Example Response

Status code: 202

Success.

```
{
  "job_id" : "bf26cf3c-d046-4080-bb45-f114be7afa5f"
}
```

Status Code

Status Code	Description
202	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.11.11 Rebooting an Instance

Function

This API is used to reboot an instance.

URI

POST /gaussdb/v3/instances/{instance_id}/restart

Table 4-846 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-847 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Response Parameters

Status code: 202

Table 4-848 Response header parameter

Parameter	Type	Description
job_id	String	Job ID for rebooting an instance.

Table 4-849 Response body parameters

Parameter	Type	Description
job_id	String	Job ID.

Status code: default

Table 4-850 Response header parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

Restarting an instance

```
POST https://{Endpoint}/rds/gaussdb/v3/instances/3d39c18788b54a919bab633874c159dfin14/restart
```

Example Response

Status code: 202

Success.

```
{
  "job_id" : "bf26cf3c-d046-4080-bb45-f114be7afa5f"
}
```

Status Code

Status Code	Description
202	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.11.12 Starting an Instance or Node

Function

This API is used to start an instance or start a node.

URI

POST /gaussdb/v3/instances/{instance_id}/db-startup

Table 4-851 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-852 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Table 4-853 Request body parameter

Parameter	Mandatory	Type	Description
node_ids	Yes	String	ID of the node to be started. The value cannot be **null**. If the value is left blank, the instance is started.

Response Parameters

Status code: 202

Table 4-854 Response header parameter

Parameter	Type	Description
job_id	String	ID of the asynchronous task for starting the instance or node.

Table 4-855 Response body parameters

Parameter	Type	Description
job_id	String	Job ID.

Status code: default

Table 4-856 Response header parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Table 4-857 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

- Start a node (if **node_ids** is configured)

```
https://{Endpoint}/rds/gaussdb/v3/instances/d8e6ca5a624745bcb546a227aa3ae1cfin14/db-startup

{
  "node_ids" : [ "a4e6ca5a624745bcb546a227aa373kcfin14" ]
}
```

- Start the instance (if **node_ids** is left blank)

```
https://{Endpoint}/rds/gaussdb/v3/instances/d8e6ca5a624745bcb546a227aa3ae1cfin14/db-startup

{
  "node_ids" : [ ]
}
```

Example Response

Status code: 202

Success.

```
{
  "job_id" : "bf26cf3c-d046-4080-bb45-f114be7afa5f"
}
```


Status Code

Status Code	Description
202	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.11.13 Checking Whether Hosts Are Associated to an Instance to be Managed

Function

This API is used to check whether all hosts in an instance to be managed have been added to GaussDB Management Platform (TPOPS). All associated hosts are returned.

URI

POST /v3/instances/management/check-hosts-manageable

Request Parameters

Table 4-858 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Table 4-859 Request body parameter

Parameter	Mandatory	Type	Description
manage_ip	No	String	Management IP address. This parameter is mandatory when the host network protocol is IPv4.
manage_ipv6	No	String	Management IPv6 address. This parameter is mandatory when the host network protocol is IPv6.

Response Parameters

Status code: 200

Table 4-860 Response body parameters

Parameter	Type	Description
host_status_info	Array of **Table 4-861** objects	Instance host information.

Table 4-861 HostStatusInfo

Parameter	Type	Description
host_name	String	Host name.
manage_ip	String	Management IP address.
is_online	String	Online status, which can be: ● **added**: The host is to be managed. ● **notAdded**: The host has been managed.
host_type	String	Host type.

Status code: default

Table 4-862 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

- When the host network protocol is IPv4:

```
https://{Endpoint}/rds/v3/instances/management/check-hosts-manageable

{
  "manage_ip" : "192.168.*.*"
}
```

- When the host network protocol is IPv6:

```
https://{Endpoint}/rds/v3/instances/management/check-hosts-manageable

{
  "manage_ipv6" : "fd00:aaaa:20:cb:7a2e:5ff6:*:*"
}
```

Example Response

Status code: 200

Success.

```
{
  "host_status_info" : [ {
    "host_name" : "host-192-168-*-*",
    "manage_ip" : "192.168.*.*",
    "is_online" : "added",
    "host_type": "BMS"
  }, {
    "host_name" : "host-192-168-*-*",
    "manage_ip" : "192.168.*.*",
    "is_online" : "added",
    "host_type": "BMS"
  }, {
    "host_name" : "host-192-168-*-*",
    "manage_ip" : "192.168.*.*",
    "is_online" : "added",
    "host_type": "BMS"
  } ]
}
```

Status Code

Status Code	Description
200	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.11.14 Managing an Instance

Function

This API is used to manage an instance.

URI

POST /v3/instances/management

Request Parameters

Table 4-863 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Table 4-864 Request body parameters

Parameter	Mandatory	Type	Description
instance_name	Yes	String	Instance name.
manage_ip_list	Yes	Array of strings	Management IP addresses.
manage_ipv6_list	No	Array of strings	Management IPv6 addresses.
manage_user	Yes	String	Administrator account.
manage_userpass	Yes	String	Administrator password.
storage_device	No	**Table 4-865** object	Storage device Information.

Table 4-865 DesignStorageDeviceInfo

Parameter	Mandatory	Type	Description
device_id	Yes	String	Device ID.
device_directory	Yes	String	Path for storing backups.
real_path	Yes	String	NAS mount point path.

Parameter	Mandatory	Type	Description
is_manually_mount	Yes	Boolean	Whether NAS can be manually mounted.

Response Parameters

Status code: 200

Table 4-866 Response body parameters

Parameter	Type	Description
instanceName	String	Instance name.
instance_id	String	Instance ID.
job_id	String	Job ID.

Status code: default

Table 4-867 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

Managing a centralized instance with 3 nodes

```
https://{Endpoint}/rds/v3/instances/management

{
  "instance_name" : "instance-a",
  "manage_ip_list" : [ "192.168.*.*", "192.168.*.*", "192.168.*.*" ],
  "manage_user" : "rdsAdmin",
  "manage_userpass" : "*",
  "storage_device" : {
    "device_id" : "nas_server_1",
    "device_directory" : "/data/nas",
    "real_path" : "/home/nfs/data1",
    "is_manually_mount" : false
  }
}
```

Example Response

Status code: 200

Success.

```
{
  "instanceName": "instance-a",
  "instance_id" : "cc6fd964d93f4003851dfc29d57d30a5in14",
  "job_id" : "3bcaff3eb858e9b94ef5a3010e7f65be"
}
```


Status Code

Status Code	Description
200	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.11.15 Rectifying an AZ Fault

Function

This API is used to rectify an AZ fault by switching, forcibly starting, resuming the AZ.

URI

POST /v3/instances/{instance_id}/repair

Table 4-868 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-869 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	Selected token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Table 4-870 Request body parameters

Parameter	Mandatory	Type	Description
availability_zone_id	Yes	String	AZ ID. This parameter is optional when **operation** is set to **resume**.
operation	Yes	String	Operation type. The value can be **switch** (switching the AZ), **forceStart** (forcibly starting the AZ), or **resume** (resuming the AZ). Enumerated values: • **switch** • **forceStart** • **resume**

Response Parameters

Status code: 200

Table 4-871 Response body parameters

Parameter	Type	Description
job_id	String	Job ID.

Example Request

```
https://{Endpoint}/rds/v3/instances/{instance_id}/repair

{
  "availability_zone_id" : "az_01",
  "operation" : "switch"
}
```

Example Response

Status code: 200

Rectifying an AZ fault

```
{
  "job_id" : "66d184f0-04cf-4fae-9336-490f3ead8efb"
}
```

Status Code

Status Code	Description
200	Normal

Error Code

For details, see **Error Codes**.

4.11.16 Performing a Primary/Standby DN Switchover in an Instance Group

Function

This API is used to perform a primary/standby DN switchover in an instance group.

URI

POST /gaussdb/v3/instances/group/switchover

Request Parameters

Table 4-872 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	Selected token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Table 4-873 Request body parameters

Parameter	Mandatory	Type	Description
availability_zone	Yes	String	AZ name.

Parameter	Mandatory	Type	Description
instance_group_ids	No	Array of String	Instance group IDs. Either **instance_group_ids** or **instance_ids** must be specified. If both are specified, **instance_group_ids** takes precedence.
instance_ids	No	Array of String	Instance IDs. Either **instance_group_ids** or **instance_ids** must be specified. If both are specified, **instance_group_ids** takes precedence.

Response Parameters

Status code: 202

Table 4-874 Response body parameters

Parameter	Type	Description
job_id	String	Job ID.

Example Request

```
https://{Endpoint}/rds/gaussdb/v3/instances/group/switchover
{
  "availability_zone" : "az_01",
  "instance_group_ids" : ["e50134ce-471e-453e-a348-4838c59d2a96"],
  "instance_ids": ["3cec6a7f88974495a7af8c0732e670ddin14"]
}
```

Example Response

Status code: 202

Performing a primary/standby DN switchover in an instance group

```
{
  "job_id" : "66d184f0-04cf-4fae-9336-490f3ead8efb"
}
```

Status Codes

Status Code	Description
202	Normal

Error Codes

For details, see **Error Codes**.

4.11.17 Adding a Node

Function

This API is used to add a node.

URI

POST /v3/instances/{instance_id}/expansion

Table 4-875 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-876 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Table 4-877 Request body parameter

Parameter	Mandatory	Type	Description
expand_cluster	Yes	**Table 4-878** object	Information about the nodes to be added.

Table 4-878 Nodes

Parameter	Mandatory	Type	Description
count	Yes	Integer	How many times the nodes of an instance are increased by.
manage_ip	Yes	Array of strings	IP addresses of the nodes to be added.

Response Parameters

Status code: 202

Table 4-879 Response body parameters

Parameter	Type	Description
job_id	String	Job ID.

Example Request

Adding nodes for a three-node instance

```
POST https://{Endpoint}/rds/v3/instances/cc6fd964d93f4003851dfc29d57d30a5in14/expansion

{
  "expand_cluster" : {
    "nodes" : {
      "count" : 1,
      "manage_ip" : [ "192.168.0.*", "192.168.0.*", "192.168.0.*" ]
    }
  }
}
```

Example Response

Status code: 202

Success.

```
{
  "job_id" : "724fdb793a798-4a9c-b4dc-68e71fed1626"
}
```

Status Code

Status Code	Description
202	Normal

Error Code

For details, see **Error Codes**.

4.11.18 Replacing a Node

Function

This API is used to replace a node. You can replace a node no matter whether the IP address of the node to be replaced is the same as that of the node used for replacement.

Constraints

- When the node status is normal, you cannot replace the node.
- You can only replace a node at a time.

URI

POST /v3/instances/{instance_id}/nodes/replacement

Table 4-880 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-881 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Table 4-882 Request body parameters

Parameter	Mandatory	Type	Description
node_ids	Yes	Array of strings	IDs of nodes to be replaced.

Parameter	Mandatory	Type	Description
manage_ips	Yes	Array of strings	Host IP addresses that can be used.
is_reset_server	No	Boolean	Whether to automatically clear the host. Default value: **false**

Response Parameters

Status code: 202

Table 4-883 Response body parameters

Parameter	Type	Description
replaceMinNodeResponses	Array of **Table 4-884** objects	Response body list.

Table 4-884 ReplaceMinNodeResponse

Parameter	Type	Description
old_node_id	String	ID of the old node.
job_id	String	Job ID.

Example Request

Replacing a node

```
https://{Endpoint}/rds/v3/instances/cc6fd964d93f4003851dfc29d57d30a5in14/nodes/replacement

{
  "node_ids" : [ "27197c6ba2e743d3bdf4f8d8bf09fbfcno14" ],
  "manage_ips" : [ "192.168.*.*" ],
  "is_reset_server" : true
}
```

Example Response

Status code: 202

Success.

```
{
  "replaceMinNodeResponses" : [ {
    "old_node_id" : "27197c6ba2e743d3bdf4f8d8bf09fbfcno14",
    "job_id" : "724fdb793a798-4a9c-b4dc-68e71fed1626"
  } ]
}
```

Status Code

Status Code	Description
202	Normal

Error Code

For details, see **Error Codes**.

4.11.19 Querying Versions That an Instance Can Be Upgraded to

Function

This API is used to query versions that an instance can be upgraded to.

URI

GET /gaussdb/v3/instances/{instance_id}/db-upgrade/candidate-versions

Table 4-885 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-886 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Response Parameters

Status code: 200

Table 4-887 Response body parameters

Parameter	Type	Description
upgrade_type_list	Array of **Table 4-888** objects	Upgrade type list.
rollback_enabled	Boolean	Whether rollback is supported.
source_version	String	Source version.
target_version	String	Target version. If the current version is the latest, **null** is returned.
roll_upgrade_progress	**Table 4-890** object	Rollback progress.
upgrade_candidate_versions	Array of strings	Versions that the instance can be upgraded to.
hotfix_upgrade_candidate_versions	Array of strings	Patch versions that can be upgraded.
hotfix_rollback_candidate_versions	Array of strings	Patch versions that can be rolled back.
hotfix_upgrade_infos	Array of **Table 4-891** objects	Hot patch versions that can be upgraded.
hotfix_rollback_infos	Array of **Table 4-891** objects	Hot patch versions that can be rolled back.

Table 4-888 upgrade_type_list

Parameter	Type	Description
upgrade_type	String	Upgrade type. ● **grey**: Gray upgrade ● **inplace**: In-place upgrade ● **hotfix**: Hot patch update
upgrade_action_list	Array of **Table 4-889** objects	Upgrade action list. If the current version is the latest, **null** is returned.
enable	Boolean	Whether the upgrade action is available.

Parameter	Type	Description
is_parallel_upgrade	Boolean	Whether intra-AZ parallel upgrade is supported. ● **true**: The current instance is in the rolling upgrade phase of the gray upgrade. The intra-AZ parallel upgrade is supported. Once this parameter is configured, it cannot be changed later. ● **false**: The current instance is being upgraded. The intra-AZ parallel upgrade is not supported. Once this parameter is configured, it cannot be changed later. ● **null**: The current instance is not in the upgrade process.

Table 4-889 upgrade_action_list

Parameter	Type	Description
upgrade_action	String	Upgrade actions. If **upgrade_type** is **grey**, **upgrade**, **upgradeAutoCommit**, **commit**, and **rollback** are returned. If **upgrade_type** is **inplace**, only **upgradeAutoCommit** is returned. If **upgrade_type** is **hotfix**, only **upgradeAutoCommit** and **rollback** are returned. ● **upgrade**: Rolling upgrade ● **upgradeAutoCommit**: Auto-commit ● **commit**: Commit ● **rollback**: Rollback
enable	Boolean	Whether the upgrade type is available. ● **true**: It is available. ● **false**: It is unavailable.

Table 4-890 roll_upgrade_progress

Parameter	Type	Description
not_fully_upgraded_az	String	AZs that are not upgraded.
already_upgraded_az	String	AZ that has been upgraded.

Parameter	Type	Description
az_description_map	Map<String,String>	**key** and **value** indicate data center aliases.

Table 4-891 hotfix_info

Parameter	Type	Description
common_patch	String	Hot patch type. ● **common**: common hot patch ● **certain**: a specific hot patch
backup_sensitive	Boolean	Whether backups are perceived. ● **true**: Backups are perceived. ● **false**: Backups are not perceived.
description	String	Description.
version	String	Kernel version.

Example Request

Obtaining versions that an instance can be upgraded to

```
GET https://{Endpoint}/rds/gaussdb/v3/instances/f54427e5af55418b89e304babf47e52ain14/db-upgrade/
candidate-versions
```

Example Response

Status code: 200

Success.


```
{
  "upgrade_type_list" : [ {
    "upgrade_type" : "grey",
    "upgrade_action_list" : [ {
      "upgrade_action" : "commit",
      "enable" : false
    }, {
      "upgrade_action" : "rollback",
      "enable" : false
    }, {
      "upgrade_action" : "upgrade",
      "enable" : true
    }, {
      "upgrade_action" : "upgradeAutoCommit",
      "enable" : true
    } ],
    "is_parallel_upgrade": null,
    "enable" : true
  }, {
    "upgrade_type" : "hotfix",
    "upgrade_action_list" : null,
    "is_parallel_upgrade": null,
```

```
  "enable" : false
 }, {
  "upgrade_type" : "inplace",
  "upgrade_action_list" : [ {
    "upgrade_action" : "upgradeAutoCommit",
    "enable" : true
  } ],
  "is_parallel_upgrade": null,
  "enable" : true
 } ],
 "rollback_enabled" : false,
 "source_version" : "3.103.0",
 "target_version" : null,
 "roll_upgrade_progress" : {
  "not_fully_upgraded_az" : "az_01,az_02,az_03",
  "already_upgraded_az" : "",
  "az_description_map" : {
    "az_03" : "az_03",
    "az_02" : "az_02",
    "az_01" : "az_01"
  }
 },
 "upgrade_candidate_versions" : [ "3.200.0" ],
 "hotfix_upgrade_candidate_versions" : [ ],
 "hotfix_rollback_candidate_versions" : [ ] ,
 "hotfix_upgrade_infos": [],
 "hotfix_rollback_infos": []
}
```


Status Code

Status Code	Description
200	Normal

Error Code

For details, see **Error Codes**.

4.11.20 (Recommended) Querying Versions That an Instance Can Be Upgraded to

Function

This API is used to query versions that an instance can be upgraded to.

URI

GET /gaussdb/v3.1/instances/{instance_id}/db-upgrade/candidate-versions

Table 4-892 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-893 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Response Parameters

Status code: 200

Table 4-894 Response body parameters

Parameter	Type	Description
upgrade_type_list	Array of **Table 4-895** objects	Upgrade type list.
rollback_enabled	Boolean	Whether rollback is supported.
source_version	String	Source version.
target_version	String	Target version. If the current version is the latest, **null** is returned.
roll_upgrade_progress	**Table 4-897** object	Rollback progress.
upgrade_candidate_versions	Array of strings	Versions that the instance can be upgraded to.
hotfix_upgrade_candidate_versions	Array of strings	Patch versions that can be upgraded.
hotfix_rollback_candidate_versions	Array of strings	Patch versions that can be rolled back.
hotfix_upgrade_infos	Array of **Table 4-898** objects	Hot patch versions that can be upgraded.
hotfix_rollback_infos	Array of **Table 4-898** objects	Hot patch versions that can be rolled back.

Table 4-895 upgrade_type_list

Parameter	Type	Description
upgrade_type	String	Upgrade type. ● **grey**: Gray upgrade ● **inplace**: In-place upgrade ● **hotfix**: Hot patch update
upgrade_action_list	Array of **Table 4-896** objects	Upgrade action list. If the current version is the latest, **null** is returned.
enable	Boolean	Whether the upgrade action is available.
is_parallel_upgrade	Boolean	Whether intra-AZ parallel upgrade is supported. ● **true**: The current instance is in the rolling upgrade phase of the gray upgrade. The intra-AZ parallel upgrade is supported. Once this parameter is configured, it cannot be changed later. ● **false**: The current instance is being upgraded. The intra-AZ parallel upgrade is not supported. Once this parameter is configured, it cannot be changed later. ● **null**: The current instance is not in the upgrade process.

Table 4-896 upgrade_action_list

Parameter	Type	Description
upgrade_action	String	Upgrade actions. If **upgrade_type** is **grey**, **upgrade**, **upgradeAutoCommit**, **commit**, and **rollback** are returned. If **upgrade_type** is **inplace**, only **upgradeAutoCommit** is returned. If **upgrade_type** is **hotfix**, only **upgradeAutoCommit** and **rollback** are returned. ● **upgrade**: Rolling upgrade ● **upgradeAutoCommit**: Auto-commit ● **commit**: Commit ● **rollback**: Rollback

Parameter	Type	Description
enable	Boolean	Whether the upgrade type is available. ● **true**: It is available. ● **false**: It is unavailable.

Table 4-897 roll_upgrade_progress

Parameter	Type	Description
not_fully_upgraded_az	String	AZs that are not upgraded.
already_upgraded_az	String	AZ that has been upgraded.
az_description_map	Map<String,String>	**key** and **value** indicate data center aliases.

Table 4-898 hotfix_info

Parameter	Type	Description
common_patch	String	Hot patch type. ● **common**: common hot patch ● **certain**: a specific hot patch
backup_sensitive	Boolean	Whether backups are perceived. ● **true**: Backups are perceived. ● **false**: Backups are not perceived.
description	String	Description.
version	String	Kernel version.

Example Request

Obtaining versions that an instance can be upgraded to

```
GET https://{Endpoint}/rds/gaussdb/v3.1/instances/f54427e5af55418b89e304babf47e52ain14/db-upgrade/candidate-versions
```

Example Response

Status code: 200

Success.

```
{
  "upgrade_type_list" : [ {
```

```
  "upgrade_type" : "grey",
  "upgrade_action_list" : [ {
    "upgrade_action" : "commit",
    "enable" : false
  }, {
    "upgrade_action" : "rollback",
    "enable" : false
  }, {
    "upgrade_action" : "upgrade",
    "enable" : true
  }, {
    "upgrade_action" : "upgradeAutoCommit",
    "enable" : true
  } ],
  "is_parallel_upgrade": null,
  "enable" : true
}, {
  "upgrade_type" : "hotfix",
  "upgrade_action_list" : null,
  "is_parallel_upgrade": null,
  "enable" : false
}, {
  "upgrade_type" : "inplace",
  "upgrade_action_list" : [ {
    "upgrade_action" : "upgradeAutoCommit",
    "enable" : true
  } ],
  "is_parallel_upgrade": null,
  "enable" : true
} ],
"rollback_enabled" : false,
"source_version" : "V2.0-3.103.0",
"target_version" : null,
"roll_upgrade_progress" : {
  "not_fully_upgraded_az" : "az_01,az_02,az_03",
  "already_upgraded_az" : "",
  "az_description_map" : {
    "az_03" : "az_03",
    "az_02" : "az_02",
    "az_01" : "az_01"
  }
},
"upgrade_candidate_versions" : [ "V2.0-3.200.0" ],
"hotfix_upgrade_candidate_versions" : [ ],
"hotfix_rollback_candidate_versions" : [ ],
"hotfix_upgrade_infos": [],
"hotfix_rollback_infos": []
}
```


Status Code

Status Code	Description
200	Normal

Error Code

For details, see **Error Codes**.

4.11.21 Upgrading an Instance

Function

This API is used to upgrade an instance.

URI

PUT /gaussdb/v3.1/instances/{instance_id}/db-upgrade

Table 4-899 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-900 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	Token for identity authentication.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Table 4-901 Request body parameters

Parameter	Mandatory	Type	Description
target_version	Yes	String	Target version.

Parameter	Mandatory	Type	Description
upgrade_action	No	String	Upgrade action. The value can be **upgrade**, **commit**, **rollback**, or **upgradeAutoCommit**. This parameter is mandatory when **upgrade_type** is **grey**. When **upgrade_type** is **inplace**, this parameter can only be set to **upgradeAutoCommit**. When **upgrade_type** is **hotfix**, this parameter can only be set to **upgradeAutoCommit** or **rollback**.
upgrade_type	Yes	String	Upgrade type (Value: **inplace**, **grey**, or **hotfix**).
upgrade_az	No	String	AZ to be upgraded. Multiple AZs must be separated by commas (,). This parameter is mandatory when upgrade action is set to **upgrade**.

Response Parameters

Status code: 202

Table 4-902 Response body parameters

Parameter	Type	Description
job_id	String	Job ID.

Example Request

- In-place upgrade

```
PUT https://{Endpoint}/rds/gaussdb/v3.1/instances/a4e6ca5a624745bcb546a227aa373kcfin14/db-upgrade
{
  "target_version" : "3.200.0",
  "upgrade_type" : "inplace"
}
```

- Gray upgrade: Performing a rolling upgrade

```
PUT https://{Endpoint}/rds/gaussdb/v3.1/instances/a4e6ca5a624745bcb546a227aa373kcfin14/db-upgrade
{
  "target_version" : "3.200.0",
  "upgrade_action" : "upgrade",
  "upgrade_az" : "az_02,az_01",
```

```
  "upgrade_type" : "grey"
}
```

- Gray upgrade: Rolling back the upgrade
```
PUT https://{Endpoint}/rds/gaussdb/v3.1/instances/a4e6ca5a624745bcb546a227aa373kcfin14/db-upgrade
{
  "target_version" : "3.200.0",
  "upgrade_action" : "rollback",
  "upgrade_type" : "grey"
}
```
- Gray upgrade: Committing the upgrade
```
PUT https://{Endpoint}/rds/gaussdb/v3.1/instances/a4e6ca5a624745bcb546a227aa373kcfin14/db-upgrade
{
  "target_version" : "3.200.0",
  "upgrade_action" : "commit",
  "upgrade_type" : "grey"
}
```
- Gray upgrade: Auto-committing the upgrade
```
PUT https://{Endpoint}/rds/gaussdb/v3.1/instances/a4e6ca5a624745bcb546a227aa373kcfin14/db-upgrade
{
  "target_version" : "3.200.0",
  "upgrade_action" : "upgradeAutoCommit",
  "upgrade_type" : "grey"
}
```
- Hot patch upgrade: Committing an upgrade
```
PUT https://{Endpoint}/rds/gaussdb/v3.1/instances/a4e6ca5a624745bcb546a227aa373kcfin14/db-upgrade
{
  "target_version" : "8.102.0.1",
  "upgrade_action" : "upgradeAutoCommit",
  "upgrade_type" : "hotfix"
}
```
- Hot patch upgrade: Rolling back an upgrade
```
PUT https://{Endpoint}/rds/gaussdb/v3.1/instances/a4e6ca5a624745bcb546a227aa373kcfin14/db-upgrade
{
  "target_version" : "8.102.0.1",
  "upgrade_action" : "rollback",
  "upgrade_type" : "hotfix"
}
```

Example Response

Status code: 202

Response for successful operation

```
{
  "job_id" : "bf26cf3c-d046-4080-bb45-f114be7afa5f"
}
```

Status Code

Status Code	Description
202	Normal

Error Code

For details, see **Error Codes**.

4.11.22 (Recommended) Upgrading an Instance

Function

This API is used to upgrade an instance.

URI

PUT /gaussdb/v3.2/instances/{instance_id}/db-upgrade

Table 4-903 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-904 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	Token for identity authentication.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Table 4-905 Request body parameters

Parameter	Mandatory	Type	Description
target_version	Yes	String	Target version.

Parameter	Mandatory	Type	Description
upgrade_action	No	String	Upgrade action. The value can be **upgrade**, **commit**, **rollback**, or **upgradeAutoCommit**. This parameter is mandatory when **upgrade_type** is **grey**. When **upgrade_type** is **inplace**, this parameter can only be set to **upgradeAutoCommit**. When **upgrade_type** is **hotfix**, this parameter can only be set to **upgradeAutoCommit** or **rollback**.
upgrade_type	Yes	String	Upgrade type (Value: **inplace**, **grey**, or **hotfix**).
upgrade_az	No	String	AZ to be upgraded. Multiple AZs must be separated by commas (,). This parameter is mandatory when upgrade action is set to **upgrade**.

Response Parameters

Status code: 202

Table 4-906 Response body parameters

Parameter	Type	Description
job_id	String	Task ID.

Example Request

- In-place upgrade
```
PUT https://{Endpoint}/rds/gaussdb/v3.2/instances/a4e6ca5a624745bcb546a227aa373kcfin14/db-upgrade
{
  "target_version" : "V2.0-3.200.0",
  "upgrade_type" : "inplace"
}
```
- Gray upgrade: Performing a rolling upgrade
```
PUT https://{Endpoint}/rds/gaussdb/v3.2/instances/a4e6ca5a624745bcb546a227aa373kcfin14/db-upgrade
{
  "target_version" : "V2.0-3.200.0",
  "upgrade_action" : "upgrade",
  "upgrade_az" : "az_02,az_01",
```

```
  "upgrade_type" : "grey"
}
```

- Gray upgrade: Rolling back the upgrade
```
PUT https://{Endpoint}/rds/gaussdb/v3.2/instances/a4e6ca5a624745bcb546a227aa373kcfin14/db-upgrade
{
  "target_version" : "V2.0-3.200.0",
  "upgrade_action" : "rollback",
  "upgrade_type" : "grey"
}
```
- Gray upgrade: Committing the upgrade
```
PUT https://{Endpoint}/rds/gaussdb/v3.2/instances/a4e6ca5a624745bcb546a227aa373kcfin14/db-upgrade
{
  "target_version" : "V2.0-3.200.0",
  "upgrade_action" : "commit",
  "upgrade_type" : "grey"
}
```
- Gray upgrade: Auto-committing the upgrade
```
PUT https://{Endpoint}/rds/gaussdb/v3.2/instances/a4e6ca5a624745bcb546a227aa373kcfin14/db-upgrade
{
  "target_version" : "V2.0-3.200.0",
  "upgrade_action" : "upgradeAutoCommit",
  "upgrade_type" : "grey"
}
```
- Hot patch upgrade: Committing an upgrade
```
PUT https://{Endpoint}/rds/gaussdb/v3.2/instances/a4e6ca5a624745bcb546a227aa373kcfin14/db-upgrade
{
  "target_version" : "V2.0-8.102.0.1",
  "upgrade_action" : "upgradeAutoCommit",
  "upgrade_type" : "hotfix"
}
```
- Hot patch upgrade: Rolling back an upgrade
```
PUT https://{Endpoint}/rds/gaussdb/v3.2/instances/a4e6ca5a624745bcb546a227aa373kcfin14/db-upgrade
{
  "target_version" : "V2.0-8.102.0.1",
  "upgrade_action" : "rollback",
  "upgrade_type" : "hotfix"
}
```

Example Response

Status code: 202

Response for successful operation

```
{
  "job_id" : "bf26cf3c-d046-4080-bb45-f114be7afa5f"
}
```

Status Code

Status Code	Description
202	Normal

Error Code

For details, see **Error Codes**.

4.11.23 Repairing a Node

Function

This API is used to repair a node.

Constraints

If some components on a node in an instance are faulty, you can reinstall the faulty components on the node. Constraints:

- To repair a node, you need to ensure that the node is in the abnormal state and a component on the node is in the abnormal state.
- Single-replica instance nodes cannot be repaired.

URI

POST /gaussdb/v3/instances/{instance_id}/nodes/repair

Table 4-907 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-908 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Table 4-909 Request body parameter

Parameter	Mandatory	Type	Description
node_ids	Yes	Array of strings	IDs of nodes to be repaired. The number of IDs is from 1 to 3.

Response Parameters

Status code: 202

Table 4-910 Response body parameters

Parameter	Type	Description
node_id	String	Node to be repaired.
job_id	String	Job ID.
error_code	String	Error code.
error_msg	String	Error message.

Example Request

```
https://{Endpoint}/rds/gaussdb/v3/instances/3393fce85e11478dbd505ae903118ac6in14/nodes/repair

{
  "node_ids" : [ "3393fce85e11478dbd505ae903118ac6no14" ]
}
```

Example Response

Status code: 202

Accepted

```
{
  "repair_nodes" : [ {
    "node_id" : "3393fce85e11478dbd505ae903118ac6no14",
    "job_id" : "c31055d0-1986-454f-b0f0-fac2cb3f3bf9"
  } ]
}
```

Status Code

Status Code	Description
202	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.11.24 Changing a Deployment Model

Function

This API is used to change the deployment model of a centralized instance.

URI

PUT /gaussdb/v3/instances/{instance_id}/solution

Table 4-911 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-912 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Table 4-913 Request body parameters

Parameter	Mandatory	Type	Description
solution	Yes	String	Deployment models that the instance can be changed to. ● **triset**: 3 nodes (1 primary + 2 standby) ● **pentaset**: 5 nodes (1 primary + 4 standby) ● **trisetnoetcd**: 3 nodes (1 primary + 2 standby, without ETCD) ● **pentasetnoetcd**: 5 nodes (1 primary + 4 standby, without ETCD) **triset** and **pentaset** can be changed to each other. **trisetnoetcd** and **pentasetnoetcd** can be changed to each other. Enumerated values: ● **triset** ● **pentaset** ● **trisetnoetcd** ● **pentasetnoetcd**
manage_ip	No	Array of strings	Host management IP addresses.
node_id	No	Array of strings	IDs of nodes to be deleted.

Response Parameters

Status code: 202

Table 4-914 Response body parameters

Parameter	Type	Description
job_id	String	Job ID.

Status code: default

Table 4-915 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

- Changing a 5-node centralized instance to a 3-node centralized instance

```
put https://{Endpoint}/rds/gaussdb/v3/instances/3393fce85e11478dbd505ae903118ac6in14/solution

 {
   "solution" : "triset",
   "node_id" : [ "50aabb8339a344d085d9dfff7e2f6835no14",
"ff1e3f175caf4d56a9156d2194c8acbbno14" ]
 }
```

- Changing a 3-node (1 primary + 2 standby) centralized instance to a 5-node centralized instance

```
put https://{Endpoint}/rds/gaussdb/v3/instances/3393fce85e11478dbd505ae903118ac6in14/solution

 {
   "solution" : "pentaset",
   "manage_ip" : [ "100.*.*.1", "100.*.*.2" ]
 }
```

Example Response

Status code: 202

Accepted

```
{
  "job_id" : "0b363a99bf863c28a05f7e90d642b265"
}
```

Status Code

Status Code	Description
202	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.11.25 Enabling Port for M Compatibility

Function

This API is used to enable port for M compatibility of a specified instance.

Constraints

The port number cannot be empty when the function is enabled.

URI

POST /gaussdb/v3/instances/{instance_id}/mysql-compatibility

Table 4-916 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-917 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Table 4-918 Request body parameters

Parameter	Mandatory	Type	Description
port	No	String	Port for M compatibility. Value range: **1024** to **39989** The following ports cannot be used: **2378**, **2379**, **2380**, **2400**, **4999**, **5000**, **5001**, **5100**, **5500**, **5999**, **6000**, **6001**, **6009**, **6010**, **6500**, **8015**, **8097**, **8098**, **8181**, **9090**, **9100**, **9180**, **9187**, **9200**, **12016**, **12017**, **20049**, **20050**, **21731**, **21732**, **32122**, **32123**, **32124**, **32125**, **32126**, **39001**, [**database port**, **database port +10**] This port and the external port of the database are mutually exclusive.

Response Parameters

Status code: 202

Table 4-919 Response body parameters

Parameter	Type	Description
job_id	String	Job ID.

Status code: default

Table 4-920 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

Enabling port for M compatibility

```
POST https://{Endpoint}/rds/gaussdb/v3/instances/d8e6ca5a624745bcb546a227aa3ae1cfin14/mysql-compatibility

{
  "port" : 3306
}
```

Example Response

Status code: 202

Success

```
{
  "job_id" : "9cb66b27111669609799e022d08d6c3a"
}
```

Status Code

Status Code	Description
202	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.11.26 Modifying or Disabling Port for M Compatibility

Function

This API is used to modify the port for M compatibility of a specified instance.

Constraints

When you modify or disable the port for M compatibility, the parameter **port** must be specified.

URI

PUT /gaussdb/v3/instances/{instance_id}/mysql-compatibility

Table 4-921 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-922 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Table 4-923 Request body parameters

Parameter	Mandatory	Type	Description
port	No	String	Port for M compatibility. This parameter is mandatory when you change the port. If this parameter is left blank, the original port is used by default. The value can be **0** or **1024** to **39989**. If the value is **0**, M compatibility port is disabled. The following ports cannot be used: **2378**, **2379**, **2380**, **2400**, **4999**, **5000**, **5001**, **5100**, **5500**, **5999**, **6000**, **6001**, **6009**, **6010**, **6500**, **8015**, **8097**, **8098**, **8181**, **9090**, **9100**, **9180**, **9187**, **9200**, **12016**, **12017**, **20049**, **20050**, **21731**, **21732**, **32122**, **32123**, **32124**, **32125**, **32126**, **39001**, [**database port**, **database port +10**] This port and the external port of the database are mutually exclusive.

Response Parameters

Status code: 202

Table 4-924 Response body parameters

Parameter	Type	Description
job_id	String	Job ID.

Status code: default

Table 4-925 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

Modifying port for M compatibility

```
PUT https://{Endpoint}/rds/gaussdb/v3/instance/e73893ef73754465a8bd2e0857bbf13ein14/mysql-compatibility

{
  "port" : 3308
}
```

Example Response

Status code: 202

Success

```
{
  "job_id" : "9cb66b27111669609799e022d08d6c3a"
}
```

Status Code

Status Code	Description
202	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.11.27 Preparing for Changing Specifications

Function

This API is used to prepare for changing specifications.

Constraints

This API is used to adjust the specifications before the scale-in.

URI

POST /gaussdb/v3/instances/{instance_id}/flavor-resize/before-execute

Table 4-926 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-927 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	Yes	String	Language. Default value: **en-us** Value: • **zh-cn** • **en-us**

Table 4-928 Request body parameters

Parameter	Mandatory	Type	Description
flavor_resizes	No	Array of **Table 4-929** objects	New specifications

Table 4-929 FlavorResize

Parameter	Mandatory	Type	Description
resize_type	No	String	Operation type. Value: • **cpu** • **memory**
target_value	No	String	Target value.

Response Parameters

Status code: 202

Table 4-930 Response body parameters

Parameter	Type	Description
job_id	String	job_id

Example Request

```
POST https://{Endpoint}/rds/gaussdb/v3/instances/ced77e4aaa8144cbbe7020b17962213ain14/flavor-resize/before-execute

{
  "flavor_resizes" : [ {
    "resize_type" : "cpu",
    "target_value" : "8"
  }, {
    "resize_type" : "memory",
    "target_value" : "64"
  } ]
}
```

Example Response

Status code: 202

OK

```
{
  "job_id" : "d3eef567-4be6-424c-860f-75684457bf72"
}
```

Status Codes

Status Code	Description
202	OK

Error Code

For details, see **Error Codes**.

4.11.28 Performing Post-Processing After Changing Specifications

Function

This API is used to perform post-change processing.

Constraints

In the scale-in scenario, ensure that the specifications change parameters transferred after the API is called are the same as those before the API is called.

URI

POST /gaussdb/v3/instances/{instance_id}/flavor-resize/after-execute

Table 4-931 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-932 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	Yes	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Table 4-933 Request body parameters

Parameter	Mandatory	Type	Description
flavor_resizes	No	Array of objects in Table 18-4	New specifications

Table 4-934 FlavorResize

Parameter	Mandatory	Type	Description
resize_type	No	String	Change type Enumerated values: ● **cpu** ● **memory**
target_value	No	String	Target value

Response Parameters

Status code: 202

Table 4-935 Response body parameters

Parameter	Type	Description
job_id	String	job_id

Example Request

```
POST https://{Endpoint}/rds/gaussdb/v3/instances/ced77e4aaa8144cbbe7020b17962213ain14/flavor-resize/
after-execute

{
  "flavor_resizes" : [ {
    "resize_type" : "cpu",
    "target_value" : "8"
  }, {
    "resize_type" : "memory",
    "target_value" : "64"
  } ]
}
```

Example Response

Status code: 202

OK

```
{
  "job_id" : "d3eef567-4be6-424c-860f-75684457bf72"
}
```

Status Codes

Status Code	Description
202	OK

Error Code

For details, see **Error Codes**.

4.11.29 Changing Disk Size

Function

This API is used to change disk size.

URI

POST /gaussdb/v3/instances/{instance_id}/volume-resize

Table 4-936 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-937 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	Yes	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Table 4-938 Request body parameters

Parameter	Mandatory	Type	Description
target_size	No	Integer	Storage space, which must always be an integral multiple of (Number of shards × 40 GB). Its value ranges from (Number of shards × 40 GB) to (Number of shards × 16 TB).

Response Parameters

None

Example Request

```
POST https://{Endpoint}/rds/gaussdb/v3/instances/ced77e4aaa8144cbbe7020b17962213ain14/volume-resize

{
  "target_size" : 300
}
```

Example Response

None

Status Codes

Status Code	Description
200	OK

Error Code

For details, see **Error Codes**.

4.11.30 Querying the Advanced Features of an Instance

Function

This API is used to query advanced features of an instance.

URI

GET /gaussdb/v3/instances/{instance_id}/advance-features

Table 4-939 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-940 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language.

Response Parameters

Status code: 200

Table 4-941 Response body parameters

Parameter	Type	Description
features	Array of **Table 4-942**	Advanced features.

Table 4-942 FeatureResult

Parameter	Type	Description
name	String	Feature name.
status	String	Whether the feature is enabled.
description	String	Feature description.
type	String	Feature value type.
value	String	Feature value.
range	String	Feature value range.
range_description	String	Feature scope description.

Status code: default

Table 4-943 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

Querying the advanced features of the current instance and checking whether each feature is enabled

```
GET https://{endpoint}/rds/gaussdb/v3/instances/e73893ef73754465a8bd2e0857bbf13ein14/advance-features
```

Example Response

Status code: 200

Success.

```
{
    "features": [
        {
            "name": "vector",
            "status": "on",
            "description": "Vector database",
            "type": "boolean",
            "range": "on|off",
            "value": "on",
            "range_description": "Specifies whether vector indexes can be created and whether vector indexes can be added, modified, and queried."
        }
    ]
}
```

Status Codes

Status Code	Description
200	Success.
default	Client or server error.

Error Code

For details, see **Error Codes**.

4.11.31 Enabling Advanced Features

Function

This API is used to enable advanced features.

URI

POST /gaussdb/v3/instances/{instance_id}/advance-features

Table 4-944 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-945 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language.

Table 4-946 Request body parameters

Parameter	Mandatory	Type	Description
params	Yes	Object	Features to be modified. The parameter value object is in the **Map<String,String>** format.

Response Parameters

Status code: 202

Table 4-947 Response body parameters

Parameter	Type	Description
workflowId	String	Job ID.

Status code: default

Table 4-948 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

Enabling advanced features

```
POST https://{endpoint}/rds/gaussdb/v3/instances/e73893ef73754465a8bd2e0857bbf13ein14/advance-features

{
  "params" : {
    "vector" : "on"
  }
}
```

Example Response

Status code: 200

Success.

```
{
   "workflowId": "884492f64f1748548ef9b9af66204ff9in29"
}
```

Status Codes

Status Code	Description
202	Success.
default	Client or server error.

Error Code

For details, see **Error Codes**.

4.11.32 Changing the Description of an Instance

Function

This API is used to change the description of an instance.

Constraints

The instance ID cannot be empty.

URI

PUT /gaussdb/v3/instances/{instance_id}/alias

Table 4-949 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-950 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Value: ● **zh-cn** ● **en-us**

Table 4-951 Request body parameters

Parameter	Mandatory	Type	Description
alias	Yes	String	The instance information to modify.

Response Parameters

Status code: 400

Table 4-952 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Status code: 500

Table 4-953 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

Changing the description of an instance

```
PUT https://{Endpoint}/rds/gaussdb/v3/instances/720fe600d21b44828303a95b4f69573cin14/alias
{
  "alias": "instance description"
}
```

Example Response

Status code: 200

API operation succeeded.

```
{ }
```

Status code: 400

API modification failed.

```
{
    "error_code": "DBS.06010019",
```

```
  "error_msg": "This is the same as the instance remarks of the current instance."
}
```


Status code: 500


```
{
  "error_code": "DBS.280005",
  "error_msg": "Server failure."
}
```


Status Codes

Status Code	Description
200	Normal
400	Duplicate instance description.
500	Longer than the upper limit of the instance description length.

Error Code

For details, see **Error Codes**.

4.12 Engine Versions and Specifications

4.12.1 Querying DB Engine Versions

Function

This API is used to query DB engine versions.

URI

GET /gaussdb/v3/datastore/versions

Request Parameters

Table 4-954 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Response Parameters

Status code: 200

Table 4-955 Response body parameters

Parameter	Type	Description
versions	Array of strings	DB version number.

Example Request

Querying DB engine versions

```
GET https://{Endpoint}/rds/gaussdb/v3/datastore/versions
```

Example Response

Status code: 200

Success.

```
{
  "versions" : [ 1.0, 1.4, 2.0 ]
}
```

Status Code

Status Code	Description
200	Normal

Error Code

For details, see **Error Codes**.

4.12.2 (Recommended) Querying DB Engine Versions

Function

This API is used to query DB engine versions.

URI

GET /gaussdb/v3.1/datastore/versions

Request Parameters

Table 4-956 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Response Parameters

Status code: 200

Table 4-957 Response body parameters

Parameter	Type	Description
versions	Array of strings	DB version number.

Example Request

Querying DB engine versions

```
GET https://{Endpoint}/rds/gaussdb/v3.1/datastore/versions
```

Example Response

Status code: 200

Success.

```
{
  "versions" : [ V2.0-1.0, V2.0-1.4, V2.0-2.0 ]
}
```

Status Code

Status Code	Description
200	Normal

Error Code

For details, see **Error Codes**.

4.13 Microservice Upgrade

4.13.1 Querying Instance Agents

Function

This API is used to query instance Agents.

URI

GET /v3/instances

Table 4-958 Query parameters

Parameter	Mandatory	Type	Description
offset	No	Integer	Index offset. If **offset** is set to *N*, the resource query starts from the *N*+1 piece of data. The value is **0** by default, indicating that the query starts from the first piece of data. The value must be a number but cannot be a negative number.
limit	No	Integer	Number of records to be queried. The default value is **10**. The value cannot be a negative number. The minimum value is **1** and the maximum value is **100**.
instance_id	No	String	Instance ID, which is used as a filter criterion for querying instances.

Parameter	Mandatory	Type	Description
instance_status	No	String	Instance status, which is used as the filter criteria for querying instances. Value: • **normal** • **abnormal** • **creating** • **createfail** • **managing** • **manage_failed** • **data_disk_full** • **shutdown**
engine_name	No	String	Engine name, which is used as a filter criterion for querying instances.
engine_version	No	String	Engine version, which is used as the filter criteria for querying instances.
upgrade_status	No	String	Upgrade status, which is used as a filter criterion for querying instances. The value can be **upgraded** (upgraded) or **unupgraded** (not upgraded).

Request Parameters

Table 4-959 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	Request token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Response Parameters

Status code: 200

Table 4-960 Response body parameters

Parameter	Type	Description
total_count	Integer	Number of instances.
instances	Array of **Table 4-961** objects	Queried instance information.

Table 4-961 Instance

Parameter	Type	Description
id	String	Instance ID.
name	String	Instance name.
mode	String	Instance mode.
created_at	String	Creation time.
engine_name	String	Engine name.
engine_version	String	DB engine version.
instance_status	String	Instance status. Value: • **normal** • **abnormal** • **creating** • **createfail** • **managing** • **manage_failed** • **data_disk_full** • **shutdown**
agent_info	**Table 4-962** object	Agent status.

Table 4-962 AgentInfo

Parameter	Type	Description
upgrade_status	String	Upgrade status.
current_version	String	Current Agent version of the instance.

Parameter	Type	Description
latest_version	String	Agent version that the instance will be upgraded to.
last_version	String	Agent version that the instance will be rolled back to.

Status code: 400

Table 4-963 Response body parameters

Parameter	Type	Description
error_code	String	Error code. Minimum length: 8 characters Maximum length: 36 characters
error_msg	String	Error message. Minimum length: 2 characters Maximum length: 512 characters

Example Request

```
GET https://{Endpoint}/rds/v3/instances
```

Example Response

Status code: 200

Success.


```
{
  "total_count" : 1,
  "instances" : [ {
    "id" : "712f425585ab42739c31ef32874b35b0in14",
    "name" : "gaussdbv5_mini_lina",
    "mode" : "Ha",
    "created_at" : "2023-08-09 11:58:22.652",
    "engine_name" : "gaussdbv5",
    "engine_version" : "1.3.1",
    "instance_status" : "normal",
    "agent_info" : {
      "upgrade_status" : "Completed",
      "current_version" : "gaussdbv5_agent_2.23.07.260",
      "latest_version" : "gaussdbv5_agent_24.1.30",
      "last_version" : "gaussdbv5_agent_2.23.07.210"
    }
  } ]
}
```

Status Code

Status Code	Description
200	Normal
400	Abnormal

Error Code

For details, see **Error Codes**.

4.13.2 (Recommended) Querying Instance Agents

Function

This API is used to query instance Agents.

URI

GET /v3.2/instances

Table 4-964 Query parameters

Parameter	Mandatory	Type	Description
offset	No	Integer	Index offset. If **offset** is set to *N*, the resource query starts from the *N*+1 piece of data. The value is **0** by default, indicating that the query starts from the first piece of data. The value must be a number but cannot be a negative number.
limit	No	Integer	Number of records to be queried. The default value is **10**. The value cannot be a negative number. The minimum value is **1** and the maximum value is **100**.
instance_id	No	String	Instance ID, which is used as a filter criterion for querying instances.

Parameter	Mandatory	Type	Description
instance_status	No	String	Instance status, which is used as the filter criteria for querying instances. Value: • **normal** • **abnormal** • **creating** • **createfail** • **managing** • **manage_failed** • **data_disk_full** • **shutdown**
engine_name	No	String	Engine name, which is used as a filter criterion for querying instances.
engine_version	No	String	Engine version, which is used as the filter criteria for querying instances.
upgrade_status	No	String	Upgrade status, which is used as a filter criterion for querying instances. The value can be **upgraded** (upgraded) or **unupgraded** (not upgraded).

Request Parameters

Table 4-965 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	Request token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Response Parameters

Status code: 200

Table 4-966 Response body parameters

Parameter	Type	Description
total_count	Integer	Total number of records.
instances	Array of **Table 4-967** objects	Queried instance information.

Table 4-967 Instance

Parameter	Type	Description
id	String	Instance ID.
name	String	Instance name.
mode	String	Instance mode.
created_at	String	Creation time.
engine_name	String	Engine name.
engine_version	String	DB engine version.
instance_status	String	Instance status. Value: • **normal** • **abnormal** • **creating** • **createfail** • **managing** • **manage_failed** • **data_disk_full** • **shutdown**
agent_info	**Table 4-968** object	Agent status.

Table 4-968 AgentInfo

Parameter	Type	Description
upgrade_status	String	Upgrade status.
current_version	String	Current Agent version of the instance.

Parameter	Type	Description
latest_version	String	Agent version that the instance will be upgraded to.
last_version	String	Agent version that the instance will be rolled back to.

Status code: 400

Table 4-969 Response body parameters

Parameter	Type	Description
error_code	String	Error Code Minimum length: 8 characters Maximum length: 36 characters
error_msg	String	Error message. Minimum length: 2 characters Maximum length: 512 characters

Example Request

```
GET https://{Endpoint}/rds/v3.2/instances
```

Example Response

Status code: 200

Success.


```
{
  "total_count" : 1,
  "instances" : [ {
    "id" : "712f425585ab42739c31ef32874b35b0in14",
    "name" : "gaussdbv5_mini_lina",
    "mode" : "Ha",
    "created_at" : "2023-08-09 11:58:22.652",
    "engine_name" : "gaussdbv5",
    "engine_version" : "V2.0-1.3.1",
    "instance_status" : "normal",
    "agent_info" : {
      "upgrade_status" : "Completed",
      "current_version" : "gaussdbv5_agent_2.23.07.260",
      "latest_version" : "gaussdbv5_agent_24.1.30",
      "last_version" : "gaussdbv5_agent_2.23.07.210"
    }
  } ]
}
```

Status Code

Status Code	Description
200	Normal
400	Abnormal

Error Code

For details, see **Error Codes**.

4.13.3 Querying Agent Information of a Specified Instance

Function

This API is used to query the Agent information of a specified instance.

URI

GET /gaussdb/v3/instances/{instance_id}/agent/status

Table 4-970 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-971 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	Request token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Response Parameters

Status code: 200

Table 4-972 Response body parameters

Parameter	Type	Description
node_list	Array of **Table 4-973** objects	Node information of a specified instance.

Table 4-973 Node

Parameter	Type	Description
node_id	String	Node ID.
node_name	String	Node name.
node_status	String	Node status.
agent_status	String	Agent status.
role	String	Role.
agent_info	**Table 4-974** object	Agent status.

Table 4-974 AgentInfo

Parameter	Type	Description
upgrade_status	String	Upgrade status.
current_version	String	Current Agent version of the node.
latest_version	String	Agent version that the node will be upgraded to.
last_version	String	Agent version that the node will be rolled back to.

Status code: 400

Table 4-975 Response body parameters

Parameter	Type	Description
error_code	String	Error code. Minimum length: 8 characters Maximum length: 36 characters

Parameter	Type	Description
error_msg	String	Error message. Minimum length: 2 characters Maximum length: 512 characters

Example Request

```
GET https://{Endpoint}/rds/gaussdb/v3/instances/4c7d8351a6bb4553a2e3a782d365d53bin14/agent/status
```

Example Response

Status code: 200

Success.


```
{
 "node_list": [{
  "node_id": "408a97321e0a4c168653b7eb87f67d8dno14",
  "node_name": "gaussdbv5_mini_lina_root_2",
  "node_status": "normal",
  "agent_status": "WORKING",
  "role": "master",
  "agent_info": {
   "upgrade_status": "Completed",
   "current_version": "gaussdbv5_agent_2.23.07.260",
   "latest_version": "gaussdbv5_agent_24.1.30",
   "last_version": "gaussdbv5_agent_2.23.07.200"
  }
 }]
}
```


Status Code

Status Code	Description
200	Normal
400	Abnormal

Error Code

For details, see **Error Codes**.

4.13.4 Upgrading the Agent of a Single Instance

Function

This API is used to upgrade the Agent of a single instance.

Constraints

- The status of each instance node is normal.
- The instance whose Agent is to be upgraded must be monitored by HA.
- The HA and Agent can communicate with each other.
- The Agent version to be upgraded must be different from the target Agent version.
- The installation package corresponding to the target Agent version has been uploaded to the SFTP server using the installation package management function.

URI

POST /gaussdb/v3/instances/{instance_id}/agent/upgrade

Table 4-976 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID. Minimum length: 32 characters Maximum length: 36 characters

Request Parameters

Table 4-977 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	Request token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Response Parameters

Status code: 200

Table 4-978 Response body parameters

Parameter	Type	Description
job_id	String	Job ID.

Status code: 400

Table 4-979 Response body parameters

Parameter	Type	Description
error_code	String	Error code. Minimum length: 8 characters Maximum length: 36 characters
error_msg	String	Error message. Minimum length: 2 characters Maximum length: 512 characters

Example Request

```
POST https://{Endpoint}/rds/gaussdb/v3/instances/725398ddd0dd41d88a609acf137698e5in14/agent/upgrade
```

Example Response

```
{
  "job_id": "bf26cf3c-d046-4080-bb45-f114be7afa5f"
}
```

Status Code

Status Code	Description
200	Normal
400	Abnormal

Error Code

For details, see **Error Codes**.

4.13.5 Batch Upgrading or Rolling Back Agents

Function

This API is used to upgrade or roll back Agents of instances in batches.

Constraints

- The status of each instance node is normal.
- The instance whose Agent is to be upgraded must be monitored by HA.
- The HA and Agent can communicate with each other.
- The Agent version to be upgraded must be different from the target Agent version.
- The Agent version to be rolled back must be different from the target Agent version.
- The installation package corresponding to the target Agent version has been uploaded to the SFTP server using the installation package management function.

URI

POST /v3/instances/agent/batch-upgrade

Request Parameters

Table 4-980 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	Request token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Table 4-981 Request body parameter

Parameter	Mandatory	Type	Description
instance_ids	Yes	Array of strings	IDs of the instances for which the Agent is to be upgraded.
grade_agent_type	No	String	Type of the operation to be performed on the Agent. The value can be **upgrade** or **rollback**. **upgrade** indicates that the Agent needs to be upgraded, and **rollback** indicates that the Agent needs to be rolled back. The default value is **upgrade**.

Response Parameters

Status code: 202

Table 4-982 Response body parameters

Parameter	Type	Description
job_id	String	Job ID.

Status code: 400

Table 4-983 Response body parameters

Parameter	Type	Description
error_code	String	Error code. Minimum length: 8 characters Maximum length: 36 characters
error_msg	String	Error message. Minimum length: 2 characters Maximum length: 512 characters

Example Request

```
POST https://{Endpoint}/rds/gaussdb/v3/instances/agent/batch-upgrade

{
  "instance_ids": ["5d3647595df04cdc8afb9563cb07130ein20","7c1256b2baa64f04b7d00bd34edc3557in14"],
  "grade_agent_type": "upgrade"
}
```

Example Response

```
{
  "job_id": "bf26cf3c-d046-4080-bb45-f114be7afa5f"
}
```

Status Code

Status Code	Description
200	Normal
400	Abnormal

Error Code

For details, see **Error Codes**.

4.14 Alarm Management

4.14.1 Testing the Alarm Interconnection

Function

This API is used to test the alarm interconnection.

Constraints

This API is only available for interworking using HTTP.

URI

POST /v3/alarm/notifications/interconnection/check

Request Parameters

Table 4-984 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Table 4-985 Request body parameters

Parameter	Mandatory	Type	Description
http	Yes	Array of **Table 4-986** objects	Information of HTTP alarm interconnection.
time_zone	No	String	Time zone information. Set this parameter based on the international standard time zone, for example, Asia/Shanghai. The default value is **UTC**.

Table 4-986 CheckNotificationRequest

Parameter	Mandatory	Type	Description
proxy	Yes	String	Proxy, a third-party platform server that supports HTTPS. The value starts with **http** and ends with a port number.
url_template	Yes	String	URL template, which is the URL of the server and starts with a slash (/).
header	Yes	String	Header template. Header content is transmitted through HTTP (in JSON format).
body	Yes	String	Body template. Body content when HTTP is transmitted using POST or PUT (only JSON object or JSON array is supported). You can declare body content in **Body Template** and the content will be filled to **Custom Script**, or directly add body content in **Custom Script**.
groovy_script	No	String	Script content. This parameter is mandatory when **groovy_enabled** is set to **true**. Only Groovy scripts can be used to modify and supplement the body template. The input parameters are as follows: ● **JSONObject body**: Body template information. ● **String message**: Generated alarm information. ● **JSONObject json**: JSON strings after alarm information is parsed. ● **List<JSONObject>**: Final message to be sent. You need to add the modified body to **List<JSONObject>**.

Parameter	Mandatory	Type	Description
groovy_enabled	No	Boolean	Custom script. If the body template does not meet platform requirements, you can use a custom script to modify or supplement the template.
ignore_duplicate_in_duration	Yes	Integer	Alarm reporting interval. If the same alarm is reported again within the interval, the alarm is ignored and no alarm notification is sent. Value: **0**, **30**, **60**, **300**, **900**, **1800**, **3600**, **10800**, **21600**, **43200**, and **86400**. The unit is second.
response_template	No	String	Validation for response value of the request. By default, regular expression matching is used. For example, if the return value of **retCode** is **0** or **1**, the request is successful. Equal-value judgment expressions are supported. For special return values such as **0|1**, enter **{"retCode_eQs":"0|1"}** to avoid misjudgment when regular expressions are directly used. Multiple conditions are supported. The validation is successful only when all conditions are met at the same time. Single-character matching is supported, for example, enter **ok**.
request_method	Yes	String	Request mode for HTTP transport. Value: • POST • GET • PUT
alarm_template	Yes	String	Alarm template, which is used for reporting alarms.
recovery_template	Yes	String	Clearance template, which is used for clearing alarms.

Parameter	Mandatory	Type	Description
alarm_level	Yes	Array of strings	Alarm severity. Only the alarms of the selected severity are reported. Select one or more of the following severities: **critical**, **major**, **minor**, and **notice**.
ssl_enabled	No	Boolean	Enable SSL. If this function is enabled, HTTPS with SSL encryption is used for alarm interconnection. If this function is disabled, HTTP is used for alarm interconnection.
is_enabled	Yes	Boolean	Enable or not.

Response Parameters

None

Example Request

```
POST https://{Endpoint}/rds/v3/alarm/notifications/interconnection/check

{
  "http" : [ {
    "is_enabled" : true,
    "header" : "{\"Content-Type\":\"application/json\"}",
    "body" : "{\"buzId\":\"GaussDBAlarm\",\"messageId\":\"$messageId\",\"params\":\"$alarmText
\",\"templateId\":\"tmp001\",\"mobile\":\"13722221111\"}",
    "alarm_template" : "DefaultJSON",
    "recovery_template" : "DefaultJSON",
    "alarm_level" : [ "critical", "major", "minor", "notice" ],
    "proxy" : "http://10.*.*.*:9528",
    "url_template" : "/{}",
    "request_method" : "POST",
    "groovy_enabled" : true,
    "groovy_script" : "import java.time.LocalDateTime; import java.time.format.DateTimeFormatter;
list.add(body); String date =
LocalDateTime.now().format(DateTimeFormatter.ofPattern(\"yyyyMMddHHmmss\")); Random random =
new Random(); StringBuilder rds = new StringBuilder();for(int i = 0; i<4;i++)
{rds.append(random.nextInt(10));}\nString messageId = date + rds.toString(); body.put(\"messageId
\",messageId);",
    "ssl_enabled" : false,
    "ignore_duplicate_in_duration": 0,
    "response_template" : "{\"retCode\":0}"
  } ],
 "time_zone":"Asia/Shanghai"
}
```

Example Response

None

Status Code

Status Code	Description
201	Normal

Error code.

For details, see **Error Codes**.

4.14.2 Adding Alarm Interconnection Configurations

Function

This API is used to add alarm interconnection configurations.

Constraints

This API is only available for interworking using HTTP.

URI

POST /v3/alarm/notifications

Request Parameters

Table 4-987 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Table 4-988 Request body parameters

Parameter	Mandatory	Type	Description
instance_ids	Yes	Array of strings	Instance IDs.

Parameter	Mandatory	Type	Description
http	Yes	Array of **Table 4-989** objects	Information of HTTP alarm interconnection.
time_zone	No	String	Time zone information. Set this parameter based on the international standard time zone, for example, Asia/ Shanghai. The default value is **UTC**.

Table 4-989 HttpNotification

Parameter	Mandatory	Type	Description
notification_id	No	String	ID of the alarm interconnection.
proxy	Yes	String	Proxy, a third-party platform server that supports HTTPS. The value starts with **https** or **http** and ends with a port number.
url_template	Yes	String	URL template, which is the URL of the server and starts with a slash (/).
header	Yes	String	Header template. Header content is transmitted through HTTP (in JSON format).
body	Yes	String	Body template. Body content when HTTP is transmitted using POST or PUT (only JSON object or JSON array is supported). You can declare body content in **Body Template** and the content will be filled to **Custom Script**, or directly add body content in **Custom Script**.

Parameter	Mandatory	Type	Description
groovy_script	No	String	Script content. This parameter is mandatory when **groovy_enabled** is set to **true**. Only Groovy scripts can be used to modify and supplement the body template. The input parameters are as follows: ● **JSONObject body**: body template information. ● **String message**: generated alarm information. ● **JSONObject json**: JSON strings after alarm information is parsed. ● **List<JSONObject>**: Final message to be sent. You need to add the modified body to **List<JSONObject>**.
groovy_enabled	No	Boolean	Custom script. If the body template does not meet platform requirements, you can use a custom script to modify or supplement the template.
response_template	No	String	Validation for response value of the request. By default, regular expression matching is used. For example, if the return value of **retCode** is **0** or **1**, the request is successful. Equal-value judgment expressions are supported. For special return values such as **0|1**, enter **{"retCode_eQs":"0|1"}** to avoid misjudgment when regular expressions are directly used. Multiple conditions are supported. The validation is successful only when all conditions are met at the same time. Single-character matching is supported, for example, enter **ok**.

Parameter	Mandatory	Type	Description
request_method	Yes	String	Request mode for HTTP transport. Value: ● POST ● GET ● PUT
ignore_duplicate_in_duration	Yes	Integer	Alarm reporting interval. If the same alarm is reported again within the interval, the alarm is ignored and no alarm notification is sent. Value: **0**, **30**, **60**, **300**, **900**, **1800**, **3600**, **10800**, **21600**, **43200**, and **86400**. The unit is second.
alarm_template	Yes	String	Alarm template, which is used for reporting alarms.
recovery_template	Yes	String	Clearance template, which is used for clearing alarms.
alarm_level	Yes	Array of strings	Alarm severity. Only the alarms of the selected severity are reported. Select one or more of the following severities: **critical**, **major**, **minor**, and **notice**.
is_enabled	Yes	Boolean	Whether to enable alarm interconnection. This parameter must be set to **true** when an alarm interconnection is added.
ssl_enabled	No	Boolean	Enable SSL. If this function is enabled, HTTPS with SSL encryption is used for alarm interconnection. If this function is disabled, HTTP is used for alarm interconnection.
is_validated	No	Boolean	Whether the connection is verified.

Response Parameters

None

Example Request

```
POST https://{Endpoint}/rds/v3/alarm/notifications

{
  "instance_ids" : [ "4fee4b5e8cef49f6b4bf90cff8868a7bin14", "20c738c85bd542fcab0ecc3331bae88ein14" ],
  "http" : [ {
    "proxy" : "http://10.*.*.*:9528",
    "url_template" : "/{}",
    "header" : "{\"Content-Type\":\"application/json\"}",
    "body" : "{\"buzId\":\"GaussDBAlarm\",\"messageId\":\"$messageId\",\"params\":\"$alarmText
\",\"templateId\":\"tmp001\",\"mobile\":\"13722221111\"}",
    "groovy_script" : "import java.time.LocalDateTime; import java.time.format.DateTimeFormatter;
list.add(body); String date =
LocalDateTime.now().format(DateTimeFormatter.ofPattern(\"yyyyMMddHHmmss\")); Random random =
new Random(); StringBuilder rds = new StringBuilder();for(int i = 0; i<4;i++)
{rds.append(random.nextInt(10));}\nString messageId = date + rds.toString(); body.put(\"messageId
\",messageId);",
    "groovy_enabled" : false,
    "response_template" : "{\"retCode\":0}",
    "request_method" : "POST",
    "ignore_duplicate_in_duration" : 0,
    "alarm_template" : "DefaultText",
    "recovery_template" : "DefaultText",
    "alarm_level" : [ "critical", "major", "minor", "notice" ],
    "is_enabled" : true,
    "ssl_enabled" : false,
    "notification_id" : "6e431181-731b-45e9-8b67-1cac34448393"
  } ],
 "time_zone":"Asia/Shanghai"
}
```

Example Response

None

Status Code

Status Code	Description
201	Normal

Error Code

For details, see **Error Codes**.

4.14.3 Modifying Alarm Interconnection Configurations

Function

This API is used to modify alarm interconnection configurations.

Constraints

This API is only available for instances that are configured with the HTTP alarm interconnection.

URI

PUT /v3/alarm/notifications

Request Parameters

Table 4-990 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Table 4-991 Request body parameters

Parameter	Mandatory	Type	Description
instance_ids	Yes	Array of strings	Instance IDs.
http	Yes	Array of **Table 4-992** objects	Information of HTTP alarm interconnection.
time_zone	No	String	Time zone information. Set this parameter based on the international standard time zone, for example, Asia/ Shanghai. The default value is **UTC**.

Table 4-992 HttpNotification

Parameter	Mandatory	Type	Description
notification_id	No	String	ID of the alarm interconnection.
proxy	Yes	String	Proxy, a third-party platform server that supports HTTPS. The value starts with **https** or **http** and ends with a port number.

Parameter	Mandatory	Type	Description
url_template	Yes	String	URL template, which is the URL of the server and starts with a slash (/).
header	Yes	String	Header template. Header content is transmitted through HTTP (in JSON format).
body	Yes	String	Body template. Body content when HTTP is transmitted using POST or PUT (only JSON object or JSON array is supported). You can declare body content in **Body Template** and the content will be filled to **Custom Script**, or directly add body content in **Custom Script**.
groovy_script	No	String	Script content. If **enable_groovy** is set to **true**, this parameter is mandatory. Only Groovy scripts can be used to modify and supplement the body template. The input parameters are as follows: ● **JSONObject body**: Body template information. ● **String message**: Generated alarm information. ● **JSONObject json**: JSON strings after alarm information is parsed. ● **List<JSONObject>**: Final message to be sent. You need to add the modified body to **List<JSONObject>**.
groovy_enabled	No	Boolean	Custom script. If the body template does not meet platform requirements, you can use a custom script to modify or supplement the template.

Parameter	Mandatory	Type	Description
response_template	No	String	Validation for response value of the request. By default, regular expression matching is used. For example, if the return value of **retCode** is **0** or **1**, the request is successful. Equal-value judgment expressions are supported. For special return values such as **0|1**, enter **{"retCode_eQs":"0|1"}** to avoid misjudgment when regular expressions are directly used. Multiple conditions are supported. The validation is successful only when all conditions are met at the same time. Single-character matching is supported, for example, enter **ok**.
request_method	Yes	String	Request mode for HTTP transport. Value: ● POST ● GET ● PUT
ignore_duplicate_in_duration	Yes	Integer	Alarm reporting interval. If the same alarm is reported again within the interval, the alarm is ignored and no alarm notification is sent. Value: **0**, **30**, **60**, **300**, **900**, **1800**, **3600**, **10800**, **21600**, **43200**, and **86400**. The unit is second.
alarm_template	Yes	String	Alarm template, which is used for reporting alarms.
recovery_template	Yes	String	Clearance template, which is used for clearing alarms.
alarm_level	Yes	Array of strings	Alarm severity. Only the alarms of the selected severity are reported. Select one or more of the following severities: **critical**, **major**, **minor**, and **notice**.
is_enabled	Yes	Boolean	Whether to enable alarm interconnection.

Parameter	Mandatory	Type	Description
ssl_enabled	No	Boolean	Enable SSL. If this function is enabled, HTTPS with SSL encryption is used for alarm interconnection. If this function is disabled, HTTP is used for alarm interconnection.
is_validated	No	Boolean	Whether the connection is verified.

Response Parameters

None

Example Request

```
POST https://{Endpoint}/rds/v3/alarm/notifications

{
  "instance_ids" : [ "4fee4b5e8cef49f6b4bf90cff8868a7bin14", "20c738c85bd542fcab0ecc3331bae88ein14" ],
  "http" : [ {
    "proxy" : "http://10.*.*.*:9528",
    "url_template" : "/{}",
    "header" : "{\"Content-Type\":\"application/json\"}",
    "body" : "{\"buzId\":\"GaussDBAlarm\",\"messageId\":\"$messageId\",\"params\":\"$alarmText
\",\"templateId\":\"tmp001\",\"mobile\":\"13722221111\"}",
    "groovy_script" : "import java.time.LocalDateTime; import java.time.format.DateTimeFormatter;
list.add(body); String date =
LocalDateTime.now().format(DateTimeFormatter.ofPattern(\"yyyyMMddHHmmss\")); Random random =
new Random(); StringBuilder rds = new StringBuilder();for(int i = 0; i<4;i++)
{rds.append(random.nextInt(10));}\nString messageId = date + rds.toString(); body.put(\"messageId
\",messageId);",
    "groovy_enabled" : false,
    "response_template" : "{\"retCode\":0}",
    "request_method" : "POST",
    "ignore_duplicate_in_duration" : 0,
    "alarm_template" : "DefaultText",
    "recovery_template" : "DefaultText",
    "alarm_level" : [ "critical", "major", "minor", "notice" ],
    "is_enabled" : false,
    "ssl_enabled" : false,
    "notification_id" : "6e431181-731b-45e9-8b67-1cac34448393"
  } ],
 "time_zone":"Asia/Shanghai"
}
```

Example Response

None

Status Code

Status Code	Description
200	Normal

Error Code

For details, see **Error Codes**.

4.14.4 Viewing Alarm Interconnection Configurations

Function

This API is used to view alarm interconnection configurations.

URI

GET /v3/instances/{instance_id}/alarm/notifications

Table 4-993 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-994 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Response Parameters

Status code: 200

Table 4-995 Response body parameters

Parameter	Type	Description
instance_id	String	Instance ID.
http	Array of **Table 4-996** objects	Information of HTTP alarm interconnection.
snmp3	Array of **Table 4-997** objects	Information of SNMP3 alarm interconnection.
syslog	Array of **Table 4-999** objects	Information of Syslog alarm interconnection.
snmp2	Array of **Table 4-998** objects	SNMP2 configuration information.

Table 4-996 HttpNotification

Parameter	Type	Description
notification_id	String	ID of the alarm interconnection.
proxy	String	Proxy, a third-party platform server that supports HTTPS. The value starts with **https** or **http** and ends with a port number.
url_template	String	URL template, which is the URL of the server and starts with a slash (/).
header	String	Header template. Header content is transmitted through HTTP (in JSON format).
body	String	Body template, which is the body content when HTTP uses the POST or PUT method for transmission.
groovy_script	String	Script content. If **enable_groovy** is set to **true**, this parameter is mandatory. Only Groovy scripts can be used to modify and supplement the body template.
groovy_enabled	Boolean	Custom script. If the body template does not meet platform requirements, you can use a custom script to modify or supplement the template.
response_template	String	Validation for response value of the request.

Parameter	Type	Description
request_method	String	Request method used by of HTTP transmission.
ignore_duplicate_in_duration	Integer	Alarm reporting interval. If the same alarm is reported again within the interval, the alarm is ignored and no alarm notification is sent.
alarm_template	String	Alarm template, which is used for reporting alarms.
recovery_template	String	Clearance template, which is used for clearing alarms.
alarm_level	Array of strings	Alarm severity. Only the alarms of the selected severity are reported.
is_enabled	Boolean	Whether to enable alarm interconnection.
ssl_enabled	Boolean	Enable SSL. If this function is enabled, HTTPS with SSL encryption is used for alarm interconnection. If this function is disabled, HTTP is used for alarm interconnection.
is_validated	Boolean	Whether the connection is verified.
time_zone	String	Time zone information. The value complies with the international standard time zone format.

Table 4-997 Snmp3Notification

Parameter	Type	Description
ip	String	IP address of the installed SNMP server.
port	String	Port number of the installed SNMP server.
notification_id	String	ID of the alarm interconnection.
is_enabled	Boolean	Whether to enable alarm interconnection.
alarm_template	String	Alarm template, which is used for reporting alarms.
recovery_template	String	Clearance template, which is used for clearing alarms.
ignore_duplicate_in_duration	Integer	If the same alarm is reported again within the interval, the alarm is ignored and no alarm notification is sent.
alarm_level	Array of strings	Alarm severity. Only the alarms of the selected severity are reported.

Parameter	Type	Description
is_validated	Boolean	Whether the connection is verified.
time_zone	String	Time zone information. The value complies with the international standard time zone format.

Table 4-998 Snmp2Conf

Parameter	Type	Description
ip	String	IP address of the SNMPv2 receiver.
port	String	Port of the SNMPv2 receiver.
notification_id	String	Alarm notification ID.
is_enabled	Boolean	Whether alarm notification is enabled.
alarm_template	String	Alarm template, which is used for reporting alarms.
recovery_template	String	Clearance template, which is used for clearing alarms.
ignore_duplicate_in_duration	String	If the same alarm is reported again within the interval, the alarm is ignored and no alarm notification is sent.
alarm_level	Array of strings	Alarm severity supported. Value: **notice**, **minor**, **major**, or **critical**.
is_validated	Boolean	Whether the connection is verified.
community_name	String	SNMPv2 community name.

Table 4-999 SyslogNotification

Parameter	Type	Description
ip	String	IP address of the remote Syslog server.
port	String	Port number of the Syslog server. Value range: **1** to **65535**.
cert_id	String	ID of the SSL certificate. You need to enable the SSL certificate to authenticate the Syslog server that interconnects with alarms. To upload an SSL certificate, choose **Platform Management** > **SSL Certificates**.

Parameter	Type	Description
ssl_enabled	Boolean	Whether to enable SSL.
alarm_count	Integer	Number of alarms that can be reported. Value range: **1** to **100**
notification_id	String	ID of the alarm interconnection.
is_enabled	Boolean	Whether to enable alarm interconnection.
alarm_template	String	Alarm template, which is used for reporting alarms.
recovery_template	String	Clearance template, which is used for clearing alarms.
ignore_duplicate_in_duration	Integer	Alarm reporting interval. If the same alarm is reported again within the interval, the alarm is ignored and no alarm notification is sent.
alarm_level	Array of strings	Alarm severity. Only the alarms of the selected severity are reported.
is_validated	Boolean	Whether the connection is verified.
time_zone	String	Time zone information. The value complies with the international standard time zone format.

Example Request

```
GET https://{Endpoint}/rds/v3/instances/{instance_id}/alarm/notifications
```

Example Response

Status code: 200

Normal


```
{
  "syslog" : [ ],
  "snmp2" : [ ],
  "snmp3" : [ ],
  "http" : [ {
    "header" : "{\"Content-Type\":\"application/json\"}",
    "body" : "{\"buzId\":\"GaussDBAlarm\",\"messageId\":\"$messageId\",\"params\":\"$alarmText
\",\"templateId\":\"tmp001\",\"mobile\":\"13722221111\"}",
    "notification_id" : "6e431181-731b-45e9-8b67-1cac34448393",
    "is_enabled" : true,
    "alarm_template" : "DefaultText",
    "recovery_template" : "DefaultText",
    "ignore_duplicate_in_duration" : 0,
    "alarm_level" : [ "critical", "major", "minor", "notice" ],
    "is_validated" : false,
    "proxy" : "http://10.*.*.*:9528",
    "url_template" : "/{}",
    "request_method" : "POST",
    "groovy_enabled" : false,
    "groovy_script" : "import java.time.LocalDateTime; import java.time.format.DateTimeFormatter;
```

```
list.add(body); String date =
LocalDateTime.now().format(DateTimeFormatter.ofPattern(\"yyyyMMddHHmmss\")); Random random =
new Random(); StringBuilder rds = new StringBuilder();for(int i = 0; i<4;i++)
{rds.append(random.nextInt(10));}\nString messageId = date + rds.toString(); body.put(\"messageId
\",messageId);",
    "ssl_enabled" : false,
    "response_template" : "{\"retCode\":0}",
    "time_zone":"Asia/Shanghai"
  } ],
  "instance_id" : "4fee4b5e8cef49f6b4bf90cff8868a7bin14"
}
```

Status Code

Status Code	Description
200	Normal

Error code.

For details, see **Error Codes**.

4.14.5 Querying Alarm Results

Function

This API is used to query alarm results.

URI

POST /v3/alarm/records

Request Parameters

Table 4-1000 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Table 4-1001 Request body parameters

Parameter	Mandatory	Type	Description
is_active	Yes	Boolean	Alarm activation status. **true** indicates querying activated alarms, and **false** indicates querying historical recovered or cleared alarms.
is_convergence	Yes	Boolean	Whether to group alarms. **true** (viewing the alarms in the alarm list) and **false** (viewing alarm details).
offset	No	Integer	Index offset. If **offset** is set to *N*, the resource query starts from the *N*+1 piece of data. The value is **0** by default, indicating that the query starts from the first piece of data. The value must be a number but cannot be a negative number. For example, if this parameter is set to **1** and **limit** is set to **10**, only the 11th to 20th records are displayed.
limit	No	Integer	Number of records to be queried. The default value is **100**. The value cannot be a negative number. The minimum value is **1** and the maximum value is **100**. For example, if this parameter is set to **10**, a maximum of 10 records can be displayed.
alarm_id	No	String	Alarm ID. When **is_convergence** is set to **false**, and this parameter and **instance_id** are both configured, you can query alarms by alarm ID and alarm instance ID.
instance_id	No	String	ID of the instance where the alarm is generated. When **is_convergence** is set to **false**, and this parameter and **alarm_id** are both configured, you can query alarms by alarm ID and instance ID.

Response Parameters

Status code: 200

Table 4-1002 Response body parameters

Parameter	Type	Description
records	Array of **Table 4-1003** objects	Alarm results.
total_count	Integer	Total number of records.

Table 4-1003 AlarmRecord

Parameter	Type	Description
id	String	Alarm sequence ID. This parameter is valid only when **is_convergence** is set to **false**.
severity	String	Alarm severity. Value: **notice**, **minor**, **major**, or **critical**.
description	String	Alarm description. This parameter is valid only when **is_convergence** is set to **false**.
status	String	Alarm status. Values: **activated**, **cleared**, and **recovered**.
alarm_id	String	Alarm ID.
alarm_name	String	Alarm name.
alarm_remarks	String	Alarm definition. This parameter is valid only when **is_convergence** is set to **false**.
alarm_category	String	Alarm source type.
instance_id	String	ID of the instance where the alarm is generated. This parameter is valid only when **alarm_type** is set to **metric** or **event**.
instance_name	String	Name of the instance where the alarm is generated. This parameter is valid only when **alarm_type** is set to **metric** or **event**.
host_ip	String	IP address of the node where the alarm is generated. This parameter is valid only when **is_convergence** is set to **false**.
alarm_detail_raise_count	Integer	Number of times that the alarm is triggered. This parameter is valid only when **is_convergence** is set to **false**.
alarm_items_sum	Integer	Total number of alarms with the same alarm ID in the same instance. This parameter is valid only when **is_convergence** is set to **true**.

Parameter	Type	Description
alarm_type	String	Alarm type. The value can be **metric** (metric alarms), **event** (event alarms), **system-metric** (system metric alarms), **system-event** (system event alarms), or **system** (system alarms).
first_alarm_time	Object	Time when an alarm is triggered for the first time.
last_alarm_time	Object	Time when the latest alarm is triggered.
alarm_duration	String	Alarm duration.
clear_alarm_time	Object	Time when an alarm is cleared. This parameter is invalid when **is_active** is set to **false**.
component_name	String	Type of the component where the alarm is generated.
impaction	String	Alarm impact. This parameter is valid only when **is_convergence** is set to **false**.
possible_causes	String	Possible cause of the alarm. This parameter is valid only when **is_convergence** is set to **false**.
processing_steps	String	Alarm clearance suggestion. This parameter is valid only when **is_convergence** is set to **false**.
notify_status	String	Alarm notification status. This parameter is valid only when **is_convergence** is set to **false**.
automatic_clear	String	Whether the alarm can be automatically cleared. This parameter is valid only when **is_convergence** is set to **false**.
node_id	String	ID of the node where the alarm is generated. This parameter is valid only when **is_convergence** is set to **false**. If the alarm is a system alarm, the value of this parameter is the IP address of the management plane node where the alarm is generated.
resource_id	String	Internal alarm ID. This parameter is valid only when **is_convergence** is set to **false**.

Status code: 400

Table 4-1004 Response body parameters

Parameter	Type	Description
error_code	String	Error code. Minimum length: 8 characters Maximum length: 36 characters
error_msg	String	Error message. Minimum length: 2 characters Maximum length: 512 characters

Example Request

```
https://{Endpoint}/rds/v3/alarm/records

{
  "is_active" : true,
  "is_convergence" : true,
  "limit" : 10,
  "offset" : 0
}
```

Example Response

Status code: 200

API operation succeeded.

```
{
  "records" : [ {
"severity": "critical",
            "status": "activated",
            "alarm_id": "5023503",
            "alarm_name": "automated full backup",
            "alarm_category": "alarm management",
            "instance_id": "dc44e2653a2c47149cfb3109692c3590in14",
            "instance_name": "ha-3-lxy",
            "alarm_items_sum": 1,
            "alarm_type": "event",
            "first_alarm_time": 1698079853000,
            "last_alarm_time": 1698079853000,
            "alarm_duration": "9 hours 26 minutes",
            "component_name": "automated full backup"
} ],
  "total_count" : 1
}
```

Status Code

Status Code	Description
200	Normal
400	Abnormal

Error Code

For details, see **Error Codes**.

4.14.6 Querying the Number of Alarms

Function

This API is used to query the number of alarms.

URI

GET /v3/alarm/records/statistic

Request Parameters

Table 4-1005 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Response Parameters

Status code: 200

Table 4-1006 Response body parameters

Parameter	Type	Description
system_severity_statistics	Array of objects	Number of system alarms of each severity. Array length: 4 characters
instance_severity_statistics	Array of objects	Number of instance alarms of each severity. Array length: 4 characters

Status code: 400

Table 4-1007 Response body parameters

Parameter	Type	Description
error_code	String	Error code. Minimum length: 8 characters Maximum length: 36 characters
error_msg	String	Error message. Minimum length: 2 characters Maximum length: 512 characters

Example Request

None

Example Response

```
{
    "system_severity_statistics": [
        {
            "severity": "notice",
            "count": 1
        },
        {
            "severity": "minor",
            "count": 3
        },
        {
            "severity": "major",
            "count": 80
        },
        {
            "severity": "critical",
            "count": 0
        }
    ],
    "instance_severity_statistics": [
        {
            "severity": "notice",
            "count": 151
        },
        {
            "severity": "minor",
            "count": 97
        },
        {
            "severity": "major",
            "count": 904
        },
        {
            "severity": "critical",
            "count": 966
        }
    ]
}
```

Status Code

Status Code	Description
200	Normal
400	Abnormal

Error Code

For details, see **Error Codes**.

4.14.7 Applying Alarm Settings to Instances

Function

- This API is used to apply alarm settings to instances.
- Only metric and event alarm configurations can be modified. Alarm interconnection configurations cannot be changed.

URI

POST /v3/alarm/update/configs

Request Parameters

Table 4-1008 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Table 4-1009 Request body parameters

Parameter	Mandatory	Type	Description
source_id	Yes	String	Original instance ID. Constraints: Before calling this API, you need to configure alarm settings for an instance. Then, you can apply the alarm settings to target instances.
target_instance_ids	Yes	Array of strings	Target instance IDs. The alarm settings of original instances will be applied to target instances.

Response Parameters

None

Example Request

```
POST https://{Endpoint}/rds/v3/alarm/update/configs

{
  "source_id" : "6970e9cf7adc41a78e0f1f50215c9d3bin14",
  "target_instance_ids" : [ "4e60721144874b67b3fc7c8f7acfc887in14" ]
}
```

Example Response

None

Status Code

Status Code	Description
200	Normal

Error Code

For details, see **Error Codes**.

4.15 Monitoring Dashboards

4.15.1 Querying Monitoring Metrics

Function

This API is used to query monitoring metrics.

URI

POST /v3/monitor/metrics

Request Parameters

Table 4-1010 Request body parameter

Parameter	Mandatory	Type	Description
start_time	Yes	Long	Query start time of a metric. The time is a timestamp in milliseconds.
end_time	Yes	Long	Query end time of a metric. The time is a timestamp in milliseconds. The query period of a metric must be fewer than 30 days.
is_latest_point_only	No	String	Whether to query the monitoring data of the most recent time point. The value can be **true** or **false**.
instances	Yes	Array of **Table 4-1011** objects	Instances that you want to query metrics of.
metric_names	Yes	Array of strings	Metric names to be queried. (To query the name of each metric, see **Monitoring Metrics**.)

Table 4-1011 InstanceReq

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.
node_ids	No	Array of strings	Node IDs of the corresponding instance.

Response Parameters

Status code: 200

Table 4-1012 Response body parameters

Parameter	Type	Description
metric_infos	Array of **Table 4-1013** objects	Metric information.

Table 4-1013 Metrics

Parameter	Type	Description
instance_id	String	Instance ID.
time_interval	Integer	Default interval. Its value depends on the selected time range.
metrics	Array of **Table 4-1014** objects	Metric information.

Table 4-1014 Metric

Parameter	Type	Description
name	String	Metric name.
node_id	String	Node ID. If **is_latest_point_only** in the request body is set to **true**, only the monitoring data of the most recent time point is returned regardless of the node quantity specified by **node_ids** in the request body. Therefore, the **node_id** field always remains empty. If **is_latest_point_only** is set to **false**, the monitoring data of the nodes specified by **node_ids** is returned based on the selected time range. The response value of **node_id** is the node IDs specified by **node_ids** in the request body. (If **node_ids** is empty, data of all nodes is queried by default.)
collection_frequency	Integer	Metric collection frequency.

Parameter	Type	Description
series	Map<String,Array< Number>	Metric dimension and value.
timestamps	Array of numbers	Timestamp.

Status code: 400

Table 4-1015 Response body parameters

Parameter	Type	Description
error_code	String	Error code. Minimum length: 8 characters Maximum length: 36 characters
error_msg	String	Error message. Minimum length: 2 characters Maximum length: 512 characters

Example Request

```
https://{Endpoint}/rds/v3/monitor/metrics
{
  "start_time": 1713230727384,
  "end_time": 1713234327384,
  "is_latest_point_only": false,
  "instances": [
    {
      "instance_id": "f9f73d520cb34a0fad6d674350012673in14"
    }
  ],
  "metric_names": [
    "sys001_cpu_usage"
  ]
}
```

Example Response

Status code: 200

API operation succeeded.

```
{
  "metric_infos": [
    {
      "metrics": [
        {
          "name": "sys001_cpu_usage",
          "series": {
            "192.168.1.*": [
              23.05,
              13.07,
              25.39,
              16.88,
              22.77
```

```
      ],
      "192.168.1.*": [
        20.69,
        20.16,
        27.97,
        24.72,
        36.31
      ],
      "192.168.1.*": [
        25.43,
        20.39,
        24.5,
        19.39,
        29.69
      ],
      "192.168.1.*": [
        21.08,
        20.32,
        29.41,
        24.41,
        29.21
      ]
     },
     "timestamps": [
      1713230727000,
      1713230737000,
      1713230747000,
      1713230757000,
      1713230767000
     ],
     "node_id":
"0a105fbc4dd3412fb37ed418a9ae8bd8no14,268d32dba3764c9ab3de999fb67d7edbno14,34d3ae97babc431e
b47edcd0c1ed510dno14,febc8345c2174cc1a05cb5e799e8fcc5no14",
     "collection_frequency": 10
    }
   ],
   "instance_id": "f9f73d520cb34a0fad6d674350012673in14",
   "time_interval": 10
  }
 ]
}
```


Status code: 400

API operation failed.


```
{
  "errCode" : "DBS.280001",
  "externalMessage" : "Parameter error. [metric_names]"
}
```


Status Code

Status Code	Description
200	Normal
400	Abnormal

Error Code

For details, see **Error Codes**.

4.15.2 Querying Monitoring Metric Groups

Function

This API is used to query monitoring metric groups.

URI

GET /v3/monitor/metrics/series

Table 4-1016 Query parameters

Parameter	Mandatory	Type	Description
name	Yes	Array	Metric group name. The value can be **CPUMEMORY**, **CONNECTION**, **SYNCSTAT**, **PROCESSRESOURCE**, **NETWORK**, **TRANSACTION**, **IOSTORAGE**, **LOCK**, **INSTATUS**, **FORDR**, or **AGENTSTATUS**.

Request Parameters

None

Response Parameters

Status code: 200

Table 4-1017 Response body parameters

Parameter	Type	Description
series	Array of **Table 4-1018** objects	Metric group information.

Table 4-1018 MetricGroup

Parameter	Type	Description
name	String	Metric group name.
metrics	Array of **Table 4-1019** objects	Metric details.

Table 4-1019 singleMetric

Parameter	Type	Description
name	String	Metric name.
unit	String	Metric unit.

Status code: 400

Table 4-1020 Response body parameters

Parameter	Type	Description
error_code	String	Error code. Minimum length: 8 characters Maximum length: 36 characters
error_msg	String	Error message. Minimum length: 2 characters Maximum length: 512 characters

Example Request

```
https://{Endpoint}/rds/v3/monitor/metrics/series?name=CPUMEMORY
```

Example Response

Status code: 200

API operation succeeded.


```
{
  "series" : [ {
    "name" : "CPUMEMORY",
    "metrics" : [ {
      "name" : "sys001_cpu_usage",
      "unit" : "%"
    }, {
      "name" : "sys010_mem_usage",
      "unit" : "%"
    }, {
      "name" : "cpu_sys_usage",
      "unit" : "%"
    }, {
      "name" : "cpu_user_usage",
      "unit" : "%"
    }, {
      "name" : "sys002_idle_usage",
      "unit" : "%"
    }, {
      "name" : "sys016_cpu_load",
      "unit" : "Count"
    }, {
      "name" : "mem_used_size",
      "unit" : "GB"
    }, {
```

```
    "name" : "buffer_size",
    "unit" : "GB"
  }, {
    "name" : "cache_size",
    "unit" : "GB"
  }, {
    "name" : "sys013_mem_free_size",
    "unit" : "GB"
  }, {
    "name" : "sys024_swap_total_size",
    "unit" : "MB"
  }, {
    "name" : "sys026_swap_used_size",
    "unit" : "MB"
  }, {
    "name" : "sys025_swap_used_ratio",
    "unit" : "%"
  }, {
    "name" : "cpu_wait_usage",
    "unit" : "%"
  } ]
 } ]
}
```


Status code: 400

API operation failed.


```
{
 "errCode" : "DBS.280270",
 "externalMessage" : "The parameter does not exist."
}
```


Status Code

Status Code	Description
200	Normal
400	Abnormal

Error Code

For details, see **Error Codes**.

4.16 Task Center

4.16.1 Querying Task Status by Job ID

Function

This API is used to query the status of asynchronous tasks in the task center based on job ID.

URI

GET /v3/jobs

Table 4-1021 Query parameters

Parameter	Mandatory	Type	Description
id	Yes	String	Job ID.

Request Parameters

Table 4-1022 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Response Parameters

Status code: 200

Table 4-1023 Response body parameters

Parameter	Type	Description
job	Table 19-4 object	Task details.

Table 4-1024 Job

Parameter	Type	Description
id	String	Task ID.
name	String	Task name.
status	String	Task execution status.
created	Long	Creation time.
ended	Long	End time.

Parameter	Type	Description
progress	String	Task execution progress.
instance_id	String	Instance ID.
fail_reason	String	Task failure information.

Status code: 400

Table 4-1025 Response body parameters

Parameter	Type	Description
error_code	String	Error code. Minimum length: 8 characters Maximum length: 36 characters
error_msg	String	Error message. Minimum length: 2 characters Maximum length: 512 characters

Example Request

Querying task details

```
https://{Endpoint}/rds/v3/jobs?id=baeb3b373d1626a37d197acf3ac6f6bb
```

Example Response

Status code: 200

Task details queried.


```
{
  "job" : {
    "instance_id" : "5d3647595df04cdc8afb9563cb07130ein20",
    "created" : 1695352115314,
    "name" : "CreateGaussDBV5MiniDBMindInstance",
    "ended" : 1695352527911,
    "id" : "baeb3b373d1626a37d197acf3ac6f6bb",
    "fail_reason" : null,
    "status" : "Completed"
  }
}
```


Status Code

Status Code	Description
200	Normal

Status Code	Description
400	Abnormal

Error Code

For details, see **Error Codes**.

4.16.2 Obtaining Asynchronous Batch Processing Task Details

Function

This API is used to query asynchronous batch processing task details.

URI

GET /gaussdb/v3/instances/batch-async-task/records

Table 4-1026 Query parameters

Parameter	Mandatory	Type	Description
job_id	Yes	String	Task ID.
offset	No	Integer	Index offset. If **offset** is set to *N*, the resource query starts from the *N*+1 data entry. The default value is **0**, indicating that the query starts from the first data entry. The value must be a number but cannot be a negative number.
limit	No	Integer	Number of queries. The default value is **10**. The value cannot be a negative number. The minimum value is **1**, and the maximum value is **100**.

Request Parameters

Table 4-1027 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Response Parameters

Status code: 200

Table 4-1028 Response body parameters

Parameter	Type	Description
batch_async_job	Object	Task details. For details, see **Table 4-1029**.
total_count	Integer	Total number of records.
running_count	Integer	Number of tasks in process.
failed_count	Integer	Failed inspection items.
notrun_count	Integer	Number of tasks not requiring execution.
completed_count	Integer	Number of completed tasks.
instance_async_tasks	Array of Object	Subtask details. For details, see **Table 4-1030**.

Table 4-1029 Task details

Parameter	Type	Description
job_id	String	Task ID.

Parameter	Type	Description
status	String	Task status. ● **Running**: The task is in progress. ● **Failed**: The task failed. ● **NoRun**: The task does not need to be executed. ● **Completed**: The task is complete.
task_type	String	Task type: **instance_group_switchover**: batch primary/ standby switchover between instance nodes within a single cluster
created_at	String	Creation time in the "yyyy-mm-ddThh:mm:ssZ" format. **T** is the separator between calendar and hourly notation of time. **Z** indicates the time zone offset. For example, in the Beijing time zone, the offset is **+0800**.
updated_at	String	Creation time in the "yyyy-mm-ddThh:mm:ssZ" format. **T** is the separator between calendar and hourly notation of time. **Z** indicates the time zone offset. For example, in the Beijing time zone, the offset is **+0800**.

Table 4-1030 Subtask information

Parameter	Type	Description
instance_id	String	Instance ID.
instance_name	String	Instance name.
support_actions	Array of String	Supported operations. This parameter is only available when **task_type** is set to **instance_group_switchover**.
status	String	Task status. ● **Running**: The task is in progress. ● **Failed**: The task failed. ● **NoRun**: The task does not need to be executed. ● **Completed**: The task is complete.

Parameter	Type	Description
created_at	String	Creation time in the "yyyy-mm-ddThh:mm:ssZ" format. **T** is the separator between calendar and hourly notation of time. **Z** indicates the time zone offset. For example, in the Beijing time zone, the offset is **+0800**.
end_at	String	End time. The format is yyyy-mm-ddThh:mm:ssZ. **T** is the separator between calendar and hourly notation of time. **Z** indicates the time zone offset. For example, in the Beijing time zone, the offset is **+0800**.
updated_at	String	Modification time in the "yyyy-mm-ddThh:mm:ssZ" format. **T** is the separator between calendar and hourly notation of time. **Z** indicates the time zone offset. For example, in the Beijing time zone, the offset is **+0800**.
error_code	String	Error code.
error_msg	String	Error message.

Status code: 400

Table 4-1031 Response body parameters

Parameter	Type	Description
error_code	String	Error code. Minimum length: 8 characters Maximum length: 36 characters
error_msg	String	Error message. Minimum length: 2 characters Maximum length: 512 characters

Example Request

Querying task details

```
https://{Endpoint}/rds/gaussdb/v3/instances/batch-async-task/records?
job_id=baeb3b373d1626a37d197acf3ac6f6bb
```

Example Response

Status code: 200

Querying task details


```
{
  "batch_async_job":
  {
    "job_id": "b8d45cab-5568-4a11-b2d6-87fb5a75d73d",
    "status": "Completed",
    "task_type": "instance_group_switchover",
    "created_at": "2025-10-10T06:41:45Z",
    "updated_at": "2025-10-10T06:41:46Z"
  },
  "total_count": 2,
  "running_count": 0,
  "failed_count": 0,
  "notrun_count": 1,
  "completed_count": 1,
  "instance_async_tasks":
  [
    {
      "status": "Completed",
      "instance_id": "81fe62ebedac4635bd811198808f5de2in14",
      "instance_name": "gaussdbv5_hcs_3CTv10_ecs_default_1LBPvkbQe",
      "support_actions":
      [],
      "created_at": "2025-10-10T06:41:45Z",
      "updated_at": "2025-10-10T06:41:46Z",
      "end_at": "2025-10-10T06:41:46Z"
    },
    {
      "status": "NoRun",
      "instance_id": "f028576f6e414527a8ed502eb760ec30in14",
      "instance_name": "gauss-zlq-az3",
      "support_actions":
      [],
      "created_at": "2025-10-10T06:41:45Z",
      "updated_at": "2025-10-10T06:41:45Z",
      "end_at": "2025-10-10T06:41:45Z"
    }
  ]
}
```


Status Codes

Status Code	Description
200	Normal
400	Abnormal

Error Codes

For details, see **Error Codes**.

4.17 DR Management

4.17.1 Editing a DR Task Name

Function

This API is used to edit the name of a DR task.

URI

PUT /gaussdb/v3.6/instances/{instance_id}/disaster-recovery/rename

Table 4-1032 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-1033 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Table 4-1034 Request body parameters

Parameter	Mandatory	Type	Description
synchronization_id	Yes	String	DR relationship ID.
dr_task_name	Yes	String	DR task name.

Response Parameters

Status code: 400

Table 4-1035 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

```
PUT https://{Endpoint}/rds/gaussdb/v3.6/instances/2ec6359607c94871888cdba0d4b727acin14/disaster-recovery/rename

{
  "synchronization_id" : "cd8c4a4a-2436-46aa-8c79-172c9019268f",
  "dr_task_name" : "dr-task-123"
}
```

Example Response

None

Status Code

Status Code	Description
200	Normal
400	Abnormal

Error Code

For details, see **Error Codes**.

4.17.2 Querying Instances That Have Established a DR Relationship

Function

This API is used to query instances that have established a DR relationship.

URI

GET /gaussdb/v3.6/disaster-recovery/relations

Table 4-1036 Query parameters

Parameter	Mandatory	Type	Description
limit	No	Integer	Number of records to be queried. The default value is **100**. The value cannot be a negative number. The minimum value is **1** and the maximum value is **100**.
offset	No	Integer	Index offset. If **offset** is set to *N*, the resource query starts from the *N*+1 piece of data. The value is **0** by default, indicating that the query starts from the first piece of data. The value must be a number but cannot be a negative number.
instance_name	No	String	Instance name. It can be used to query and filter primary instance name.
instance_id	No	String	Instance ID. It can be used to query and filter primary instance ID.

Request Parameters

Table 4-1037 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Response Parameters

Status code: 200

Table 4-1038 Response body parameters

Parameter	Type	Description
total	Integer	Total number of query results.
relations	Array of **Table 4-1039** objects	DR relationship details.

Table 4-1039 DisasterRelations

Parameter	Type	Description
disaster_type	String	DR type.
name	String	DR task name.
disaster_role	String	Instance role in the DR relationship. Enumerated values: • **disaster** • **master**
created	String	Creation time in the "yyyy-mm-dd hh:mm:ss" format.
updated	String	Update time in the "yyyy-mm-dd hh:mm:ss" format.
slave_region_instance_info	**Table 4-1040** object	DR instance information.
master_region_instance_info	**Table 4-1040** object	Primary instance information.
id	String	DR task ID.
synchronization_id	String	Unique ID of the DR relationship.
instance_id	String	Primary instance ID.
instance_name	String	Primary instance name.
instance_status	String	Primary instance status.
status	String	DR task status.
precheck_failed_reason	String	Pre-verification failure cause.
actions	Array of strings	Current actions of the instance.

Parameter	Type	Description
skip_monitor	Boolean	Whether monitoring is skipped.
bind_disaster_recovery_relations	Array of **Table 4-1041** objects	Other DR relationships of a primary instance.

Table 4-1040 RegionInstanceInfo

Parameter	Type	Description
instance_id	String	Instance ID.
project_id	String	Instance project ID.
project_name	String	Instance project name.
region_code	String	Region code.
ip_address	String	Data IP addresses, which are separated by commas (,).

Table 4-1041 BindDisasterRelations

Parameter	Type	Description
disaster_type	String	DR type.
name	String	DR task name.
disaster_role	String	Instance role in the DR relationship. Enumerated values: ● **disaster** ● **master**
created	String	Creation time in the "yyyy-mm-dd hh:mm:ss" format.
updated	String	Update time in the "yyyy-mm-dd hh:mm:ss" format.
slave_region_instance_info	**Table 4-1040** object	DR instance information.
master_region_instance_info	**Table 4-1040** object	Primary instance information.
id	String	DR task ID.

Parameter	Type	Description
synchronization_id	String	Unique ID of the DR relationship.
instance_id	String	Primary instance ID.
instance_name	String	Primary instance name.
instance_status	String	Primary instance status.
status	String	DR task status.
precheck_failed_reason	String	Pre-verification failure cause.
actions	Array of strings	Current actions of the instance.
skip_monitor	Boolean	Whether monitoring is skipped.

Status code: 400

Table 4-1042 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

```
GET https://{Endpoint}/rds/gaussdb/v3.6/disaster-recovery/relations?limit=100&offset=0
```

Example Response

Status code: 200

Normal


```
{
  "total" : 1,
  "relations" : [ {
    "name" : "DR-task-0962",
    "created" : "2022-06-16 09:39:51",
    "updated" : "2022-06-16 09:44:55",
    "id" : "30b74120-4b50-495d-8174-7eeddf7feac5",
    "synchronization_id" : "9446f822-ccd2-43dc-929c-0b78ba7fdf64",
    "status" : "normal",
    "precheck_failed_reason" : "",
    "disaster_type" : "stream",
    "disaster_role" : "disaster",
    "slave_region_instance_info" : {
      "region_code" : "sa-fb-1",
      "instance_id" : "d1060faf0d3743dd8830b39e7423bc9ain14",
      "project_id" : "4a52a8a8f63d4c6d966dff0e20b69738",
      "project_name" : "sa-fb-1_GaussDB",
      "ip_address" : "172.*.*.*,172.*.*.*,172.*.*.*"
```

```
  },
  "master_region_instance_info" : {
    "region_code" : "sa-fb-1",
    "instance_id" : "54d4b9fb131745fcba32cb90a05cabb8in14",
    "project_id" : "4a52a8a8f63d4c6d966dff0e20b69738",
    "project_name" : "sa-fb-1_GaussDB",
    "ip_address" : "172.*.*.*,172.*.*.*,172.*.*.*"
  },
  "instance_id" : "d1060faf0d3743dd8830b39e7423bc9ain14",
  "instance_name" : "hly-0609-02",
  "action" : [ "STREAM_DISASTER_RELATION_MASTER_TAG" ] ,
  "skip_monitor" : false,
  "bind_disaster_recovery_relations": null

 } ]
}
```

Status Code

Status Code	Description
200	Normal
400	Abnormal

Error Code

For details, see **Error Codes**.

4.17.3 Querying Historical DR Operation Records

Function

This API is used to query historical DR operation records. Currently, only the records of rectifying DR switchover failures can be queried.

URI

GET /gaussdb/v3/instances/{instance_id}/disaster-recovery/records

Table 4-1043 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Table 4-1044 Query parameters

Parameter	Mandatory	Type	Description
limit	No	Integer	Number of records to be queried. The default value is **100**. The value cannot be a negative number. The minimum value is **1** and the maximum value is **100**.
offset	No	Integer	Index offset. If **offset** is set to *N*, the resource query starts from the *N*+1 piece of data. The value is **0** by default, indicating that the query starts from the first piece of data. The value must be a number but cannot be a negative number.
entity_type	Yes	String	Entity type. Enumerated value: **dr**.
entity_id	Yes	String	DR relationship ID.

Request Parameters

Table 4-1045 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Response Parameters

Status code: 200

Table 4-1046 Response body parameters

Parameter	Type	Description
records	Array of **Table 4-1047** objects	Historical DR operation records.

Table 4-1047 DisasterInfoRecords

Parameter	Type	Description
id	String	Primary key ID.
entity_id	String	Entity ID.
entity_type	String	Entity type.
job_id	String	Workflow ID.
instance_id	String	Instance ID.
action	String	Current action of the instance.
status	String	Operation status.
message	String	Detailed information.
created_at	String	Creation time, for example, 2024-03-01T18:50:20Z.
updated_at	String	Modification time, for example, 2024-03-01T18:50:20Z.
extended_info	**Table 6** object	Extended information of a DR operation record.

Table 4-1048 DisasterRecordExtendedInfo

Parameter	Type	Description
extraKeys	Array of strings	Extend flag.
allExtras	**Table 4-1049** object	Detailed extended description.
extra	**Table 4-1050** object	Extended description.

Table 4-1049 DisasterRecordExtendedAllDesInfo

Parameter	Type	Description
DISASTER_RECOVERY_CLUSTER_POSTCHECK	String	Post-check details of a DR switchover.

Table 4-1050 DisasterRecordExtendedDesInfo

Parameter	Type	Description
DISASTER_RECOVERY_CLUSTER_POSTCHECK	String	Post-check description of a DR switchover.

Status code: 400

Table 4-1051 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

```
GET https://{Endpoint}/rds/gaussdb/v3/instances/02b7a8fb063a4c46801f57c482950347in14/disaster-recovery/records?limit=100&offset=0&entity_type=dr&entity_id=8c2347c8-0e0d-4758-a580-cb9ead8a29ac
```

Example Response

Status code: **200**

Normal

```
{
    "records": [
        {
            "id": "1125b778-f349-45a8-8f13-c4ed7c6d3cbf",
            "action": "dr_stream_switchover_to_postcheck",
            "status": "success",
            "message": "Check result after active/standby switchover, instance status: [normal], DR status: 
[Incremental synchronization], ID of the node that is not involved in switchover: []",
            "entity_id": "8c2347c8-0e0d-4758-a580-cb9ead8a29ac",
            "entity_type": "dr",
            "job_id": "53a5898e-fe0c-4208-b3ef-bb06c40723a3",
            "instance_id": "68565c5b424a40748d6484d13c62b5e4in14",
            "created_at": "2024-06-29T02:21:27Z",
            "updated_at": "2024-06-29T02:21:27Z",
            "extended_info": {
                "extraKeys": [
                    "DISASTER_RECOVERY_CLUSTER_POSTCHECK"
                ],
                "allExtras": {
                    "DISASTER_RECOVERY_CLUSTER_POSTCHECK": "{\"isOmKernelMutualExclusion
```

```
\":false,\"clusterState\":\"NORMAL\",\"hadrClusterState\":\"ARCHIVE\",\"notFailoverInfo\":
[],\"notSwitchoverInfo\":[],\"incorrectParamsInfo\":[]}"
                },
                "extra": {
                    "DISASTER_RECOVERY_CLUSTER_POSTCHECK": "{\"isOmKernelMutualExclusion
\":false,\"clusterState\":\"NORMAL\",\"hadrClusterState\":\"ARCHIVE\",\"notFailoverInfo\":
[],\"notSwitchoverInfo\":[],\"incorrectParamsInfo\":[]}"
                }
            }
        }
    ]
}
```

Status Code

Status Code	Description
200	Normal
400	Abnormal

Error Code

For details, see **Error Codes**.

4.17.4 Querying Instance DR Status

Function

This API is used to query the real-time DR status of an instance.

URI

GET /gaussdb/v3.6/instances/{instance_id}/disaster-recovery/monitor

Table 4-1052 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Table 4-1053 Query parameters

Parameter	Mandatory	Type	Description
disaster_type	Yes	String	DR type. Enumerated values: • **stream** • **dorado**

Parameter	Mandatory	Type	Description
synchronization_id	No	String	DR relationship ID. This parameter is optional for dual-instance streaming DR and mandatory for three-instance streaming DR.

Request Parameters

Table 4-1054 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Response Parameters

Status code: 200

Table 4-1055 Response body parameters

Parameter	Type	Description
instance_id	String	Instance ID.
status	String	DR status.
rpo	String	RPO.
rto	String	RTO.
rpo_threshold	String	RPO threshold.
rto_threshold	String	RTO threshold.
switchover_progress	String	Switchover progress (%), for example, **40%**.
failover_progress	String	Failover progress (%), for example, **40%**.

Status code: 400

Table 4-1056 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

```
GET https://{Endpoint}/rds/gaussdb/v3.6/instances/2ec6359607c94871888cdba0d4b727acin14/disaster-recovery/monitor?disaster_type=stream
```

Example Response

Status code: 200

Normal

```
{
  "rpo" : "0",
  "rto" : "0",
  "status" : "archive",
  "instance_id" : "b3692282395340c5b5421ecf3d7bd6adin14",
  "rto_threshold" : "900",
  "rpo_threshold" : "10",
  "failover_progress" : "",
  "switchover_progress" : ""
}
```

Status Code

Status Code	Description
200	Normal
400	Abnormal

Error Code

For details, see **Error Codes**.

4.17.5 Creating a DR Task

Function

This API is used to establish a DR relationship (initiated from the primary instance).

URI

POST /gaussdb/v3.6/instances/{instance_id}/disaster-recovery/construct

Table 4-1057 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-1058 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Table 4-1059 Request body parameters

Parameter	Mandatory	Type	Description
disaster_type	Yes	String	DR type. Enumerated values: • **stream** • **dorado**
dr_ip	Yes	String	DR instance IP address.
dr_user_name	Yes	String	Account name of the DR instance.
dr_user_password	Yes	String	Account password of the DR instance.
dr_task_name	No	String	DR task name.

Response Parameters

Status code: 202

Table 4-1060 Response body parameters

Parameter	Type	Description
job_id	String	Job ID.

Status code: 400

Table 4-1061 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

```
POST https://{Endpoint}/rds/gaussdb/v3.6/instances/2ec6359607c94871888cdba0d4b727acin14/disaster-recovery/construct

{
  "disaster_type" : "stream",
  "dr_ip" : "127.*.*.*",
  "dr_user_name" : "root",
  "dr_user_password" : "***",
  "dr_task_name" : "dr-task-123"
}
```

Example Response

Status code: 202

Normal

```
{
  "job_id" : "f88479d2-e2af-4585-a0b8-6e66966d7484"
}
```

Status Code

Status Code	Description
202	Normal
400	Abnormal

Error Code

For details, see **Error Codes**.

4.17.6 Promoting the DR instance to Primary

Function

This API is used to trigger a DR failover to promote the DR instance to primary (initiated from the DR instance).

URI

POST /gaussdb/v3.6/instances/{instance_id}/disaster-recovery/failover

Table 4-1062 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-1063 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Table 4-1064 Request body parameters

Parameter	Mandatory	Type	Description
is_support_restore	No	Boolean	Whether to re-establish a DR relationship. (The instance version must be V2.0-3.100 or later.) • **true**: supported • **false** (default value): Not supported

Parameter	Mandatory	Type	Description
disaster_type	Yes	String	DR type. Enumerated values: • **stream** • **dorado**

Response Parameters

Status code: 202

Table 4-1065 Response body parameters

Parameter	Type	Description
job_id	String	Job ID.

Status code: default

Table 4-1066 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

```
POST https://{Endpoint}/rds/gaussdb/v3.6/instances/2ec6359607c94871888cdba0d4b727acin14/disaster-recovery/failover

{
  "is_support_restore" : true,
  "disaster_type" : "stream"
}
```

Example Response

Status code: 202

Normal

```
{
  "job_id" : "8bf0175e-562e-45f6-942a-2e2c72c577d7"
}
```

Status Code

Status Code	Description
202	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.17.7 Stopping a DR Relationship

Function

This API is used to stop a DR relationship (initiated from the primary instance).

URI

POST /gaussdb/v3.6/instances/{instance_id}/disaster-recovery/release

Table 4-1067 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-1068 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Table 4-1069 Request body parameters

Parameter	Mandatory	Type	Description
disaster_type	Yes	String	DR type. Enumerated values: ● **stream** ● **dorado**
synchronization_id	No	String	DR relationship ID. This parameter is optional for dual-instance streaming DR and mandatory for three-instance streaming DR.
not_stop_oceanstor_dorado	No	Boolean	Whether to remove the OceanStor Dorado replication relationship. The default value is **true**. Enumerated values: ● **true**: Remove the replication relationship. ● **false**: Retain the replication relationship.

Response Parameters

Status code: 202

Table 4-1070 Response body parameters

Parameter	Type	Description
job_id	String	Job ID.

Status code: default

Table 4-1071 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

```
POST https://{Endpoint}/rds/gaussdb/v3.6/instances/2ec6359607c94871888cdba0d4b727acin14/disaster-recovery/release
```

```
{
  "synchronization_id" : "cd8c4a4a-2436-46aa-8c79-172c9019268f",
  "disaster_type" : "stream"
}
```


Example Response

Status code: 202

Normal


```
{
  "job_id" : "8bf0175e-562e-45f6-942a-2e2c72c577d7"
}
```


Status Codes

Status Code	Description
202	Normal
default	Abnormal

Error Codes

For details, see **Error Codes**.

4.17.8 Switching Roles of Primary and DR Instances

Function

This API is used to switch the roles of the primary and DR instances (initiated from primary or DR instance).

CAUTION

The switching task can be delivered only after the **DR pre-check** is successful.

URI

POST /gaussdb/v3.6/instances/{instance_id}/disaster-recovery/switchover

Table 4-1072 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-1073 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Table 4-1074 Request body parameters

Parameter	Mandatory	Type	Description
disaster_type	Yes	String	DR type. Enumerated values: ● **stream** ● **dorado**
synchronization_id	No	String	DR relationship ID. This parameter is optional for dual-instance streaming DR and mandatory for three-instance streaming DR.
post_process_config	No	String	Self-healing is supported. If the value of this parameter is **AUTO**, self-healing is supported. If it is **MANUAL**, self-healing is not supported. If this parameter is left blank, the default value **MANUAL** is used. To use this function, see "DR Management" > "Performing a Primary/ Standby Switchover" in *User Guide*. Enumerated values: ● AUTO ● MANUAL

Response Parameters

Status code: 202

Table 4-1075 Response body parameters

Parameter	Type	Description
job_id	String	Job ID.

Status code: default

Table 4-1076 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

```
POST https://{Endpoint}/rds/gaussdb/v3.6/instances/2ec6359607c94871888cdba0d4b727acin14/disaster-recovery/switchover

{
  "synchronization_id" : "cd8c4a4a-2436-46aa-8c79-172c9019268f",
  "disaster_type" : "stream"
}
```

Example Response

Status code: 202

Normal

```
{
  "job_id" : "8bf0175e-562e-45f6-942a-2e2c72c577d7"
}
```

Status Code

Status Code	Description
202	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.17.9 Re-establishing a DR Relationship

Function

This API is used to re-establish a DR relationship during the streaming DR.

URI

POST /gaussdb/v3.6/instances/{instance_id}/disaster-recovery/restore

Table 4-1077 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-1078 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Table 4-1079 Request body parameters

Parameter	Mandatory	Type	Description
disaster_type	Yes	String	DR type. Enumerated values: • **stream** • **dorado**
synchronization_id	No	String	DR relationship ID. This parameter is optional for dual-instance streaming DR and mandatory for three-instance streaming DR.

Response Parameters

Status code: 202

Table 4-1080 Response body parameters

Parameter	Type	Description
job_id	String	Job ID.

Status code: default

Table 4-1081 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

```
POST https://{Endpoint}/rds/gaussdb/v3.6/instances/2ec6359607c94871888cdba0d4b727acin14/disaster-recovery/restore

{
  "synchronization_id" : "cd8c4a4a-2436-46aa-8c79-172c9019268f",
  "disaster_type" : "stream"
}
```

Example Response

Status code: 202

Normal

```
{
  "job_id" : "8bf0175e-562e-45f6-942a-2e2c72c577d7"
}
```

Status Code

Status Code	Description
202	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.17.10 Enabling a DR Drill

Function

This API is used to enable a DR drill. Currently, this API is only suitable for dual-instance streaming DR.

URI

POST /gaussdb/v3.6/instances/{instance_id}/disaster-recovery/simulation-start

Table 4-1082 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-1083 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Table 4-1084 Request body parameters

Parameter	Mandatory	Type	Description
xlog_keep_ratio	No	Integer	Percentage of the log cache space to the available disk space. The value ranges from 10 to 100. The default value is 10.
disaster_type	Yes	String	DR type. Enumerated value: **stream**.

Response Parameters

Status code: 202

Table 4-1085 Response body parameters

Parameter	Type	Description
job_id	String	Job ID.

Status code: default

Table 4-1086 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

Enabling a DR drill

```
POST https://{Endpoint}/rds/gaussdb/v3.6/instances/02b7a8fb063a4c46801f57c482950347in14/disaster-recovery/simulation-start

{
  "xlog_keep_ratio" : 10,
  "disaster_type" : "stream"
}
```

Example Response

Status code: 202

Normal

```
{
  "job_id" : "9c242441-fdcc-404f-ba78-f7856aed3edf"
}
```

Status Code

Status Code	Description
202	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.17.11 Disabling a DR Drill

Function

This API is used to disable a DR drill for a DR instance. Currently, this API is only suitable for streaming DR.

URI

POST /gaussdb/v3.6/instances/{instance_id}/disaster-recovery/simulation-stop

Table 4-1087 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-1088 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Table 4-1089 Request body parameters

Parameter	Mandatory	Type	Description
disaster_type	Yes	String	DR type. Enumerated value: **stream**.

Response Parameters

Status code: 202

Table 4-1090 Response body parameters

Parameter	Type	Description
job_id	String	Job ID.

Status code: default

Table 4-1091 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

Disabling a DR drill

```
POST https://{Endpoint}/rds/gaussdb/v3.6/instances/02b7a8fb063a4c46801f57c482950347in14/disaster-recovery/simulation-stop

{
  "disaster_type" : "stream"
}
```

Example Response

Status code: 202

Normal

```
{
  "job_id" : "9c242441-fdcc-404f-ba78-f7856aed3edf"
}
```

Status Code

Status Code	Description
202	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.17.12 Enabling Log Cache

Function

This API is used to enable Log Cache for the primary instance. Currently, this API is only suitable for streaming DR.

URI

POST /gaussdb/v3.6/instances/{instance_id}/disaster-recovery/keep-log-start

Table 4-1092 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-1093 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Table 4-1094 Request body parameters

Parameter	Mandatory	Type	Description
xlog_keep_ratio	No	Integer	Percentage of the log cache space to the available disk space. The value ranges from 10 to 100. The default value is 10.
disaster_type	Yes	String	DR type. Enumerated value: **stream**.

Response Parameters

Status code: 202

Table 4-1095 Response body parameters

Parameter	Type	Description
job_id	String	Job ID.

Status code: default

Table 4-1096 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

Enabling Log Cache

```
POST https://{Endpoint}/rds/gaussdb/v3.6/instances/02b7a8fb063a4c46801f57c482950347in14/disaster-recovery/keep-log-start

{
  "xlog_keep_ratio" : 10,
  "disaster_type" : "stream"
}
```

Example Response

Status code: 202

Normal

```
{
  "job_id" : "9c242441-fdcc-404f-ba78-f7856aed3edf"
}
```

Status Code

Status Code	Description
202	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.17.13 Disabling Log Cache of the Primary Instance

Function

This API is used to disable Log Cache. Currently, this API is only suitable for streaming DR.

URI

POST /gaussdb/v3.6/instances/{instance_id}/disaster-recovery/keep-log-stop

Table 4-1097 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-1098 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Table 4-1099 Request body parameters

Parameter	Mandatory	Type	Description
disaster_type	Yes	String	DR type. Enumerated value: **stream**.
is_force_release	No	Boolean	Forcible or not. The value is **false** by default when this parameter is not transferred. Enumerated values: ● **true**: forcibly disabling Log Cache ● **false**: not forcibly disabling Log Cache

Response Parameters

Status code: 202

Table 4-1100 Response body parameters

Parameter	Type	Description
job_id	String	Job ID.

Status code: default

Table 4-1101 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

Disabling Log Cache

```
POST https://{Endpoint}/rds/gaussdb/v3.6/instances/02b7a8fb063a4c46801f57c482950347in14/disaster-recovery/keep-log-stop

{
  "disaster_type" : "stream"
}
```

Example Response

Status code: 202

Normal

```
{
  "job_id" : "9c242441-fdcc-404f-ba78-f7856aed3edf"
}
```

Status Code

Status Code	Description
202	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.17.14 Manually Deleting Failed DR Tasks

Function

This API is used to manually delete failed DR tasks.

URI

POST /gaussdb/v3/instances/{instance_id}/disaster-recovery/construct/cleanup

Table 4-1102 URI parameters

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-1103 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: • **zh-cn** • **en-us**

Table 4-1104 Request body parameters

Parameter	Mandatory	Type	Description
disaster_type	Yes	String	DR type. Enumerated values: • **stream** • **dorado**
synchronization_id	Yes	String	DR relationship ID.

Response Parameters

Status code: 202

Table 4-1105 Response body parameters

Parameter	Type	Description
job_id	String	Job ID.

Status code: default

Table 4-1106 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

Manually deleting failed DR tasks

```
POST https://{Endpoint}/rds/gaussdb/v3/instances/02b7a8fb063a4c46801f57c482950347in14/disaster-recovery/construct/cleanup

{
  "disaster_type" : "stream",
  "synchronization_id" : "fb063a4c46801f57c48295034734"
}
```

Example Response

Status code: 202

Normal

```
{
  "job_id" : "9c242441-fdcc-404f-ba78-f7856aed3edf"
}
```

Status Code

Status Code	Description
202	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.17.15 Promoting a DR Instance to Primary or Demoting a Primary Instance to DR After a DR Switchover Failure

Function

This API is used to promote a DR instance to primary or demote a primary instance to DR after a DR switchover failure.

URI

POST /gaussdb/v3/instances/{instance_id}/disaster-recovery/switchover/repair

Table 4-1107 URI parameters

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-1108 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. Default value: **en-us** Enumerated values: ● **zh-cn** ● **en-us**

Table 4-1109 Request body parameters

Parameter	Mandatory	Type	Description
target_role	Yes	String	Target role. Enumerated values: ● **master** ● **disaster**
synchronization_id	Yes	String	DR relationship ID.

Response Parameters

Status code: 202

Table 4-1110 Response body parameters

Parameter	Type	Description
job_id	String	Job ID.

Status code: default

Table 4-1111 Response body parameters

Parameter	Type	Description
error_code	String	Error code.
error_msg	String	Error message.

Example Request

Manually promoting a DR instance to primary or demoting a primary instance to DR after a DR switchover failure

```
POST https://{Endpoint}/rds/gaussdb/v3/instances/02b7a8fb063a4c46801f57c482950347in14/disaster-recovery/switchover/repair

{
  "target_role" : "master",
  "synchronization_id" : "fb063a4c46801f57c48295034734"
}
```

Example Response

Status code: 202

Normal

```
{
  "job_id" : "9c242441-fdcc-404f-ba78-f7856aed3edf"
}
```

Status Code

Status Code	Description
202	Normal
default	Abnormal

Error Code

For details, see **Error Codes**.

4.17.16 Querying Instances That Can Be Used as a Primary Instance in a DR Relationship

Function

This API is used to query instances that can be used as a primary instance in a DR relationship.

URI

POST /gaussdb/v3/instances/disaster/instanceList

Request Parameters

Table 4-1112 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. The default value is **en-us**. Enumerated values: • **zh-cn** • **en-us**

Table 4-1113 Request body parameters

Parameter	Mandatory	Type	Description
disaster_type	Yes	String	DR type. Minimum length: 2 characters Maximum length: 64 characters

Response Parameters

Status code: 200

Table 4-1114 Response body parameters

Parameter	Type	Description
instance_list	Object	Instance list.

Table 4-1115 instance_list

Parameter	Type	Description
instance_id	String	Instance ID.
instance_name	String	Instance name.
instance_status	String	Instance status.
instance_type	String	Instance type.
engine_version	String	DB engine version.
consistency_protocol	String	Consistency protocol of a DR instance.
network_protocol	String	Network protocol of a DR instance.

Status code: 400

Table 4-1116 Response body parameters

Parameter	Type	Description
error_code	String	Error code. Minimum length: 8 characters Maximum length: 36 characters
error_msg	String	Error message. Minimum length: 2 characters Maximum length: 512 characters

Example Request

```
POST https://{Endpoint}/rds/gaussdb/v3/instances/disaster/instanceList

{
  "disaster_type" : "stream"
}
```


Example Response

Status code: 200

reponseBody


```
{
  "instance_list" : [ {
  "instance_id" : "436f8521862140c6ac51571b69157a88in14",
  "instance_name" : "gauss-001",
  "instance_status" : "normal",
  "instance_type" : "Ha",
  "engine_version" : 3.2,
  "consistency_protocol" : "quorum",
  "network_protocol" : "IPv4"
 } ]
}
```


Status Code

Status Code	Description
200	Normal
400	Abnormal

Error Code

For details, see **Error Codes**.

4.17.17 (Recommended) Querying Instances That Can Be Used as a Primary Instance in a DR Relationship

Function

This API is used to query instances that can be used as a primary instance in a DR relationship.

URI

POST /gaussdb/v3.1/instances/disaster/instance-list

Request Parameters

Table 4-1117 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.

Parameter	Mandatory	Type	Description
X-Language	No	String	Language. The default value is **en-us**. Enumerated values: • **zh-cn** • **en-us**

Table 4-1118 Request body parameters

Parameter	Mandatory	Type	Description
disaster_type	Yes	String	DR type. Minimum length: 2 characters Maximum length: 64 characters

Response Parameters

Status code: 200

Table 4-1119 Response body parameters

Parameter	Type	Description
instance_list	Object	Instance list.

Table 4-1120 instance_list

Parameter	Type	Description
instance_id	String	Instance ID.
instance_name	String	Instance name.
instance_status	String	Instance status.
instance_type	String	Instance type.
engine_version	String	DB engine version.
consistency_protocol	String	Consistency protocol of a DR instance.

Parameter	Type	Description
network_protocol	String	Network protocol of a DR instance.

Status code: 400

Table 4-1121 Response body parameters

Parameter	Type	Description
error_code	String	Error Code Minimum length: 8 characters Maximum length: 36 characters
error_msg	String	Error message. Minimum length: 2 characters Maximum length: 512 characters

Example Request

```
POST https://{Endpoint}/rds/gaussdb/v3.1/instances/disaster/instanceList

{
  "disaster_type" : "stream"
}
```

Example Response

Status code: 200

reponseBody

```
{
    "instance_list" : [ {
    "instance_id" : "436f8521862140c6ac51571b69157a88in14",
    "instance_name" : "gauss-001",
    "instance_status" : "normal",
    "instance_type" : "Ha",
    "engine_version" : V2.0-3.2,
    "consistency_protocol" : "quorum",
    "network_protocol" : "IPv4"
  } ]
}
```

Status Code

Status Code	Description
200	Normal
400	Abnormal

Error Code

For details, see **Error Codes**.

4.17.18 Deleting a DR Record

Function

This API is used to delete a DR record.

URI

POST /gaussdb/v3/instances/disaster/record/delete

Request Parameters

Table 4-1122 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. The default value is **en-us**. Enumerated values: • **zh-cn** • **en-us**

Table 4-1123 Request body parameters

Parameter	Mandatory	Type	Description
id	Yes	String	DR relationship ID. Minimum length: 1 character Maximum length: 64 characters

Response Parameters

Status code: 200

Table 4-1124 Response body parameters

Parameter	Type	Description
job_id	String	Main task ID.

Status code: 400

Table 4-1125 Response body parameters

Parameter	Type	Description
error_code	String	Error code. Minimum length: 8 characters Maximum length: 36 characters
error_msg	String	Error message. Minimum length: 2 characters Maximum length: 512 characters

Example Request

```
POST https://{Endpoint}/rds/gaussdb/v3/instances/disaster/record/delete

{
  "id" : "a6bbafa3-1f83-4d00-80ef-d3bcced64e68"
}
```

Example Response

Status code: 200

reponseBody

```
{
  "job_id" : "27582df7-bff6-4aec-a64a-b128da3c166"
}
```

Status Code

Status Code	Description
200	Normal
400	Abnormal

Error Code

For details, see **Error Codes**.

4.17.19 Changing the Primary Instance of a DR Relationship

Function

This API is used to change the primary instance of the DR instance in a DR relationship.

URI

POST /gaussdb/v3/instances/{instance_id}/disaster-recovery/change-primary

Table 4-1126 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-1127 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. The default value is **en-us**. Enumerated values: • **zh-cn** • **en-us**

Table 4-1128 Request body parameters

Parameter	Mandatory	Type	Description
synchronization_id	Yes	String	ID of the DR relationship between the current instance and the primary instance to be changed.
disaster_type	Yes	String	DR type. Enumerated values: • **stream** • **dorado**

Response Parameters

Status code: 200

Table 4-1129 Response body parameters

Parameter	Type	Description
job_id	String	Primary task ID.

Status code: 400

Table 4-1130 Response body parameters

Parameter	Type	Description
error_code	String	Error code. Minimum length: 8 characters Maximum length: 36 characters
error_msg	String	Error message. Minimum length: 2 characters Maximum length: 512 characters

Example Request

```
POST https://{Endpoint}/rds/gaussdb/v3/instances/{instanceId}/disaster-recovery/change-primary

{
  "disaster_type" : "stream",
  "synchronization_id" : "acf48e79-0ef5-48b7-99ad-1b4ca1bbb424"
}
```

Example Response

Status code: 200

reponseBody

```
{
  "job_id" : "27582df7-bff6-4aec-a64a-b128da3c166"
}
```

Status Code

Status Code	Description
200	Normal
400	Abnormal

Error Code

For details, see **Error Codes**.

4.17.20 Pre-checking DR Operations

Function

This API is used to pre-check the operations of establishing a DR relationship, switching roles of primary and DR instances, promoting a DR instance to primary, and re-establishing a DR relationship.

URI

POST /gaussdb/v3.5/instances/{instance_id}/disaster-recovery/pre-process

Table 4-1131 URI parameter

Parameter	Mandatory	Type	Description
instance_id	Yes	String	Instance ID.

Request Parameters

Table 4-1132 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. The default value is **en-us**. Value: • **zh-cn** • **en-us**

Table 4-1133 Request body parameters

Parameter	Mandatory	Type	Description
dr_type	Yes	String	DR type. Value: • **stream** • **dorado**

Parameter	Mandatory	Type	Description
dr_action	Yes	String	DR operation type. Value: • **switchover** • **failover** • **restore** • **construct**
synchronization_id	No	String	DR relationship ID. This parameter is optional for dual-instance streaming DR. For three-instance streaming DR, when **dr_action** is set to **switchover**, **failover**, or **restore**, this parameter is mandatory.
dr_ip	No	String	DR IP address of the DR cluster during the establishment of the DR relationship. This parameter is mandatory when the value of **dr_action** is **construct**.
dr_user_name	No	String	Database account of the DR cluster during the establishment of the DR relationship. This parameter is mandatory when the value of **dr_action** is **construct**.
dr_user_password	No	String	Database password of the DR cluster during the establishment of the DR relationship. This parameter is mandatory when the value of **dr_action** is **construct**.

Response Parameters

Status code: 200

Table 4-1134 Response body parameters

Parameter	Type	Description
job_id	String	Main task ID.

Status code: 400

Table 4-1135 Response body parameters

Parameter	Type	Description
error_code	String	Error code. Minimum length: 8 characters Maximum length: 36 characters
error_msg	String	Error message. Minimum length: 2 characters Maximum length: 512 characters

Example Request

```
POST https://{Endpoint}/rds/gaussdb/v3.5/instances/02b7a8fb063a4c46801f57c482950347in14/disaster-recovery/pre-process

{
  "dr_type" : "stream",
  "synchronization_id" : "acf48e79-0ef5-48b7-99ad-1b4ca1bbb424",
  "dr_action" : "switchover"
}
```

Example Response

Status code: 200

reponseBody

```
{
  "job_id" : "27582df7-bff6-4aec-a64a-b128da3c166"
}
```

Status Codes

Status Code	Description
200	Normal
400	Abnormal

Error Code

For details, see **Error Codes**.

4.18 License Management

4.18.1 Obtaining a Revocation Code

Function

- This function can only be used to obtain the revocation code of a commercial license control item.
- Performing this operation will change the ESN. Keep the revocation code secure after obtaining it.

URI

POST /gaussdb/v3/license/invalidation

Request Parameters

Table 4-1136 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. The default value is **en-us**. Value: • **zh-cn** • **en-us**

Response Parameters

Status code: 200

Table 4-1137 Response body parameters

Parameter	Type	Description
revocation_codes	Array of strings	License revocation code list.

Status code: 400

Table 4-1138 Response body parameters

Parameter	Type	Description
error_code	String	Error code. Minimum length: 8 characters Maximum length: 36 characters

Parameter	Type	Description
error_msg	String	Error message. Minimum length: 2 characters Maximum length: 512 characters

Example Request

```
POST https://{Endpoint}/rds/gaussdb/v3/license/invalidation
```

Example Response

Status code: 200

reponseBody


```
{
  "revocation_codes" : ["LIC202***R525T:EF381CEA7******3454F8A9F" ]
}
```


Status Codes

Status Code	Description
200	Normal
400	Abnormal

Error Code

For details, see **Error Codes**.

4.19 Log Management

4.19.1 Querying Operation Logs

Function

This API is used to query operation logs.

URI

POST /v3/operation-traces

Request Parameters

Table 4-1139 Request header parameters

Parameter	Mandatory	Type	Description
X-Mini-Token	Yes	String	User token.
X-Language	No	String	Language. The default value is **en-us**. Value: • **zh-cn** • **en-us**

Table 4-1140 Request body parameters

Parameter	Mandatory	Type	Description
offset	No	Integer	Index offset. If **offset** is set to *N*, the resource query starts from the *N*+1 data entry. The default value is **0**, indicating that the query starts from the first data entry. The value must be a number but cannot be a negative number.
limit	No	Integer	Number of records to be queried. The default value is **10**. The value cannot be a negative number. The minimum value is **1**, and the maximum value is **100**.
start_time	No	String	Time when the query for operation logs starts, in the format of yyyy-mm-ddThh:mm:ssZ (UTC time).
end_time	No	String	Time when the query for operation logs ends, in the format of yyyy-mm-ddThh:mm:ssZ (UTC time).
resource_type	No	String	Resource type.
resource_name	No	String	Resource name.
trace_name	No	String	Operation name.

Parameter	Mandatory	Type	Description
trace_rating	No	String	Operation result. ● **normal**: The operation is successful. ● **warning**: The operation failed.
user_name	No	String	Operator.
source_ip	No	String	IP address.
sort_key	No	String	Sorting field.
sort_dir	No	String	Sorting order, which can be: ● **asc**: The query results are displayed in ascending order. ● **desc**: The query results are displayed in descending order.

Response Parameters

Status code: 200

Table 4-1141 Response body parameters

Parameter	Type	Description
total_count	Integer	Total number of records.
traces	**Table 4-1142** object	Private key pairs.

Table 4-1142 OperationTracesRecord

Parameter	Type	Description
service_type	String	Service type.
source_ip	String	IP address.
occur_time	String	Occurrence time.
resource_type	String	Resource type.
resource_name	String	Resource name.
code	String	Status code.

Parameter	Type	Description
message	String	Error message.
location_info	String	Additional information required for fault locating after a request error.
user_id	String	User ID.
user_name	String	Username.
resource_id	String	Resource ID.
trace_name	String	Operation name.
trace_rating	String	Operation result. ● **normal**: The operation is successful. ● **warning**: The operation failed.
trace_type	String	Audit type.

Status code: 400

Table 4-1143 Response body parameters

Parameter	Type	Description
error_code	String	Error Code Minimum length: 8 characters Maximum length: 36 characters
error_msg	String	Error message. Minimum length: 2 characters Maximum length: 512 characters

Example Request

```
POST https://{Endpoint}/rds/v3/operation-traces

{
    "end_time": "2024-12-11T03:56:31Z+0800",
    "limit": "10",
    "offset": "0",
    "service_type": "GaussDB",
    "resource_type": "user",
    "sort_dir": "occur_time",
    "sort_key": "desc",
    "start_time": "2024-12-11T02:56:31Z+0800",
    "trace_rating": "normal",
    "trace_name": "loginForOpenApi",
    "user_name": "a",
    "source_ip": "29.5.135.147"
}
```

Example Response

Status code: 200

reponseBody


```
{
    "traces": [
        {
            "code": 200,
            "message": null,
            "service_type": "GaussDB",
            "source_ip": "29.5.135.147",
            "occur_time": "2024-12-11T03:04:03Z",
            "resource_type": "user",
            "resource_name": "a",
            "location_info": null,
            "user_id": "f0c78038c207456eb88863b0aa532673",
            "user_name": "z30034744",
            "resource_id": "f0c78038c207456eb88863b0aa532673",
            "trace_name": "loginForOpenApi",
            "trace_rating": "normal",
            "trace_type": "ApiCall"
        }
    ],
    "total_count": 1
}
```


Status Codes

Status Code	Description
200	Normal
400	Abnormal

Error Code

For details, see **Error Codes**.

5 Permissions

This section describes the permissions required for GaussDB Management Platform (TPOPS) APIs.

Table 5-1 Platform user management

Function	API	Permission
Generating a token	POST /v3/auth/mini-token	None
Logging in to the system	POST /v3/login	None
Redirecting a login request	GET /v3/login/redirect	None
Logging out of the system	POST /v3/logout	None
Querying users	GET /v3/users	dbs:user:list
Adding a user	POST /v3/users	dbs:user:addOrUpdate
Deleting a user	DELETE /v3/users/{user_id}	dbs:user:delete
Querying Basic Information About a User	GET /v3/users/{user_id}	dbs:base
Updating information about a user	PUT /v3/users/{user_id}	dbs:user:addOrUpdate
Checking whether a username is valid	POST /v3/users/name-validation	dbs:user:addOrUpdate
Letting a user reset their own password	PUT /v3/users/{user_id}/password	dbs:base

Function	API	Permission
Letting a user reset another user's password	PUT /v3/users/{user_id}/password/others	dbs:user:addOrUpdate
Querying the instances allocated to a user	GET /v3/users/{user_id}/instances	dbs:user:list
Allocating instances to a user	PUT /v3/users/{user_id}/instances	dbs:user:addOrUpdate
Querying roles of a user	GET /v3/users/{user_id}/roles	dbs:user:list
Assigning roles to a user	PUT /v3/users/{user_id}/roles	dbs:user:addOrUpdate
Querying user groups that a user has been added to	GET /v3/users/{user_id}/user-groups	dbs:user:list
Adding a user to user groups	PUT /v3/users/{user_id}/user-groups	dbs:user:addOrUpdate
Locking users	POST /v3/users/lock	dbs:user:addOrUpdate
Unlocking users	POST /v3/users/unlock	dbs:user:addOrUpdate
Checking whether a password is a weak password	POST /v3/users/weak-password-verification	dbs:base
Querying user groups	GET /v3/user-groups	dbs:userGroup:list
Adding a user group	POST /v3/user-groups	dbs:userGroup:addOrUpdate
Updating information about a user group	PUT /v3/user-groups/{user_group_id}	dbs:userGroup:addOrUpdate
Deleting a user group	DELETE /v3/user-groups/{user_group_id}	dbs:userGroup:delete
Querying user roles of a user group	GET /v3/user-groups/{user_group_id}/roles	dbs:userGroup:list
Updating role information about a user group	PUT /v3/user-groups/{user_group_id}/roles	dbs:userGroup:addOrUpdate
Querying users in a user group	GET /v3/user-groups/{user_group_id}/users	dbs:userGroup:list
Updating user information about a user group	PUT /v3/user-groups/{user_group_id}/users	dbs:userGroup:addOrUpdate
Querying roles	GET /v3/roles	dbs:userRole:list

Function	API	Permission
Adding a role	POST /v3/roles	dbs:userRole:addOrUpdate
Updating information about a role	PUT /v3/roles/{role_id}	dbs:userRole:addOrUpdate
Deleting a role	DELETE /v3/roles/{role_id}	dbs:userRole:delete
Querying the permissions of a role	GET /v3/roles/{role_id}/permissions	dbs:userRole:list
Querying permissions	GET /v3/permissions	dbs:userRole:list

Table 5-2 Backup management

Function	API	Permission
Configuring an automated backup policy	PUT /gaussdb/v3/instances/{instance_id}/backups/policy	gaussdb:instance:modifyBackupPolicy
Stopping a backup	POST /gaussdb/v3/instances/{instance_id}/backups/stop	gaussdb:backup:stopBackup
Configuring extended backup policies	PUT /gaussdb/v3/instances/{instance_id}/backups/config	gaussdb:backup:setConfiguration
Querying an automated backup policy	GET /gaussdb/v3/instances/{instance_id}/backups/policy	gaussdb:backup:list gaussdb:instance:list
Querying extended backup policies	GET /gaussdb/v3/instances/{instance_id}/backups/config	gaussdb:backup:showConfiguration
Creating a manual backup	POST /gaussdb/v3/backups	gaussdb:backup:create
Deleting a manual backup	DELETE /gaussdb/v3/backups/{backup_id}	gaussdb:backup:delete
Querying information about the original DB instance based on a specific point of time or a backup file	GET /gaussdb/v3/instance-snapshot	gaussdb:instance:list gaussdb:backup:list

Function	API	Permission
Querying information about the original DB instance based on a specific point of time or a backup file (recommended)	GET /gaussdb/v3.1/instance-snapshot	gaussdb:instance:list gaussdb:backup:list
Querying instances that can be used for backups and restorations	GET /gaussdb/v3/restorable-instances	gaussdb:instance:list gaussdb:backup:list
Querying instances that can be used for backups and restorations (recommended)	GET /gaussdb/v3.1/restorable-instances	gaussdb:instance:list gaussdb:backup:list
Querying the restoration time range	GET /gaussdb/v3/instances/{instance_id}/restore-time	gaussdb:instance:list gaussdb:backup:list
Querying differential backups	GET /gaussdb/v3/instances/{instance_id}/differential-backups	gaussdb:backup:list
Restoring data to the original instance or an existing instance	POST /gaussdb/v3/instances/recovery	gaussdb:instance:restoreInPlace
Confirming data integrity after a backup is restored	POST /gaussdb/v3/instances/{instance_id}/confirm-restore-data	gaussdb:backup:confirm
Querying backups	GET /gaussdb/v3.1/backups	gaussdb:backup:list
Querying backups (recommended)	GET /gaussdb/v3.2/backups	gaussdb:backup:list
Querying the archive log path of an instance	GET /gaussdb/v3/instances/{instance_id}/archive-log-path	gaussdb:backup:getArchiveLogPath
Synchronizing metadata using storage media	POST /gaussdb/v3/backups/sync-metadata	gaussdb:backup:syncMetadata
Deleting backup data	DELETE /gaussdb/v3/backups/batch-delete	gaussdb:backup:delete
Installing an SSL certificate	POST /gaussdb/v3/instances/{instance_id}/backups/ssl-certs	gaussdb:backup:create

Function	API	Permission
Changing a backup method	PUT /gaussdb/v3.1/instances/{instance_id}/change-backup-media	gaussdb:backup:changeBackupMedia
Querying backup details	GET /gaussdb/v3/instances/{instance_id}/backups/detail	gaussdb:backup:queryBackupInfoDetail
Querying restoration details	GET /gaussdb/v3/instances/{instance_id}/backups/detail	gaussdb:backup:queryBackupInfoDetail
Sending backup or restoration results	PUT /gaussdb/v3/instances/{instance_id}/backups/notify	gaussdb:backup:notifyBackupAndRestoreResult

Table 5-3 Parameter configuration

Function	API	Permission
Obtaining the parameter template of a specified DB instance	GET /gaussdb/v3.1/instances/{instance_id}/configurations	gaussdb:param:list gaussdb:instance:list
Obtaining the parameter template of a specified DB instance (recommended)	GET /gaussdb/v3.2/instances/{instance_id}/configurations	gaussdb:param:list gaussdb:instance:list
Modifying parameters of a specified DB instance	PUT /gaussdb/v3/instances/{instance_id}/configurations	gaussdb:param:modify

Table 5-4 Database and user management

Function	API	Permission
Creating a database	POST /gaussdb/v3/instances/{instance_id}/database	gaussdb:instance:createDatabase gaussdb:instance:modify
Creating a database schema	POST /gaussdb/v3/instances/{instance_id}/schema	gaussdb:instance:createDatabaseSchema gaussdb:instance:modify
Querying database schemas	GET /gaussdb/v3/instances/{instance_id}/schemas	gaussdb:instance:list gaussdb:instance:listSchemas

Function	API	Permission
Creating a database user	POST /gaussdb/v3/instances/{instance_id}/db-user	gaussdb:instance:createDatabaseUser gaussdb:instance:modify
Deleting a database user	DELETE /gaussdb/v3/instances/{instance_id}/db-user	gaussdb:databaseUser:drop gaussdb:databaseUser:drop
Granting permissions to and revoking permissions from a database user	POST /gaussdb/v3/instances/{instance_id}/db-user/privileges	gaussdb:privileges:grantRevoke
Modifying attributes of a database user	POST /gaussdb/v3/instances/{instance_id}/db-user/attributes	gaussdb:instance:modifyDatabaseUser
Initializing the password of a built-in database user	PUT /gaussdb/v3/instances/{instance_id}/db-user/init-password	gaussdb:sysDatabaseUser:initPassword
Resetting the password of a database user	PUT /gaussdb/v3/instances/{instance_id}/db-user/password	gaussdb:instance:modifyDatabasePasswd gaussdb:instance:modify
Locking a database user	PUT /gaussdb/v3/instances/{instance_id}/db-user/lock	gaussdb:instance:modifyDatabaseUser
Unlocking a database user	PUT /gaussdb/v3/instances/{instance_id}/db-user/unlock	gaussdb:instance:modifyDatabaseUser
Querying databases	GET /gaussdb/v3/instances/{instance_id}/databases	gaussdb:instance:list gaussdb:instance:listDatabases
Querying database users	GET /gaussdb/v3/instances/{instance_id}/db-users	gaussdb:instance:list gaussdb:instance:listDatabaseUsers
Querying connection users, applications, table statistics, and lock information of a database	GET /gaussdb/v3/instances/{instance_id}/database/relation-and-lock	gaussdb:dbRelationAndLockInfo:list

Function	API	Permission
Flashing back a table	POST /gaussdb/v3/instances/{instance_id}/db-table/flashback	gaussdb:sysDatabaseTable:flashBack
Creating a database role	POST /gaussdb/v3/instances/{instance_id}/db-role	gaussdb:instance:createDbRole
Deleting a database role	DELETE /gaussdb/v3/instances/{instance_id}/db-role	gaussdb:instance:deleteDbRole
Granting permissions to and revoking permissions from a database user or a database role	POST /gaussdb/v3.1/instances/{instance_id}/db-user/privileges	gaussdb:instance:grantDatabaseUserPrivileges

Table 5-5 Database diagnosis and optimization

Function	API	Permission
Querying node and component information on database key views	GET /gaussdb/v3/instances/{instance_id}/key-view-execute-node	gaussdb:instance:listRealTimeSession
Querying real-time sessions of a component on a node	POST /gaussdb/v3/instances/{instance_id}/real-time-session	gaussdb:instance:listRealTimeSession
Killing specified sessions	POST /gaussdb/v3/instances/{instance_id}/kill-session	gaussdb:instance:killSession
Killing all idle sessions	POST /gaussdb/v3/instances/{instance_id}/kill-free-session	gaussdb:instance:killFreeSession
Querying table analysis information	POST /gaussdb/v3/instances/{instance_id}/database-table-analysis	gaussdb:instance:getDatabasesTableAnalysis
Querying index analysis information	POST /gaussdb/v3/instances/{instance_id}/database-index-analysis	gaussdb:instance:getDatabasesIndexAnalysis
Querying WDR snapshot status	GET /gaussdb/v3/instances/{instance_id}/wdr-snapshot/status	gaussdb:instance:list gaussdb:instance:listWdrSnapshot

Function	API	Permission
Enabling or disabling WDR snapshots	PUT /gaussdb/v3/instances/{instance_id}/wdr-snapshot/status	gaussdb:instance:modify gaussdb:instance:operateWdrSnapshot
Querying collection results for WDR snapshots	GET /gaussdb/v3/instances/{instance_id}/wdr-snapshots	gaussdb:instance:list gaussdb:instance:listWdrSnapshot
Generating a WDR snapshot immediately	POST /gaussdb/v3/instances/{instance_id}/wdr-snapshots	gaussdb:instance:modify gaussdb:instance:operateWdrSnapshot
Collecting a WDR snapshot	POST /gaussdb/v3/instances/{instance_id}/wdr-snapshots/collect	gaussdb:instance:modify gaussdb:instance:list gaussdb:instance:operateWdrSnapshot
Downloading a WDR snapshot	POST /gaussdb/v3/instances/{instance_id}/wdr-snapshots/download	gaussdb:instance:modify gaussdb:instance:operateWdrSnapshot
Querying top SQL data	POST /v3/instances/{instance_id}/top-sqls	gaussdb:monitor:topSql:list
Querying nodes where slow SQL statements are executed	POST /gaussdb/v3/instances/{instance_id}/slow-sql-execute-node	gaussdb:monitor:slowSql:list
Querying slow SQL statements	POST /gaussdb/v3/instances/{instance_id}/slow-sql-list	gaussdb:monitor:slowSql:list
Querying details about a slow SQL statement	POST /gaussdb/v3/instances/{instance_id}/slow-sql-detail	gaussdb:monitor:slowSql:list
Creating a SQL patch	POST /gaussdb/v3/instances/{instance_id}/sql-patch	gaussdb:instance:operateSqlPatch
Querying SQL patch information	GET /gaussdb/v3/instances/{instance_id}/sql-patch	gaussdb:instance:getSqlPatch
Enabling or disabling a SQL Patch or deleting a SQL patch	POST /gaussdb/v3/instances/{instance_id}/handle-sql-patch	gaussdb:instance:operateSqlPatch

Table 5-6 Intelligent O&M

Function	API	Permission
Managing an instance or canceling the management for an instance	POST /v3/db-assist/manage-instance	gaussdb:instance:dbmindManage
Querying instances that have been managed or can be managed by a DBMind instance	GET /v3/db-assist/manageable-instances	gaussdb:instance:list
Querying instances that have been managed or can be managed by a DBMind instance (recommended)	GET /v3.1/db-assist/manageable-instances	gaussdb:instance:list
Querying DBMind instances	GET /v3/db-assist/instances	gaussdb:instance:list
Querying DBMind instances (recommended)	GET /v3.1/db-assist/instances	gaussdb:instance:list

Table 5-7 SQL throttling

Function	API	Permission
Creating a throttling task	POST /gaussdb/v3/instances/{instance_id}/limit-task	gaussdb:instance:flowlimitAddOrUpdate gaussdb:instance:createSQLLimitRule
Deleting a throttling task	DELETE /gaussdb/v3/instances/{instance_id}/limit-task/{task_id}	gaussdb:instance:flowlimitDelete gaussdb:instance:deleteSQLLimitRule
Querying throttling tasks based on search criteria	GET /gaussdb/v3/instances/{instance_id}/limit-task-list	gaussdb:instance:listFlowlimit gaussdb:instance:getSQLLimitRules
Querying SQL templates of a specified node	GET /gaussdb/v3/instances/{instance_id}/list-node-limit-sql-model	gaussdb:instance:listFlowlimit gaussdb:instance:getSQLLimitRuleTemplate

Table 5-8 Host key management

Function	API	Permission
Generating a key pair for host connection	POST /gaussdb/v3/hosts/key-pair	gaussdb:keyPair:generate
Querying key pair information	GET /gaussdb/v3/hosts/key-pair	gaussdb:dataCenter:list
Modifying key information	PUT /gaussdb/v3/hosts/key-pair	gaussdb:keyPair:modify
Deleting a key pair	DELETE /gaussdb/v3/hosts/key-pair	gaussdb:keyPair:delete
Downloading a public key	GET /gaussdb/v3/hosts/public-key	gaussdb:keyPair:download

Table 5-9 Data center management

Function	API	Permission
Querying hosts	GET /gaussdb/v3/hosts	gaussdb:host:list
Adding hosts	POST /gaussdb/v3/hosts	gaussdb:host:add
Modifying static host information	PUT /gaussdb/v3/hosts	gaussdb:host:modify
Deleting a host	DELETE /gaussdb/v3/hosts	gaussdb:host:delete
Querying the standardized inspection result of a host	GET /gaussdb/v3/hosts/{host_id}/detection-results	gaussdb:host:list
Downloading a host import template	GET /gaussdb/v3/hosts/template-file	gaussdb:host:add
Batch importing host files for check	POST /gaussdb/v3/hosts/file-verification	gaussdb:host:add
Querying available host specifications	GET /gaussdb/v3/hosts/specifications	gaussdb:host:list
Querying data centers	GET /gaussdb/v3/data-centers	gaussdb:dataCenter:list
Adding a data center	POST /gaussdb/v3/data-centers	gaussdb:dataCenter:add
Editing data center information	PUT /gaussdb/v3/data-centers	gaussdb:dataCenter:modify

Function	API	Permission
Deleting a data center	DELETE /gaussdb/v3/data-centers	gaussdb:dataCenter:delete

Table 5-10 Routine inspection

Function	API	Permission
Querying inspection tasks	GET /gaussdb/v3/inspections	gaussdb:instance:getInspection
Creating an inspection task	POST /gaussdb/v3/inspections	gaussdb:instance:createInspection
Deleting inspection tasks in batches	POST /gaussdb/v3/inspections/batch-delete	gaussdb:instance:batchDeleteInspection
Executing an inspection task	POST /gaussdb/v3/inspections/{inspection_id}/start	gaussdb:instance:startInspection
Querying details of an inspection task	GET /gaussdb/v3/inspections/{inspection_id}	gaussdb:instance:getInspection
Modifying an inspection task	PUT /gaussdb/v3/inspections/{inspection_id}	gaussdb:instance:modifyInspection
Querying inspection result statistics	GET /gaussdb/v3/inspections/records/{record_id}/statistic	gaussdb:instance:getInspection
Querying execution details of inspection items	GET /gaussdb/v3/inspections/records/{record_id}	gaussdb:instance:getInspection
Querying inspection items	GET /gaussdb/v3/inspections/items	gaussdb:instance:getInspection
Querying inspection records	POST /gaussdb/v3/inspections/records	gaussdb:instance:getInspection
Deleting inspection records in batches	POST /gaussdb/v3/inspections/records/batch-delete	gaussdb:instance:deleteInspectionRecord
Querying inspection scenarios	GET /gaussdb/v3/inspections/scenes	gaussdb:instance:getInspection
Adding an inspection scenario	POST /gaussdb/v3/inspections/scenes	gaussdb:instance:createInspection

Function	API	Permission
Deleting an inspection scenario	DELETE /gaussdb/v3/inspections/scenes	gaussdb:instance:batchDeleteInspection
Modifying an inspection scenario	PUT /gaussdb/v3/inspections/scenes	gaussdb:instance:modifyInspection

Table 5-11 Instance management

Function	API	Permission
Configuring the extended information of an instance	PUT /gaussdb/v3/instances/{instance_id}/extend-info	gaussdb:instance:setInstanceExtendInfo
Querying the extended information of an instance	GET /gaussdb/v3/instances/{instance_id}/extend-info	gaussdb:instance:list
Querying components of an instance	GET /gaussdb/v3/instances/{instance_id}/components	gaussdb:instance:list
Performing a primary/standby switchover	POST /gaussdb/v3/instances/{instance_id}/switch-shard	gaussdb:instance:switchShard gaussdb:instance:modify
Deleting an instance	DELETE /gaussdb/v3/instances/{instance_id}/management	gaussdb:instance:delete
Creating an instance or restoring data to a new instance	POST /gaussdb/v3/instances	gaussdb:instance:create
Creating an instance or restoring data to a new instance (recommended)	POST /gaussdb/v3.1/instances	gaussdb:instance:create
Querying instances or instance details	GET /gaussdb/v3.1/instances	gaussdb:instance:list
Querying instances or instance details (recommended)	GET /gaussdb/v3.2/instances	gaussdb:instance:list
Stopping an instance or node	POST /gaussdb/v3/instances/{instance_id}/db-stop	gaussdb:instance:stop
Restarting a database	POST /gaussdb/v3/instances/{instance_id}/restart	gaussdb:instance:modify gaussdb:instance:restart

Function	API	Permission
Starting an instance or node	POST /gaussdb/v3/instances/{instance_id}/db-startup	gaussdb:instance:start
Checking whether hosts are associated with an instance to be managed	POST /v3/instances/management/check-hosts-manageable	gaussdb:instance:manage
Managing instances	POST /v3/instances/management	gaussdb:instance:manage
Rectifying AZ faults	POST /v3/instances/{instance_id}/repair	gaussdb:instance:repair
Adding a node	POST /v3/instances/{instance_id}/expansion	gaussdb:instance:modifySpec
Replacing a node	POST /v3/instances/{instance_id}/nodes/replacement	gaussdb:instance:replaceNode
Querying versions that an instance can be upgraded to	GET /gaussdb/v3/instances/{instance_id}/db-upgrade/candidate-versions	gaussdb:instance:upgradeDatabaseVersion
Querying versions that an instance can be upgraded to (recommended)	GET /gaussdb/v3.1/instances/{instance_id}/db-upgrade/candidate-versions	gaussdb:instance:upgradeDatabaseVersion
Upgrading an instance	PUT /gaussdb/v3.1/instances/{instance_id}/db-upgrade	gaussdb:instance:upgradeDatabaseVersion gaussdb:instance:modify
Upgrading an instance (recommended)	PUT /gaussdb/v3.2/instances/{instance_id}/db-upgrade	gaussdb:instance:upgradeDatabaseVersion gaussdb:instance:modify
Repairing a node	POST /gaussdb/v3/instances/{instance_id}/nodes/repair	gaussdb:instance:repairNode
Changing a deployment model	PUT /gaussdb/v3/instances/{instance_id}/solution	gaussdb:instance:changeDeploymentSolution
Enabling the M compatibility port	POST /gaussdb/v3/instances/{instance_id}/mysql-compatibility	gaussdb:instance:startMySQLCompatibility

Function	API	Permission
Changing or disabling M compatibility port	PUT /gaussdb/v3/instances/{instance_id}/mysql-compatibility	gaussdb:instance:updateMySQLCompatibility
Preparing for changing specifications	POST /gaussdb/v3/instances/{instance_id}/flavor-resize/before-execute	gaussdb:instance:resizeFlavor
Performing post-processing after changing specifications	POST /gaussdb/v3/instances/{instance_id}/flavor-resize/after-execute	gaussdb:instance:resizeFlavor
Changing disk size	POST /gaussdb/v3/instances/{instance_id}/volume-resize	gaussdb:instance:expandVolume
Querying the advanced features of an instance	GET /gaussdb/v3/instances/{instance_id}/advance-features	gaussdb:instance:listFeatures
Enabling advanced features	POST /gaussdb/v3/instances/{instance_id}/advance-features	gaussdb:instance:updateFeatures
Changing the description of an instance	PUT /gaussdb/v3/instances/{instance_id}/alias	gaussdb:instance:updateDescription
Performing a primary/standby DN switchover in an instance group	POST /gaussdb/v3/instances/group/switchover	gaussdb:instance:switchoverInstanceGroupItems

Table 5-12 Engine versions and specifications

Function	API	Permission
Querying DB engine versions	GET /gaussdb/v3/datastore/versions	gaussdb:instance:list
Querying DB engine versions (recommended)	GET /gaussdb/v3.1/datastore/versions	gaussdb:instance:list

Table 5-13 Microservice upgrade

Function	API	Permission
Querying instance Agents	GET /v3/instances	dbs:agent:getStatus

Function	API	Permission
Querying instance Agents (recommended)	GET /v3.2/instances	dbs:agent:getStatus
Querying Agent information about a specified instance	GET /gaussdb/v3/instances/{instance_id}/agent/status	dbs:agent:getStatus
Upgrading the Agent of a single instance	POST /gaussdb/v3/instances/{instance_id}/agent/upgrade	dbs:agent:upgrade
Batch upgrading or rolling back Agents	POST /v3/instances/agent/batch-upgrade	dbs:agent:getStatus

Table 5-14 Alarm management

Function	API	Permission
Testing alarm interconnection	POST /v3/alarm/notifications/interconnection/check	gaussdb:base
Adding alarm interconnection configurations	POST /v3/alarm/notifications	dbs:alarm:addNotice
Modifying alarm interconnection configurations	PUT /v3/alarm/notifications	dbs:alarm:updateNotice
Viewing alarm interconnection configurations	GET /v3/instances/{instance_id}/alarm/notifications	dbs:alarm:listNotices
Querying alarm results	POST /v3/alarm/records	dbs:alarm:list
Querying the number of alarms	GET /v3/alarm/records/statistic	dbs:alarm:list
Applying alarm settings to instances	POST /v3/alarm/update/configs	dbs:alarm:updateConfig

Table 5-15 Monitoring dashboards

Function	API	Permission
Querying monitoring metrics	POST /v3/monitor/metrics	dbs:monitor:list

Function	API	Permission
Querying monitoring metric groups	GET /v3/monitor/metrics/series	dbs:monitor:listMetrics

Table 5-16 Task center

Function	API	Permission
Querying task status by job ID	GET /v3/jobs	dbs:task:list
Querying asynchronous batch processing task details	GET /gaussdb/v3/instances/batch-async-task/records	gaussdb:instance:listBatchAsyncTaskRecords

Table 5-17 DR management

Function	API	Permission
Editing a DR task name	PUT /gaussdb/v3.6/instances/{instance_id}/disaster-recovery/rename	gaussdb:disasterRecovery:update
Querying instances that have established a DR relationship	GET /gaussdb/v3.6/disaster-recovery/relations	gaussdb:disasterRecovery:list
Querying historical DR operation records	GET /gaussdb/v3/instances/{instance_id}/disaster-recovery/records	gaussdb:instance:listRecord
Querying instance DR status	GET /gaussdb/v3.6/instances/{instance_id}/disaster-recovery/monitor	gaussdb:disasterRecovery:list
Creating a DR task	POST /gaussdb/v3.6/instances/{instance_id}/disaster-recovery/construct	gaussdb:disasterRecovery:construct
Promoting the DR instance to primary	POST /gaussdb/v3.6/instances/{instance_id}/disaster-recovery/failover	gaussdb:disasterRecovery:failover
Stopping a DR relationship	POST /gaussdb/v3.6/instances/{instance_id}/disaster-recovery/release	gaussdb:disasterRecovery:release

Function	API	Permission
Switching roles of primary and DR instances	POST /gaussdb/v3.6/instances/{instance_id}/disaster-recovery/switchover	gaussdb:disasterRecovery:switchover
Re-establishing a DR relationship	POST /gaussdb/v3.6/instances/{instance_id}/disaster-recovery/restore	gaussdb:disasterRecovery:construct gaussdb:instance:create
Enabling a DR drill	POST /gaussdb/v3.6/instances/{instance_id}/disaster-recovery/simulation-start	gaussdb:disasterRecovery:simulation
Disabling a DR drill	POST /gaussdb/v3.6/instances/{instance_id}/disaster-recovery/simulation-stop	gaussdb:disasterRecovery:simulation
Enabling Log cache	POST /gaussdb/v3.6/instances/{instance_id}/disaster-recovery/keep-log-start	gaussdb:disasterRecovery:keeplog
Disabling log cache of the primary instance	POST /gaussdb/v3.6/instances/{instance_id}/disaster-recovery/keep-log-stop	gaussdb:disasterRecovery:keeplog
Manually deleting failed DR tasks	POST /gaussdb/v3/instances/{instance_id}/disaster-recovery/construct/cleanup	gaussdb:disasterRecovery:constructCleanup
Promoting a DR instance to primary or demoting a primary instance to DR after a DR switchover failure	POST /gaussdb/v3/instances/{instance_id}/disaster-recovery/switchover/repair	gaussdb:disasterRecovery:switchoverRepair
Querying instances that can be used as a primary instance in a DR relationship	POST /gaussdb/v3/instances/disaster/instanceList	gaussdb:disasterRecovery:list
Querying instances that can be used as a primary instance in a DR relationship (recommended)	POST /gaussdb/v3.1/instances/disaster/instance-list	gaussdb:disasterRecovery:list

Function	API	Permission
Deleting a DR record	POST /gaussdb/v3/instances/disaster/record/delete	gaussdb:disasterRecovery:delete
Changing the primary instance of a DR relationship	POST /gaussdb/v3/instances/{instance_id}/disaster-recovery/change-primary	gaussdb:disasterRecovery:changePrimary
Pre-checking DR operations	POST /gaussdb/v3.5/instances/{instance_id}/disaster-recovery/pre-process	gaussdb:disasterRecovery:check

Table 5-18 License management

Function	API	Permission
Obtaining a revocation code	POST /gaussdb/v3/license/invalidation	gaussdb:instance:license:operate

Table 5-19 Log management

Function	API	Permission
Querying operation logs	POST /v3/operation-traces	dbs:trace:list

A Appendixes

A.1 Abnormal Request Results

Table A-1 Abnormal parameters

Parameter	Type	Description
error_code	String	Error code returned when a task submission exception occurs
error_msg	String	Error description returned when a task submission exception occurs

Example response:

```
{
    "error_msg": "***",
    "error_code": "***"
}
```

A.2 Status Codes

Table A-2 Status Codes

Status Code	Message	Description
100	Continue	The client should continue with its request. This interim response is used to inform the client that the initial part of the request has been received and has not yet been rejected by the server.

Status Code	Message	Description
101	Switching Protocols	The protocol should be switched. The protocol can only be switched to a more advanced protocol. For example, the current HTTP protocol is switched to a later version.
200	OK	Request succeeded.
201	Created	The request for creating a resource or task has been fulfilled.
202	Accepted	The request has been accepted, but the processing has not been completed.
203	Non-Authoritative Information	Unauthorized information. The request is successful.
204	NoContent	The server has successfully processed the request, but has not returned any content. The status code is returned in response to an HTTP OPTIONS request.
205	Reset Content	The server has fulfilled the request, but the requester is required to reset the content.
206	Partial Content	The server has processed certain GET requests.
300	Multiple Choices	There are multiple options for the location of the requested resource. The response contains a list of resource characteristics and addresses from which the user or user agent (such as a browser) can choose the most appropriate one.
301	Moved Permanently	The requested resource has been assigned a new permanent URI, and the new URI is contained in the response.
302	Found	The requested resource was temporarily moved.
303	See Other	The response to the request can be found under a different URI. It should be retrieved using a **GET** or **POST** method.
304	Not Modified	The requested resource has not been modified. In such a case, there is no need to retransmit the resource since the client still has a previously-downloaded copy.
305	Use Proxy	The requested resource must be accessed through a proxy.

Status Code	Message	Description
306	Unused	The HTTP status code is no longer used.
400	BadRequest	Invalid request. The client should not repeat the request without modifications.
401	Unauthorized	The status code is returned after the client provides the authentication information, indicating that the authentication information is incorrect or invalid.
402	Payment Required	This status code is reserved for future use.
403	Forbidden	The server understood the request, but is refusing to fulfill it. The client should not repeat the request without modifications.
404	NotFound	The requested resource cannot be found. The client should not repeat the request without modifications.
405	MethodNotAllowed	The method specified in the request is not supported for the requested resource. The client should not repeat the request without modifications.
406	Not Acceptable	The server cannot fulfill the request according to the content characteristics of the request.
407	Proxy Authentication Required	This status code is similar to 401, but indicates that the client must first authenticate itself with the proxy.
408	Request Time-out	The server timed out waiting for the request. The client may repeat the request without modifications at any later time.
409	Conflict	The request could not be processed due to a conflict. This status code indicates that the resource that the client attempts to create already exists, or the request fails to be processed because of the update of the conflict request.
410	Gone	The requested resource is no longer available. The requested resource has been deleted permanently.

Status Code	Message	Description
411	Length Required	The server refuses to process the request without a defined Content-Length.
412	Precondition Failed	The server does not meet one of the preconditions that the requester puts on the request.
413	Request Entity Too Large	The request is larger than that a server is able to process. The server may close the connection to prevent the client from continuing the request. If the server temporarily cannot process the request, the response will contain a Retry-After header field.
414	Request-URI Too Large	The URI provided was too long for the server to process.
415	Unsupported Media Type	The server is unable to process the media format in the request.
416	Requested range not satisfied	The requested range is invalid.
417	Expectation Failed	The server fails to meet the requirements of the Expect request-header field.
422	UnprocessableEntity	The request is well-formed but is unable to be processed due to semantic errors.
429	TooManyRequests	The client has sent more requests than its rate limit is allowed within a given amount of time, or the server has received more requests than it is able to process within a given amount of time. In this case, it is advisable for the client to re-initiate requests after the time specified in the Retry-After header of the response expires.
500	InternalServerError	The server is able to receive the request but it could not understand the request.
501	Not Implemented	The server does not support the requested function.
502	Bad Gateway	The server acting as a gateway or proxy receives an invalid response from a remote server.
503	ServiceUnavailable	The requested service is invalid. The client should not repeat the request without modifications.
504	ServerTimeout	The request cannot be fulfilled within a given time. The response will reach the client only if the request carries a timeout parameter.

Status Code	Message	Description
505	HTTP Version not supported	The server does not support the HTTP protocol version used in the request.

A.3 Error Codes

Table A-3 Error codes

Status Code	Error Code	Description
400	DBS.00100007	The alarm query parameter is invalid.
400	DBS.00101005	Alarm interconnection verification failed.
400	DBS.00110006	Invalid access.
400	DBS.00110007	You do not have the permissions needed to perform this operation.
400	DBS.05000008	The current version is the latest. No updates are available.
400	DBS.06010013	Parameter error.
400	DBS.06013001	The collection type of WDR snapshots is invalid.
400	DBS.06013002	This operation is not supported for DR instances.
400	DBS.06013003	WDR snapshot collection failed.
400	DBS.06013004	WDR snapshot is disabled.
400	DBS.06015011	The instance types are inconsistent.
400	DBS.06015013	The replica consistency protocols are inconsistent.
400	DBS.06015021	The backup media is not supported.
400	DBS.06025010	The number of selected hosts is incorrect.
400	DBS.06025011	The selected host OS is not supported.
400	DBS.06025012	The selected host type is not supported.
400	DBS.06025013	The selected heterogeneous attribute of the host is not supported.
400	DBS.06025014	The selected CPU vendor of the host is not supported.

Status Code	Error Code	Description
400	DBS.06025015	The selected host is in use.
400	DBS.06025016	No upgrade is required because the source and target Agent versions of the current instance are the same.
400	DBS.06025017	Agent upgrade failed. Check the instance and node status and try again.
400	DBS.06025024	Invalid host or data center name length.
400	DBS.06025025	Invalid host or data center description length.
400	DBS.06025026	Invalid host SSH port number.
400	DBS.06025027	Some hosts to add, modify, or delete at a time are duplicate.
400	DBS.06025028	The selected host cannot be modified.
400	DBS.06025029	The selected host cannot be deleted.
400	DBS.06025030	There are hosts in the selected data center.
400	DBS.06025031	Selected host not found.
400	DBS.06025043	NIC mapping failed.
400	DBS.06025044	The host failed to go online.
400	DBS.06025046	Invalid host disk type.
400	DBS.06025047	The host was brought online on another platform or failed to connect to the Agent.
400	DBS.06025048	Failed to download the OS patch installation package.
400	DBS.06025056	Flashback failed.
400	DBS.06025058	Selected data center not found.
400	DBS.06025059	The selected data center already exists.
400	DBS.06025060	The selected data center cannot be modified.
400	DBS.06025061	The host failed the standardization check.
400	DBS.06025062	Host network connection failed.
400	DBS.06025063	Failed to connect to the host using SSH. Check the password or SSHD.
400	DBS.06025064	Failed to install the Agent on the host.
400	DBS.06025067	Check whether the AZ name is the same as the data center name.

Status Code	Error Code	Description
400	DBS.06025088	The key name already exists.
400	DBS.06025089	The key algorithm is not supported.
400	DBS.06025090	Invalid key name length.
400	DBS.06025091	Invalid key description length.
400	DBS.06025092	A maximum of 20 key pairs can be created.
400	DBS.06025093	The SSH private key does not exist. Check the ID.
400	DBS.06025096	The online and offline hosts do not have credentials. Select a password or SSH private key.
400	DBS.06025097	The SSH private key has been bound to a host. Ensure that the host is no longer used and forcibly delete it.
400	DBS.06025101	The OS PATCH package does not exist.
400	DBS.06025119	There is no available Agent process in the container.
400	DBS.06025120	Failed to delete Agent information.
400	DBS.06025127	There are duplicate data centers to be added.
400	DBS.06025132	The number of selected data centers is incorrect.
400	DBS.06025187	Static host information fails to be modified.
400	DBS.06025188	The host network protocol does not support IPv6.
400	DBS.06025192	The host using the sandbox installation mode does not support three-plane networks.
400	DBS.06025193	The host installation mode is not supported.
400	DBS.06025194	The management IP address and data IP address of the host using the two-plane network are different.
400	DBS.06280008	The schema already exists.
400	DBS.06280028	Too many instances are managed.
400	DBS.200001	Invalid parameters.
400	DBS.200002	DB instance not found.

Status Code	Error Code	Description
400	DBS.200006	The request body was left blank.
409	DBS.200019	Another operation is being performed on the instance or the instance is faulty.
400	DBS.200021	Invalid DB instance name.
400	DBS.200025	Invalid value of the AZ.
400	DBS.200029	Invalid username or password.
400	DBS.200044	Resource not found or permission denied.
400	DBS.200052	Invalid database root password.
400	DBS.200062	The username does not meet rules or is same as the database administrator account.
400	DBS.200068	Password too weak. Enter a stronger password.
400	DBS.200079	The primary/standby switchover cannot be performed because the instance is abnormal.
400	DBS.200475	New password must be different from the old one.
400	DBS.200476	The new password must be different from the old password spelled backwards.
400	DBS.200480	The node topology of the selected DB instance is different from that of the original DB instance.
400	DBS.200602	Instance not found.
400	DBS.200751	Backup file not found.
400	DBS.200818	Database user deletion failed.
400	DBS.200821	Modification for database user permissions failed.
400	DBS.200823	The database does not exist.
400	DBS.200824	Database account not found.
400	DBS.200856	The DB engine version is not supported.
400	DBS.201003	Resource not found or permission denied.
400	DBS.201004	The backup type does not exist.
400	DBS.201010	Backup file not found.
400	DBS.201015	Another operation is being performed on the DB instance.

Status Code	Error Code	Description
400	DBS.201016	Operation cannot be performed on nodes in the current state.
400	DBS.201019	The restoration task does not exist.
400	DBS.201032	Invalid restoration point in time.
400	DBS.201106	Invalid retention days.
400	DBS.201109	Invalid backup flow control.
400	DBS.201110	The number of differential prefetch pages is invalid.
400	DBS.201111	The shard size must be within the range and a multiple of 4.
409	DBS.201202	Another operation is being performed on the instance or the instance is faulty.
400	DBS.201203	Backup file not found.
400	DBS.201204	The backup type is not allowed.
400	DBS.201209	The backup name already exists.
400	DBS.201210	Invalid backup name.
400	DBS.201218	The backup is not a manual backup.
400	DBS.216009	Operation cannot be performed on instances in the current state.
400	DBS.216011	Operation cannot be performed on nodes in the current state.
400	DBS.216016	The Instance is not found.
400	DBS.216026	Node not found.
400	DBS.216030	The current node does not belong to the instance.
400	DBS.280001	Invalid parameters.
500	DBS.280005	Server error.
400	DBS.280011	The DB instance is abnormal.
400	DBS.280012	Upgrade failed. Other operations are being performed on the instance.
400	DBS.280110	Selected DB instance does not exist.
400	DBS.280124	The backup file ID is invalid.
400	DBS.280125	Invalid backup policy.

Status Code	Error Code	Description
400	DBS.280215	Invalid backup cycle.
400	DBS.280216	Invalid backup start time.
400	DBS.280236	Invalid DB engine version.
400	DBS.280251	Invalid backup cycle.
400	DBS.280404	Invalid instance ID.
400	DBS.280416	Invalid backup end time.
400	DBS.280439	Invalid value. Enter a positive integer of no more than 100.
400	DBS.280440	Invalid offset. Enter a positive integer or zero.
400	DBS.280472	Invalid time range.
400	DBS.280602	Invalid HA consistency.
400	DBS.280618	Differential backup cycle invalid.
400	DBS.280631	Invalid database name.
400	DBS.280637	The number of differential prefetch pages is invalid.
400	DBS.280657	Invalid upgrade parameter. Check the upgrade type and upgrade action.
400	DBS.280658	Invalid AZ for gray upgrade AZ.
400	DBS.280659	Invalid number of shards for gray upgrade.
400	DBS.280660	Instance gray upgrade check failed. Check the instance status.
400	DBS.280670	The query date and time are out of range. ● Check whether the date and time range is valid. ● Check whether the date and time is out of range.
400	DBS.301147	The node cannot be started because more than half of the nodes are stopped.
400	DBS.6013005	There must be more than one available AZ.
400	DBS.6013035	The patch name already exists.
400	DBS.6013036	Invalid patch name.
400	DBS.6013037	Invalid hint.
400	DBS.6013038	Incorrect SQL patches.

Status Code	Error Code	Description
400	DBS.6013063	The database tablespace is not found.
400	DBS.6025033	Invalid AZ fault operation type.
400	DBS.6025117	Invalid database password.
400	DBS.6025177	A Dorado device must be bound to the instance.
400	DBS.280801	Invalid DR type.
400	DBS.06013088	Operation not supported. templatem database not found and replica consistency protocol is Paxos.
400	DBS.00120000	The parameter format is incorrect. Check the input parameter format.
400	DBS.06030135	WDR snapshot not found. Snapshot ID: ${snapshotId}
400	DBS.06030047	The instance engine type does not match the target engine type. The selected instance type does not support the current operation. Generally, only GaussDB database instances support the operation. Check the instance type.
500	DBS.06030079	Failed to deliver the command. Check the background logs.
500	DBS.06030080	Failed to run the common O&M command. Check the Agent background log for further analysis. Error msg: ${errorMsg}.
400	DBS.06030081	Verification failed. The parameter may be invalid or does not meet the scenario requirements. More troubleshooting information: - validation item: ${paraName}, - error msg: ${errorMsg}
500	DBS.06030092	Export data file stream failed. File Info=${fileName}. Log Info: ${errorMsg}. ● Check whether the file exists or whether the file status supports download. ● For ASP download, check whether the job ID is valid and whether the corresponding job is successfully completed.
400	DBS.06020162	This operation is not supported.
400	DBS.06015069	The instance group does not support this operation.

Status Code	Error Code	Description
400	DBS.00110005	Login expired.
400	DBS.06025217	Operations are not allowed on the metadata database managed by TPOPS.
400	DBS.201506	Failed to access Agent.

A.4 Monitoring Metrics

The following table lists the monitoring metrics.

CPU/ Memory **Group name: CPUMEMORY**	CPU Usage	sys001_cpu_usage
	CPU Idle Time Percentage	sys002_idle_usage
	CPU Kernel Mode Time Percentage	cpu_sys_usage
	CPU User Mode Time Percentage	cpu_user_usage
	CPU I/O Wait Time Percentage	cpu_wait_usage
	1-Minute CPU Load	sys016_cpu_load
	5-Minute CPU Load	
	15-Minute CPU Load	
	Used Memory Size	mem_used_size
	Buffer Memory	buffer_size
	Cache Memory	cache_size
	Memory Usage	sys010_mem_usage
	Available Memory Size	sys013_mem_free_size
	Total Swap Memory Size	sys024_swap_total_size
	Swap Memory Usage	sys025_swap_used_ratio
	Used Swap Memory Size	sys026_swap_used_size

Disk/ Storage **Group name: IOSTORAGE**	Semaphore	sys014_signal_num
	Inode Usage	sys023_file_handler_num
	System Disk Usage	sys041_sys_disk_usage
	Used Data Disk Size	sys063_disk_used_size
	Total Data Disk Size	sys064_disk_total_size
	Data Disk Usage	sys065_disk_usage
	Disk IOPS	sys069_iops
	Data Disk Write Throughput	sys071_disk_write_throughput
	Data Disk Read Throughput	sys072_disk_read_throughput
	Time Required for per Data Disk Write	sys075_avg_disk_ms_per_write
	Time Required for per Data Disk Read	sys076_avg_disk_ms_per_read
	Average Service Time for Data Disk I/O Requests	sys081_svctm
	Leaked Handles	rds122_gaussv5_fd_leak_num
	NAS Capacity Usage	nas_usage
	NVMe Cache Usage	rds228_nvme_cache_usage
	Used NVMe Cache	rds229_nvme_cache_used_size
Network **Group name: NETWORK**	Network Receive Rate	sys011_bytes_in
	Network Transmit Rate	sys012_bytes_out
	Received Error Packets Rate	rx_errors
	Receive Dropped Packets Rate	rx_dropped
	Transmit Error Packets Rate	tx_errors
	Transmit Dropped Packets Rate	tx_dropped

Connections **Group name: CONNECTION**	User Logins per Second	rds081_gaussv5_login_counter
	User Logouts per Second	rds082_gaussv5_logout_counter
	Lock-Waiting Session Rate	rds085_gaussv5_wait_ratio
	Active Session Rate	rds086_gaussv5_active_ratio
	CN Connections	rds089_gaussv5_inuse_counter
	Active Sessions	rds117_gaussv5_active_session
	Waiting Sessions	rds118_gaussv5_wait_session
	Online Sessions	rds143_gaussv5_online_session
	Online Session Rate	rds144_online_ratio
	Sessions Waiting for Locks	rds195_gaussv5_locks_session
	Percentage of Sessions Waiting for Locks	rds196_gaussv5_locks_ratio
	Peak Active Sessions	rds416_gaussv5_max_active_session
Workloads **Group name: TRANSACTION**	User Committed Transactions per Second	rds092_gaussv5_commit_counter
	User Rollback Transactions per Second	rds093_gaussv5_rollback_counter
	Background Committed Transactions per Second	rds094_gaussv5_bg_commit_counter
	Background Rollback Transactions per Second	rds095_gaussv5_bg_rollback_counter
	Average Response Time of User Transactions	rds096_gaussv5_resp_avg
	User Transaction Rollback Rate	rds097_gaussv5_rollback_ratio
	Background Transaction Rollback Rate	rds098_gaussv5_bg_rollback_ratio
	Data Definition Language/s	rds099_gaussv5_ddl_count
	Data Manipulation Language/s	rds100_gaussv5_dml_count
	Data Control Language/s	rds101_gaussv5_dcl_count
	DDL and DCL Rate	rds102_gaussv5_ddl_dcl_ratio

	Response Time of 80% SQL Statements	rds103_gaussv5_P80
	Response Time of 95% SQL Statements	rds104_gaussv5_P95
	Percentage of SELECT Statements	rds129_select_distribution
	Percentage of UPDATE Statements	rds130_update_distribution
	Percentage of INSERT Statements	rds131_insert_distribution
	Percentage of DELETE Statements	rds132_delete_distribution
	SQL Statements	rds145_gaussdbv5_statement
	Maximum Execution Duration of Database Transactions	rds190_long_running_transaction_exectime
	Transactions Longer Than 60 Minutes	rds311_long_transactions_num_60min
	Transactions Longer Than 120 Minutes	rds312_long_transactions_num_120min
	Total User Transactions per Second (TPS)	rds400_gaussv5_total_counter
	User Committed Transactions Rate	rds401_gaussv5_commit_ratio
	Average SQL Statements per Second	rds402_gaussv5_ddl_dml_dcl_count
	INSERT Statements per Second	rds403_gaussv5_insert_sql_count
	SELECT Statements per Second	rds404_gaussv5_select_sql_count
	DELETE Statements per Second	rds405_gaussv5_delete_sql_count
	UPDATE Statements per Second	rds406_gaussv5_update_sql_count
Locks **Group name: LOCK**	Deadlocks	rds091_gaussv5_deadlocks
	Total Locks	rds407_gaussv5_total_locks_count
	Table-level Locks	rds408_gaussv5_relation_locks_count
	Row-level Locks	rds409_gaussv5_tuple_locks_count

	Other Locks	rds410_gaussv5_other_locks_count
Synchronization status **Group name: SYNCSTAT**	Primary Node Flow Control Duration	rds079_gaussv5_current_sleep_time
	Standby Node RTO	rds080_gaussv5_current_rto
	Standby Node Redo Progress	rds083_gaussv5_standby_delay
	Data Volume to Flush to Disks	rds105_gaussv5_ckpt_delay
	Xlog Rate	rds123_xlog_lsn
	Candidate Slots	rds124_candidate_slots_num
	Dirty Pages Not Flushed to Disks	rds125_dirty_page_num
	WAL Log Size in the Replication Slot	rds192_replication_slot_wal_log_size
Process resources **Group name: PROCESSRESOURCE**	Buffer Hit Rate	rds090_gaussv5_buffer_hit_ratio
	Physical Reads per Second	rds106_gaussv5_phyrds
	Physical Writes per Second	rds107_gaussv5_phywrts
	Thread Pool Usage	rds108_gaussdbv5_thread_pool
	Instance Memory Usage Threshold	rds109_max_process_memory
	Process Used Memory	rds110_process_used_memory
	Process Memory Usage	rds111_process_used_memory_usage
	Dynamic Memory Usage Threshold	rds112_max_dynamic_memory
	Used Dynamic Memory	rds113_dynamic_used_memory
	Dynamic Memory Usage	rds114_dynamic_used_memory_usage
	Used Other Memory	rds115_other_used_memory
	Used Shared Memory	rds116_shared_used_memory
	Xlogs	rds126_xlog_num
	System Database Size	rds127_sys_database_size
	User Database Total Size	rds128_user_database_size

	Used Undo Pages	rds170_undo_size
	Undo Page Generation Rate	rds171_undo_rate
	Damaged Data Pages	rds184_bad_block_num
	Disk-Read Blocks per Second	rds411_gaussv5_blks_read_counter
	Cache-Read Blocks per Second	rds412_gaussv5_blks_hit_counter
Instance information **Group name: INSTATUS**	Instance Status	gaussdb_instance_status
	Primary/Standby Status of DNs	datanode_switch_status
	CN Start	rds133_cn_startup_time
	DN Start	rds134_dn_startup_time
	GTM Start	rds135_gtm_startup_time
	CMS Start	rds136_cms_startup_time
	ETCD Start	rds137_etcd_startup_time
	CMA Start	rds138_cma_startup_time
	GTM Memory Usage	rds139_gtm_process_mem_usage
	ETCD Memory Usage	rds140_etcd_process_mem_usage
	CMS Memory Usage	rds141_cms_process_mem_usage
	CMA Memory Usage	rds142_cma_process_mem_usage
	CN Temporary Directory Size	rds119_cn_temp_dir_size
	DN Temporary Directory Size	rds120_dn_temp_dir_size
	ETCD Data Directory Size	rds121_etcd_data_dir_size
	ERROR Logs of the OM Module	rds148_gaussdbv5_error_om
	ERROR Logs of the CM Module	rds149_gaussdbv5_error_cm
	ERROR Logs of the Kernel	rds150_gaussdbv5_error_kernel
	om_agent Status	rds150_om_agent_process
	om_monitor Status	rds151_om_monitor_process

	Offset from Valid Clock Source	sys084_clock_synchronization_diff
Management Agent Status **Group name: AGENTSTATUS**	dbmanager Status	rds075_dbmanager_process_code
	watchdog Status	rds076_watchdog_process_code
	agent_monitor Status	rds077_agentmonitor_process_code
DR status **Group name: FORDR**	Shard Log Gap of DR Instance	rds172_streaming_dr_xlog_gap
	Replay Rate of Shard Logs in DR Instance	rds175_streaming_dr_xlog_replay_rate
	Size of Shard Logs to Be Replayed in DR Instance	rds173_streaming_dr_xlog_to_be_replayed
	Flushing Rate of Shard Logs in DR Instance	rds174_streaming_dr_xlog_flushing_rate
	Required Log Synchronization Time per DR Instance Shard (RPO)	rds176_streaming_dr_rpo
	Required Log Replay Time per DR Instance Shard (RTO)	rds177_streaming_dr_rto
	Estimated Log Replay Time Between Primary and Standby Dorado-based Clusters (RTO)	rds183_sharestore_local_rto

B Change History

Released On	Change Description
2025-09-30	The modification is as follows: • Added the **job_id**, **status**, **wdr_type**, **job_start_time**, and **job_end_time** parameters to **Querying Collection Results for WDR Snapshot Reports**. • Added the **start_snapshot_id**, **end_snapshot_id**, and **mode** parameters to **Collecting a WDR Snapshot Report**. • Added the **table_list** parameter to **Creating Manual Backups**. • Added the **table_list** and **schema_type** parameters to **Restoring Data to the Original Instance or an Existing Instance**. • Added the **schema_type** parameter to **Creating an Instance or Restoring Data to a New Instance** and **(Recommended) Creating an Instance or Restoring Data to a New Instance**. • Added the **dir_settings** parameter to **Adding Hosts**. • Added the **dir_settings** parameter as a response parameter to **Querying Hosts** and **Batch Importing Host Files for Check**. • Added the **backup_type** parameter to **Querying Differential Backups**. Added the following APIs: • **Querying Database Tables** • **Querying Available WDR Snapshot Groups** • **Querying Common O&M Commands (SQL and Python Commands)** • **Delivering Common O&M Commands (Synchronous and Asynchronous Commands)** • **Querying the Execution Results of Asynchronous O&M Commands**

Released On	Change Description
	• **Downloading the Execution Result File of an Asynchronous O&M Command** • **Performing a Primary/Standby DN Switchover in an Instance Group** • **Obtaining Asynchronous Batch Processing Task Details** • **Restoring Data from a Backup Set**
2025-06-30	Modified the following APIs: • Added the **backup_parallel_degree** parameter to **Configuring an Automated Backup Policy**, **Configuring an Extended Backup Policy**, **Querying an Automated Backup Policy**, and **Querying an Extended Backup Policy**. • Added the **restore_parallel_degree** parameter to **Restoring Data to the Original Instance or an Existing Instance**, **Creating an Instance or Restoring Data to a New Instance**, and **(Recommended) Creating an Instance or Restoring Data to a New Instance**. • Added the **dorado_shared_volume** parameter to **Creating an Instance or Restoring Data to a New Instance** and **(Recommended) Creating an Instance or Restoring Data to a New Instance**. Added the following APIs: • **Changing a Backup Method** • **Querying Backup Details** • **Querying Restoration Details** • **Sending Backup or Restoration Results**
2025-04-30	Modified the following APIs: • Added the **schema_name**, **db_name**, and **query_plan_optimize** parameters to the response parameters of **Querying Slow SQL Statements** and **Querying Details About a Slow SQL Statement**. • Added the **node_id** parameter to the response parameters of **Querying Throttling Tasks Based on Search Criteria**.

Released On	Change Description
2025-03-30	Added the following APIs: • Querying Operation Logs • Logging In to the System • Redirecting a Login Request • Logging Out of the System • Querying Users • Adding a User • Deleting a User • Querying Basic Information About a User • Updating Information About a User • Checking Whether a Username Is Valid • Letting a User Reset Their Own Password • Letting a User Reset Another User's Password • Querying the Instances Allocated to a User • Allocating Instances to a User • Querying Roles of a User • Assigning Roles to a User • Querying User Groups That a User Has Been Added To • Adding a User to User Groups • Locking Users • Unlocking Users • Checking Whether a Password Is a Weak Password • Querying User Groups • Adding a User Group • Updating Information About a User Group • Deleting a User Group • Querying User Roles of a User Group • Updating Role Information About a User Group • Querying Users in a User Group • Updating User Information About a User Group • Querying Roles • Adding a Role • Updating Information About a Role • Deleting a Role • Querying the Permissions of a Role • Querying Permissions • Preparing for Changing Specifications

Released On	Change Description
2024-12-30	Added the following APIs: ● **Querying the Advanced Features of an Instance** ● **Enabling Advanced Features** ● **Changing the Description of an Instance** ● **Obtaining a Revocation Code** ● **Pre-checking DR Operations** Modified the following APIs: ● Added the **data_center_id**, **data_center**, **instance_id**, **instance_name**, and **manage_ip** parameters to **Querying Hosts**. ● Added the **is_force_rm_host** parameter to **Deleting a Host**.